ÉCOLE NATIONALE DES PONTS ET CHAUSSÉES

IP PARIS

THÈSE DE DOCTORAT

de l'École nationale des ponts et chaussées

Mathematical study of mean-field models for moiré materials

École doctorale N°531, MSTIC

Mathématiques

Thèse préparée au CERMICS, ENPC
Équipe Matherials, Inria

Thèse soutenue le 30 septembre 2026, par

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Abstract

Version française

Ce manuscrit traite de l'étude des matériaux moirés en physique quantique, d'un point de vue mathématique. Il considère deux approches, chacune associée à une méthode.

La première consiste à modéliser un système physique expérimental, comme une bicouche de graphène 2D encapsulée entre deux électrodes, grâce à ses spécificités géométriques. Ceci fait apparaître les notion d'ergodicité et d'opérateur stationnaire, et permet de formuler un modèle de Théorie de la Fonctionnelle de la Densité, pour lequel ce travail démontre qu'il existe un état fondamental associé à la minimisation de la fonctionnelle d'énergie, pour le modèle de Hartree-Fock réduit.

La seconde approche consiste à changer d'échelle et à considérer une méthode qui s'appliquerait à l'échelle du moiré : la Théorie du Champ Moyen Dynamique. Cette méthode permet de calculer une approximation de la fonction de Green à un corps d'un système tel qu'un modèle de Hubbard sur le réseau moiré et donc d'investiguer les corrélations électroniques du système. Cette thèse apporte une formalisation mathématique de cette théorie lorsqu'elle est couplée à un solveur d'impureté particulier (la Théorie Perturbative Itérative), une démonstration de l'existence d'une solution aux équations qui en découlent, ainsi que l'analyse numérique de la méthode, en pointant ses limites.

Mots-clés

Physique mathématique ; Calcul de structure électronique ; Calcul variationnel ; Théorie de la fonctionnelle de la densité ; Modèle de Hartree-Fock réduit ; Opérateurs stationnaires ; Théorie du champ moyen dynamique ; Fonction de Green quantique à un corps ; Fonctions de Pick ; Théorie perturbative itérée ; Analyse numérique.

English version

This manuscript addresses the study of moiré materials in quantum physics from a mathematical perspective. It considers two approaches, each associated with a specific method.

The first approach involves modeling an experimental physical system—such as a 2D graphene bilayer encapsulated between two electrodes—based on its geometric properties. This introduces the concepts of ergodicity and stationary operators, and allows us to formulate a Density Functional Theory model, for which this work demonstrates that there exists a ground state associated with the minimization of the energy functional for the reduced Hartree-Fock model.

The second approach involves changing the scale and considering a method applicable at the moiré scale: Dynamical Mean-Field Theory. This method allows us to calculate an approximation of the one-body Green's function for a system such as a Hubbard model on the moiré lattice and thus to investigate the system's electronic correlation. This thesis provides a mathematical formalization of this theory when coupled with a specific impurity solver (Iterated Perturbative Theory), a proof of the existence of a solution to the resulting equations, and a numerical analysis of the method, highlighting its limitations.

Keywords

Mathematical physics ; Electronic structure calculations ; Variational calculus ; Density functional theory ; Reduced Hartree–Fock model ; Stationary operators ; Dynamical mean-field theory ; One-body quantum Green's function ; Pick functions ; Iterated perturbative theory ; Numerical analysis.

Remerciements

Bienvenue dans mon chapitre préféré, de loin. J'ai commencé à le rédiger bien avant de penser à rédiger ma thèse, ce qui explique peut-être en partie le retard pris sur celle-ci. Dans ce chapitre, je vais essayer de remercier tout le monde qui a compté pour moi avant et surtout pendant ma thèse, et qui a fait partie de la trajectoire de vie qui m'a amené jusqu'à écrire ces lignes (et je prie par avance celles et ceux que j'aurai oublié·es de m'excuser, ce n'est pas une erreur, c'est une faute). Les prérequis pour comprendre cette partie sont une maîtrise sommaire de la langue française (quelques notions d'anglais et d'italien peuvent faciliter la lecture), de la curiosité ou l'espoir de ne pas s'ennuyer pendant ma soutenance, ainsi qu'un peu de temps et de courage si votre objectif est de terminer ce chapitre avant d'aller vous coucher.

Je ne peux que commencer cette section par remercier le jury, d'avoir accepté de prendre le temps d'évaluer mon travail, mais également d'être venu jusqu'à Champs-sur-Marne. Iels seront récompensé·es par le pot, qui, je l'espère, sera à la hauteur de leur engagement. I would especially like to thank Huajie Chen and Reinhold Schneider for taking on the thankless task of serving as rapporteurs, and for showing unparalleled flexibility regarding the numerous postponements of the defense date. Merci Antoine, Thomas, Salma et Isabelle d'avoir accepté de faire partie de mon jury. Merci Antoine d'avoir été un joyeux feu follet (scientifiquement parlant, bien entendu) que je suis de loin depuis le début de ma thèse. Merci Thomas d'avoir accepté de te pencher sur une thèse de mathématiques, qui bénéficiera de ton recul sur la physique. Merci Salma et Isabelle d'avoir pavé mon chemin scientifique de vos travaux, qui m'ont grandement inspiré.

Dans un second temps, il me faut prendre la place nécessaire pour remercier mes directeurs de thèse, David et Éric. Vous avez été de formidables encadrants, et je vous serai éternellement reconnaissant pour tout ce que vous avez pu m'apporter, à la fois mathématiquement et humainement. Vous avez mille qualités (et certainement moins de défauts), la première d'entre elles étant bien sûr d'être tombés amoureux de la Bretagne.

Merci David de m'avoir pris sous ton aile dans la période d'incertitude de la pandémie. Ton énergie, ta confiance et tes encouragements m'ont guidé de mon stage de M1 à aujourd'hui, et ont été fondamentaux dans mon engagement dans la recherche. J'ai toujours adoré les moments que tu prenais à m'expliquer ce que tu avais compris la veille et ceux passés à réfléchir ensemble.

Merci Éric de m'avoir donné la chance de faire cette thèse avec toi, en ce double sens où tu m'as accordé la confiance de commencer en cherchant un financement pour que je puisse travailler avec toi, et où être ton étudiant en thèse est vraiment une chance. Tu as toujours été à l'écoute, tu m'as permis de travailler avec Alfred, un scientifique de renom, ce qui fut pour moi la plus belle aventure scientifique de ma thèse. Tu es à la fois un puits de science et d'une humilité scientifique noble (si, si, c'est bien ce que j'ai écrit).

Ce fut un plaisir d'avoir travaillé avec vous deux, et c'est avec autant de plaisir que je continuerai de le faire. C'est enfin avec un plaisir au moins aussi grand que j'ai partagé avec vous des moments en dehors du travail, dans le bon air breton de Roscoff ou ailleurs en Europe, et j'ai pu compter sur vous dans les moments joyeux comme difficiles.

J'aimerais ensuite reprendre le fil d'une partie de mon éducation académique, qui a été le long fleuve presque tranquille qui m'a bercé par ses flots dans les méandres de la science pour échouer ici sur les berges de la connaissance (je crains malheureusement d'être enlisé dans le sable mouvant desdites berges, ce qui constitue une autre explication du retard mentionné plus haut).

Parmi mes guides spirituels, je dois dire merci à mes profs de prépa, notamment Yves Dutrieux, Laure-Hélène Reydellet et Hervé Gianella, qui m'ont guidé sur les rails de la science et m'ont accordé leur confiance. Une grande part de mon appétit de mathématiques et de physique vient de vous. J'aimais tellement l'enseignement que vous prodiguiez que je ne savais que choisir dans la suite de mes études parmi les sujets que vous rendiez tous captivants, au point de rester un an de plus qu'initialement prévu sur les bancs de l'école.

La prépa a aussi été l'occasion de se faire des ami·es, voyez-vous, que j'aimerais également remercier pour tout ce qu'iels m'ont apporté depuis maintenant une décennie. Je pense notamment à Alex, Brus, Claire, Clara, Clémence, Dodo, Elsa, Loïc, Marc, Paul & Paul, Pauline, Philippe, Tim et Victor.

Merci également aux personnes que j'ai pu croiser et côtoyer à Cachan. Je pense notamment à Freddy P (prononcé "frédi pi"), qui m'a toujours soutenu malgré mes tergiversations académiques, et a été d'une précieuse aide dans mon parcours à l'ENS.

Je pense bien sûr à tous·tes mes ami·es de l'ENS, du département de math et d'ailleurs. Les nombreuses rencontres de la riche vie étudiante ont été une pierre angulaire de mon parcours dans la vie, et la personne que je suis aujourd'hui vous doit beaucoup. Je pense en particulier aux matheux (que j'écris ici malheureusement sans point médian), Cyril, Louis, Dylan, Hugo, Nicolas, Arthur, Siwan, Angelo, Clément ; aux ami·es du BDE, Tom, Chanus, Baptman, Schnikou, Yanis, Niels ; du rugby, Fred, l'Aspique et les Kyottes ; et de toutes les assos, clubs, et autres mouvements politiques où j'ai fourré le nez.

J'ai une pensée spéciale pour Blandine et Rafael, avec qui j'ai adoré partager une page de notre vie et qui m'ont vraisemblablement plus apporté du point de vue mathématique que l'inverse, la différence ayant toutefois été remboursée en gâteaux au chocolat.

Le Cermics est un endroit magnifique pour faire une thèse. C'est un labo plein de vie, doté d'une très bonne ambiance, d'un cadre idéal pour s'évader dans le parc de Noisiel

ou pour faire un plouf dans la piscine (ou dans la Marne), et d'un jardin pontanique qui permet aux plus motivé·es de cultiver un petit potager entre deux démonstrations. Si ce n'est pas la belle vie, ça ? Vous aurez du mal à me croire, sans doute, mais il est de plus peuplé de personnes sympathiques, à propos desquelles j'aimerais faire une légère digression dithyrambique.

Alfred, si un dieu de la générosité existait, tu serais son prophète. Tout ce que j'ai pu t'apporter, tu me l'as rendu au centuple. Je suis très fier et heureux de te compter parmi mes amis. Tu as été le soleil qui m'a guidé lorsque les nuages de la thèse ont pesé sur moi. Tu m'as aidé à aimer la recherche, et le bout de chemin que nous avons parcouru ensemble mérite de continuer à tracer sa route aussi longtemps que possible. Merci pour les fonctions de Pick, merci pour le potager, merci pour les rires et les virées sportives, merci pour tout.

Alicia, je pense que tu sais à quel point tu comptes pour moi, et je ne peux pas m'étaler sur ce sujet ici, il y a des gens qui regardent. Sache néanmoins que de t'avoir rencontrée pendant ma thèse a été d'une richesse infinie pour moi, et je ne cherche qu'à te rendre la pareille tant que je le pourrai. Bon courage pour la dernière ligne droite, tu n'auras pas réussi à soutenir avant moi, na.

Éloïse (j'ai toujours trouvé qu'avec un accent c'était plus joli). Ah, Éloïse. Que dire ? Mais surtout, que faire ? Et que penser ? Qu'ouïr ? Queer ? Pardon, je m'égare, une fois n'est pas coutume. Je me trouve ici pour vous remercier de l'infléchissement sur mon âme dont vous avez fait preuve, très chère, par le verbe et par la grâce dont vous manipulez savamment les ficelles en tout instant. Dame Élo, avoir été votre compagnon de thèse et vous retrouver sur mon chemin, comme on bute sur une racine en forêt, fut une source de joie sans commune mesure, soyez-en assurée. J'espère par ailleurs qu'il fait beau le jour de ma soutenance, afin qu'il soit plutôt con de donner une journée pareille à un patron.

Laurent, ton talent d'improvisation et ta poésie m'ont beaucoup inspiré, d'un point de vue strictement scientifique. J'espère avoir l'honneur de laisser transparaître par instants ton enseignement. À vrai dire, ta personnalité entière m'inspire beaucoup, et continuera de le faire, à n'en point douter.

Régis, je suis ravi d'avoir fait ta connaissance, et la connaissance d'une partie des tes connaissances, qui sont infinies, d'après le fascicule de description. Merci pour les discussions interminables, les courses terminables et je m'arrête là car le prochain qualificatif est "minable".

Simon et Albéric, merci d'avoir été les compagnons de thèse les plus gentils du monde, et d'être devenus les amis les plus gentils du monde. J'ai adoré évoluer ces trois (quatre, *ndlr*) ans auprès de vous, à la fois au labo et surtout en dehors.

Zoé, je suis désolé de te consacrer un paragraphe entier qui sera lu par des millions de personnes, mais je te dois bien ça. C'était tellement évident de devenir ton ami, car tu transpires la joie de vivre, l'ouverture d'esprit et l'amour du partage. Bien que je déteste ces valeurs, il paraît que les opposés s'attirent. Je suis heureux de t'avoir rencontrée,

d'avoir commencé à partager tant de choses avec toi et même de pouvoir te voir à ma soutenance alors que tu aurais dû être quelque part en Asie.

Bien sûr, la liste ne s'arrête pas là et je remercie également les autres doctorant·es et post-docs que j'ai pu croiser au labo et avec qui j'ai beaucoup aimé partager les moments mathématiques et les moments informels. Je pense à Gaspard qui collectionne les canards vivants et avec qui je pouvais vibrer rugby, à Hélène avec qui je peux trouver en une minute top chrono un sujet de débat sur lequel disserter des heures en étant (presque) parfaitement d'accord, et à Antonin, Antoine, Charlotte, Clément, Coco, Cyrille, Élise, Fabian, Épiphane, Grégoire, Guillaume, Héloïse, Henri, Kacem, Léo, Louis, Martin, Mathis, Nadège, Nail, Noé, Oscar, Owen, Paul, Raphaël, Renato, Rutger, Seta, Shiva, Sofiane, Thomas, Vincent et Yann.

Merci aussi aux permanent·es du labo qui le font vivre et avancer, merci Frédéric, Gabriel, Tony, Julien, Virgine, Loucas et Urbain, notamment.

Grazie alla squadra italiana del lab: Edoardo, Emanuele, Roberta, Annamaria, Giulia, Laura e Gabriele. Siete i migliori, grazie per il vostro buon umore in laboratorio, per le lezioni di italiano improvvisate e per tutto ciò che il vostro paese ha apportato alla cucina.

Merci à la team Éric d'être la meilleure équipe, Gaspard, Alfred, Laurent, Élo, Alicia, Théo, Maxime, Henri, Laura et Yukuan, avec qui j'ai aimé vivre de nombreuses conférences et profiter du bon air breton de Roscoff pendant ces quelques années.

Enfin, je tiens à remercier tout particulièrement celles qui font (ou ont fait) tenir debout le labo, car sans elles, rien ne serait possible. Un énorme merci à Isabelle et à Stéphanie pour leur soutien et leur expertise tant de fois opportune. Par ailleurs, je remercie très chaleureusement Dalida pour son aide précieuse et son investissement pour me permettre de soutenir malgré quelques déboires administratives.

Pendant la thèse, j'ai aussi eu l'occasion de rencontrer d'autres personnes formidables, comme Thaddeus, dont la curiosité et la gentillesse se marient à merveille ; Jibouille, qui trouve le courage de m'écouter raconter mes histoires, sûrement en remerciement de mon intérêt non feint pour Sainte-Beuve ; Pablo, Oussama et Timothée qui sont de redoutables chasseurs de bouquetins ; Mi-Song, Hassan, César, JZ, Diata, Édouard et consort(s), avec qui j'ai adoré profiter des couchers de soleil bretons à Roscoff ; et Marguerite et Jakob, avec qui j'ai adoré partager la folle aventure des RJMI.

La thèse a également été l'occasion d'enseigner pendant trois années à l'ENS à Paris. Je remercie David et Djalil pour m'avoir offert cette opportunité d'affronter les coriaces ulmites, contre qui ma seule arme était mon âge vénérable, qui dépassait le leur de quatre ou cinq unités. Merci Bertrand pour ta sagesse, ta patience et ta pédagogie. Ce fut pour moi une très belle et enrichissante expérience, sur le plan de l'enseignement donc, mais aussi car j'y ai côtoyé de nombreuses personnes qui m'ont encore une fois marqué.

Pour commencer, je remercie mes collègues de bureau Théo, Martin, Lucas, puis Samuel, pour la bonne ambiance qui a régné dans notre bureau (souvent peu garni, il faut l'avouer) lorsque nous nous retrouvions, et la richesse des discussions qui ponctu-

aient ces moments. Merci à Lucas et Martin de vous poser les questions que trop de scientifiques, à mon instar, rechignent souvent à se poser.

Merci Alexis, Arthur, Corentin, Daniel, Florent, Matja, Louis-Pierre, Lino, Ons, Paul, Quentin d'avoir rendu chacune de mes venues à l'ENS plus vivantes.

Merci enfin à Cyril, Fabienne, Claire, Marion, Yue et Julien pour votre gestion exemplaire du labo et la bonne ambiance qui est conséquence directe de votre bonne humeur.

Je dois aussi dire merci à toutes les personnes qui m'ont permis d'atterrir à Lyon en douceur. Je pense notamment à Marie-Pierre et Jérôme, qui ont eu l'immense gentillesse de m'accueillir, et que j'ai pris un grand plaisir à connaître. Je pense aussi à mon ancien coloc très (trop) temporaire Greg, qui a été un coup de foudre amical au premier regard. Merci Mad pour ton efficacité et ta persévérance pour nous trouver un appart afin de fonder notre petite coloc. Et enfin, je pense à Baptiste et Thomas, avec qui j'ai adoré partager des soirées lyonnaises à débattre sans fin, d'ailleurs je pense qu'il nous reste quelques conversations à terminer.

Merci également à toutes les personnes qui m'ont si gentiment accueilli à l'UMPA, et qui m'ont aidé à gérer cette dernière année de thèse en parallèle de mes nouvelles responsabilités à Lyon. Je pense notamment à Adrien, Amine, Antoine, Baptiste, Benjamin, Clairon, Corentin, Ella, Gabriela, Isao, Laurent, Mohamed, Sasha, Séverin, Simon, Tanguy, Victor et tous·tes les autres bien entendu.

Merci à Alice et Céline de m'avoir adopté si rapidement à mon arrivée à Lyon, et d'être les formidables personnes que je suis très heureux d'avoir rencontrées.

Merci à tous·tes mes ami·es de longue date de m'avoir encouragé et apporté votre soutien aux moments où j'en avais besoin, voire même d'avoir fait semblant d'être intéressé·es par mes histoires, être l'objet de votre attention est un de mes passe-temps favoris. Je pense notamment aux ami·es du lycée ou d'avant, Adrien, Béa, Nassim, Nico, Nicolas, Teddy, Thibault et Lucile, dont le courage est une source d'inspiration quotidienne. Un grand merci à la secte (ce n'est pas une vraie secte, *ndlr*), qui est aussi proche qu'une famille. Merci à Kiki, Lulu, Mad, Q et Zaz d'être les plus drôles.

Merci à ma famille justement, aux Perrin comme aux Roussel qui ont fait de moi un authentique Perrin-Roussel, premier du nom. Merci notamment à Daniel et Marie-Jeanne pour avoir activé le bouton magique de la reproduction sociale, qui m'a mené jusqu'ici. Merci à eux surtout pour toute l'éducation dont j'ai profité, des brimbelles des ballons vosgiens aux dîners studieux en prépa. Merci à Tadig et Mahig pour m'avoir toujours poussé à donner le meilleur de moi-même, mais de toujours m'avoir regardé avec tendresse quand je préférais faire du vélo que travailler. Merci à Simon et Lou-Anne, mes hobbits préférés. Merci infiniment à maman, qui est la meilleure des mamans (fait scientifiquement établi[1]), à qui je dois évidemment tant de choses. Enfin, merci à papa, à qui j'aurais tellement aimé pouvoir montrer ces lignes.

[1] C'est écrit dans le dernier rapport du GIEC, allez le lire si vous ne me croyez pas.

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I Introduction

Paul Dirac wrote in 1929: *"The general theory of quantum mechanics is now almost complete, the imperfections that still remain being in connection with the exact fitting in of the theory with relativity ideas. These give rise to difficulties only when high-speed particles are involved, and are therefore of no importance in the consideration of atomic and molecular structure and ordinary chemical reactions, in which it is, indeed, usually sufficiently accurate if one neglects relativity variation of mass with velocity and assumes only Coulomb forces between the various electrons and atomic nuclei. The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are thus completely known, and the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble. It there fore becomes desirable that approximate practical methods of applying quantum mechanics should be developed, which can lead to an explanation of the main features of complex atomic systems without too much computation."* [Dir29].

At the end of the nineteenth century, several prominent physicists, including Maxwell, Michelson, and Thomson (better known as Lord Kelvin), held a view of science that could be badly caricatured as follows: "*we already understand most of the fundamental concepts of physics; what remains to be done is to improve the precision of measurements by a few decimal places.*" This is, of course, not a faithful representation of their actual views [Mon20]. Nevertheless, their message emphasized the importance of quantitative accuracy: a deeper understanding of physical phenomena requires increasingly precise measurements. In turn, advances in experimental precision often pave the way for major theoretical breakthroughs.

It is in this spirit that Paul Dirac, in 1929, expressed the view quoted above regarding quantum physics, which had emerged in the meantime. Although the basic principles of quantum mechanics were already fairly well understood, solving the Schrödinger equation exactly was impossible except in the simplest cases, and even approximate calculations quickly became intractable for sufficiently large systems. Dirac therefore stressed the need for simpler models and effective theories capable of providing approximate predictions for quantities of physical interest, making it possible to investigate complex systems such as large molecules or crystalline solids.

Nearly one hundred years later, in 2026, many new theories have been developed and countless phenomena have been discovered. Yet the fundamental challenge identified by

Dirac remains essentially unchanged: to design practical models that allow the effective computation of physically relevant quantities. This thesis is part of that broader endeavor. It approaches the problem from the perspective of mathematics, joining a collective effort in which physicists, mathematicians, chemists, and computational scientists all play a role. Indeed, one of the beauties of quantum science lies in the fruitful and necessary interaction between researchers from diverse backgrounds, working together to better understand the complexity of the physical world.

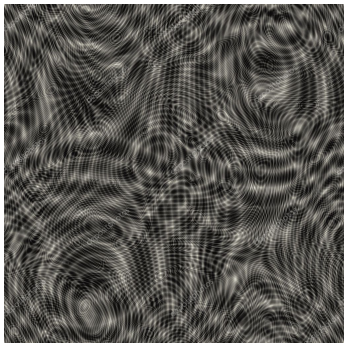
In this thesis, we focus more specifically on the understanding of so-called *moiré materials*, that will be introduced in Sec. I.1, which is an active research field of the last decades, both in physics and in mathematical physics, since such materials illustrate the need of the emergence of effective models. The mathematical study of such models is the purpose of this work. In particular, the focus on the properties of moiré materials led to the study of two methods, namely Density Functional Theory in Chap. II, and Dynamical Mean-Field Theory in Chap. III and IV.

I.1 Moiré materials

I.1.1 A new experimental and theoretical framework

What is a *moiré* material? The term *moiré* originally refers to watered patterns of silk or *mohair* cloth, a textile that was used by aristocrats for dresses, capes, or sashes (see the painting of Peter the Great wearing a moire ribbon sash in Fig. I.1b). By analogy, it refers to the pattern that emerges when two periodic structures are superimposed with a slight mismatch in orientation or spacing. For instance, moiré patterns may appear when photographing a finely textured fabric, when two semi-transparent grids are overlaid with a small relative rotation, or when two digital pixel arrays interfere due to differences in sampling frequencies. Similar effects can also be observed macroscopically by looking through two slightly shifted fences or gratings. In all these situations, the superposition of two microscopic periodic structures generates a larger-scale modulation pattern, often visually striking and characterized by a new emergent spatial scale, see Fig. I.1.

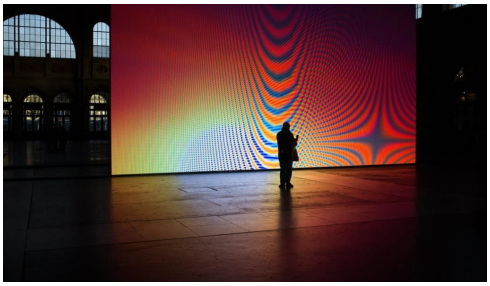
From a geometric point of view, we will consider generic situations when a moiré structure appears as two periodic media with close but distinct periodicities are superimposed. More precisely, we will focus on the context of condensed matter physics, and more specifically on so-called *moiré materials*. Moiré materials are typically artificial layered systems constructed by stacking atomically thin crystals with a controlled relative twist angle or lattice mismatch. The resulting interference pattern defines a new geometric structure whose characteristic length scale is typically much larger than the original microscopic lattice spacing. This *moiré superlattice* (see Fig. I.3) dramatically modifies the electronic properties of the system and gives rise to a rich variety of emergent quantum phenomena.



(a) A *moiré* pattern created by superimposing two structures of smaller scale.



(b) Peter the Great, painted by Paul Delaroche, is wearing a *moiré* sash.



(c) The resolution of the photograph creates a *moiré* pattern.

Figure I.1: Different moiré patterns in (almost) everyday life.

A paradigmatic example consists in stacking two identical two-dimensional crystals with a small relative rotation angle θ. When θ is small, the interference between the two atomic lattices generates a moiré pattern whose characteristic length scale behaves approximately like

$$L_{\mathrm{M}} \sim \frac{a}{\theta}, \tag{I.1.1}$$

where a denotes the microscopic lattice spacing and θ is measured in radians. Consequently, a very small twist angle may generate a mesoscopic spatial structure several orders of magnitude larger than the atomic scale, as it is highlighted for twisted bilayer graphene in Fig. I.3.

Arising physical phenomena. This new emergent scale strongly influences the behavior of electrons inside the material. In particular, the moiré superlattice acts as an effective periodic environment for electronic motion, thereby altering electronic transport, spectral properties, localization phenomena, and interaction effects [And+21].

The study of moiré systems has therefore become a central topic in modern condensed matter physics, especially because these materials provide highly tunable platforms for investigating strongly correlated quantum phenomena. The study of moiré materials with twist angle has even become a discipline in its own right: *twistronics* [Jor22]. Several important classes of moiré materials have been investigated experimentally and theoretically

over the last decade. An important family consists of transition metal dichalcogenides (TMDs), such as MoS_2, WS_2, $MoSe_2$, or WSe_2. These materials are layered semiconductors that can be assembled into twisted bilayers or heterobilayers, also called van der Waals heterostructures [GG13], as the layers are bonded by the van der Waals force, see Fig. I.2. These materials exhibit moiré excitons, correlated insulating states, and topological phenomena [Ken+21; MS16].

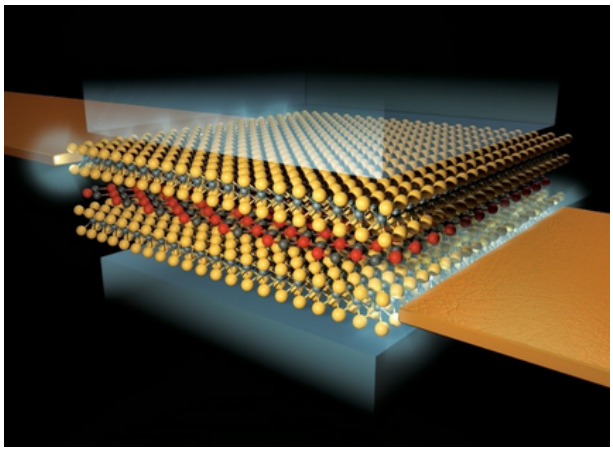


Figure I.2: Computer-generated image of a TMD heterostructure created by the University of Texas at Austin.

Among all moiré materials, twisted bilayer graphene (TBG) has undoubtedly become the most emblematic example. Graphene itself, isolated experimentally in 2004 by Novoselov et al. [Nov+04], consists of a single layer of carbon atoms arranged on a honeycomb lattice. The natural form associated with graphene is graphite, a crystalline allotrope used, among other things, for pencil leads. When two graphene layers are stacked with a small relative twist angle, a moiré superlattice emerges (Fig. I.3) and profoundly modifies the electronic structure of the system. The theoretical possibility that

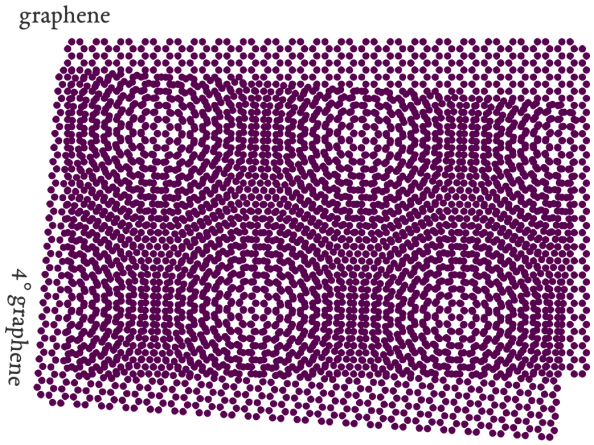


Figure I.3: Twisted Bilayer Graphene (TBG) : two layers of graphene (carbon atoms on a honeycomb lattice) are stacked on top of the other, with a $\theta = 4^\circ$ twist angle.

small twist angles could lead to strong electronic renormalization effects was first investigated by Lopes dos Santos et al. [LPC07]. A major breakthrough occurred in 2011, when Bistritzer and MacDonald predicted the existence of so-called *magic angles*, at which the electronic bands become nearly flat [BM11b]. At these special angles, the suppression of kinetic energy dramatically enhances interaction effects.

A few years later in 2018, two landmark experiments by Cao et al. reported the observation of correlated insulating states and unconventional superconductivity in magic-angle twisted bilayer graphene [Cao+18a; Cao+18b]. These discoveries triggered enormous interest in moiré materials and initiated an extensive experimental and theoretical effort devoted to understanding the interplay between geometry, topology, and strong correlations in these systems (more than 40 000 publications in the last decade with the keywords *twisted bilayer graphene*[1]).

In conventional crystalline solids, the kinetic energy of electrons is typically dominant compared to electron-electron interactions. In contrast, in moiré materials — near the so-called magic angles for TBG — flat bands appear in the band diagram, so that Coulomb interaction effects become the leading energy scale. Since the kinetic energy scale becomes extremely small, even moderate electron-electron interactions may drive the system into strongly correlated phases.

The observed insulating phases are often interpreted as examples of *correlated insulators*, namely insulating states that cannot be explained solely by single-particle band theory. In several situations, these phases exhibit similarities with Mott insulators known from strongly correlated condensed matter systems such as cuprates [Po+18; Oh+21]. However, moiré systems offer an important additional advantage: their effective parameters can often be tuned experimentally through twist angle, carrier density, external electric fields, or substrate engineering [Ada+24; Bre+20].

From a broader perspective, moiré materials occupy an intermediate regime between traditional strongly correlated electronic materials and ultracold atomic systems in optical lattices, as displayed in Fig I.4. Compared with cuprates, moiré systems possess much larger effective lattice spacings and significantly enhanced tunability. On the other hand, they remain genuine electronic materials with long-range Coulomb interactions and nontrivial band topology, unlike many cold atom realizations.

Besides correlation effects, topology also plays a fundamental role in the physics of moiré systems. The moiré band structure may exhibit nontrivial topological properties characterized by Berry curvature effects and nonzero Chern numbers [BCZ20; Led+20]. Such topological features influence electronic transport properties and may lead to anomalous Hall responses, orbital magnetism, and topologically protected edge states. These phenomena illustrate the fact that geometry and topology become inseparable from many-body physics in the moiré context. However, since the topological structure of moiré

[1] Google Scholar search on June 15, 2026.

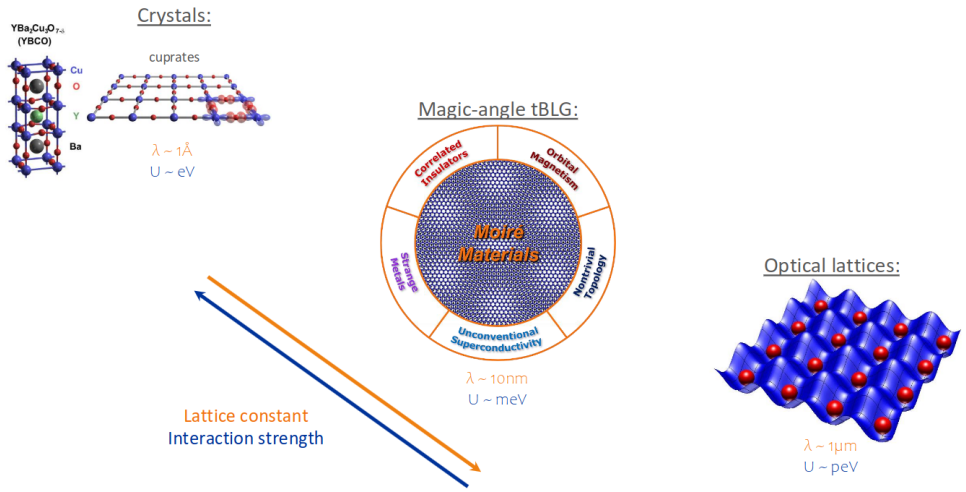


Figure I.4: Comparison scale of cuprates, moiré materials and optical lattices in terms of interaction strength and effective lattice constant. Image by Dmitri Efetov [Efe26].

materials is not studied in this thesis, we refer the reader to the very rich literature for more insight on the subject, like the very good review by Cayssol and Fuchs [CF21].

Understanding these phenomena theoretically remains a major challenge. Even at the level of simplified effective models, the interplay between geometry, topology, and strong correlations leads to highly nonperturbative regimes that are only partially understood mathematically. This motivates the development of effective many-body descriptions capable of capturing both large-scale electronic structures and interaction-driven phenomena.

I.1.2 Mathematical challenges for moiré materials

Although the physics literature on moiré materials has expanded enormously over the last decade, the corresponding rigorous mathematical analysis remains comparatively recent and still largely incomplete. As mentioned previously, the continuum model introduced by Bistritzer and MacDonald [BM11b] predicted the appearance of nearly flat bands at the so-called magic angles and provided a remarkably successful effective description of low-energy electronic states in twisted bilayer graphene.

From a mathematical perspective, two major directions have emerged in the study of moiré Hamiltonians. The first concerns the spectral theory of the Bistritzer–MacDonald model and, in particular, the existence of perfectly flat bands in its chiral limit. Building on the work of Tarnopolsky, Kruchkov, and Vishwanath, Becker, Embree, Wittsten, Zworski, and collaborators established rigorous results on magic angles using semiclassical, microlocal, and analytic methods, proving the existence of flat bands and uncovering remarkable structures such as infinitely many magic angles, band degeneracies, and hidden integrability properties of the chiral model [Bec+21; BHZ23; BHZ25]. Independently, Watson, Luskin, and collaborators developed complementary approaches leading to rigor-

ous proofs of the existence of magic angles and perfectly flat bands in the chiral continuum model [WL21; Wat+23].

A second line of research focuses on the mathematical derivation and justification of continuum moiré models. On the one hand, Cancès, Garrigue, and Gontier derived Bistritzer–MacDonald-type Hamiltonians directly from Density Functional Theory through asymptotic and variational arguments, thereby connecting first-principles electronic structure models to effective moiré-scale descriptions [CGG23]. On the other hand, Watson, Luskin, Kong, Liu, MacDonald, and collaborators rigorously derived and justified the Bistritzer–MacDonald model and its higher-order extensions from atomistic tight-binding dynamics, providing quantitative error estimates and precise regimes of validity for continuum approximations of twisted bilayer graphene [Wat+23; Kon+24; Qui+25].

More generally, moiré materials belong to the broader class of non-periodic quantum systems, as their incommensurate structures naturally lead to aperiodic or ergodic Schrödinger operators. Their mathematical analysis therefore draws on the rich spectral theory of ergodic operators developed by Pastur, Figotin, Carmona, Lacroix, and others [PF92; CL12], as well as on methods originating from the study of quasi-crystals and aperiodic order [Sen96; BG13].

In this section, we focus on two main mathematical questions related to moiré materials, namely the geometry of the models and the need for effective approximate models, and their mathematical analysis.

Geometrical questions. One of the major challenges originates from the geometry of the moiré superlattice itself. In certain situations, the superposition of two lattices remains globally periodic. Such systems are called *commensurate*. For twisted bilayer graphene, commensurability occurs only for a zero-Lebesgue measure set of twist angles satisfying a condition of the form

$$\tan\frac{\theta}{2} \in \frac{1}{\sqrt{3}}\,\mathbb{Q}, \tag{I.1.2}$$

up to equivalent parametrizations of the commensurate angles [Mel10; LPC12]. In these cases, the system may formally be described by a periodic supercell.

However, even when periodicity survives, the corresponding moiré unit cell rapidly becomes extremely large as the twist angle decreases, see (I.1.1). Near the magic-angle regime, a single moiré cell may contain several thousands or even tens of thousands of carbon atoms. Consequently, the associated electronic problem involves a prohibitively large number of interacting electrons and degrees of freedom. This multiscale explosion makes direct *ab initio* approaches essentially intractable in experimentally relevant regimes [CFK20]. Even density functional calculations become computationally demanding when the moiré supercell grows excessively large.

More fundamentally, since generically, twisted layered systems are *incommensurate* because of (I.1.2), the two underlying lattices possess no common period, and the re-

sulting structure is genuinely aperiodic. In such situations, mathematically speaking, the standard framework of periodic solid-state physics breaks down. In particular, one can no longer directly rely on Bloch–Floquet theory, used in the periodic setting, to decompose the electronic Hamiltonian into independent momentum sectors. Since translational invariance is absent, many fundamental notions become nontrivial:

- how should one define observables such as the density of states?
- what is the appropriate notion of thermodynamic limit?
- how can one characterize equilibrium electronic states?
- what replaces the usual Brillouin-zone description?

In the periodic setting, the thermodynamic limit is comparatively well understood for several mean-field models. In particular, Catto, Le Bris, and Lions established the convergence of thermodynamic limits for Thomas–Fermi and Thomas–Fermi–von Weizsäcker type theories toward periodic infinite-volume models [CLL98; CLL01].

For genuinely incommensurate systems, however, alternative frameworks are required. One possible approach consists in replacing periodicity by ergodicity, which is the approach developed in Chap. II, following [CLL13]. The ergodicity then comes precisely from the incommensurability of the two lattices. In this context, the Birkhoff-von Neumann theorem [Moo15] allows one define mathematical objects via thermodynamic limits. Such approaches have already proved successful in the study of disordered crystals and ergodic defects [BLL07; CDL08; CLL13].

This point of view naturally leads to the use of operator-algebraic methods, including crossed-product algebras and von Neumann algebras, in order to define observables and traces in the absence of periodicity. They also provide a unified framework capable of treating both periodic and aperiodic systems within the same formalism, see Chap. II. This mathematical framework was developed by Bellissard and collaborators [Bel02; BvS94], based on ideas of Connes [Con94]. From a conceptual perspective, this corresponds to choosing an appropriate representation of the electronic system adapted to the geometry under consideration. Depending on the physical problem, different representations may become more natural or more efficient analytically. In this sense, the mathematical study of moiré materials shares several structural similarities with the analysis of ergodic Schrödinger operators, quasi-crystals, and defective crystals.

Effective models. Apart from these geometric questions, solving the full many-body Schrödinger equation remains far beyond reach for realistic moiré systems. One must therefore introduce effective models capable of capturing the relevant physics while remaining mathematically and computationally tractable, in the spirit of Paul Dirac's citation.

A first important class of approaches is provided by density functional theory (DFT), which replace the full many-body wave function by simpler collective quantities such as the electronic density or the one-body density matrix [CF23]. These mean-field descriptions form the basis of most modern electronic structure calculations and are particularly well suited to large electronic systems. Furthermore, it has a tractable mathematical structure, that allowed a wide range of studies since it was introduced by Hohenberg, Kohn and Sham [HK64; KS65] in the 1960s, see [Lie83]. DFT and its application to moiré materials is introduced in a more detailed fashion in Sec. I.2.

Nevertheless, the strongly correlated nature of many moiré materials introduces further difficulties. As discussed previously, narrow moiré bands drastically reduce the kinetic energy scale, so that electron-electron interactions become dominant. In such regimes, simple mean-field approximations like DFT may fail to capture essential physical phenomena such as Mott localization, dynamical correlations, or unconventional superconductivity. This motivates the introduction of simplified effective lattice models designed to focus on the low-energy correlated degrees of freedom. In many situations, moiré materials may indeed be described through tight-binding approximations, where electrons hop between effective localized orbitals associated with the moiré superlattice. The mathematical justification of tight-binding limits for graphene-type systems has been investigated rigorously in several works, notably by Fefferman and Weinstein and collaborators [FW12; FLW18].

Within this perspective, Hubbard-type Hamiltonians provide natural effective models for strongly correlated moiré systems. Such models retain the competition between kinetic delocalization and local Coulomb repulsion while remaining significantly simpler than the full electronic problem. The Hubbard model is briefly introduced in Sec. I.3.1. However, even these simplified many-body lattice models remain highly nontrivial mathematically. The Hubbard model in one dimension has been thoroughly studied by Lieb and Wu [LW03], but in general, many mathematical questions remain open, and from a numerical point of view, the dimension of the system grows exponentially with the size of the systems, which is an obvious obstacle to direct computations.

One of the most successful approaches developed in condensed matter physics for treating strong local correlations, in particular for Hubbard models, is Dynamical Mean-Field Theory (DMFT) [Geo+96]. Instead of describing the system solely through static electronic densities, DMFT introduces frequency-dependent Green's functions and self-energies capable of encoding dynamical correlation effects. Roughly speaking, DMFT maps the Hubbard lattice many-body problem onto an effective collection of quantum impurity models coupled self-consistently to so-called external baths (for more details, see Sec. I.3.1 and Chap. III). The central object becomes the electronic *Green's function*, whose analytic structure contains detailed information about spectral and correlation properties of the system. The Green's function is sometimes also called two-points function or correlation function in Quantum Field Theory [LM99].

The mathematical analysis of DMFT also raises many difficult questions, which are

detailed in Sec. I.3.2. In particular, this method introduced by physicists in the 1990s has led to very little mathematical study, and the first challenge was to formulate the equations of DMFT in a rigorous mathematical setting, before even tackling the questions of existence, uniqueness, etc. One notable contribution is the thesis of Lindsey [Lin19], which investigates mathematical aspects of Green's functions and self-consistent impurity problems arising in DMFT. The appendix of Chap. III offers a precise mathematical formulation of DMFT in an adapted framework described in Sec. I.3.1, in the continuation of Lindsey's work.

This manuscript is devoted to the study of these two complementary approaches. On the one hand, we investigate mean-field electronic models adapted to large and possibly incommensurate systems. On the other hand, we study DMFT, a theory designed to capture the strong correlation effects arising in moiré materials, for effective Hubbard models.

I.2 Density functional theory for encapsulated moiré materials

I.2.1 Density Functional Theory and electronic structure models

Among the many theoretical approaches used to investigate the electronic properties of two-dimensional and moiré materials, Density Functional Theory (DFT) occupies a central position. Since the foundational works of Hohenberg, Kohn, and Sham [HK64; KS65] (leading to the 1998 Nobel Prize in Chemistry for Walter Kohn), DFT has become the standard framework for first-principles calculations in condensed matter physics, materials science, and quantum chemistry [OPN09]. Note that actually, even if the method is widely used and studied, the mathematical derivation of DFT is actually not complete, and is only well understood in one dimension [CL25].

The fundamental idea of DFT is to replace the many-body electronic wave function, which depends on all electronic coordinates simultaneously, by simpler collective quantities such as the electronic density or the (Kohn-Sham) one-body density matrix. This reduction dramatically decreases the complexity of the quantum many-body problem while retaining a substantial amount of physically relevant information [CF23].

In the context of moiré materials, DFT plays a somewhat different role from that of strongly correlated approaches such as Dynamical Mean-Field Theory, which is the topic of Sec. I.3. Indeed, standard Kohn–Sham DFT is primarily designed to describe ground-state properties, including total energies, charge distributions, and effective one-particle electronic structures. As a consequence, it does not directly provide access to dynamical correlation functions, which are often regarded as the primary signatures of strongly correlated electronic behavior. For this reason, DFT alone is generally insufficient to

quantitatively reproduce the rich phase diagrams observed in twisted multilayer systems, including correlated insulating phases, unconventional superconductivity, or Mott transitions. Nevertheless, it remains an indispensable tool for understanding many aspects of moiré materials.

In particular, DFT calculations have proved extremely successful for describing atomic relaxation effects in twisted multilayers [NK17; Car+19; Car+18b; Car+18a]. These relaxations significantly modify the local stacking configurations and therefore strongly influence the electronic structure of the resulting moiré superlattice. DFT is also widely used to investigate phonon modes [Lu+22], estimate electron-phonon coupling parameters [Bar+01], and derive effective low-energy Hamiltonians for subsequent many-body studies [Fan+18; CGG23].

From a mathematical perspective, DFT is particularly appealing because it provides a systematic hierarchy of models ranging from orbital-free approximations to sophisticated Kohn–Sham theories. In this thesis, we mainly focus on the reduced Hartree–Fock (rHF) approximation, sometimes referred to as the Hartree or Random Phase Approximation (RPA) level of theory. Within Perdew's Jacob's ladder of density-functional approximations [PS01], this corresponds to the lowest rung, where the exchange-correlation functional is neglected. We also consider the LDA (Local Density Approximation) exchange functional energy, but only in the periodic setting, see Chap. II. Although quantitatively less accurate than modern exchange-correlation approximations, the rHF model retains the fundamental self-consistent structure of Kohn–Sham theory while enjoying a crucial mathematical advantage: convexity. This property considerably simplifies the analysis and allows one to establish existence as well as partial uniqueness results that are often inaccessible for more realistic but non-convex models.

One of the contributions of this thesis and of Chap. II is to consider a particular geometry: the moiré material is *encapsulated*, as will be described in the next section. It is a physically more relevant situation than to consider a 2D material in a free 3D space.

I.2.2 Encapsulated two-dimensional materials

From the experimental point of view, the study of Twisted Bilayer Graphene (TBG) relies crucially on encapsulation techniques using typically hexagonal boron nitride (hBN) [Cao+18b]. The layers are encapsulated between two hBN sheets as in Fig. I.5, which protect the sample from disorder and environmental perturbations while significantly improving electronic mobility [Dea+10]. Additional metallic gates allow researchers to tune carrier densities and apply external displacement fields, thereby providing an exceptional degree of control over the system [Lon+23]. Actually, the active material is even typically embedded within a complex device consisting of insulating substrates, metallic gates, and dielectric layers, which we partially take in account in our analysis in this thesis.

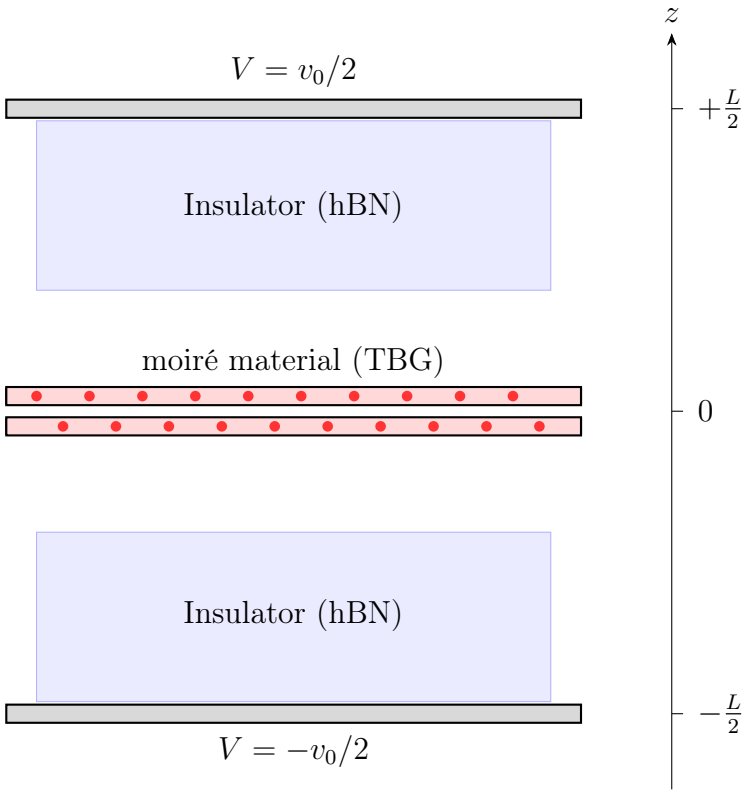


Figure I.5: Schematic diagram of the physical system.

Physically speaking, the encapsulating layers act primarily as dielectric media. Since the Fermi level of the system lies within the band gap of the insulator, and since the low-energy phenomena of interest involve energies much smaller than this gap, the electronic degrees of freedom of the encapsulating material may be integrated out. Only the electrons belonging to the active two-dimensional structure need to be treated explicitly [Wan+25b]. This approximation is closely related to implicit solvent models commonly used in quantum chemistry [TMC05]. The surrounding environment is not described microscopically but through effective dielectric parameters entering the electrostatic equations.

As a consequence, the electronic system remains intrinsically three-dimensional, but its charge density is strongly localized around a two-dimensional plane. The resulting geometry combines features of both two-dimensional and three-dimensional electronic systems. In particular, the Coulomb interaction is deeply modified. The presence of metallic gates imposes effective boundary conditions on the electrostatic potential, leading to a substantial screening of long-range interactions. This screening mechanism plays an important role both physically and mathematically, as it changes the structure of the Poisson equation governing the electrostatic potential. It also implies that electrons cannot escape to infinity along the direction orthogonal to the material. This confinement eliminates several difficulties encountered in the study of standalone two-dimensional materials embedded in three-dimensional space [GLM21; GLM23] and provides a more tractable framework for mathematical analysis. The mathematical analysis of the Coulomb potential within

this encapsulated setting can be found in Chap. II.

For these reasons, encapsulated systems constitute a particularly attractive setting for developing mathematical theories of moiré materials while remaining closely connected to experimentally relevant devices.

I.2.3 Geometry of the bilayer

The mathematical analysis of density-functional models for moiré materials combines several distinct difficulties. The first originates from the geometry of the underlying structure, as we have seen in Sec. I.1.2. Indeed, generic twisted structures are incommensurate. In this regime, no global lattice periodicity exists and Bloch–Floquet theory is no longer available. The usual notions of Brillouin zone, crystal momentum, and periodic unit cell cease to provide an adequate description of the system.

These difficulties naturally lead to questions familiar from the theory of disordered and aperiodic media:

- How should one define electronic observables?
- What is the appropriate notion of thermodynamic limit?
- How can one formulate self-consistent electronic structure models without periodicity?

A large mathematical literature has been devoted to linear aperiodic systems, including ergodic Schrödinger operators, quasi-crystals, and disordered media [PF92; CL12; BG13]. Considerably less is known for nonlinear mean-field models arising in density functional theory. For Thomas–Fermi and Thomas–Fermi–von Weizsäcker theories, ergodic formulations were developed by Blanc, Le Bris, and Lions [BLL07]. Beyond these orbital-free models, rigorous results for electronic structure theories involving one-body density matrices have mainly been obtained in the framework of reduced Hartree–Fock models for disordered crystals [CLL13; Lah14]. In this thesis, we extend these ideas to the setting of encapsulated moiré materials. Our main objective is to construct and analyze reduced Hartree–Fock models for periodic and incommensurate multilayer structures relevant to modern moiré devices.

To treat incommensurate systems, we adopt the configuration-space formalism introduced by Bellissard and collaborators [BvS94; Bel02], itself rooted in the noncommutative geometric viewpoint developed by Connes [Con94]. This framework allows one to replace the missing periodicity by an ergodic structure and to define physically meaningful quantities through averaged observables and traces on suitable operator algebras. The formalism is detailed in Chap. II.

The main idea is described in Fig. I.6. Consider only the transversal direction x (the direction of the 2D moiré material), and suppose that each layer is described by an atomic

charge density $m_i(x)$, $i = 1, 2$, periodic with respect to a lattice $\mathbb{L}_i$. Then a configuration $\omega = (\omega_1, \omega_2) \in \Omega := \mathbb{R}^2/\mathbb{L}_1 \times \mathbb{R}^2/\mathbb{L}_2$ corresponds to a shift of each layer by ω_1 and ω_2, respectively, as in the Fig. I.6. Letting $m : \Omega \times \mathbb{R}^2 \to \mathbb{R}$ be the function defined by

$$m(\omega, x) := m_1(x - \omega_1) + m_2(x - \omega_2), \tag{I.2.1}$$

m corresponds to the total atomic charge density at x, in the configuration ω. Introducing the group action $\alpha : \mathbb{R}^2 \ni x \mapsto \alpha_x$, with

$$\alpha_x : \omega \mapsto (\omega_1 - x, \omega_2 - x) \in \Omega,$$

we see that the relation (I.2.1) implies that the function m is *stationary*, in the sense that

$$m(\omega, x) = m(\alpha_x \omega, 0), \tag{I.2.2}$$

so m is actually determined by $m(\cdot, 0)$. The advantage is the compactness of the set Ω, while the apparent loss is the fact that Ω is 4-dimensional (5 if we add the z-direction), as we started with a 2-dimensional problem (3 if we add the z-direction).

We can now distinguish several mathematical structures, depending on the relation between the lattices $\mathbb{L}_1$ and $\mathbb{L}_2$.

- If $\mathbb{L}_1$ and $\mathbb{L}_2$ are commensurate, that is when $\mathbb{L} = \mathbb{L}_1 \cap \mathbb{L}_2$ is also a $2d$-lattice of $\mathbb{R}^2$, then $m(\omega, \cdot)$ is $\mathbb{L}$-periodic. The physical properties will depend on the configuration (think of AA and AB stacking for TBG). In this case, the usual Bloch-Floquet theory can be used to described the electronic properties of the material.

- If the lattices are incommensurate, in the sense that $\mathbb{L}_1^* \cap \mathbb{L}_2^* = \{0\}$ (with $\mathbb{L}_i^*$ the reciprocal lattice of $\mathbb{L}_i$), then the action α is *ergodic*. We expect that the physical properties will not depend on the configuration, and we can use in this case the formalism of von Neumann algebras to define our quantities of interest. See Chap. II for more details.

- There are intermediate cases, which could be treated by a mixture of techniques. We do not tackle this issue in this thesis.

Within this setting, we establish the existence of reduced Hartree–Fock ground states for quasi-periodic encapsulated multilayers, including physically relevant examples such as twisted bilayer graphene at generic twist angles.

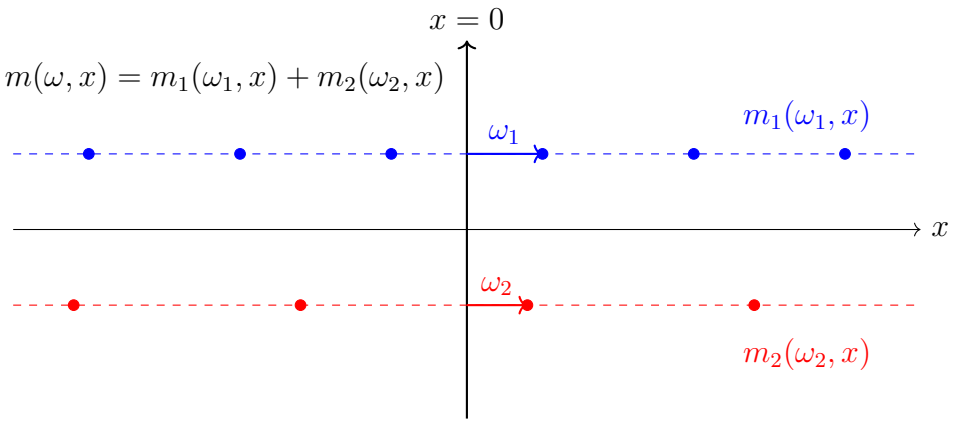


Figure I.6: Aperiodic bilayer system in one dimension at the configuration $\omega = (\omega_1, \omega_2)$. The total atomic charge density m is stationary: $m(\omega, x) = m(\alpha_x \omega, 0)$.

I.3 IPT-DMFT from a mathematical perspective

In this section, introducing Chap. III and IV, we study Dynamical Mean-Field Theory from a mathematical point of view, with the choice of Iterated Perturbative Theory as impurity solver, as it will be detailed below.

Let us first give a quick historical background of the Dynamical mean-field theory (DMFT). DMFT emerged as a powerful approach to study strongly correlated electron systems by exploiting the simplifications arising in the limit of large spatial dimensions. In their seminal 1989 paper, Metzner and Vollhardt [MV89] demonstrated that for the Hubbard model on a d-dimensional hypercubic lattice, the self-energy becomes purely local when $d \to \infty$. This result mirrors the behavior of classical models such as the Ising model, where mean-field theory becomes exact in the same limit. Building upon these insights, Müller-Hartmann [Mül89] analyzed the scaling of hopping amplitudes and diagrammatic contributions in infinite dimensions, solidifying the theoretical foundation of the approach.

In 1992, Georges and Kotliar [GK92] suggested a concrete implementation of this large-dimensionality simplification by mapping the Hubbard model onto a self-consistent Anderson Impurity Model (AIM). In their formulation, the lattice problem is approximated by an impurity embedded in an effective medium described by an AIM, whose hybridization function is determined self-consistently so that the local Green's function of the impurity coincides with that of the lattice. This framework was initially referred to as the *Local Impurity Self-Consistent Approximation* (LISA) [GK92] to emphasize its structure as an impurity problem embedded in a self-consistently determined bath.

The approach was thoroughly formalized and reviewed in the 1996 article by Georges, Kotliar, Krauth, and Rozenberg [Geo+96], establishing DMFT as a central method in condensed matter physics. This review highlighted that DMFT is a "mean-field" theory in the sense that it becomes exact in infinite dimensions, yet it retains the full temporal

(frequency-dependent) dynamics of local Green's functions, hence the name *dynamical* mean-field theory.

Subsequent works extended DMFT to include for example its combination with electronic structure methods such as Density Functional Theory (DFT+DMFT) [AL14; Hel+06].

DMFT and moiré materials. Moiré materials, formed by stacking two or more atomically thin layers with a slight lattice mismatch or twist angle (see Sec. I.1), have recently emerged as an exciting platform for studying strong correlation phenomena. These materials exhibit long-range geometric moiré patterns that, in particular, drastically reduce electronic kinetic energy, resulting in narrow bands with energy scales often below tens of meV [AM20; Bal+20]. This kinetic energy suppression enhances the effects of Coulomb interactions, leading to a wealth of correlated electronic phases [Cao+18a].

From a model perspective, these systems can be thought as moiré Hubbard models, as briefly evoked in Sec. I.1.2 [Wu+18]. The tunability of electronic structure parameters, such as the effective hopping amplitude and interaction strength, makes moiré materials ideal realizations of the Hubbard model with adjustable parameters in situ via gating, twist angle modification, or choice of substrate dielectric environment [Wu+18; PWD20].

One of the central phenomena accessible in these systems is the correlation-driven Mott metal–insulator transition [BK94], characterized by a change in the electronic state from a metallic phase with delocalized electrons to an insulating phase with localized electrons due to strong electron-electron interactions. Theoretical understanding of such transitions, particularly in two-dimensional systems, requires methods capable of capturing the strong local dynamical correlations inherent in the Mott phenomenon. This is where dynamical mean-field theory (DMFT) plays a central role. DMFT approximates the lattice problem by a self-consistent impurity model, capturing local quantum fluctuations while neglecting nonlocal dynamical correlations. Single-site DMFT provides insight into local Mott physics, while cluster extensions (such as DCA or CDMFT) incorporate short-range spatial correlations crucial for capturing magnetic ordering or pairing instabilities [Bri+16; PHK08].

Many other applications arose, DMFT has become a practical tool and a theoretical framework for studying strong correlation in moiré materials. However, there is still very little mathematical work on the subject, and one purpose of this thesis is to contribute to addressing this gap.

I.3.1 Dynamical mean-field theory

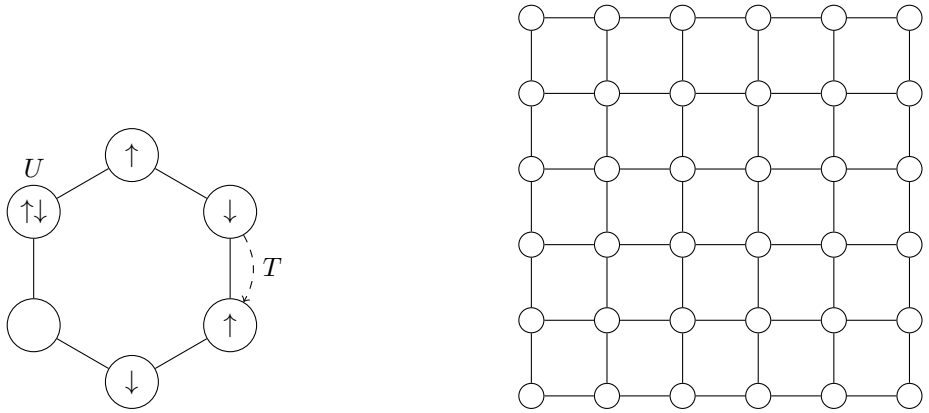
In this section, we briefly introduce the framework of DMFT. More details and a rigorous mathematical formulation are to be found in Chap. III, which corresponds to a joint work with Éric Cancès and Alfred Kirsch [CKP26].

Hubbard model. The theory is primarily motivated by lattice models of interacting electrons, most notably the Hubbard model [GK92]. In its simplest form, the Hubbard Hamiltonian may be written in second quantization as

$$H = \sum_{(i,j),\sigma} T_{i,j} a^\dagger_{i\sigma} a_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow},$$

where $T_{i,j}$ denotes the hopping amplitude between neighboring lattice sites (i, j), U measures the strength of the on-site Coulomb interaction, the operators $a^\dagger_{i\sigma}$, $a_{i\sigma}$ create and annihilate electrons of spin σ at site i, and $n_{i\sigma} = a^\dagger_{i\sigma} a_{i\sigma}$ denotes the number operator at the site i and spin σ.

At a qualitative level, the Hubbard model describes the competition between two opposite physical tendencies: on the one hand, kinetic delocalization, driven by the hopping term, and on the other hand, electronic localization induced by Coulomb repulsion. This competition gives rise to many of the strongly correlated phenomena discussed previously in Sec. I.1. The model can be defined on a general graph, see for example Fig. I.7.



(a) The Pariser-Parr-Pople model of benzene C_6H_6: the cyclic graph C_6.

(b) A periodic lattice with nearest neighbors of degree $d = 2$.

Figure I.7: The Hubbard model for different graphs. DMFT was first developed for graph (b), to understand the limit $d \to \infty$ [MV89].

However, the direct analysis of interacting lattice many-body systems rapidly becomes prohibitively difficult due to the exponential growth of the Hilbert space with the system size. Dynamical Mean-Field Theory addresses this difficulty by replacing the original lattice problem with an effective local impurity problem embedded in a self-consistent environment.

Anderson Impurity Model (AIM). More precisely, DMFT approximates the lattice system by a collection of Anderson Impurity Models (AIM, represented in Fig. I.8), in which a finite number of interacting impurity orbitals are coupled to a non-interacting electronic bath [And61]. Roughly speaking, the impurity degrees of freedom represent a

local portion of the original lattice, while the bath models the influence of the surrounding electronic environment.

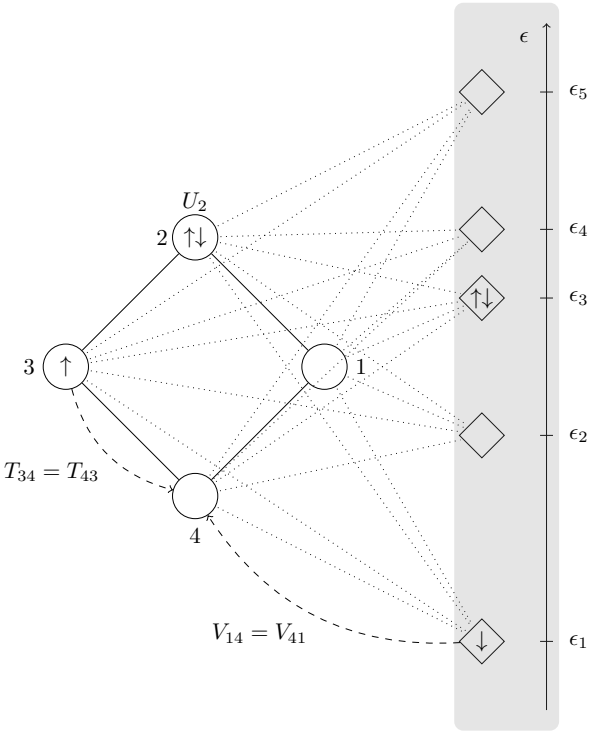


Figure I.8: The AIM of an impurity (Hubbard graph on the l.h.s.) and a *bath* (energies ϵ represented on the r.h.s.) in a 6-electron state.

The central idea of DMFT is that local dynamical correlations can be treated accurately at the impurity level while the surrounding lattice is incorporated through an effective mean-field description.

Green's function, Self-energy, Hybridization function. From a mathematical perspective, the fundamental objects of DMFT are no longer static electronic densities as in DFT, but rather frequency-dependent Green's functions and self-energies. More precisely, we introduce several quantities of interest.

The first central quantity is the quantum Green's function G that we see as an operator-valued function, acting on some Hilbert space $\mathcal{H}$:

$$G : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H}),$$

which depends on a complex frequency variable z, of positive imaginary part: $z \in \mathbb{C}_+ := \{w \in \mathbb{C} \mid \Im(w) > 0\}$. Physically, the Green's function encodes spectral and dynamical properties of the electronic system, including excitation energies, density of states, and more [BM61; TC90].

A second key object is the self-energy

$$\Sigma(z) = G^0(z)^{-1} - G(z)^{-1}, \tag{I.3.1}$$

which describes the effective interaction corrections to the non-interacting electronic dynamics. Here, G^0 denotes the Green's function of the system without taking electronic interaction into account. In many respects, the self-energy may be viewed as an effective dynamical potential generated by electron-electron interactions. Indeed, it will be shown in Chap. III that $G^0(z) = (z - h)^{-1}$, with h the one-body Hamiltonian describing the non-interacting system, so that $G(z) = (z - h - \Sigma(z))^{-1}$, thanks to (I.3.1).

Finally, the impurity model is characterized by a hybridization function

$$\Delta : z \in \mathbb{C}_+ \mapsto \Delta(z),$$

which describe the coupling between the impurity and the surrounding bath (function V and energies ϵ in Fig. I.8, respectively). These functions encode how electrons may enter and leave the impurity subsystem through the effective environment.

The philosophy of DMFT is based on a self-consistent determination of these objects, through the partition of the initial lattice (or graph) into smaller systems (impurities), as described in Fig. I.9.

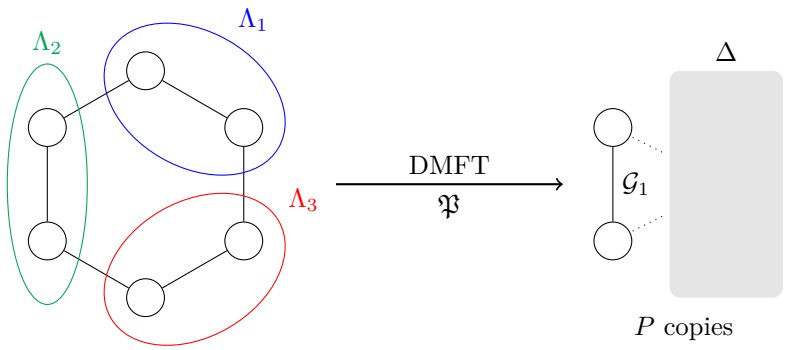


Figure I.9: Partition of the total Hubbard graph into smaller graphs (vertices Λ_i), called clusters. For the sake of simplicity, we consider here a translation-invariant setting, in which all clusters are described by the same AIM, represented by the hybridization function Δ.

At a schematic level, the theory may be viewed as a system of coupled nonlinear functional equations. The first equation takes the form

$$\Sigma = \mathrm{Solv}(\Delta),$$

where Solv denotes an *impurity solver*. Given a hybridization function Δ, the impurity solver computes the associated interacting self-energy Σ by solving the effective Anderson impurity problem. As a matter of fact, Solv is a formal function and is not computable, and must be approximated. The precise definition of the impurity solver Solv hence depends on the chosen approximation or numerical method. In practice, a wide variety of impurity solvers may be used, including perturbative methods, quantum Monte

Carlo approaches, exact diagonalization, or numerical renormalization techniques. In the present work, we shall focus on one particular family of impurity solvers, namely Iterated Perturbation Theory (IPT), which will be introduced in Sec. I.3.2.

Once the self-energy has been computed, one reconstructs an approximate lattice Green's function through the Dyson equation, following from (I.3.1): $G^{-1} := (G^0)^{-1} - \mathrm{Solv}(\Delta)$. The second equation of DMFT is the self-consistency condition. Its role is to impose compatibility between the impurity model and the effective lattice description. Roughly speaking, the DMFT ansatz assumes that the local blocks of the lattice Green's function can be approximated by the impurity Green's function. More precisely, the local components associated with the impurity degrees of freedom are required to coincide with the corresponding local components extracted from the effective lattice Green's function constructed using the self-energy Σ. The precise derivation of the mathematical equations is developed in Chap. III.

This leads to a nonlinear fixed-point problem coupling the impurity model and the effective lattice description, where the variable are either the local self-energies Σ or the hybridization functions Δ. At an abstract level, the DMFT iteration scheme used in practice may therefore be summarized schematically as follows:

$$\Delta \longmapsto \Sigma = \mathrm{Solv}(\Delta) \longmapsto G \longmapsto \Delta_{\mathrm{new}},$$

until convergence is reached. The AIM impurity models are solved by the impurity solver, and then combined in the self-consistency equation to define the new parameters of the AIMs: Δ_{new} is computed from the approximated Green's function G thanks to a Schur complement, see the appendix of Chap. III for more details.

As we will see in the next section, the mathematical structure of the equations at stake is nontrivial. The unknowns are matrix-valued (or operator-valued) analytic functions of a complex variable satisfying strong positivity and causality constraints. Furthermore, the impurity solver itself remains rather poorly understood and there is no rigorous mathematical general definition at this point.

I.3.2 Mathematical study of DMFT

Although Dynamical Mean-Field Theory has become one of the central computational tools for strongly correlated electronic systems, its mathematical foundations remain only partially understood. Even at a formal level, DMFT involves highly nontrivial nonlinear functional equations coupling analytic operator-valued functions through self-consistency conditions. As a consequence, many fundamental mathematical questions remain open.

One of the first difficulties concerns the very definition of the DMFT self-consistency map. At a schematic level, the theory may be viewed as an iterative procedure based on the impurity solver on the one-hand, and the Dyson-type self-consistent equation.

Unfortunately, the mathematical properties of the map

$$\mathrm{Solv} : \Delta \longmapsto \Sigma$$

are in general poorly understood. In particular, even defining the impurity solver rigorously is highly nontrivial. In its exact formulation, the impurity problem remains an interacting quantum many-body system whose solution requires the computation of dynamical correlation functions. Questions of existence, uniqueness, regularity, and preservation of analyticity naturally arise.

At a more formal level, these issues are closely related to the Luttinger–Ward functional formalism [LW60], where the self-energy is viewed as a functional derivative of a universal interacting free-energy functional. While this formalism plays a major conceptual role in many-body physics, its rigorous mathematical status remains delicate, especially outside perturbative regimes. See [LL18] for a mathematical investigation of the perturbative viewpoint.

Because of these difficulties, practical implementations of DMFT typically rely on approximate impurity solvers. In the present work, we focus on one of the most classical and widely used approximations, namely Iterated Perturbation Theory (IPT) [GK92; KK96], that we introduce shortly. A more detailed formulation can be found in Chap. III. In other words, we work with the approximation

$$\mathrm{Solv} \approx F^{\mathrm{IPT}},$$

where $F^{\mathrm{IPT}} : \Delta \mapsto \Sigma$ is some function that associates a self-energy to a given hybridization function, described below.

Iterative Perturbation Theory (IPT). Roughly speaking, IPT approximates the self-energy through a low-order perturbative expansion in the interaction parameter U, while preserving certain important physical constraints. The resulting approximation remains computationally inexpensive and often reproduces qualitatively correct physical behaviors, especially near half-filling and in moderately correlated regimes. However, from a mathematical perspective, IPT raises several fundamental questions.

First, the IPT construction is generally formulated in terms of Matsubara frequencies

$$\omega_n := \frac{(2n+1)\pi}{\beta}, \quad n \in \mathbb{Z}, \tag{I.3.2}$$

that is, discrete frequencies arising from finite-temperature (β is the inverse temperature) quantum statistical mechanics. The self-energy is initially defined only on the discrete set of points formed by the imaginary frequencies ($i\omega_n$) through nonlinear expressions involving the hybridization functions of the AIM.

One must therefore address several nontrivial interpolation problems, such as:

- does there exist an analytic function extending the Matsubara data?

- is this extension unique?
- what regularity or growth properties does it satisfy?
- does the interpolation preserve the physical constraints of the problem?

These questions lead to complex-analytic and operator-theoretic considerations. Many results can be found in the theory of analytic interpolation, also called the Analytic Continuation Problem (ACP) [Pic15; Nev19; Wal35; Nic94; Nic16]. Some of these issues are addressed in this manuscript in Chap. III.

Pick functions. A central role is actually played by the class of Pick functions (or more generally Pick matrices, Pick operators), also known as Herglotz or Nevanlinna functions [GT97; Nic94]. Recall that a scalar Pick function is an analytic function mapping the complex upper half-plane into itself, that is, a function f such that

$$f : \mathbb{C}_+ := \{z \in \mathbb{C} \mid \mathrm{Im}(z) > 0\} \to \overline{\mathbb{C}}_+, \quad \text{s.t. } f \text{ is analytic.} \tag{I.3.3}$$

In the matrix-valued setting, one similarly considers analytic matrix-valued functions whose imaginary part remains positive semidefinite in the upper half-plane, that is

$$(M - M^\dagger)/(2i)$$

is positive semidefinite. An important property of Pick functions is that they have an integral representation, stated by the Nevanlinna-Riesz Theorem [GT97]. To each Pick function can be associated a unique positive Borel measure, called the Nevanlinna-Riesz measure of the Pick function. A typical example of Pick functions are the Cauchy-Stieltjes transforms of probability measures [Eck22; GH03], which corresponds to the scalar version of (I.3.4) (up to a minus sign).

These functions appear naturally in quantum many-body theory because physical Green's functions admit spectral representations of the form

$$G(z) = \int_{\mathbb{R}} \frac{d\mu(\varepsilon)}{z - \varepsilon}, \tag{I.3.4}$$

where the Nevanlinna-Riesz measure μ is a positive matrix-valued measure, called the *spectral measure* in this framework [MRC16]. For finite systems (a Hubbard graph with a finite number of sites for example), this is known as the Källén-Lehmann formula [Käl52]. As a consequence, the associated Green's functions satisfy strong positivity and causality properties, because it follows from (I.3.4) that $-G$ is a Pick matrix (operator). More precisely, the physically admissible quantities satisfy:

$$\boxed{-G, \quad -\Sigma, \quad -\Delta \quad \text{are (operator-) matrix-valued Pick functions.}} \tag{I.3.5}$$

We can then denote by $\mu_G, \mu_\Sigma, \mu_\Delta$ their respective Nevalinna-Riesz measure.

A fundamental question then becomes: *does the IPT impurity solver preserve the Pick property?* More precisely, if the hybridization function Δ encodes a localized AIM, does the corresponding IPT self-energy Σ remain a Pick function? Under which assumptions? In which function spaces? These issues are addressed in Chap. III. Closely related questions concern the appropriate formulation of the DMFT equations themselves. Since the unknowns are analytic matrix-valued functions satisfying positivity constraints, several possible formulations coexist: equations on Matsubara frequencies (usual framework for numerical approach, see Sec. I.3.3), functional equations, etc. Chap. III includes the derivation of the IPT-DMFT equations in terms of μ_G, μ_Σ and μ_Δ, the Nevanlinna-Riesz measures of G, Σ and Δ, and addresses the question of the existence of a solution as a fixed-point problem in a suitable space of operator-valued functions.

Once a suitable formulation has been established, one may ask classical analytical questions:

- does the IPT-DMFT equations admit a solution?
- is the solution physically admissible?
- is it unique?
- how does it depend on the parameters?
- what regularity and asymptotic properties does it possess?

Even for simplified impurity solvers such as IPT, these questions are far from obvious. The nonlinear structure of the equations, combined with the analytic and positivity constraints imposed by physicality, creates a highly delicate functional-analytic problem. We investigate several of these questions within the framework of IPT-DMFT in Chap. III. In particular, we have established a formalization of the IPT self-energy as well as the IPT-DMFT equations, for which we were able to prove the preservation of Pick-type properties, and the existence of self-consistent IPT-DMFT solutions, as well as a regularity property of such solution(s). This work can be thought in the continuity of Michael Lindsey's thesis [Lin19] and of his collaboration with Lin Lin and Reinhold Schneider. However, as mentioned in Sec. I.3.4, we did not manage to prove uniqueness to the solution(s) of the IPT-DMFT equations, so it remains a challenge for future works on the subject.

I.3.3 Study of the discretized IPT-DMFT equations

As mentioned in the previous section, the DMFT equations, and more precisely the IPT-DMFT equations in our context, are self-consistent equations, that can be seen as a fixed-point problem. In Chap. III, the existence of solutions to the functional equations

are proven. Yet, from a practical and numerical perspective, it is natural to solve the problem using an iterative scheme.

In the case of the IPT approximation, that is when IPT is chosen to approximate the solver of the localized impurity models, the common numerical tools are based on the Matsubara frequencies formalism, as it is done in the TRIQS library [Par+15].

Recall that the Matsubara frequencies (I.3.2) are defined by $i\omega_n = i(2n+1)\frac{\pi}{\beta}$, where $n \in \mathbb{Z}$ and $\beta > 0$ is the inverse temperature. In the IPT approximation, the values of the self-energy of the AIM at the Matsubara frequencies $\Sigma(i\omega_n)$ are approximated by some values Σ_n defined by a formula explicited in Chap. IV using only the values of the hybridization at the Matsubara frequencies as well. In the first place, we do not consider the question of the analytic continuation, once these values are computed.

Hence, the IPT solver can be seen as a function

$$\text{Solv} := F^{\text{IPT}} : (\Delta(i\omega_n))_{n\in\mathbb{Z}} \mapsto \Sigma_n. \tag{I.3.6}$$

In order to perform numerical computations, the usual discretization considers a truncation of the number of Matsubara frequencies taken in account. Let $N_\omega \in \mathbb{N}$ be this cut-off. For symmetry reasons, the negative Matsubara frequencies do not carry relevant information, so the frequencies taken in account are $i\omega_0 = i\pi/\beta, \dots, i\omega_{N_\omega}$, see Fig. I.10.

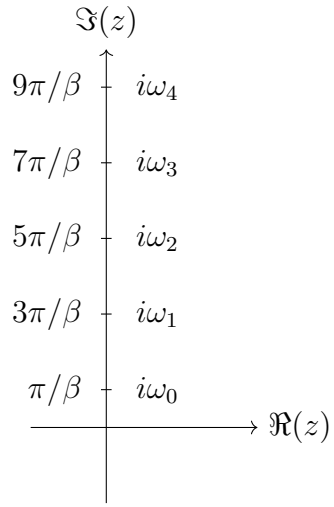


Figure I.10: The set $(i\omega_n)_{n\in[\![0,N_\omega]\!]}$ of points in $\mathbb{C}_+$ used for the discretization of Δ and Σ with $N_\omega = 4$.

From this naive truncation procedure, multiple mathematical questions arise, both from a theoretical and numerical point of view. This study is the topic of Chap. IV.

Existence and uniqueness. First, this formulation gives rise to a new set of equations, which we call the IPT-DMFT discretized equations, that are of the following form.

$$\begin{cases} (\Sigma_n)_{n\in[\![0,N_\omega]\!]} &= F^{\text{IPT}}_{N_\omega}\left((\Delta_n)_{n\in[\![0,N_\omega]\!]}\right) \\ (\Delta_n)_{n\in[\![0,N_\omega]\!]} &= F^{\text{SC}}\left((\Sigma_n)_{n\in[\![0,N_\omega]\!]}\right), \end{cases}$$

where the function F^{SC} derives from the self-consistency condition. The natural theoretical questions are then the following.

- Is the problem well-defined? In what spaces?
- Is there a solution?
- Is it unique?
- Are there conditions on the parameters U and T of the underlying Hubbard model?

Answer to these questions are provided in Chap. IV. We are able to prove that there is indeed a solution, for a certain range of parameters, via compactness arguments, and that it is unique for another range of parameters, thanks to a Picard fixed-point Theorem. We also investigated numerically and theoretically these questions for a specific Hubbard graph, namely the simplest, the Hubbard dimer.

Hubbard dimer. A particularly important class of systems is obtained when the underlying Hubbard graph $\mathcal{G}$ is bipartite. This framework encompasses several models of central interest in condensed matter physics, including graphene, and more generally regular trees (or Bethe lattices), the Lieb lattice, and hypercubic lattices in arbitrary dimensions [Geo+96]. It also covers periodic supercell approximations constructed from Lieb or hypercubic lattices, provided that the number of sites is even in each spatial direction.

The relevance of bipartite graphs stems from a fundamental symmetry of the Hubbard model. Since DMFT is designed to approximate the Green's function associated with the Gibbs state of the Hubbard Hamiltonian, any symmetry of the latter is expected to be reflected in the DMFT equations. In the case of a bipartite graph at half-filling, the Hubbard model enjoys a particle–hole symmetry [Lie04]. As shown in Chap. IV, this symmetry implies that the Green's function satisfies

$$G(z) = -G(-\overline{z})^{\dagger}, \qquad z \in \mathbb{C}_{+}.$$

In particular, the values of the Green's function at the Matsubara frequencies are anti-Hermitian matrices: $G(i\omega_n) + G(i\omega_n)^{\dagger} = 0$. This additional structure plays an important role in both the analysis of the DMFT equations and their numerical approximation, and motivates the study conducted in Chap. IV. Within the special case of bipartite graphs, the Hubbard dimer is the simplest non trivial case [LW68]. It corresponds to a graph with 2 vertices and one edge. Chap. IV focuses in particular on this simple model (for which it is actually possible to compute analytically the exact Green's function), which already provides insight into the structure of solutions to discretized IPT-DMFT equations, particularly in terms of the physicality of the solutions, the number of solutions (which are not always unique), the convergence rate of the iterative process, and so on.

Pick criterion. As mentioned previously, one of the objective of the analysis is to determine if the iterative scheme used in practice is compatible with the mathematical structure of the problem, namely that (Δ, Σ and G) must be the negative of Pick functions. Since, at this point, the algorithms provides with the values of these functions only at the imaginary Matsubara frequencies, the natural question becomes: is there (the negative of) a Pick function which interpolates these values? In other words, with $(\Delta_n^\infty)_{0\leq n\leq N_\omega}$ the output of the iterative scheme, is there a solution to the Analytic Continuation Problem (ACP)

$$\text{Find } -\Delta : \mathbb{C}_+ \to \overline{\mathbb{C}}_+ \text{ analytic, such that } \Delta(i\omega_n) = \Delta_n^\infty,\ 0 \leq n \leq N_\omega. \tag{I.3.7}$$

The questions around the ACPs have been studied since more than a hundred years, as mentioned previously. In particular, Georg Pick established the so-called Pick criterion [Pic15; Nic16], which consists in checking whether a finite collection of Hermitian matrices are positive semidefinite or not. Hence it suffices to compute the lowest (real) eigenvalue to check if it is negative, in which case there is no solution to (I.3.7). In Chap. IV, we are able to show that the Pick criterion is not satisfied by the converged solution for some range of parameters, even for the simple model of the Hubbard dimer. This motivates to develop new techniques and algorithms to take this in account, as mentioned in the next section. This may also explain the plurality of IPT-DMFT solutions already observed by physicists: some of these solutions may not correspond to Pick functions and therefore may not satisfy the causality condition.

I.3.4 Mathematical challenges in DMFT

The mathematical challenges associated with DMFT that remain to be addressed are numerous and varied. Some of the key challenges include the following.

- A natural extension of our results will be to establish in which cases there is uniqueness of the solution to the IPT-DMFT equations, whose existence is proved in this manuscript.

- A better understanding of the application of DMFT to the Hubbard model on lattices in the infinite dimensional limit, which was one of the first successes of the early development of the theory [Geo+96].

- Also in the spirit of Michael Lindsey's thesis [Lin19] and the work of Antoine Georges and collaborators [KFG15], it would be interesting to have a rigorous formulation of the Luttinger–Ward functional and the diagrammatic expansion of the quantum Green's function, and the consequences in terms of the formulation of DMFT.

- A rigorous mathematical formulation of the DMFT equations in the context of moiré materials, particularly in relation to the Hubbard model and its extensions,

is another problem that has to be addressed in the future. This would build the bridge bonding the different questions tackled in this thesis. This question is already addressed by the community from a numerical perspective, as mentioned in Sec. I.3.

- Develop a new algorithm to solve the IPT-DMFT discretized equations which takes into account the physicality of the problem, that is which respects the fact that $-G, -\Sigma$ and $-\Delta$ are Pick functions. This could be done by using the mathematical formulation proposed in Chap. III based on the representation of the physical objects by their Nevanlinna-Riesz measure. This could also circumvent the theoretical and practical problems associated with analytic interpolation.

II Kohn–Sham models for encapsulated two-dimensional materials

This chapter corresponds to a preprinted article [CGP26], written in collaboration with Éric Cancès[1] *and David Gontier*[1] *:* https://doi.org/10.48550/arXiv.2606.11785 .

II.1 Introduction

The study of 2D and moiré materials is one of the most vibrant fields in today's condensed matter physics. It is also a rich source of exciting mathematical and numerical problems. The physics of 2D materials blossomed after the seminal paper by Geim and Novoselov reporting experimental transport measurements on graphene [Nov+04], made possible by the famous scotch-tape exfoliation technique used to isolate monolayer graphene. In the following years, new experimental techniques were developed to exfoliate or grow high-quality monolayer graphene, as well as many other single-layer 2D materials, such as hexagonal boron nitride (hBN) or transition metal dichalcogenides (TMDs) [Tsc+24; Mia+21; Zan+22]. In parallel, multilayer 2D materials were synthesized by stacking several identical or different monolayers on top of one another, in aligned or twisted configurations. Lattice mismatches in hetero-bilayers (e.g. graphene on top of hBN), or rotation angles in twisted homo-bilayers (e.g. twisted bilayer graphene - TBG), lead to the appearance of moiré patterns, hence the name moiré materials.

For small lattice mismatches or small twist angles, the apparent moiré periodicity is much larger than the period of the underlying monolayer periodic structures and gives rise to new physical phenomena at the nanoscale. What makes 2D moiré materials so special is that, in contrast to bulk 3D materials or surfaces, the electronic structure of a 2D moiré material can be finely controlled, to some extent, by tuning different parameters: twist angle, mechanical strain, longitudinal and/or transverse electric and/or magnetic fields. In particular, the filling factor, that is the number of extra or missing electrons per moiré cell with respect to charge neutrality, can be controlled by placing the moiré material between two metallic electrodes and imposing a potential difference between

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these electrodes (a procedure called *gating*). To experimentally realize gating, the 2D material under study must first be encapsulated between two thin insulating layers (e.g. several layers of hBN) in order to avoid short circuits (see Fig. II.1).

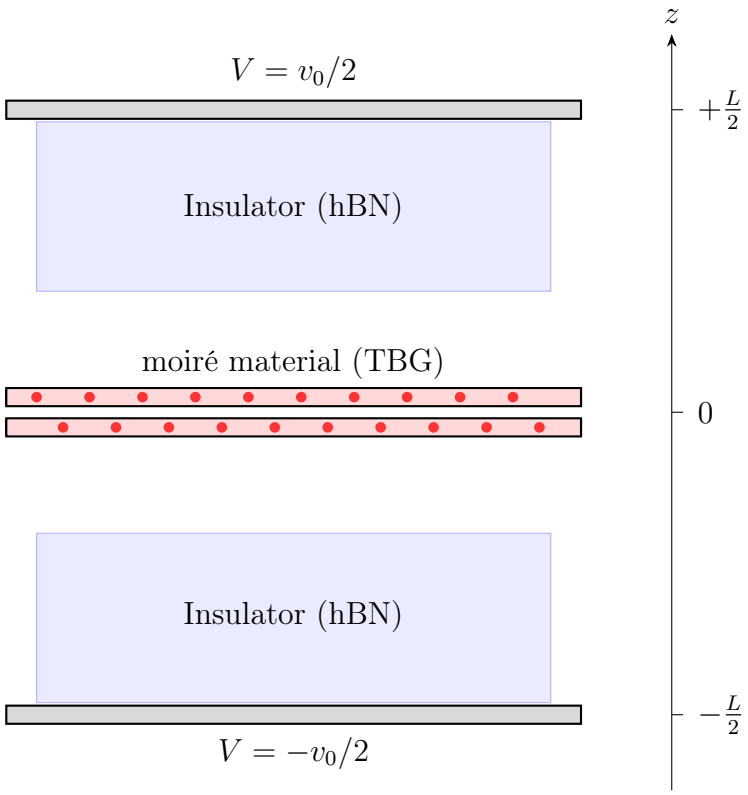


Figure II.1: Schematic diagram of the physical system.

A theoretical breakthrough in the study of moiré materials was made in 2011 by Bistritzer and MacDonald [BM11b; BM11a], who predicted the presence of so-called magic angles in TBG, at which strongly correlated electronic effects should emerge. This prediction was confirmed by experiments conducted in 2018 in Jarillo-Herrero's group [Cao+18a; Cao+18b], where strongly correlated insulating phases and, even more interestingly, *unconventional* superconducting phases (which cannot be explained by the classical BCS theory of superconductivity), were observed in the phase diagram of twisted bilayer graphene close to the first magic angle $\theta \simeq 1.1°$, for sufficiently low temperatures and suitable values of the filling factor. This discovery generated enormous interest in the condensed-matter community, because it opened new perspectives for understanding unconventional superconductivity and, ultimately, the still not fully understood high-temperature superconductivity of cuprates. Since then, correlated and superconducting phases have been observed in several other moiré materials, including twisted trilayer graphene (TTG) [Par+21], twisted double bilayer graphene (TDBG) [Liu+20], and moiré transition metal dichalcogenide (TMD) bilayers [Wan+20], as well as in non-moiré multilayer 2D materials such as rhombohedral graphene [**ZhoXieTan-21**].

In this field, perhaps more than in any other area of physics, modeling and experiment

are closely intertwined. Throughout the development of the field, the modeling of 2D and moiré materials has provided indispensable guidance for experimental studies, while at the same time being necessary to understand the rich and unusual electronic properties of these materials.

Long before monolayer graphene was experimentally realized, the electronic structure of graphene was investigated in 1947 by Wallace [Wal47]. Using a minimal model describing non-interacting electrons on a hexagonal lattice, Wallace was able to predict the semimetallic character of monolayer graphene at charge neutrality, as well as the fact that, for low-energy excitations, charge carriers (electrons and holes) behave as massless relativistic particles with Fermi velocity v_{F} ($v_{\mathrm{F}} \approx 1 \times 10^6$ m/s for graphene, which is 300 times slower than the true speed of light). These properties follow from the presence of Dirac cones at the K and K' points of the band diagram of graphene. The Bistritzer–MacDonald (BM) model is a remarkably successful minimal effective non-interacting model describing low-energy electronic excitations in TBG at the moiré scale. The BM Hamiltonian H_{BM} is a periodic self-adjoint operator on $L^2(\mathbb{R}^2;\mathbb{C}^4)$ ($L^2(\mathbb{R}^2;\mathbb{C}^8)$ if both the K and K' valleys are included), with unit cell given by the (mesoscopic) moiré cell. It depends on the twist angle θ and only three empirical parameters: the Fermi velocity v_{F} and two parameters w_0 and w_1 describing interlayer interactions at AA and AB stacking respectively. Bistritzer and MacDonald observed the emergence of almost flat bands at the Fermi level in the band diagram of H_{BM} for specific "magic" twist angles. Flat bands at the Fermi level suggest the existence of partially occupied localized electronic states, and the possibility of strongly correlated insulating or superconducting phases. While non-interacting one-particle models such as BM have proven to be invaluable in guiding physical intuition, they are not sufficient to make quantitative predictions on the phase diagram itself. To push the analysis further and numerically compute an approximate phase diagram, it is necessary to construct effective *interacting* moiré-scale models going beyond the non-interacting ones, and to simulate these models using, for instance, Dynamical Mean-Field Theory [Geo+96; Rai+24; Dat+23; Zan+22] (DMFT, see [CKP26] for some mathematical insights on this method).

The modeling of 2D and moiré materials has also stimulated mathematical research. Regarding graphene, Wallace's non-interacting lattice model was extended in two directions. On the one hand, the existence of Dirac cones at K and K' was shown by Fefferman and Weinstein [FW12] (see also [BC18]) to be a generic property of continuous 2D Schrödinger operators with honeycomb symmetries. On the other hand, Giuliani, Mastropietro and Porta [**GiuMas-09**; GMP10] studied an *interacting* model of electrons on hexagonal lattices using exact renormalization group techniques. Other mathematical works focused on the modeling of TBG. The BM model was derived from atomic-scale non-interacting tight-binding models by Watson and Luskin [**WatLus-21**], and a variant of it directly from Density Functional Theory (DFT) by Cancès, Garrigue and Gontier [CGG23]. The study of the BM model also gave rise to several mathe-

matical works. Building on previous works by Vishwanath and collaborators [TKV19], the existence of magic angles for the chiral approximation of the BM model was established in a series of works by Becker, Zworski et al. [Bec+22], and Watson, Kong, MacDonald and Luskin [Wat+23]. Extending previous works by Zaletel and collaborators [Bul+20], Becker, Lin, and Stubbs [BLS25] proved that the ground states of some interacting Bistritzer–MacDonald (IBM) models at charge neutrality are Hartree–Fock states with specific symmetries.

Among the wide variety of electronic structure models used in the study of 2D and moiré materials, Density Functional Theory (DFT) [HK64; KS65] plays an important role. DFT alone is generally unable to quantitatively reproduce the phase diagrams of moiré materials, as its purpose is only to compute the ground-state energy and one-particle density, quantities that do not *a priori* provide any direct insight into correlation functions, from which strongly correlated effects can be inferred (note, however, that a DFT theory for BCS superconductors has been developed [OGK88; Lüd+05]). On the other hand, DFT is arguably the best model for quantitatively studying atomic relaxation [Car+18b], an effect reported to play an important role in the electronic properties of multilayer 2D materials [NK17; Car+19; Car+18a], as well as phonon modes [Lu+22], and for providing reasonable approximations of electron-phonon coupling parameters [Bar+01]. It is also a valuable tool for parameterizing non-interacting and interacting models of 2D and moiré materials [Fan+18; CGG23].

In this article, we provide a mathematical analysis of Kohn–Sham models for encapsulated 2D and moiré materials (experimental setting sketched in Fig. II.1) at the reduced Hartree–Fock (rHF), also called Hartree, or RPA (Random Phase Approximation), level of theory. On the Jacob's ladder of DFT approximations proposed by Perdew [PS01], this setting corresponds to the ground level, for which the exchange-correlation functional is chosen to be zero. Although this model is not sufficiently accurate to obtain quantitative results, it is very interesting from a mathematical viewpoint, as its mathematical structure is similar to that of practically used Kohn–Sham models, with the additional property of being convex, which greatly simplifies its analysis. Note that the exact DFT model based on Lieb's density functional [Lev79; Lie83] is convex, and that non-convexities are in fact induced by (semi-)local approximations of the exchange-correlation functional. Within the rHF approximation, it is possible to prove the stability of neutral molecules [LS77; CL93], the bulk limit for certain perfect crystals [CLL98; CLL01], the existence of electronic ground states for crystals with local defects [CDL08], disordered crystals such as doped semiconductors [CLL13], and standalone homogeneous 2D materials [GLM21], and many other results.

In this contribution, we focus on periodic or quasi-periodic 2D structures, for which an *rHF energy per unit area* can be defined. Examples of periodic 2D materials include monolayers, homo-bilayers (such as bilayer graphene in AA or AB stacking), homo-multilayers (such as rhombohedral graphene), and twisted homo-bilayers or homo-multilayers with

commensurate twist angles. The geometric structure of these physical systems (with a focus on bilayers), for both the periodic and quasi-periodic settings, as well as the concept of stationary function, are recalled in Section II.2.

As in most DFT calculations performed in the encapsulated setting, the insulating encapsulating material is modeled as a continuous dielectric medium. In this approach, only the electrons of the 2D material under study (shown in red in Fig. II.1) are explicitly taken into account. In Section II.3, we provide useful results on electrostatics in the encapsulated setting. A key feature of this setting is that the conducting electrodes impose Dirichlet boundary conditions on the electrostatic potential, which effectively screen the Coulomb interaction. Most of the results presented are classical and follow from the theory of linear elliptic equations in divergence form, but we recall them for the sake of completeness. We also provide formulae for solving the Poisson equation in the special case where the dielectric permittivity only depends on the z variable, which are useful for numerical simulations.

Section II.4 deals with the quasi-periodic case. It covers twisted homo-bilayers, notably TBG, at generic (incommensurate) twist angles, as well as quasi-periodic multilayer structures such as generic twisted trilayer graphene (TTG) configurations. A large body of the mathematical literature is devoted to the study of linear quasi-periodic Hamiltonians [Shu78; AJ09; Sim82; Eli92; PF92; LSZ26], see also [CCL17; ZCZ19; LSP22; Ett+20; Wan+25a; Can+25a] for numerical methods. But to the authors' knowledge, very few contributions deal with nonlinear DFT-like models for quasi-periodic systems. The only works we are aware of are the studies of the quasi-periodic Thomas–Fermi (TF) and Thomas–Fermi–von Weizsäcker (TFW) models by Blanc, Le Bris and Lions [BLL07], which fall within the scope of orbital-free DFT. To handle quasi-periodic Kohn–Sham models, for which the main variable is the one-body density matrix (1-RDM), we rely on the C^* and von Neumann algebra framework introduced by Bellissard and collaborators [Bel02; BvS94], based on ideas of Connes [Con94]. We adapt it to the encapsulated setting and prove the existence of an rHF ground state for quasi-periodic systems in Section II.4. The main arguments are inspired from those in [CLL13].

Finally, for completeness, the encapsulated Kohn–Sham model for periodic systems is detailed and analyzed in Section II.5. The existence of an rHF ground state in the periodic setting is a simple exercise, and this result can be easily extended to some non-convex models such as Kohn–Sham LDA (local density approximation).

II.2 Geometric description of the physical systems under consideration

We begin by describing in detail the geometry of the systems in the two settings covered by this analysis: the periodic one, and the quasi-periodic one. To simplify the presentation,

we focus on bilayer systems, but our arguments obviously extend to generic periodic monolayer and multilayer systems, as well as to stationary ergodic multilayer systems.

II.2.1 Encapsulated 2D materials

As in implicit solvent models commonly used in quantum chemistry [TMC05], only the electrons of the 2D material are dealt with explicitly. This is justified by the fact that the Fermi level of the system is in the band gap of the encapsulating insulator and that we are only interested in low-energy phenomena which cannot create electron-hole pairs in the insulator.

This means that we will study a three dimensional system, but in which the electronic density is localized around the plane $z = 0$ (Fig. II.1). We denote by $L > 0$ the (fixed) distance between the two insulating layers, and set

$$I_L := \left(-\frac{L}{2}, \frac{L}{2}\right) \quad \text{and} \quad S_L := \mathbb{R}^2 \times I_L.$$

We often write $(x, z) \in \mathbb{R}^2 \times I_L$, with $x = (x_1, x_2) \in \mathbb{R}^2$ the two longitudinal coordinates, and z the one orthogonal to the system.

Let us point out that, since I_L is bounded, electrons cannot escape to $\pm\infty$ in the z-direction. This is one of the key advantages of the encapsulated setting for the mathematical analysis. Unfortunately, most of our results do not extend to standalone 2D materials (see [GLM21; GLM23] for the case of homogeneous 2D materials in 3D space).

II.2.2 Bilayer 2D materials

Still for the sake of simplicity, we assume that each layer is a perfectly periodic monolayer but our results extend to the case when atomic relaxation is taken into account (see Remark II.2.2). In this simple case considered, the nuclear charge density m is of the form

$$m(x, z) = m_1(x, z) + m_2(x, z),$$

where, for all $j = 1, 2$ and $z \in I_L$, the map $x \mapsto m_j(x, z)$ is $\mathbb{L}_j$–periodic, with $\mathbb{L}_j$ a lattice of $\mathbb{R}^2$. We set

$$\Omega_1 := \mathbb{R}^2/\mathbb{L}_1, \quad \Omega_2 := \mathbb{R}^2/\mathbb{L}_2 \quad \text{and} \quad \Omega := \Omega_1 \times \Omega_2.$$

The sets Ω_1 and Ω_2 are two 2D-tori, and the set Ω is a 4D-torus called the configuration space.

For a **configuration** $\omega = (\omega_1, \omega_2) \in \Omega = \Omega_1 \times \Omega_2$, we set

$$m_\omega(x, z) := m_1(x - \omega_1, z) + m_2(x - \omega_2, z). \tag{II.2.1}$$

In other words, ω_j encodes the shift by ω_j of the j-th lattice in the longitudinal direction. We introduce the group action $\mathbb{R}^2 \ni x \mapsto \alpha_x$ defined by

$$\alpha_x : \Omega \to \Omega, \qquad \alpha_x(\omega_1, \omega_2) := (\omega_1 - x, \omega_2 - x) \in \Omega,$$

where $\omega_1 - x$ and $\omega_2 - x$ are seen as elements of the torus Ω_1 and Ω_2 respectively, that is are taken modulo $\mathbb{L}_1$ and $\mathbb{L}_2$ respectively. Note that α_x is measure preserving (for the Lebesgue measure for Ω).

In what follows, we consider functions $f = f(\omega, x, z) : \Omega \times \mathbb{R}^2 \times I_L \to \mathbb{C}$, and we often write $f_\omega(x, z)$ for $f(\omega, x, z)$. We say that such a function f is **stationary** if

$$\forall x' \in \mathbb{R}^2, \qquad f(\omega, x, z) = f(\alpha_{x'}\omega, x - x', z).$$

In particular, taking $x' = x$ leads to $f(\omega, x, z) = f(\alpha_x\omega, 0, z)$. In fact, stationary functions are of the form $f(\omega, x, z) := \mathfrak{f}(\alpha_x\omega, z) = \mathfrak{f}(\omega_1 - x, \omega_2 - x, z)$ for some $\mathfrak{f} : \Omega \times I_L \to \mathbb{C}$, called the **generating function** of f. For instance, $m(\omega, x, z)$ is a stationary function with generating function

$$m_\omega(x, z) := \mathfrak{m}(\alpha_x\omega, z), \quad \text{where} \quad \mathfrak{m}(\omega, z) := m_1(-\omega_1, z) + m_2(-\omega_2, z).$$

Remark II.2.1. *We restrain our study to bilayer 2D materials, but the case of multi-layers can be tackled the same way, with no additional mathematical difficulty.*

II.2.3 Periodic vs incommensurate systems

In the modeling of bilayer materials, two cases are possible depending on the lattices $\mathbb{L}_1$ and $\mathbb{L}_2$.

The periodic case (commensurate lattices). Consider first the case where $\mathbb{L} := \mathbb{L}_1 \cap \mathbb{L}_2$ is also a discrete lattice of $\mathbb{R}^2$. In this case, for all $\omega \in \Omega$, the charge m_ω is a $\mathbb{L}$–periodic function. In this case, the physical properties of the system depend on the configuration $\omega \in \Omega$. For instance, in the case of two parallel graphene sheets with $\mathbb{L}_1 = \mathbb{L}_2 = \mathbb{L} = \mathbb{Z}a_1 + \mathbb{Z}a_2$ representing the same triangular lattice, the AA stacking corresponds to $\omega_1 = \omega_2$, while the AB stacking (the equilibrium configuration for standalone untwisted bilayer graphene) corresponds to $\omega_1 = \omega_2 + \frac{1}{3}(a_1 + a_2)$. These two configurations have very different electronic properties.

In the general case of twisted materials with commensurate twist angle, it may happen that the unit cell of $\mathbb{L}$ contains a large number of atoms, so that numerical computations can be very challenging. But from a mathematical viewpoint, we are in a periodic setting, which has been studied extensively in the literature [CLL01; CLL02; GL16]. The main difference with previous works comes from the encapsulated setting, which changes the electrostatic properties of the Coulomb interaction. We refer to Section II.5 below for our results in the periodic case, and additional comments on the structural difference with the quasi-periodic case.

The quasi-periodic case (incommensurate lattices). The group action α is called **ergodic** on Ω (for the Lebesgue measure) if the only measurable functions $\mathfrak{f} : \Omega \to \mathbb{C}$ that are α-invariant, that is $\mathfrak{f} \circ \alpha_x = \mathfrak{f}$ for all $x \in \mathbb{R}^2$, are the constant functions. In our case, Fourier decomposing $\mathfrak{f} : \Omega = \Omega_1 \times \Omega_2 \to \mathbb{C}$ as

$$\mathfrak{f}(\omega) = \sum_{G \in \mathbb{L}^*} \widehat{\mathfrak{f}}_G \mathrm{e}^{\mathrm{i}(G_1\omega_1 + G_2\omega_2)}, \quad \text{so that} \quad \mathfrak{f}(\alpha_x\omega) = \sum_{G \in \mathbb{L}^*} \widehat{\mathfrak{f}}_G \mathrm{e}^{-\mathrm{i}x(G_1+G_2)} \mathrm{e}^{\mathrm{i}(G_1\omega_1 + G_2\omega_2)},$$

(the sum runs over $G = (G_1, G_2) \in \mathbb{L}_1^* \times \mathbb{L}_2^* =: \mathbb{L}^*$), we see that $\mathfrak{f} \circ \alpha_x = \mathfrak{f}$ for all x iff $\widehat{\mathfrak{f}}_G = 0$ for all G with $G_1 + G_2 \neq 0$. We deduce that the action is ergodic if and only if the two lattices are **incommensurate**, in the sense $\mathbb{L}_1^* \cap \mathbb{L}_2^* = \{0\}$, that is

$$\forall\, G_1 \in \mathbb{L}_1^*, \quad \forall\, G_2 \in \mathbb{L}_2^*, \qquad G_1 + G_2 = 0 \quad \Longleftrightarrow \quad G_1 = G_2 = 0.$$

The Birkhoff-Von Neumann ergodic theorem [Moo15; PF92; CL12] states that if $f(\omega, x) = \mathfrak{f}(\alpha_x\omega, 0)$ is stationary, and if $\omega \mapsto f(\omega, x_0)$ is integrable for some $x_0 \in \mathbb{R}^2$, then we have, setting $B_R := \{x \in \mathbb{R}^2 : \|x\| < R\}$,

$$\lim_{R\to\infty} \fint_{B_R} f(\omega_0, x)\mathrm{d}x = \fint_{\Omega} f(\omega, x_0)\mathrm{d}\omega = \fint_{\Omega} \mathfrak{f}(\omega)\,\mathrm{d}\omega, \tag{II.2.2}$$

for almost every $x_0 \in \mathbb{R}^2$ and almost every $\omega_0 \in \Omega$. Here, we use the notation $\fint_{\mathcal{A}} := \frac{1}{|\mathcal{A}|}\int_{\mathcal{A}}$ for normalized integrals on the specified domain. In other words, the *averaged integral per unit area* is well-defined for stationary functions, and independent of the configuration $\omega_0 \in \Omega$.

In view of the Birkhoff–Von Neumann ergodic theorem, we expect that, in the incommensurate case, the physical properties of the material are independent of the configuration $\omega \in \Omega$.

Note that **intermediate cases** are also possible. For instance when $\mathbb{L}_1$ and $\mathbb{L}_2$ are commensurate in one direction but incommensurate in the other (take $\mathbb{L}_1 = \mathbb{Z} \oplus \mathbb{Z}$ and $\mathbb{L}_2 = \mathbb{Z} \oplus \sqrt{2}\mathbb{Z}$). We do not consider such situations in the present article.

Remark II.2.2 (Atomic relaxation)**.** *As already mentioned in the introduction, atomic relaxation plays an important role in the physics of multilayer 2D materials. For unrelaxed periodic configurations, and in the absence of translation symmetry breaking, the relaxed configuration is also periodic and our arguments are unchanged. In the incommensurate case, the configuration space formalism, used in this contribution to study the electronic structure of the system, can be also used to study atomic relaxation in multilayer 2D materials [CLM19] (here also under a non-symmetry-breaking assumption on atomic displacements). After relaxation, the nuclear distribution is no longer the sum of two periodic functions, but still is a stationary function of the form* $m_{\mathrm{relaxed}}(\omega, x, z)$*. For these reasons, all our results also apply to relaxed configurations (in the absence of symmetry breaking).*

II.3 Encapsulated electrostatics

The encapsulated setting modifies the effective Coulomb interaction between the charged particles (nuclei and electrons) of the 2D material. As will be seen below, the encapsulated Coulomb potential shares properties which are closer to a Yukawa potential (exponential decay of the Green's kernel) than a full space Coulomb potential (polynomial decay of the Green's kernel). This will allow us to define the encapsulated Coulomb potential for densities which do not decay at infinity (such as periodic or stationary densities).

Recall that the thin layers of insulating material are modeled by a polarizable continuum medium. We model this with dielectric permittivity ϵ depending solely on the transverse coordinate z such that

$$1 \leq \epsilon(z) \leq \epsilon_{\mathrm{r}} \quad \text{a.e.}, \tag{II.3.1}$$

where ϵ_{r} is the dielectric permittivity of the insulating material. The potential difference v_0 between the lower and upper electrodes is tunable. Without loss of generality, we can assume that the potentials of the lower and upper electrodes are $v_0/2$ and $-v_0/2$ respectively (see Fig. II.1).

II.3.1 Finite charge distributions

The results in this section are classical and are just recalled for the sake of completeness. Let $f \in H^{-1}(S_L)$ be an electronic charge density (in what follows, $f = \rho - m$ is the difference between the electronic density $\rho \geq 0$ and the charge density $m \geq 0$). The electrostatic potential V_{f,v_0} generated by this charge density between the electrodes is solution to the inhomogeneous Dirichlet problem

$$\begin{cases} -\mathrm{div}(\epsilon \nabla V_{f,v_0}) = 4\pi f & \text{in} \quad S_L, \\ V_{f,v_0}(\cdot, \pm L/2) = \pm v_0/2. \end{cases} \tag{II.3.2}$$

We will mainly focus on the case $v_0 = 0$, and study the Dirichlet problem

$$\text{Find} \quad V_f \in H^1_0(S_L) \quad \text{solution to} \quad -\mathrm{div}(\epsilon \nabla V_f) = 4\pi f \quad \text{in} \quad S_L. \tag{II.3.3}$$

In this case, the boundary condition $V_f(x, \pm L/2) = 0$ has been encoded through the Sobolev space $H^1_0(S_L)$. The reduction to the case $v_0 = 0$ is justified by the following decomposition

$$V_{f,v_0} = V_f + v_0 g_0, \tag{II.3.4}$$

where $g_0 = g_0(z) \in H^1(I_L)$ is the function

$$g_0(z) := \frac{\int_{-L/2}^{z} \frac{\mathrm{d}t}{\epsilon(t)}}{\int_{-L/2}^{L/2} \frac{\mathrm{d}t}{\epsilon(t)}} - \frac{1}{2}, \quad \text{which solves} \quad \begin{cases} -\partial_z(\epsilon \partial_z g_0) & = 0 \quad \text{in} \quad I_L \\ g_0(\pm L/2) & = \pm 1/2 \end{cases}. \tag{II.3.5}$$

Equation (II.3.3) is a classical problem. First, we recall that the weak formulation of (II.3.3) is

$$\forall h \in H^1_0(S_L), \qquad \int_{S_L} \epsilon \nabla V_f \cdot \nabla h = 4\pi \langle f, h \rangle_{H^{-1}(S_L), H^1_0(S_L)}. \tag{II.3.6}$$

Recall Lax–Milgram theory, which states the following.

Proposition II.3.1. *Let $f \in H^{-1}(S_L)$. There exists a unique $V_f \in H^1_0(S_L)$ solution to (II.3.3).*

For brevity, we skip the elementary proof of this result, but emphasize that it relies on the Poincaré inequality in the z–direction, which implies that

$$\forall V \in H^1_0(S_L), \quad \|V\|_{L^2(S_L)} \leq \frac{L}{\pi} \|\nabla V\|_{L^2(S_L)}. \tag{II.3.7}$$

In particular, the homogeneous norm $V \mapsto \|\nabla V\|_{L^2(S_L)}$ is equivalent to the $H^1(S_L)$ norm, and we have the embedding $H^1_0(S_L) \hookrightarrow L^2(S_L)$. We also have $H^1_0(S_L) \hookrightarrow L^6(S_L)$ by Sobolev inequality.

For any $f \in H^{-1}(S_L)$, we define the **encapsulated Hartree energy** of f by

$$\mathcal{D}(f, f) := \langle f, V_f \rangle_{H^{-1}(S_L), H^1_0(S_L)} = \frac{1}{4\pi} \int_{S_L} \epsilon |\nabla V_f|^2 \geq 0.$$

The last inequality, together with (II.3.7), shows that the quadratic form $f \mapsto \mathcal{D}(f, f)$ is positive definite. In particular, it is strictly convex. We also recall a well-known result.

Lemma II.3.2. *If $f \geq 0$, then $V_f \geq 0$ as well.*

Proof. We take $h = V_- = \max\{0, -V_f\} \in H^1_0(S_L)$ in the weak-formulation (II.3.6), and get

$$0 \geq -\int_{S_L} \epsilon |\nabla V_-|^2 = 4\pi \langle f, V_- \rangle_{H^{-1}(S_L), H^1_0(S_L)} \geq 0.$$

So both terms are null. Hence $\nabla V_- = 0$, so that, in view of (II.3.7), $V_- = 0$ almost everywhere. So $V_f \geq 0$. □

Actually, using Stampacchia's theory [Sta65] (see also [GT01, Chapter 8]), we can prove the following

Lemma II.3.3. *If $f \in H^{-1}(S_L) \cap L^p(S_L)$ with $p > \frac{3}{2}$, then $V_f \in L^\infty(S_L)$, and there is a constant $C_p > 0$ such that*

$$\forall f \in H^{-1}(S_L) \cap L^p(S_L), \quad \|V_f\|_{L^\infty(S_L)} \leq C_p \|f\|^{\frac{p}{5p-6}}_{L^p(S_L)} \|f\|^{\frac{4p-6}{5p-6}}_{H^{-1}(S_L)}.$$

In addition, for all $\frac{3}{2} < p \leq 2$, there is a constant $C_{p,L} > 0$ such that

$$\forall f \in L^p(S_L), \quad \|V_f\|_{L^\infty(S_L)} \leq C_{p,L} \|f\|_{L^p(S_L)} \qquad \textit{(case } \tfrac{3}{2} < p \leq 2\textit{)}. \tag{II.3.8}$$

Let us provide a proof for completeness, which only uses the weak-formulation (II.3.6), and works in particular in the case where $\epsilon(\cdot)$ is not a smooth function.

Proof. For $t > 0$, we define

$$\phi_t := (V_f - t)_+ \in H^1_0(S_L), \quad A(t) := \{V_f > t\} \subset S_L \quad \text{and} \quad \mu(t) := |A(t)|.$$

Taking ϕ_t as a test function in (II.3.6) using $\epsilon \geq 1$, and Hölder's inequality, we get

$$\|\nabla\phi_t\|^2_{L^2(S_L)} \leq \int_{S_L} \epsilon|\nabla\phi_t|^2 = 4\pi \int_{S_L} f\phi_t \leq 4\pi\|f\|_{L^p(S_L)}\|\phi_t\|_{L^{p'}(S_L)}. \tag{II.3.9}$$

Since $p > \frac{3}{2}$, we have $p' < 3$ (and in particular, $1 \leq p' \leq 6$). On the other hand, using that $\phi_t \in H^1_0(A(t))$ with $A(t)$ a bounded set, we can use Hölder's inequality and the Sobolev inequality $H^1_0(S_L) \hookrightarrow L^6(S_L)$ so get

$$\|\phi_t\|_{L^{p'}(S_L)} \leq \|\phi_t\|_{L^6(S_L)}|A(t)|^\alpha \leq C\|\nabla\phi_t\|_{L^2(S_L)}\mu(t)^\alpha \quad \text{with} \quad \frac{1}{6} + \frac{\alpha}{1} = \frac{1}{p'} = 1 - \frac{1}{p}.$$

Injecting into (II.3.9) gives

$$\|\nabla\phi_t\|_{L^2(S_L)} \leq C\|f\|_{L^p(S_L)}\,\mu(t)^\alpha. \tag{II.3.10}$$

Finally, for $s > t$, since $V_f - t \geq s - t$ on $A(s) \subset A(t)$, we also have the Markov type inequality

$$(s-t)\,\mu(s) \leq \int_{A(s)} (V_f - t) \leq \|\phi_t\|_{L^6(S_L)}\,\mu(s)^{5/6} \leq C\|\nabla\phi_t\|_{L^2(S_L)}\,\mu(s)^{5/6}.$$

Combined with (II.3.10), this gives

$$\mu(s) \leq \frac{C\|f\|^6_{L^p(S_L)}}{(s-t)^6}\,\mu(t)^\beta, \qquad \beta := 6\alpha = 5 - \frac{6}{p}. \tag{II.3.11}$$

Since $p > 3/2$, we have $\beta > 1$.

In the original works by Stampacchia, which was restricted to bounded domains B, the iteration can be started at $t_0 = 0$ (using $\mu(t_0) \leq |B|$). In our case however, S_L is unbounded, and we need to start at $t_0 > 0$. To control $\mu(t_0)$, we use that $V_f \in H^1_0(S_L) \hookrightarrow L^6(S_L)$ with $\|V_f\|_{L^6(S_L)} \leq C\|f\|_{H^{-1}(S_L)}$, and get the Markov inequality

$$\mu(t_0) \leq \frac{\|V_f\|^6_{L^6(S_L)}}{t_0^6} \leq \frac{C\|f\|^6_{H^{-1}(S_L)}}{t_0^6} < \infty.$$

The Stampacchia iteration lemma [Sta65; KS00] applied to (II.3.11) yields $\mu(t_0 + T) = 0$ with $T = C\|f\|_{L^p}\mu(t_0)^{(\beta-1)/6}$. So, for all $t_0 > 0$, we have

$$\|V_f\|_{L^\infty(S_L)} \leq t_0 + T \leq t_0 + C\|f\|_{L^p}\mu(t_0)^{\frac{\beta-1}{6}} \leq t_0 + C'\|f\|_{L^p}\|f\|^{\beta-1}_{H^{-1}}\frac{1}{t_0^{\beta-1}}.$$

Optimizing over $t_0 > 0$ and taking $t_0 = \left((\beta-1)C'\|f\|_{L^p}\|f\|_{H^{-1}}^{\beta-1}\right)^{1/\beta}$ yields, as wanted,

$$\|V_f\|_{L^\infty(S_L)} \leq C''\|f\|_{L^p(S_L)}^{\frac{1}{\beta}}\|f\|_{H^{-1}(S_L)}^{\frac{\beta-1}{\beta}} = C''\|f\|_{L^p(S_L)}^{\frac{p}{5p-6}}\|f\|_{H^{-1}(S_L)}^{\frac{4p-6}{5p-6}}.$$

Applying the same argument to $-V_f$ concludes the proof of the first inequality of Lemma II.3.3. To prove the second one, we recall the Sobolev embedding $H_0^1(S_L) \hookrightarrow L^2(S_L) \cap L^6(S_L)$, see (II.3.7), so that $L^2(S_L) + L^{6/5}(S_L) \hookrightarrow H^{-1}(S_L)$, and in particular, $L^p(S_L) \hookrightarrow H^{-1}(S_L)$ for all $\frac{3}{2} < p \leq 2$.

□

For encapsulated 2D materials, the electronic density is not in $H^{-1}(S_L)$, since it does not decay in the longitudinal directions. In the physical settings considered here, we expect it to be either periodic or quasi-periodic. In order to prove that the corresponding potential is a bounded function, we need to study the Green's function $\mathcal{G}$ of the operator $-\mathrm{div}(\epsilon\nabla\cdot)$ in S_L with homogeneous Dirichlet boundary conditions.

II.3.2 Green's function

We denote by $\mathcal{L}$ the self-adjoint operator

$$\mathcal{L}u := -\mathrm{div}(\epsilon\nabla u),$$

on $L^2(S_L)$ with domain $H^2(S_L) \cap H_0^1(S_L)$ and form domain $H_0^1(S_L)$. Since ϵ only depends on the z-variable, we have

$$\mathcal{L} := -\Delta_x \otimes \epsilon(z) + \mathbb{I}_{L^2(\mathbb{R}^2)} \otimes \widetilde{\mathcal{L}}, \tag{II.3.12}$$

where $\widetilde{\mathcal{L}}$ is the self-adjoint operator $L^2(I_L)$ with domain $H^2(I_L) \cap H_0^1(I_L)$ and form domain $H_0^1(I_L)$, defined by $\widetilde{\mathcal{L}}(u) := -\partial_z(\epsilon\partial_z u)$. The operator $\widetilde{\mathcal{L}}$ is coercive and has a compact resolvent.

Lemma II.3.4. *We have the operator bound* $\mathcal{L} \geq \frac{\pi^2}{L^2} > 0$.

Proof. Let μ_1 denote the lowest eigenvalue of $\widetilde{\mathcal{L}}$. Since $\epsilon \geq 1$, we have $\widetilde{\mathcal{L}} \geq -\partial_{zz}^2$ on $H_0^1(I_L)$. Considering the lowest eigenvalue of both operators, we get $\mu_1 \geq \frac{\pi^2}{L^2}$. Finally, we have $\mathcal{L} \geq \mathbb{I} \otimes \widetilde{\mathcal{L}}$, so $\mathcal{L} \geq \mu_1 \geq \frac{\pi^2}{L^2}$ as well. □

The fact that $\mathcal{L} \geq \mu_1 > 0$, hence has a spectral gap at energy 0, shows that the encapsulated Coulomb potential shares properties which are closer to those of a Yukawa potential than to the usual 3D Coulomb potential. In some sense, the Dirichlet boundary conditions screen the electrostatic potential.

Let us define the Green's function $\mathcal{G}(x,z;x',z') := \mathcal{G}_{x',z'}(x,z)$ as the distributional solution to

$$-\mathrm{div}(\epsilon\nabla\mathcal{G}_{x',z'}) = 4\pi\delta_{x',z'}, \qquad \mathcal{G}_{x',z'}(x,\pm L/2) = 0.$$

This function can be seen as the kernel of the operator $\mathcal{L}^{-1}$ from $L^2(S_L)$ to itself. By longitudinal-translation invariance, we have $\mathcal{G}(x,z;x',z') = \mathcal{G}(x-x';z,z')$. By the maximum principle, the function $\mathcal{G}$ is nonnegative.

II.3.3 Explicit expression of the Green's function

The function $\mathcal{G}$ is the kernel of the operator $\mathcal{L}^{-1}$ with

$$\mathcal{L} := -\Delta_x \otimes \epsilon(z) + \mathbb{I}_{L^2(\mathbb{R}^2)} \otimes \widetilde{\mathcal{L}},$$

and where $\widetilde{\mathcal{L}}$ is the self-adjoint operator on $L^2(I_L)$ with form domain $H^1_0(I_L)$ defined by $\widetilde{\mathcal{L}}(u) := -\partial_z(\epsilon\partial_z u)$. Because of the ϵ factor in the first term, we consider the modified 1D operator

$$\widetilde{\mathcal{L}}_\epsilon u := -\frac{1}{\epsilon}\partial_z(\epsilon\partial_z u),$$

acting on the weighted Hilbert space $L^2_\epsilon := L^2(I_L, \epsilon(z)\mathrm{d}z)$, with form domain $H^1_0(I_L)$. We denote by

$$\langle f, g\rangle_{L^2_\epsilon} := \int_{I_L} \overline{f(z)} g(z)\epsilon(z)\mathrm{d}z.$$

It can be checked that $\widetilde{\mathcal{L}}_\epsilon$, with form domain $H^1_0(I_L)$, is self-adjoint on L^2_ϵ, positive and with compact resolvent. We consider its spectral decomposition of the form

$$\widetilde{\mathcal{L}}_\epsilon = \sum_{k=1}^{\infty} \lambda_k |g_k\rangle\langle g_k|, \qquad 0 < \lambda_1 \leq \lambda_2 \leq \cdots, \qquad \langle g_k, g_\ell\rangle_{L^2_\epsilon} := \int_{I_L} \epsilon g_k g_\ell = \delta_{k\ell}.$$

In particular, the eigenstates g_k are solutions to the generalized elliptic eigenproblem

$$-\partial_z\left(\epsilon\partial_z g_k\right) = \lambda_k \epsilon g_k, \qquad g_k \in H^1_0(I_L).$$

Consider a charge distribution $f \in C^\infty_0(S_L)$ and the corresponding potential V_f, so that

$$\epsilon(z)(-\Delta_x V_f(x,z)) - \partial_z\big(\epsilon(z)\partial_z V_f(x,z)\big) = 4\pi f(x,z) = 4\pi\epsilon(z)\left(\epsilon(z)^{-1} f(x,z)\right). \quad \text{(II.3.13)}$$

For each $x \in \mathbb{R}^2$, we expand $\epsilon^{-1}f(x,\cdot)$ and $V_f(x,\cdot)$ in the (g_k) basis (for the L^2_ϵ inner product), giving an expansion of the form

$$f(x,z) = \epsilon(z)\sum_{k=1}^{\infty} f_k(x) g_k(z), \qquad V_f(x,z) = \sum_{k=1}^{\infty} V_k(x) g_k(z). \quad \text{(II.3.14)}$$

We multiply (II.3.13) by $g_k(z)$ and integrate on I_L. Using the orthonormality of the (g_k) basis for the L^2_ϵ inner product, we get

$$-\Delta_x V_k(x) + \lambda_k V_k(x) = 4\pi f_k(x), \quad \text{(II.3.15)}$$

from which we infer that

$$V_k = 2 f_k \star Y_{\sqrt{\lambda_k}},$$

where Y_κ denotes the 2D Yukawa potential, that is the Green's function of the Poisson equation

$$-\Delta Y_\kappa + \kappa^2 Y_\kappa = 2\pi\delta_0 \quad \text{in } \mathcal{S}'(\mathbb{R}^2).$$

Recall that we have

$$\forall x \in \mathbb{R}^2 \setminus \{0\}, \quad Y_\kappa(x) = Y_1(\kappa x), \quad Y_1(x) = K_0(|x|),$$

where K_0 is the modified Bessel function of the second kind. We finally get

$$\mathcal{G}(x,z,z') = 2\sum_{k=1}^{\infty} Y_{\sqrt{\lambda_k}}(x) g_k(z) g_k(z') = 2\sum_{k=1}^{\infty} K_0\left(\sqrt{\lambda_k}|x|\right) g_k(z) g_k(z'). \tag{II.3.16}$$

This formula allows to prove the exponential decay of the Green's kernel.

Lemma II.3.5. *For all compact $\Gamma \subset \mathbb{R}^2$ containing a neighborhood of 0, there exist $C_\Gamma > 0$ and $\mu > 0$ so that*

$$\forall (x,z),(x',z') \in S_L, \qquad (x-x') \notin \Gamma \implies |\mathcal{G}(x,z;x',z')| \leq C_\Gamma \, \mathrm{e}^{-\sqrt{\mu}|x-x'|}.$$

Proof. The functions g_k are in $H^1_0(I_L)$, are L^2_ϵ-normalized, i.e. $\int_{I_L} \varepsilon(z)|g_k|^2(z)\,dz = 1$, and satisfy

$$-\partial_z\left(\epsilon \partial_z g_k\right) = \lambda_k \epsilon g_k. \tag{II.3.17}$$

It is a classical result that $\lambda_k \sim_{k\to\infty} \frac{\pi^2 k^2}{L^2}$ [Zet05, Thm. 4.3.1]. Multiplying (II.3.17) by g_k and integrating shows that

$$\|g_k'\|^2_{L^2(I_L)} \leq \int_{I_L} \epsilon |g_k'|^2 = \lambda_k.$$

A variant of the Gagliardo–Nirenberg inequalities then shows that $g_k \in L^\infty(I_L)$ with $\|g_k\|_\infty \leq \sqrt{2}\,\lambda_k^{1/4}$. Indeed, we have, for any $z \in I_L$, since $\epsilon \geq 1$,

$$g_k^2(z) = \int_{-\frac{L}{2}}^{z} 2 g_k(s) g_k'(s) \mathrm{d}s \leq 2\|g_k\|_{L^2(I_L)} \|g_k'\|_{L^2(I_L)} \leq 2\sqrt{\lambda_k}.$$

Recall that the Bessel function K_0 is a smooth positive function on $(0,+\infty)$, satisfying

$$K_0(s) \underset{s\to 0^+}{\sim} -\ln(s), \qquad K_0(s) \underset{s\to+\infty}{\sim} \sqrt{\frac{\pi}{2s}} \mathrm{e}^{-s}.$$

In particular, for all $s_0 > 0$, there is $C \geq 0$ such that

$$\forall s \geq s_0, \quad 0 < K_0(s) \leq C\mathrm{e}^{-s}, \quad |K_0'(s)| \leq C\mathrm{e}^{-s}.$$

Together with the formula (II.3.16) for the Green's function and the bound on g_k, we obtain that, for all $|x| \geq R_0 := s_0/\sqrt{\lambda_1}$,

$$|\mathcal{G}(x,z,z')| \leq C\sum_{k=1}^{\infty} \|g_k\|^2_\infty \mathrm{e}^{-\sqrt{\lambda_k}|x|} = C\left(\sum_{k=1}^{\infty} \sqrt{\lambda_k} \mathrm{e}^{-(\sqrt{\lambda_k}-\sqrt{\lambda_1})|x|}\right) \mathrm{e}^{-\sqrt{\lambda_1}|x|}.$$

The last series converges in view of the asymptotics $\lambda_k \sim_{k\to\infty} \frac{\pi^2 k^2}{L^2}$. This proves Lemma II.3.5 with $\mu = \lambda_1$. □

II.3.4 Charge distributions in uniform Lebesgue spaces.

As we already mentioned, the electronic density of an encapsulated 2D material is not in $H^{-1}(S_L)$, so we also need to extend the definition of V_f to larger classes of charge distributions f. For instance, in the periodic setting, the electronic density is a locally-integrable $\mathbb{L}$-periodic function on S_L. For completeness, let us work in the larger $L^p_{\mathrm{unif}}(S_L)$ class, defined for $1 \leq p \leq \infty$ by

$$L^p_{\mathrm{unif}}(S_L) := \left\{ f \in L^p_{\mathrm{loc}}(S_L), \quad \|f\|_{L^p_{\mathrm{unif}}(S_L)} := \sup_{x \in \mathbb{R}^2} \|f\|_{L^p((x+[0,1]^2)\times I_L)} < \infty \right\}.$$

Note that $L^\infty_{\mathrm{unif}}(S_L) = L^\infty(S_L)$, that $L^q_{\mathrm{unif}} \subset L^p_{\mathrm{unif}}$ if $p \leq q$ (this comes from Hölder's inequality), and that L^p_{loc} functions which are $\mathbb{L}$-periodic with respect to the longitudinal variable x are in $L^p_{\mathrm{unif}}(S_L)$ as well.

Lemma II.3.6. *If $f \in L^p_{\mathrm{unif}}(S_L)$ for some $p > \frac{3}{2}$, then for all $(x,z) \in S_L$, the integral*

$$V_f(x,z) := \int_{S_L} \mathcal{G}(x,z;x',z') f(x',z') \mathrm{d}x' \mathrm{d}z'$$

is well-defined. The function V_f satisfies the Dirichlet problem (II.3.2) with $v_0 = 0$. In addition, $V_f \in L^\infty(S_L)$, and there is a constant $C^{\mathrm{unif}}_{p,L} > 0$ such that

$$\forall f \in L^p_{\mathrm{unif}}(S_L), \qquad \|V_f\|_{L^\infty(S_L)} \leq C^{\mathrm{unif}}_{p,L} \|f\|_{L^p_{\mathrm{unif}}(S_L)}.$$

In particular, V_f belongs to all $L^q_{\mathrm{unif}}(S_L)$ spaces for all $1 \leq q \leq \infty$.

Proof. By longitudinal-translation invariance of the problem, it is enough to bound $V_f(x_0, z_0)$ for $x_0 = 0$. Consider a compact subset Γ of $\mathbb{R}^2$ containing a neighbourhood of $x_0 = 0$. We write $f = f_1 + f_2$ with

$$f_1(x,z) = f(x,z)\mathbb{1}(x \in \Gamma), \qquad f_2(x,z) = f(x,z)\mathbb{1}(x \notin \Gamma).$$

We have similarly $V_f = V_{f_1} + V_{f_2}$. We bound the two terms independently. Since $f_1 \in L^p(S_L)$, Lemma II.3.3 shows that $V_{f_1} \in L^\infty(S_L)$, with

$$\|V_{f_1}\|_{L^\infty(S_L)} \leq C_p \|f_1\|^{\frac{p}{5p-6}}_{L^p(S_L)} \|f_1\|^{\frac{4p-6}{5p-6}}_{H^{-1}(S_L)} \leq C_{p,\Gamma,L} \|f_1\|_{L^p(S_L)} \leq \widetilde{C}_{p,\Gamma,L} \|f_1\|_{L^p_{\mathrm{unif}}(S_L)}.$$

The last inequality comes from the fact that f_1 has compact support in $\Gamma \times I_L$, so that $L^p(\Gamma \times I_L) \hookrightarrow L^{6/5}(S_L) \hookrightarrow H^{-1}(S_L)$.

To bound $V_{f_2}(0, z_0)$, we use the exponential decay of $\mathcal{G}$ in Lemma II.3.5. We have

$$|V_{f_2}(0,z_0)| \leq C_\Gamma \int_{S_L} \mathrm{e}^{-\sqrt{\mu}|x'|} |f(x',z')| \mathrm{d}x' \mathrm{d}z'.$$

We decompose $\mathbb{R}^2$ as $\mathbb{R}^2 = \bigcup_{k \in \mathbb{Z}^2} Q_k$ with $Q_k = k + [0,1]^2$, and write

$$|V_{f_2}(0,z_0)| \leq C' \sum_{k \in \mathbb{Z}^2} \mathrm{e}^{-\sqrt{\mu}|k|} \int_{Q_k \times I_L} |f| \leq C' \|f\|_{L^1_{\mathrm{unif}}} \sum_{k \in \mathbb{Z}^2} \mathrm{e}^{-\sqrt{\mu}|k|}.$$

The last sum is convergent. Finally, Hölder's inequality shows that

$$\|f\|_{L^1_{\mathrm{unif}}(S_L)} \leq L^{1/p'} \|f\|_{L^p_{\mathrm{unif}}(S_L)},$$

and the result follows. □

In the case where f is a $\mathbb{L}$–periodic function in the x variable, then so is V_f. In this case, we can define the **periodic Hartree energy per unit cell** as

$$\mathcal{D}_{\mathrm{per}}(f,f) := \int_{(\mathbb{R}^2/\mathbb{L})\times I_L} V_f(x,z) f(x,z) \mathrm{d}x\mathrm{d}z = \frac{1}{4\pi} \int_{(\mathbb{R}^2/\mathbb{L})\times I_L} \epsilon(z) |\nabla V_f|^2 \mathrm{d}x\mathrm{d}z \quad \geq 0. \tag{II.3.18}$$

In the case where $f \in L^p_{\mathrm{per}}(S_L)$ for some $p > \frac{3}{2}$, we have $V_f \in L^\infty(S_L)$, and the first integral is well defined (recall that $L^p_{\mathrm{per}}(S_L) \hookrightarrow L^1_{\mathrm{per}}(S_L)$).

II.3.5 Stationary charge distributions

We now focus on the stationary case. Recall that $f_\omega(x,z)$ is stationary if it is of the form $\mathfrak{f}(\alpha_x\omega, z)$ for some generating function $\mathfrak{f} : \Omega \times I_L \to \mathbb{C}$, that is

$$f_\omega(x,z) = \mathfrak{f}(\alpha_x\omega, z) = \mathfrak{f}(\omega_1 - x, \omega_2 - x, z).$$

If $\mathfrak{f}$ is a continuous function (hence bounded) in $C^0(\Omega \times I_L)$, then so is f_ω. In this case, $f_\omega \in L^\infty(S_L)$, and one can define the potential V_{f_ω} as in Lemma II.3.6, which is bounded. By translation invariance of $\mathcal{G}$ along the two first dimensions, V_f is also stationary, of the form

$$V_f(\omega, x, z) := \mathcal{V}_{\mathfrak{f}}(\alpha_x\omega, z).$$

In the sequel, we work with generating functions $\mathfrak{f}$ in the Lebesgue space $L^p(\Omega \times I_L)$, which are not necessarily continuous. In this case, $f_\omega(x,z) := \mathfrak{f}(\alpha_x\omega, z)$ may not belong to any $L^p_{\mathrm{unif}}(S_L)$ space, and it is unclear at this point that the corresponding potential V_f is still well-defined. The spaces $L^{r,p}$ introduced below are precisely designed to handle this difficulty

For $1 \leq r, p < \infty$, we introduce the Lebesgue spaces

$$L^{r,p} := \left\{ \nu \in L^1(\Omega \times I_L), \quad \|\nu\|^r_{L^{r,p}} := \int_{\Omega_2} \left(\int_{\Omega_1 \times I_L} |\nu(\omega_1, \omega_1 + \omega_2, z)|^p \, \mathrm{d}\omega_1 \, \mathrm{d}z \right)^{r/p} \mathrm{d}\omega_2 < \infty \right\},$$

and analogous definitions for $r = \infty$ or $p = \infty$. After a change of variable $\omega_2' := \omega_1 + \omega_2$, this is also the mixed Lebesgue space

$$L^{r,p} \simeq L^r(\Omega_2, L^p(\Omega_1 \times I_L)).$$

When $r = p$, we have $L^{p,p} = L^p(\Omega \times I_L)$. In addition, since $\Omega_1 \times I_L$ is bounded, Hölder's inequality shows that $L^{r,q} \subset L^{r,p}$ if $p \leq q$, and since Ω_2 is compact, we also have $L^{s,p} \subset L^{r,p}$ if $r \leq s$. In particular, $L^{r,p} \subset L^{\min(r,p)}(\Omega \times I_L) \subset L^1(\Omega \times I_L)$.

The key result of this section is the following, which is the stationary equivalent to Lemma II.3.6.

Theorem II.3.7. *If $\mathfrak{f} \in L^{r,p}$ with $1 \le r \le \infty$ and $\frac{3}{2} < p \le \infty$ and $f_\omega(x,z) := \mathfrak{f}(\alpha_x\omega, z)$. Then for all $(x,z) \in S_L$ and almost all $\omega \in \Omega$, the integral*

$$V_{f_\omega}(x,z) := \int_{S_L} \mathcal{G}(x,z;x',z') f_\omega(x',z') \mathrm{d}x'\mathrm{d}z'$$

is well-defined. The function V_{f_ω} is solution to the Dirichlet problem (II.3.2) for $f = f_\omega$ and $v_0 = 0$. The function $V_f(\omega,x,z) := V_{f_\omega}(x,z)$ is stationary, of the form $V_f(\omega,x,z) = \mathcal{V}_\mathfrak{f}(\alpha_x\omega,z)$. In addition, $\mathcal{V}_\mathfrak{f} \in L^{r,\infty}$, and there exists a constant $C^{\mathrm{stat}}_{r,p,L} > 0$ such that

$$\forall \mathfrak{f} \in L^{r,p}, \qquad \|\mathcal{V}_\mathfrak{f}\|_{L^{r,\infty}} \le C^{\mathrm{stat}}_{r,p,L} \|\mathfrak{f}\|_{L^{r,p}}.$$

Proof. By linearity, it is enough to prove the result for $\mathfrak{f}$ a nonnegative measurable function in $L^{r,p}$. In this case, f_ω and V_{f_ω} are well-defined for almost $\omega \in \Omega$ and nonnegative since $\mathcal{G} \ge 0$ by the maximum principle.

It is useful to work with a configuration of the form $(\omega_1, \omega_1+\omega_2)$. We have

$$\mathcal{V}_\mathfrak{f}(\omega_1,\omega_1+\omega_2,z) := V_{f_{\omega_1,\omega_1+\omega_2}}(x=0,z) = \int_{S_L} \mathcal{G}(-x',z,z') f_{\omega_1,\omega_1+\omega_2}(x',z') \mathrm{d}x'\mathrm{d}z'.$$

As in the proof of Lemma II.3.6, we write

$$f_{\omega_1,\omega_1+\omega_2} = f^{(1)}_{\omega_1,\omega_1+\omega_2} + f^{(2)}_{\omega_1,\omega_1+\omega_2},$$

with

$$f^{(1)}_{\omega_1,\omega_1+\omega_2}(x,z) := f_{\omega_1,\omega_1+\omega_2}(x,z)\mathbb{1}(x \in \Gamma), \quad \text{and} \quad f^{(2)}_{\omega_1,\omega_1+\omega_2}(x,z) := f_{\omega_1,\omega_1+\omega_2}(x,z)\mathbb{1}(x \notin \Gamma).$$

Here Γ is a compact subset of $\mathbb{R}^2$ containing a neighborhood of 0, that will be specified later. We denote by $V^{(1)}$ and $V^{(2)}$ the corresponding potentials. To control $V^{(1)}$, we use that $f^{(1)}_{\omega_1,\omega_1+\omega_2}$ is compactly-supported. As in the proof of Lemma II.3.6, we get

$$\|V^{(1)}\|^p_{L^\infty(S_L)} \le C\|f^{(1)}\|^p_{L^p(S_L)} = \int_{S_L} \mathbb{1}(x' \in \Gamma)\mathfrak{f}^p(\omega_1 - x', \omega_1+\omega_2-x', z')\mathrm{d}x'\mathrm{d}z'.$$

Taking Γ a shifted version of Ω_1 containing a neighbourhood of 0, we can make the change of variable $\omega_1' := -x' + \omega_1$, and get

$$\left|\mathcal{V}^{(1)}_\mathfrak{f}(\omega_1,\omega_1+\omega_2,z)\right|^p = \left|V^{(1)}(0,z)\right|^p \le C\int_{\Omega_1} \mathfrak{f}^p(\omega_1', \omega_1'+\omega_2, z')\mathrm{d}\omega_1'\mathrm{d}z'.$$

The right-hand side no longer depends on ω_1, so we can take the supremum in ω_1 on the left-hand side. Integrating first in z (and using that $|I_L| = L$), raising to the power r/p, and integrating in ω_2 gives

$$\left|\mathcal{V}^{(1)}_\mathfrak{f}\right|_{L^{r,\infty}} \le CL\|\mathfrak{f}\|_{L^{r,p}}.$$

We now focus on the term $V^{(2)}$. This time, we use the exponential decay of $\mathcal{G}$ in Lemma II.3.5, and get that there is a constant $C_\Gamma > 0$ such that

$$|V^{(2)}(0,z)| \le C\int_{S_L} \mathrm{e}^{-\sqrt{\mu}|x'|} f_{\omega_1,\omega_1+\omega_2}(x',z')\mathrm{d}x'\mathrm{d}z' = C\int_{S_L} \mathrm{e}^{-\sqrt{\mu}|x'|}\mathfrak{f}(\omega_1-x', \omega_1+\omega_2-x', z')\mathrm{d}x'\mathrm{d}z'.$$

We write $x' = \omega_1' + \ell$ with $\omega_1' \in \Omega_1$ and $\ell \in \mathbb{L}_1$ and get

$$|V^{(2)}(0,z)| \leq C \sum_{\ell \in \mathbb{L}_1} \int_{\Omega_1 \times I_L} \mathrm{e}^{-\sqrt{\mu}|\omega_1'+\ell|} \mathfrak{f}(\omega_1 - \omega_1', \omega_1 + \omega_2 - \omega_1' - \ell, z') \mathrm{d}\omega_1' \mathrm{d}z'.$$

Using that $\mathrm{e}^{-\sqrt{\mu}|\omega_1'+\ell|} \leq \mathrm{e}^{-\sqrt{\mu}|\ell|} \mathrm{e}^{\sqrt{\mu}|\omega_1'|} \leq C\mathrm{e}^{-\sqrt{\mu}|\ell|}$, and making the change of variable $\omega_1' = -\omega_1'' + \omega_1$ gives

$$\left| \mathcal{V}_{\mathfrak{f}}^{(2)}(\omega_1, \omega_1 + \omega_2, z) \right| =: \left| V^{(2)}(0,z) \right| \leq C' \sum_{\ell \in \mathbb{L}_1} \mathrm{e}^{-\sqrt{\mu}|\ell|} \underbrace{\int_{\Omega_1 \times I_L} \mathfrak{f}(\omega_1'', \omega_1'' + \omega_2 - \ell, z') \mathrm{d}\omega_1'' \mathrm{d}z'}_{f_\ell(\omega_2)}.$$

Again, the right-hand side no longer depends on ω_1, and we can take the supremum in ω_1 on the left-hand side. On the other hand, Jensen's inequality with the convex function $t \mapsto t^r$ shows that

$$\left(\sum_{\ell \in \mathbb{L}_1} w_\ell f_\ell \right)^r \leq W^{r-1} \left(\sum_{\ell \in \mathbb{L}_1} w_\ell f_\ell^r \right), \qquad W := \sum_{\ell \in \mathbb{L}_1} w_\ell.$$

With $w_\ell := \mathrm{e}^{-\sqrt{\mu}|\ell|}$ and $f_\ell = f_\ell(\omega_2)$, we deduce that

$$\sup_{\omega_1 \in \Omega_1} \left| \mathcal{V}_{\mathfrak{f}}^{(2)}(\omega_1, \omega_1 + \omega_2, z) \right|^r \leq (C')^r W^{r-1} \sum_{\ell \in \mathbb{L}_1} \mathrm{e}^{-\sqrt{\mu}|\ell|} \left(\int_{\Omega_1 \times I_L} \mathfrak{f}(\omega_1', \omega_1' + \omega_2 - \ell, z') \mathrm{d}\omega_1' \mathrm{d}z' \right)^r.$$

We integrate in $\omega_2 \in \Omega_2$ (the integral no longer depends on ℓ), take the supremum in z, and get

$$\left\| V_{\mathfrak{f}}^{(2)} \right\|_{L^{r,\infty}} \leq C' W \|\mathfrak{f}\|_{L^{r,1}} \leq C'' \|\mathfrak{f}\|_{L^{r,p}},$$

where we used that $L^{r,p} \subset L^{r,1}$ with continuous embedding by Hölder's inequality. The proof follows. □

We are now in position to define the **stationary Hartree energy per unit area**. Denoting by $(e_{G_1,G_2})_{(G_1,G_2) \in \mathbb{L}_1^* \times \mathbb{L}_2^*}$ the orthonormal Fourier basis of $L^2(\Omega)$, we set for all $\mathfrak{f} \in L^1(\Omega \times I_L)$

$$\mathcal{D}_{\mathrm{erg}}(\mathfrak{f}, \mathfrak{f}) := \frac{4\pi}{|\Omega|} \sum_{(G_1,G_2,k) \in \mathbb{L}_1^* \times \mathbb{L}_2^* \times \mathbb{N}^*} \frac{|\widehat{\mathfrak{f}}_{G_1,G_2,k}|^2}{|G_1+G_2|^2 + \lambda_k}, \tag{II.3.19}$$

where

$$\widehat{\mathfrak{f}}_{G_1,G_2,k} := \int_{\Omega \times I_L} \mathfrak{f}(\omega, z) \, \overline{e_{G_1,G_2}}(\omega) \, g_k(z) \, \mathrm{d}\omega \, \mathrm{d}z.$$

Lemma II.3.8. *The map $L^1(\Omega \times I_L) \ni \mathfrak{f} \mapsto \mathcal{D}_{\mathrm{erg}}(\mathfrak{f}, \mathfrak{f}) \in \mathbb{R}_+ \cup \{+\infty\}$ is convex. It is finite and continuous on $L^{r,p}$ for all $\frac{3}{2} < p \leq \infty$ and $2 \leq r \leq \infty$, and weakly lower-semicontinuous on $L^p(\Omega \times I_L)$ for all $1 < p < \infty$. In addition, for all $0 \leq \mathfrak{f}, \mathfrak{g} \in L^1(\Omega \times I_L)$ satisfying $\mathcal{D}_{\mathrm{erg}}(\mathfrak{f}, \mathfrak{f}) < \infty$ and $\mathcal{D}_{\mathrm{erg}}(\mathfrak{g}, \mathfrak{g}) < \infty$, we have*

$$\mathcal{D}_{\mathrm{erg}}(\mathfrak{f}, \mathfrak{g}) = \int_{I_L} \fint_{\Omega} \mathcal{V}_{\mathfrak{f}}(\omega, z) \mathfrak{g}(\omega, z) \, \mathrm{d}\omega \mathrm{d}z = \int_{I_L} \fint_{\Omega} \mathfrak{f}(\omega, z) \mathcal{V}_{\mathfrak{g}}(\omega, z) \, \mathrm{d}\omega \mathrm{d}z.$$

Proof. It follows from (II.3.19) that $\mathcal{D}_{\mathrm{erg}}$ is a convex positive quadratic form on $L^1(\Omega\times I_L)$, and weakly lower-semicontinuous on $L^p(\Omega\times I_L)$ for all $1<p<\infty$ by Fatou's lemma. To prove continuity on $L^{r,p}$ for suitable r,p, we recall that the dual space of $L^{r,p}$ is $L^{r',p'}$ with $\frac{1}{r}+\frac{1}{r'}=1$ and $\frac{1}{p}+\frac{1}{p'}=1$. So, using that $p<\infty$ and $r'\leq 2\leq r$, we get

$$\mathcal{V}_{\mathfrak{f}}\in L^{r,\infty}\subset L^{r,p'}\subset L^{r',p'}=(L^{r,p})^*.$$

In particular, we have the inequality

$$\mathcal{D}_{\mathrm{erg}}(\mathfrak{f},\mathfrak{f})\leq C\|\mathfrak{f}\|_{L^{r,p}}\|\mathcal{V}_{\mathfrak{f}}\|_{L^{r,\infty}}\leq C'\|\mathfrak{f}\|^2_{L^{r,p}},$$

so the quadratic form $\mathfrak{f}\mapsto\mathcal{D}_{\mathrm{erg}}(\mathfrak{f},\mathfrak{f})$ is strongly continuous for the $L^{r,p}$ topology, for all $\frac{3}{2}<p\leq\infty$ and $2\leq r\leq\infty$. To prove the equality when $0\leq\mathfrak{f},\mathfrak{g}\in L^1(\Omega\times I_L)$, we note that we have equality for $\mathfrak{f}_N:=\min\{N,\mathfrak{f}\}$ by Lemma II.3.9, and conclude by monotone convergence. Since $\mathcal{D}_{\mathrm{erg}}$ is a positive quadratic form, we have the Cauchy-Schwarz inequality

$$\mathcal{D}_{\mathrm{erg}}(\mathfrak{f},\mathfrak{g})^2\leq\mathcal{D}_{\mathrm{erg}}(\mathfrak{f},\mathfrak{f})\,\mathcal{D}_{\mathrm{erg}}(\mathfrak{g},\mathfrak{g}).$$ □

Lemma II.3.9. *Let $f_\omega(x,z)$ be a stationary function with generating function $\mathfrak{f}\in L^{r,p}$ with $\frac{3}{2}<p\leq\infty$ and $2\leq r\leq\infty$, and let V_{f_ω} be its associated potential, with generating function $\mathcal{V}_{\mathfrak{f}}\in L^{r,\infty}$. Then, for almost all $\omega\in\Omega$, the Hartree energy per unit area*

$$\lim_{R\to\infty}\int_{I_L}\fint_{B_R}V_{f_\omega}(x,z)f_\omega(x,z)\mathrm{d}x\mathrm{d}z \tag{II.3.20}$$

is well defined.

1. *If the two lattices $\mathbb{L}_1$ and $\mathbb{L}_2$ are* **incommensurate**, *then this energy is almost surely independent of ω, and we have for almost all $\omega\in\Omega$,*

$$\lim_{R\to\infty}\int_{I_L}\fint_{B_R}V_\omega(x,z)f_\omega(x,z)\mathrm{d}x\mathrm{d}z=\int_{I_L}\fint_{\Omega}\mathcal{V}_{\mathfrak{f}}(\omega',z)\mathfrak{f}(\omega',z)\mathrm{d}\omega'\mathrm{d}z=\mathcal{D}_{\mathrm{erg}}(\mathfrak{f},\mathfrak{f}). \tag{II.3.21}$$

2. *If the two lattices $\mathbb{L}_1$ and $\mathbb{L}_2$ are* **commensurate**, *f_ω is $\mathbb{L}$-periodic, with $\mathbb{L}=\mathbb{L}_1\cap\mathbb{L}_2$, for each fixed configuration $\omega\in\Omega$, and we are back to the case of periodic material. However, there is no relation between the different configurations, and the Hartree energy depends in general on the configuration ω.*

Proof. The first equality in (II.3.21) is a direct consequence of the Birkhoff–Von Neumann ergodic theorem, and we skip it for brevity. A direct computation shows that the second equality holds for $\mathfrak{f}\in L^2(\Omega\times I_L)$. We conclude by density. □

The original equation $-\mathrm{div}(\epsilon\nabla V_{f_\omega})=4\pi f_\omega$ can be recast in terms of the generating functions $\mathfrak{f}$ and $V_{\mathfrak{f}}$. Indeed, Since $f_\omega(x,z)=\mathfrak{f}(\omega_1-x,\omega_2-x,z)$, we have

$$(\nabla f_\omega)(0,z)=(\Lambda\mathfrak{f})(\omega_1,\omega_2,z),\qquad\text{with}\quad\Lambda=\begin{pmatrix}-\partial_{\omega_{1,1}}-\partial_{\omega_{2,1}}\\-\partial_{\omega_{1,2}}-\partial_{\omega_{2,2}}\\\partial_z\end{pmatrix}. \tag{II.3.22}$$

The usual gradient operator therefore becomes the Λ operator for the corresponding generating functions. The two first components of Λ are also $\nabla_{\omega_1} + \nabla_{\omega_2}$, and describe a derivation operator in a specific direction, induced by the group action α. In particular, the operator $\Lambda^*\Lambda$ is not coercive. Nevertheless, we deduce that

$$\Lambda^*(\epsilon\Lambda\mathcal{V}_\mathfrak{f}) = 4\pi\mathfrak{f} \quad \text{on} \quad \Omega \times I_L.$$

Multiplying by $\mathcal{V}_\mathfrak{f}$ and integrating shows that (compare with (II.3.18))

$$\mathcal{D}_{\mathrm{erg}}(\mathfrak{f},\mathfrak{f}) = \int_{I_L} \fint_\Omega \mathcal{V}_\mathfrak{f}(\omega', z)\mathfrak{f}(\omega', z)\mathrm{d}\omega'\mathrm{d}z = \frac{1}{4\pi}\int_{I_L} \fint_\Omega \epsilon(z)|\Lambda\mathcal{V}_\mathfrak{f}|^2(\omega, z)\mathrm{d}\omega\,\mathrm{d}z \qquad \geq 0.$$

II.4 Reduced Hartree–Fock models for encapsulated quasi-periodic 2D materials

In this section, we introduce a reduced Hartree–Fock (rHF) model for encapsulated quasi-periodic 2D materials, prove the existence of ground-state density matrix and the uniqueness of the ground-state density, and derive the corresponding rHF equations.

The rHF model is convex and strictly convex in the density, which greatly simplifies its analysis. It has been extensively studied in the mathematical literature [LS77; Sol91; CL93; CLL98; CLL01; CDL08; CLL13; GL16]. In the periodic setting (Section II.5), convexity is not so essential, and existence results can be established for non-convex models actually used in practice in condensed-matter physics, such as Kohn–Sham LDA.

In what follows, we consider two incommensurate lattices $\mathbb{L}_1$ and $\mathbb{L}_2$, the ergodic group action $\mathbb{R}^2 \ni x \mapsto \alpha : \Omega \to \Omega$ on $\Omega = (\mathbb{R}^2/\mathbb{L}_1) \times (\mathbb{R}^2/\mathbb{L}_2)$ defined by $\alpha_x(\omega_1, \omega_2) = (\omega_1 - x, \omega_2 - x)$, and a nuclear charge distribution $m : \Omega \times S_L \to \mathbb{R}$, stationary in the sense that

$$\forall \omega \in \Omega, \quad m_\omega(x, z) = \mathfrak{m}(\alpha_x\omega, z),$$

for some generating function $\mathfrak{m} : \Omega \times I_L \to \mathbb{R}$. Our goal is to define an rHF energy functional for a stationary one-body density operator γ describing electrons interacting with the nuclear charge distribution m, in the encapsulated setting.

II.4.1 Stationary operators and trace per unit area

We first need to extend the notion of stationary functions to operators. We introduce the von Neumann algebra of (bounded) **fibered stationary operators**

$$\mathcal{A} := \left\{ A = \int_\Omega^\oplus A_\omega \mathrm{d}\omega \in L^\infty(\Omega, \mathcal{B}(L^2(S_L))) \text{ such that } \forall x \in \mathbb{R}^2, U_x A_\omega U_x^* = A_{\alpha_{-x}\omega} \right\}, \tag{II.4.1}$$

where U_x denotes the usual translation operator on $L^2(S_L)$ defined by $U_x(f)(x',z) = f(x'-x,z)$. This algebra can in fact [**tak2010theoryII**; Tak01] be identified with the tensor product

$$\mathcal{A} = \left(L^\infty(\Omega) \rtimes_\alpha \mathbb{R}^2\right) \overline{\otimes} \mathcal{B}(L^2(I_L)).$$

Such operators act on the Hilbert space $L^2(\Omega, L^2(S_L))$ as follows: if $f(\omega,x,z) =: f_\omega(x,z)$ is in $L^2(\Omega, L^2(S_L))$, then

$$Af = g \quad \text{means} \quad A_\omega f_\omega = g_\omega \quad \text{in } L^2(S_L), \text{ for a.a. } \omega \in \Omega.$$

A fibered self-adjoint operator $A = \int_\Omega^\oplus A_\omega \,\mathrm{d}\omega$ is called *affiliated* to $\mathcal{A}$ if its resolvent is in $\mathcal{A}$. A typical example of self-adjoint operator affiliated to $\mathcal{A}$ is the ergodic Schrödinger operator $H = \int_\Omega^\oplus H_\omega \,\mathrm{d}\omega$ with $H_\omega := -\frac{1}{2}\Delta_0 + V_\omega$, where Δ_0 is the Dirichlet Laplacian on S_L and $V_\omega(x) = V(\omega,x)$ a stationary potential in a suitable function space.

The von Neumann algebra $\mathcal{A}$ is a type II_∞ factor [Tak10, Chap. XIII] and therefore has a unique semifinite faithful normal trace (up to a scaling factor), which we denote by τ. In what follows, we set, for $1 \le p < \infty$,

$$L^p(\mathcal{A},\tau) := \{A \in \widetilde{\mathcal{A}} \ : \ \tau(|A|^p) < \infty\}, \quad \text{and} \quad L^\infty(\mathcal{A},\tau) := \mathcal{A}. \tag{II.4.2}$$

The non-commutative $L^p(A,\tau)$ spaces share properties which are similar to the usual Lebesgue spaces, see [Seg53; Nel74]. The following result highlights the fact that one can fix the scaling factor in τ so that it corresponds to the notion of trace per unit area. For a function $\eta \in L^\infty(\mathbb{R}^2)$, we denote by M_η the multiplication operator by $\eta(x)$ (independent of z).

Proposition II.4.1. *Let $A \in L^1(\mathcal{A},\tau)$. Then, for a.e. $\omega \in \Omega$, A_ω is locally trace-class on $L^2(S_L)$. If ρ_{A_ω} denotes the density of A_ω, then $\rho(\omega,x,z) := \rho_{A_\omega}(x,z)$ is a stationary function. The corresponding generating function ϱ_A belongs to $L^1(\Omega \times I_L)$, and is called the* **generalized density** *of A.*

For any $\eta \in L^\infty(\mathbb{R}^2) \cap L^2(\mathbb{R}^2)$, with $\|\eta\|_2 = 1$, we have $\int_\Omega \mathrm{Tr}_{L^2(S_L)}(|M_\eta A_\omega M_\eta|)\mathrm{d}\omega < \infty$, and the formulas

$$\tau(A) = \fint_\Omega \mathrm{Tr}_{L^2(S_L)}(M_\eta A_\omega M_\eta)\mathrm{d}\omega, \quad \text{and} \quad \tau(A) = \int_{I_L}\fint_\Omega \varrho_A(\omega,z)\mathrm{d}\omega\mathrm{d}z. \tag{II.4.3}$$

Proof. Denote by ρ_ω the density of A_ω, so that

$$\forall \eta \in C_0^\infty(S_L), \quad \mathrm{Tr}_{L^2(S_L)}(M_\eta A_\omega M_\eta) = \int_{S_L} \rho_\omega |\eta|^2. \tag{II.4.4}$$

Since $U_x A_\omega U_x^* = A_{\alpha_{-x}\omega}$, we have $\rho_\omega(x'-x,z) = \rho_{\alpha_{-x}(\omega)}(x',z)$, so $\rho(\omega,x,z) := \rho_\omega(x,z)$ is stationary. In particular, since α_x is measure preserving, the quantity

$$\int_{I_L}\fint_\Omega \rho(\omega,x,z)\mathrm{d}\omega\mathrm{d}z = \int_{I_L}\fint_\Omega \rho(\alpha_x\omega,0,z)\mathrm{d}\omega\mathrm{d}z = \int_{I_L}\fint_\Omega \rho(\omega,0,z)\mathrm{d}\omega\mathrm{d}z = \int_{I_L}\fint_\Omega \varrho(\omega,z)\mathrm{d}\omega\mathrm{d}z$$

is independent of $x \in \mathbb{R}^2$. We define $\tau(A)$ the trace per unit surface of A as this common quantity. We can check that $\tau(\cdot)$ is linear and that $\tau(A) \geq 0$ if $A \geq 0$, so this is the von Neumann trace (up to a scaling factor). We now integrate (II.4.4) in $\omega \in \Omega$, and get that

$$\fint_\Omega \mathrm{Tr}_{L^2(S_L)}(M_\eta A_\omega M_\eta)\mathrm{d}\omega = \int_{S_L} \left(\fint_\Omega \rho(\omega, x, z)\mathrm{d}\omega \mathrm{d}z \right) |\eta|^2(x)\mathrm{d}x = \tau(A) \int_{\mathbb{R}^2} |\eta|^2(x)\mathrm{d}x = \tau(A).$$

□

Taking $\eta = |B_R|^{-1/2}\mathbb{1}_{B_R}$ and the limit $R \to +\infty$, we get

$$\tau(A) = \lim_{R\to+\infty} \fint_\Omega \frac{1}{|B_R|}\mathrm{Tr}_{L^2(S_L)}(M_{\mathbb{1}_{B_R}} A_\omega M_{\mathbb{1}_{B_R}})\mathrm{d}\omega = \lim_{R\to\infty} \fint_\Omega \int_{I_L} \fint_{B_R} \rho_{A_\omega}(x, z)\mathrm{d}x\mathrm{d}z\mathrm{d}\omega.$$

In the ergodic case, Birkhoff theorem implies that we also have

$$\tau(A) = \lim_{R\to+\infty} \frac{1}{|B_R|}\mathrm{Tr}\,(M_{\mathbb{1}_{B_R}} A_\omega M_{\mathbb{1}_{B_R}}) = \lim_{R\to\infty} \int_{I_L} \fint_{B_R} \rho_{A_\omega}(x, z)\mathrm{d}x\mathrm{d}z \quad \text{for a.a. } \omega \in \Omega.$$

Example II.4.2 (Fourier multipliers)**.** *As an example, consider a convolution operator J acting on $L^2(\mathbb{R}^2)$ with kernel $J(x, x') = j(x - x')$. Then J is a Fourier multiplier by $\widehat{j}(k)$, and we can write $J = \widehat{j}(-\mathrm{i}\nabla_x)$. We set*

$$A := \int_\Omega^\oplus A_\omega \mathrm{d}\omega, \quad \textit{with} \quad A_\omega := J \otimes |h\rangle\langle h|,$$

where $h \in L^2(I_L)$ is normalized $\|h\|_{L^2(I_L)} = 1$. The function h in the z-variable is only here to handle the z-direction. Since A_ω is independent of ω and commutes with all translations, A is a fibered stationary operator affiliated to $\mathcal{A}$. A simple computation shows that $\varrho_A(\omega, z) = j(0)|h(z)|^2$. In particular, ϱ_A is independent of $\omega \in \Omega$. Similarly, we get

$$\tau(|A|^p) = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} |\widehat{j}(k)|^p \mathrm{d}k.$$

We deduce that $A \in L^p(\mathcal{A}, \tau)$ iff $\widehat{j} \in L^p(\mathbb{R}^2)$. In particular, $A \in L^p(\mathcal{A}, \tau)$ is not necessarily a bounded operator in $\mathcal{A} = L^\infty(\mathcal{A}, \tau)$. Note that the usual Schatten classes $\mathfrak{S}_p := L^p(\mathcal{B}(L^2), \mathrm{Tr}\,)$ consist of compact -hence bounded- operators.

II.4.2 The encapsulated kinetic energy

We also define the subspace $W_0^{1,1}(\mathcal{A}, \tau)$ of stationary operators of finite kinetic energy (per unit area) by

$$W_0^{1,1}(\mathcal{A}, \tau) := \left\{ A \in L^1(\mathcal{A}, \tau) \mid \sqrt{-\Delta_0}|A|\sqrt{-\Delta_0} \in L^1(\mathcal{A}, \tau) \right\}, \tag{II.4.5}$$

where $-\Delta_0$ is the Dirichlet Laplacian on S_L, that is with domain $H^2(S_L) \cap H_0^1(S_L)$ (we put a 0 subscript to emphasize Dirichlet boundary conditions). With a slight abuse of notation, we extended the operator $-\Delta_0$ as an affiliated fibered stationary operator

by setting $\Delta_0 = \int_\Omega^\oplus \Delta_0 \mathrm{d}\omega$ on $L^2(\Omega \times S_L)$, that is with an action independent of the configuration $\omega \in \Omega$.

The condition in (II.4.5) is equivalent to $\sqrt{-\Delta_0}|A|^{1/2} \in L^2(\mathcal{A}, \tau)$. It implies that all eigenspaces of A are in the form domain of $-\Delta_0$, hence satisfy Dirichlet boundary conditions. It implies for instance that its generalized density ϱ_A satisfies the boundary conditions

$$\varrho_A\left(\cdot, \pm\tfrac{L}{2}\right) = 0 \quad \text{in the sense of the trace.}$$

For $A \in W_0^{1,1}(\mathcal{A}, \tau)$, we define the **encapsulated kinetic energy per unit area** of A by

$$\frac{1}{2}\tau(-\Delta_0 A) := \frac{1}{2}\tau\left(\sqrt{-\Delta_0} A \sqrt{-\Delta_0}\right). \tag{II.4.6}$$

Using the spectral decomposition of the 1D Dirichlet Laplacian given by

$$-\partial_{zz,0}^2 = \sum_{k=1}^\infty \mu_k |h_k\rangle\langle h_k|, \quad \mu_k := \frac{\pi^2 k^2}{L^2}, \quad h_k(z) := \sqrt{\frac{2}{L}} \sin\left(\pi k \left[\frac{z}{L} + \frac{1}{2}\right]\right),$$

we have

$$-\Delta_0 = (-\Delta_x) \otimes \mathbb{I}_{L^2(I_L)} + \mathbb{I}_{L^2(\mathbb{R}^2)} \otimes (-\partial_{zz,0}^2) = \sum_{k=1}^\infty (-\Delta_x + \mu_k) \otimes |h_k\rangle\langle h_k|. \tag{II.4.7}$$

Again, we see the effect of the boundary conditions, which implies the spectral gap $(-\Delta_0) \geq \mu_1 > 0$, as in Lemma II.3.4.

II.4.3 Stationary one-body density matrices, and kinetic inequalities

In what follows, we are interested in **stationary one-body density matrices**. We therefore set

$$\mathcal{K} := \left\{\gamma \in W_0^{1,1}(\mathcal{A}, \tau), \quad 0 \leq \gamma = \gamma^* \leq 1\right\}.$$

If we write $\gamma = \int_\Omega^\oplus \gamma_\omega \mathrm{d}\omega$ with γ_ω acting on $L^2(S_L)$, then we have

$$0 \leq \gamma_\omega = \gamma_\omega^* \leq 1, \quad \text{and} \quad \forall x \in \mathbb{R}^2, \quad U_x \gamma_\omega U_x^* = \gamma_{\alpha_{-x}\omega},$$

where U_x is the longitudinal-translation operator introduced in Section II.4.1. Proposition II.4.1 above shows that any such one-body density matrix admits a generalized density $\varrho_\gamma \in L^1(\Omega \times I_L)$. We record several inequalities for stationary operators, including Lieb–Thirring inequality [LT75; LT76], and Hoffmann–Ostenhof inequality [HH77]. Recall that the operator Λ has been defined in (II.3.22).

Lemma II.4.3. *For all $\gamma \in \mathcal{K}$, we have*

$$\tau(-\Delta_0\gamma) \geq \frac{\pi^2}{L^2}\tau(\gamma) = \frac{\pi^2}{L^2}\int_{I_L} \fint_\Omega \varrho_\gamma \qquad \textit{(Poincaré's inequality)}, \tag{II.4.8}$$

$$\tau(-\Delta_0\gamma) \geq C_{\mathrm{LT}} \int_{I_L} \fint_\Omega \varrho_\gamma^{5/3} \qquad \textit{(Lieb–Thirring inequality)}, \tag{II.4.9}$$

$$\tau(-\Delta_0\gamma) \geq \int_{I_L} \fint_\Omega |\Lambda\sqrt{\varrho_\gamma}|^2 \qquad \textit{(Hoffmann–Ostenhof inequality)}. \tag{II.4.10}$$

The kinetic energy of γ therefore controls the L^p-norm of ϱ_γ for all $1 \leq p \leq 5/3$, by interpolation. Note that the Poincaré inequality is specific to our encapsulated setting, while the two other inequalities also hold for standalone 2D materials.

Proof. Poincaré's inequality comes from $\gamma \geq 0$ together with the operator inequality $-\Delta_0 \geq \mu_1$.

In order to prove the Lieb–Thirring inequality in the encapsulated stationary case, we use two tools. To handle the boundary conditions in the z-direction, we introduce the extension-by-0 operator $i : L^2(S_L) \to L^2(\mathbb{R}^3)$, and set

$$\widetilde{\gamma}_\omega := i\gamma_\omega i^*, \qquad \text{acting on } L^2(\mathbb{R}^3).$$

Since γ_ω satisfies Dirichlet boundary conditions, we have

$$\sqrt{-\Delta_0}\gamma_\omega\sqrt{-\Delta_0} = \sqrt{-\Delta}\widetilde{\gamma}_\omega\sqrt{-\Delta},$$

where $-\Delta$ denotes the usual three–dimensional Laplacian.

The second tool we use is a thermodynamic procedure to control the fact that the operators γ_ω and $\widetilde{\gamma}_\omega$ do not decay at infinity in the longitudinal directions. For $R > 0$, we set $\chi_R(x,z) := \chi\left(\frac{x}{R}\right)$ (independent of z), where $\chi : \mathbb{R}^2 \to [0,1]$ is a smooth cut-off radial nonincreasing function, with $\chi(x) = 1$ for $|x| < 1$ and $\chi(x) = 0$ for $|x| > 2$. We set

$$\gamma_{\omega,R} := \chi_R\widetilde{\gamma}_\omega\chi_R = \chi_R i\,\gamma_\omega\, i^*\chi_R.$$

The operator $\gamma_{\omega,R}$ satisfies $0 \leq \gamma_{\omega,R} \leq 1$ and is trace class. We denote by $\rho_{\omega,R}$ its density. The usual Lieb–Thirring inequality [LT75; LT76] shows that $\rho_{\omega,R} \in L^{5/3}(\mathbb{R}^3)$ with compact support, and that

$$C_{\mathrm{LT}}\int_{S_L} \rho_\omega^{5/3}\chi_R^{10/3} = C_{\mathrm{LT}}\int_{\mathbb{R}^3} \rho_{\omega,R}^{5/3} \leq \mathrm{Tr}_{\,L^2(\mathbb{R}^3)}\left(-\Delta\gamma_{\omega,R}\right) := \sum_{j=1}^{3} \mathrm{Tr}\,\left(P_j\gamma_{\omega,R}P_j\right),$$

where $P_j := -\mathrm{i}\partial_{x_j}$ the momentum operator along the x_j-direction. We have, for $j \in \{1,2\}$,

$$P_j\gamma_{\omega,R}P_j = \chi_R(P_j\widetilde{\gamma}_\omega P_j)\chi_R + [P_j,\chi_R]\widetilde{\gamma}_\omega\chi_R + \chi_R\widetilde{\gamma}_\omega[P_j,\chi_R],$$

with $[P_j,\chi_R] = (-\mathrm{i}\partial_j\chi_R)(x) = \frac{1}{R}(-\mathrm{i}\partial_j\chi)(x/R)$. Integrating in $\omega \in \Omega$ and taking the trace shows that

$$\fint_\Omega \mathrm{Tr}\,\left(P_j\gamma_{\omega,R}P_j\right) = \fint_\Omega \mathrm{Tr}\,\left(\chi_R(P_j\widetilde{\gamma}_\omega P_j)\chi_R\right) + O\left(R\right).$$

Using Proposition II.4.1 with $\eta := \frac{\chi_R}{\|\chi_R\|_2}$ gives

$$C_{\mathrm{LT}} \left(\int_{I_L} \fint_{\Omega} \varrho^{\frac{5}{3}} \right) \frac{\|\chi_R\|_{\frac{10}{3}}^{\frac{10}{3}}}{\|\chi_R\|_2^2} \leq \tau(-\Delta_0 \gamma) + O(R^{-1}).$$

Note that the ratio in the left-hand side is independent of R. Taking $R \to \infty$ shows that the remainder is actually null. For the previous computation to be valid, we need χ to be differentiable. However, since the result is valid for all χ, one can consider a sequence χ_n converging to $\mathbb{1}(|x| < 1)$, and we get, as wanted,

$$C_{\mathrm{LT}} \left(\int_{I_L} \fint_{\Omega} \varrho^{\frac{5}{3}} \right) \leq \tau(-\Delta_0 \gamma),$$

with the same 3D Lieb–Thirring constant.

We proceed similarly for the proof of Hoffmann–Ostenhof inequality. We get

$$\mathrm{Tr}\,(-\Delta \gamma_{\omega,R}) \geq \int_{\mathbb{R}^3} |\nabla \sqrt{\rho_{\omega,R}}|^2. \tag{II.4.11}$$

We conclude as for the proof of the Lieb–Thirring inequality. The passage from $|\nabla \sqrt{\rho}|$ to $|\Lambda \sqrt{\varrho}|$ comes from (II.3.22). □

Remark II.4.4 (Lieb–Thirring inequality for Neumann boundary conditions)**.** *The bound* (II.4.9) *does not hold if we replace the Dirichlet boundary conditions by the Neumann ones. Indeed, consider a smooth compactly supported 2D function $j \in C_0^\infty(\mathbb{R}^2)$ with $0 \leq \widehat{j}(k) \leq 1$ pointwise. For $\lambda > 0$ a scaling factor, we denote by J_λ the convolution operator acting on $L^2(\mathbb{R}^2)$, and with kernel $J_\lambda(x, x') = \lambda^2 j(\lambda(x - x'))$. The scaling is chosen so that J_λ is the Fourier multiplier by $\widehat{j}(k/\lambda)$. We finally consider the operator (see Example II.4.2)*

$$\gamma_\lambda := J_\lambda \otimes \left(L^{-1} |\mathbb{1}\rangle\langle \mathbb{1}| \right).$$

Since $0 \leq \widehat{j}(k/\lambda) \leq 1$, we have $0 \leq \gamma_\lambda = \gamma_\lambda^* \leq 1$. A computation shows that the density of γ_λ is constant, independent of $(x, z) \in S_L$, given by

$$\rho_\lambda(x) = \gamma_\lambda(x, x) = \lambda^2 \rho_{\lambda=1}(x) = \frac{1}{L} \lambda^2 j(0).$$

On the other hand, the kinetic energy per unit area scales as the two-dimensional one, namely

$$\tau(-\Delta_{\mathrm{N}} \gamma_\lambda) = \lambda^4 \tau(-\Delta_{\mathrm{N}} \gamma_{\lambda=1}) = \lambda^4 \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} |k|^2 \widehat{j}(k) \mathrm{d}k.$$

The scaling in λ of $\tau(-\Delta_{\mathrm{N}} \gamma_\lambda)$ therefore matches the one of ρ_λ^2 rather than $\rho_\lambda^{5/3}$. We recover the 2D Lieb–Thirring inequality in this example.

II.4.4 Reduced Hartree–Fock model

We define the rHF energy (per unit area) functional of a trial stationary density-matrix $\gamma \in \mathcal{K}$ as

$$\boxed{\mathcal{E}^{\mathrm{rHF}}(\gamma) := \frac{1}{2}\tau(-\Delta_0\gamma) + \frac{1}{2}\mathcal{D}_{\mathrm{erg}}(\varrho_\gamma - \mathfrak{m}, \varrho_\gamma - \mathfrak{m}) + v_0 \fint_\Omega \int_{I_L} g_0 \varrho_\gamma.} \tag{II.4.12}$$

Here $\mathfrak{m}$ is the generating function of the charge density m, that is $\mathfrak{m}(\alpha_x\omega, z) = m(\omega, x, z)$. The first term is the kinetic energy introduced in Section II.4.2, and the second term is the Hartree energy, introduced in Section II.3.5. The last term is the potential energy induced by the gating potential v_0 between the two electrodes, see (II.3.4).

Our main result is contained in the following theorem. For brevity, we focus on the microcanonical formulation of the rHF ground-state problem, but similar results can be obtained for the grand-canonical formulation

$$\min\left\{\mathcal{E}^{\mathrm{rHF}}(\gamma) - \mu\tau(\gamma),\ \gamma \in \mathcal{K}\right\},$$

where $\mu \in \mathbb{R}$ is a given arbitrary chemical potential.

Theorem II.4.5. *Let $\mathfrak{m} \in L^{\infty,p}$, with $\frac{3}{2} < p \leq \infty$ and $\lambda > 0$. The minimization problem*

$$\min\left\{\mathcal{E}^{\mathrm{rHF}}(\gamma),\ \gamma \in \mathcal{K},\ \tau(\gamma) = \lambda\right\} \tag{II.4.13}$$

admits a minimizer and all the minimizers share the same density ρ_*. Let γ_* be a minimizer of (II.4.13), and ϱ_* the generating function of ρ_*. Then $\varrho_* \in L^\infty(\Omega \times I_L)$. In addition, for almost all $\omega \in \Omega$, we have the Euler–Lagrange equations

$$\gamma_{*,\omega} = \mathbb{1}(H_{*,\omega} < \varepsilon_F) + \delta_\omega, \tag{II.4.14}$$

where $\varepsilon_F \in \mathbb{R}$ is some Fermi level, i.e. the Lagrange multiplier of the constraint $\tau(\gamma) = \lambda$, and $0 \leq \delta_\omega \leq \mathbb{1}(H_{*,\omega} = \varepsilon_F)$. Here, $H_{*,\omega}$ is the mean-field operator, defined by

$$H_{*,\omega} := -\frac{1}{2}\Delta_0 + V_{\rho_{*,\omega} - m_\omega} + v_0 g_0.$$

where $V_{\rho_{*,\omega}-m_\omega} \in L^\infty(S_L)$ is the encapsulated mean-field potential generated by the total charge density $\rho_{*,\omega} - m_\omega$, as defined in Section II.3.5. Finally, it holds that the generating function $\mathcal{V}_{\varrho_* - \mathfrak{m}}$ is in $L^\infty(\Omega \times I_L)$ as well.

Note that the assumption $\mathfrak{m} \in L^{\infty,p}$ with $p > \frac{3}{2}$ is quite natural, in the context of smeared nuclei. Indeed, this assumption is satisfied if the atomic charge density m_i corresponding to the lattice $\mathbb{L}_i$ belongs to $L^p_{\mathrm{unif}}(S_L)$, $p > \frac{3}{2}$. Recall that $\omega \in \Omega$ corresponds physically to a translation in space of the lattice. From (II.2.1), it follows that $m_\omega \in L^p_{\mathrm{unif}}(S_L)$ as well, and $\|m_\omega\|_{L^p_{\mathrm{unif}}(S_L)}$ is bounded uniformly in ω, hence the generating function verifies $\mathfrak{m} \in L^{\infty,p}$.

Remark II.4.6. *Theorem II.4.5 shows that the ground-state problem* (II.4.13) *has a minimizer for* any *positive value λ of the electronic charge per unit area. However, the model only makes sense when the electronic density of the moiré material penetrates the dielectric medium only slightly, and decays very fast away from the moiré material. This is expected to be the case around charge neutrality, which is the physically most relevant case (and confirmed with preliminary numerical simulations). This also motivates the use of Dirichlet boundary conditions in our definition of the kinetic energy operator $-\frac{1}{2}\Delta_0$. When λ increases, the ground-state density starts to penetrate the dielectric continuum modeling the insulating substrate, a phenomenon reminiscent of the escaped charge problem encountered in quantum chemistry when using implicit solvent models for describing solvated ions [***CM2001***]. For λ too large, the implicit assumption underlying the model, that the moiré material electrons can somehow be decoupled from the insulating substrate electrons, breaks down.*

Proof of Theorem II.4.5. We split the proof into several steps for clarity.

Step 1: The minimization set is non empty. Let us first prove that one can find $\gamma \in \mathcal{K}$ with $\tau(\gamma) = \lambda$ and $\mathcal{E}^{\mathrm{rHF}}(\gamma) < \infty$. Consider a smooth function $h \in H^1_0(I_L)$ with $\|h\|_{L^2(I_L)} = 1$, a smooth function $j : \mathbb{R}^2 \to \mathbb{R}$ satisfying $\widehat{j}(k) \in C^\infty_0(\mathbb{R}^2)$ and $0 \leq \widehat{j}(k) \leq 1$ pointwise, and the operator (see also Example II.4.2)

$$\gamma = \int_\Omega^\oplus \gamma_\omega \mathrm{d}\omega, \quad \gamma_\omega := \widehat{j}(-\mathrm{i}\nabla_x) \otimes |h\rangle\langle h|. \tag{II.4.15}$$

Note that γ_ω is in fact independent of $\omega \in \Omega$, and acts as a Fourier multiplier in the longitudinal variables. Since γ_ω is translation-invariant in the longitudinal directions, γ is a fibered stationary operator. The density of γ is

$$\rho_\gamma(x,z) = \left(\frac{1}{(2\pi)^2}\int_{\mathbb{R}^2} \widehat{j}(k)\mathrm{d}k\right)|h(z)|^2 = j(0)|h(z)|^2,$$

and in particular is independent of $x \in \mathbb{R}^2$. So

$$\varrho(\omega, z) = j(0)|h(z)|^2 \quad \text{and} \quad \tau(\gamma) = j(0).$$

Since $h \in H^1_0(I_L)$, it is continuous by the Sobolev embedding, so that ϱ is continuous as well. In particular, it is bounded, so that $\varrho - \mathfrak{m} \in L^{\infty,p}$. Since $p > \frac{3}{2}$ by assumption, the Hartree energy $\mathcal{D}_{\mathrm{erg}}(\varrho - \mathfrak{m}, \varrho - \mathfrak{m})$ is finite. Similarly, the kinetic energy of γ is

$$\frac{1}{2}\tau(-\Delta_0\gamma) = \frac{1}{2}\left(\frac{1}{(2\pi)^2}\int_{\mathbb{R}^2} |k|^2\widehat{j}(k)\mathrm{d}k\right) + \frac{1}{2}j(0)\|\nabla h\|^2_{L^2(I_L)},$$

and in particular is finite, since $\widehat{j} \in C^\infty_0(\mathbb{R}^2)$. This proves that the minimization set is non empty.

Step 2: The energy $\mathcal{E}^{\mathrm{rHF}}$ (II.4.12) is bounded from below. This follows directly from the fact that the first two terms are nonnegative, and the last one is bounded from below

by $-|v_0|\,\|g_0\|_{L^\infty}\|\varrho\|_{L^1}$, where $g_0(z)$ is defined in (II.3.5). Actually, since g_0 is strictly increasing with $g_0(\pm L/2) = \pm 1/2$, we have $\|g_0\|_\infty = \frac{1}{2}$. This gives $\mathcal{E}^{\mathrm{rHF}} \geq -\frac{1}{2}|v_0|\lambda$ on $\mathcal{K}$.

Step 3: Extraction of a subsequence. Consider (γ_n) a minimizing sequence for (II.4.13). As

$$0 \leq \frac{1}{2}\tau(-\Delta_0\gamma_n) \leq \mathcal{E}^{\mathrm{rHF}}(\gamma_n) + |v_0|\,\|g_0\|_{L^\infty}\lambda,$$

the sequence $(\tau(-\Delta_0\gamma_n))$ is bounded. Since $\gamma_n \in \mathcal{K}$ satisfies $0 \leq \gamma_n \leq 1$ and $\tau(\gamma_n) = \lambda$, we deduce that the sequence (γ_n) is bounded in $L^1(\mathcal{A},\tau)\cap L^\infty(\mathcal{A},\tau)$, hence in all $L^p(\mathcal{A},\tau)$ spaces.

For $p > 1$, we may use the Banach–Alaoglu theorem. Up to a subsequence (undisplayed), we may assume that

$$\begin{cases} \gamma_n \rightharpoonup \gamma_* & \text{weakly in } L^p(\mathcal{A},\tau) \text{ for all } 1 < p < \infty \\ \gamma_n \rightharpoonup_* \gamma_* & \text{weakly-}* \text{ in } L^\infty(\mathcal{A},\tau). \end{cases}$$

In particular, we have $0 \leq \gamma_* \leq 1$ and $\tau(\gamma_*) \leq \lambda$. Finally, Fatou's Lemma implies that

$$\tau(-\Delta_0\gamma_*) = \tau(\sqrt{-\Delta_0}\gamma_*\sqrt{-\Delta_0}) \leq \liminf_{n\to\infty} \tau(\sqrt{-\Delta_0}\gamma_n\sqrt{-\Delta_0}) \leq C$$

is finite. So $\gamma_* \in W_0^{1,1}(\mathcal{A},\tau)$.

Step 4: Convergence of the densities. We follow the approach in [CLL13]. This time, we use the bound

$$\tau(\sqrt{-\Delta_0}\gamma_n\sqrt{-\Delta_0}) \leq C.$$

Equivalently, $\sqrt{-\Delta_0}\gamma_n^{1/2}$ is bounded in $L^2(\mathcal{A},\tau)$. Let us focus on the sequence

$$T_n := \sqrt{-\Delta_0}\gamma_n.$$

Since $T_n = \sqrt{-\Delta_0}\gamma_n = \sqrt{-\Delta_0}\gamma_n^{1/2}\gamma_n^{1/2}$ with $\sqrt{-\Delta_0}\gamma_n^{1/2}$ bounded in $L^2(\mathcal{A},\tau)$ and $\gamma_n^{1/2}$ bounded in $L^2(\mathcal{A},\tau)\cap L^\infty(\mathcal{A},\tau)$, we deduce that T_n is bounded in $L^1(\mathcal{A},\tau)\cap L^2(\mathcal{A},\tau)$. According to the Banach–Alaoglu theorem, we may extract a subsequence (undisplayed) and a limit T_* so that

$$T_n \rightharpoonup T_* \quad \text{in } L^p(\mathcal{A},\tau) \text{ for all } 1 < p \leq 2.$$

Finally, we set $\widetilde{\gamma}_* := \sqrt{-\Delta_0}^{-1}T_*$.

Let us prove that γ_n converges weakly to $\widetilde{\gamma}_*$ in $L^p(\mathcal{A},\tau)$ spaces for all $1 \leq p < 2$ (this will eventually prove that $\widetilde{\gamma}_* = \gamma_*$).

Lemma II.4.7. *We have $(-\Delta_0)^{-1} \in L^q(\mathcal{A},\tau)$ for all $q > 3/2$, and*

$$\tau((-\Delta_0)^{-q}) = \frac{L^{2(q-1)}}{4(q-1)\pi^{2q-1}}\zeta(2q-2),$$

where $\zeta(\cdot)$ is the Riemann zeta function.

Proof. From Eqn. (II.4.7), we get

$$-\Delta_0 = \sum_{k=1}^{\infty}(-\Delta_x + \mu_k) \otimes |h_k\rangle\langle h_k|, \quad \text{hence} \quad \frac{1}{(-\Delta_0)^q} = \sum_{k=1}^{\infty} \frac{1}{(-\Delta_x + \mu_k)^q} \otimes |h_k\rangle\langle h_k|.$$

This is the sum of operators of the forms considered in Example II.4.2. We deduce that

$$\tau\left(\frac{1}{(-\Delta_x + \mu_k)^q} \otimes |h_k\rangle\langle h_k|\right) = \frac{1}{(2\pi)^2}\int_{\mathbb{R}^2} \frac{\mathrm{d}\xi}{(|\xi|^2 + \mu_k)^q} = \frac{1}{(2\pi)^2}\mu_k^{1-q}\int_{\mathbb{R}^2} \underbrace{\frac{\mathrm{d}\xi}{(|\xi|^2+1)^q}}_{=\frac{\pi}{q-1}}.$$

Recall that $\mu_k = \frac{\pi^2 k^2}{L^2}$. We obtain that for $q > \frac{3}{2}$,

$$\tau((-\Delta_0)^{-q}) = \frac{1}{4\pi(q-1)}\sum_{k=1}^{\infty}\left(\frac{L^2}{\pi^2 k^2}\right)^{q-1} = \frac{L^{2(q-1)}}{4(q-1)\pi^{2q-1}}\sum_{k=1}^{\infty}\frac{1}{k^{2q-2}} = \frac{L^{2(q-1)}}{4(q-1)\pi^{2q-1}}\zeta(2q-2),$$

which concludes the proof. □

Since $(-\Delta_0)^{-1/2}$ is in $L^q(\mathcal{A},\tau)$ for all $q > 3$, then for all $A \in L^r(\mathcal{A},\tau)$ with $r > 1$, we have $A(-\Delta_0)^{-1/2}$ in $L^s(\mathcal{A},\tau)$ for all $\frac{3r}{r+3} < s \leq r$. In particular, for all $2 < r \leq \infty$, we can tune q so that $s > 2$ as well, in which case $A(-\Delta_0)^{-1/2}$ is a valid test operator for the weak-convergence of T_n.

We deduce that, for all $A \in L^r(\mathcal{A},\tau)$ with $2 < r \leq \infty$, we have

$$\tau(A\gamma_n) = \tau(A(-\Delta_0)^{-1/2}T_n) \xrightarrow[n\to\infty]{} \tau(A(-\Delta_0)^{-1/2}T_*) = \tau(A\widetilde{\gamma}_*).$$

This proves that γ_n converges weakly to $\widetilde{\gamma}_*$ in $L^p(\mathcal{A},\tau)$ for all $1 \leq p < 2$. In particular, we have $\gamma_* = \widetilde{\gamma}_*$.

We denote by ϱ_n and ϱ_* the generalized density of γ_n and γ_* respectively. Consider any positive bounded function $\Phi(\omega, z) : \Omega \times I_L \to \mathbb{R}$, and set

$$M_\Phi := \int_\Omega (M_\phi)_\omega \mathrm{d}\omega, \qquad [(M_\phi)_\omega u](x,z) := \Phi(\alpha_x\omega, z)u(x,z).$$

In other words, M_Φ represents the multiplication operator by a stationary function $\phi_\omega(x,z) := \Phi(\alpha_x\omega, z)$. Since $\phi \in L^\infty$, we have $M_\Phi \in L^\infty(\mathcal{A},\tau)$. Taking $A = M_\Phi$ in the previous computation shows that

$$\int_{\Omega\times I_L}\Phi\varrho_n = \tau(M_\Phi\gamma_n) \xrightarrow[n\to\infty]{} \tau(M_\Phi\gamma_*) = \int_{\Omega\times I_L}\Phi\varrho_*.$$

We deduce that ϱ_n converges weakly to ϱ_* in $L^1(\Omega\times I_L)$ (recall that $(L^1)^* = L^\infty$ and that $L^1 \subsetneq (L^\infty)^*$. Recall also that the unit ball of L^1 is not compact for the weak topology). Taking $\Phi = 1$ shows that

$$\tau(\gamma_*) = \int_{\Omega\times I_L}\varrho_* = \lambda.$$

On the other hand, recall that the sequence $\tau(-\Delta_0\gamma_n)$ is bounded. Using Lemma II.4.3, we obtain that (ϱ_n) is a bounded sequence in $L^q(\Omega\times I_L)$ for all $1\leq q\leq 5/3$. In particular, up to another (undisplayed) extraction, we may assume that (ϱ_n) converges weakly to some $\widetilde{\varrho}_*$ in $L^q(\Omega\times I_L)$ for all $1<q\leq 5/3$. Actually, we have $\widetilde{\varrho}_*=\varrho_*$. Indeed, consider any $\varphi\in L^\infty(\Omega\times I_L)$. Then $\varphi\in L^q(\Omega\times I_L)$ as well, by Hölder inequality (the set $\Omega\times I_L$ is bounded), so $\int_{\Omega\times I_L}\varphi\rho_n$ converges to both $\int_{\Omega\times I_L}\varphi\rho_*$ and $\int_{\Omega\times I_L}\varphi\widetilde{\rho}_*$. We deduce that $\int\varphi\rho_*=\int\varphi\widetilde{\rho}_*$ for all such φ. So $\widetilde{\varrho}_*=\varrho_*$.

Step 5: Convergence of the energy. At this point, we proved the following convergences:

$$\begin{cases}\gamma_n\rightharpoonup\gamma_* & \text{weakly in } L^p(\mathcal{A},\tau) \text{ for all } 1<p<\infty,\\ \gamma_n\rightharpoonup_*\gamma_* & \text{weakly-}* \text{ in } L^\infty(\mathcal{A},\tau),\\ \varrho_n\rightharpoonup\varrho_* & \text{weakly in } L^p(\Omega\times I_L) \text{ for all } 1<p\leq 5/3.\end{cases}$$

In addition, we have $\tau(\gamma_*)=\lambda$ and

$$\tau(-\Delta_0\gamma_*)\leq\liminf_{n\to\infty}\tau(-\Delta_0\gamma_n).$$

According to Lemma II.3.8, the map $\varrho\mapsto\mathcal{D}_{\mathrm{erg}}(\varrho-\mathfrak{m},\varrho-\mathfrak{m})$ is weakly lower-semicontinuous in $L^{5/3}(\Omega\times I_L)$. We directly deduce that

$$\mathcal{E}^{\mathrm{rHF}}(\gamma_*)\leq\liminf_{n\to\infty}\mathcal{E}^{\mathrm{rHF}}(\gamma_n).$$

In addition, since $\gamma_*\in\mathcal{K}$ with $\tau(\gamma_*)=\lambda$, we get that γ_* is a minimizer of (II.4.13).

Step 6: Uniqueness of the ground-state density. The kinetic term $\tau(-\Delta_0\gamma)$ is linear in γ, while the Hartree term $\gamma\mapsto\mathcal{D}_{\mathrm{erg}}(\varrho_\gamma-\mathfrak{m},\varrho_\gamma-\mathfrak{m})$ is convex in γ and strictly convex in the density. Hence all minimizers must share the same density, which we denote by ϱ_*.

Step 7: Euler–Lagrange equations. The set $\{\gamma\in\mathcal{K},\ \tau(\gamma)=\lambda,\ \mathcal{D}_{\mathrm{erg}}(\varrho_\gamma,\varrho_\gamma)<\infty\}$ is convex. For any γ in this set with density ϱ, and any $0\leq t\leq 1$, we have

$$\begin{aligned}\mathcal{E}^{\mathrm{rHF}}(\gamma_*)&\leq\mathcal{E}^{\mathrm{rHF}}((1-t)\gamma_*+t\gamma)\\&\leq\mathcal{E}^{\mathrm{rHF}}(\gamma_*)+t\left[\frac{1}{2}\tau(-\Delta_0(\gamma-\gamma_*))+\mathcal{D}_{\mathrm{erg}}(\varrho_*-\mathfrak{m},\varrho-\varrho_*)+v_0\fint_\Omega\int_{I_L}g_0(\varrho-\varrho_*)\right]\\&\quad+\frac{t^2}{2}\mathcal{D}_{\mathrm{erg}}(\varrho-\varrho_*,\varrho-\varrho_*).\end{aligned}$$

Taking the limit $t\to 0^+$ shows that for all such γ, we have

$$\frac{1}{2}\tau(-\Delta_0\gamma)+\mathcal{D}_{\mathrm{erg}}(\varrho_*-\mathfrak{m},\varrho)+v_0\fint_\Omega\int_{I_L}g_0\varrho\geq\frac{1}{2}\tau(-\Delta_0\gamma_*)+\mathcal{D}_{\mathrm{erg}}(\varrho_*-\mathfrak{m},\varrho_*)+v_0\fint_\Omega\int_{I_L}g_0\varrho_*.$$

In other words, γ_* is also the minimizer of the linearized problem

$$\min\left\{\frac{1}{2}\tau(-\Delta_0\gamma)+\mathcal{D}_{\mathrm{erg}}(\varrho_*-\mathfrak{m},\varrho_\gamma)+v_0\fint_\Omega\int_{I_L}g_0\varrho_\gamma\ ;\ \gamma\in\mathcal{K},\ \tau(\gamma)=\lambda\right\}.\qquad\text{(II.4.16)}$$

Recall that ϱ_* is independent of the minimizer. Let us set $\rho_{*,\omega} := \rho_*(\omega, x, z) := \varrho_*(\alpha_x\omega, z)$, and let us introduce the mean-field operator

$$H_* := \int_\Omega H_{*,\omega}\mathrm{d}\omega, \quad \text{with} \quad H_{*,\omega} := -\frac{1}{2}\Delta_0 + V_{*,\omega} + v_0 g_0, \quad \text{with} \quad V_{*,\omega} := V_{\rho_{*,\omega}} - V_{m_\omega}.$$

Since $\mathfrak{m} \in L^{\infty,p}$ with $p > 3/2$ and $\varrho_* \in L^{5/3}$, we infer from Theorem II.3.7 that the generating functions $\mathcal{V}_\mathfrak{m}$ and $\mathcal{V}_{\varrho_\star}$ of the stationary potentials V_{m_ω} and $V_{\rho_{*,\omega}}$ belong to $L^\infty(\Omega \times I_L)$ and $L^{5/3,\infty}$, respectively. It follows that for almost all ω, V_{m_ω} is nonnegative and essentially bounded, while $V_{\rho_{*,\omega}}$ is nonnegative and in $L^1_{\mathrm{loc}}(S_L)$. In particular, $0 \le \mathcal{V}_*^- \le \mathcal{V}_\mathfrak{m}^+$, and $V_{*,\omega}^- \in L^\infty(S_L)$ with uniform bound in ω:

$$\|V_{*,\omega}^-\|_{L^\infty(S_L)} \le C^{\mathrm{stat}}_{\infty,p,L}\|\mathfrak{m}\|_{L^{\infty,p}}. \tag{II.4.17}$$

As a consequence, $H_{*,\omega}$ is a well-defined bounded-from-below self-adjoint operator on $L^2(S_L)$ [**Lew-24**]. In addition, $V_{*,\omega}(x,z)$ is a stationary function by Theorem II.3.7, so that the operator H_* is a fibered stationary (unbounded) self-adjoint operator affiliated to $\mathcal{A}$, since $(H_* + i)^{-1} \in \mathcal{A}$ and the map $\omega \mapsto H_{*,\omega}$ is measurable.

Since the density of any $\gamma \in \mathcal{K}$ satisfies $\varrho_\gamma \in L^{5/3}(\Omega \times I_L)$ by the Lieb–Thirring inequality (II.4.9), $\mathcal{V}_\mathfrak{m} \in L^\infty(\Omega \times I_L)$, $\mathcal{V}_{\varrho_*} \in L^{5/3,\infty}$, and $\mathcal{V}_\mathfrak{m}, \mathcal{V}_{\varrho_*} \ge 0$, we deduce from Lemma II.3.8 that

$$\begin{aligned}\mathcal{D}_{\mathrm{erg}}(\varrho_* - \mathfrak{m}, \varrho_\gamma) = \mathcal{D}_{\mathrm{erg}}(\varrho_*, \varrho_\gamma) - \mathcal{D}_{\mathrm{erg}}(\mathfrak{m}, \varrho_\gamma) &= \int_{\Omega\times I_L} \mathcal{V}_{\varrho_*}\varrho_\gamma - \int_{\Omega\times I_L} \mathcal{V}_\mathfrak{m}\varrho_\gamma \\ &= \tau(\mathcal{V}_{\varrho_*}\gamma) - \tau(\mathcal{V}_\mathfrak{m}\gamma) = \tau(\mathcal{V}_*\gamma).\end{aligned}$$

So the linearized problem (II.4.16) also reads

$$\min\left\{\tau(H_*\gamma),\ \gamma \in \mathcal{K},\ \tau(\gamma) = \lambda\right\}. \tag{II.4.18}$$

Lemma II.4.8. *For all $\mu \in \mathbb{R}$, the projector $\Pi_\mu := \mathbb{1}_{(-\infty,\mu)}(H_*)$ is a fibered stationary operator in $W_0^{1,1}(\mathcal{A},\tau)$ and all the minimizers of the problem*

$$\inf\left\{\tau(H_*\gamma) - \mu\tau(\gamma),\ \gamma \in \mathcal{K}\right\}, \tag{II.4.19}$$

are of the form

$$\gamma_\mu = \Pi_\mu + \delta, \quad \textit{with} \quad \delta \in W_0^{1,1}(\mathcal{A},\tau)\ \textit{self-adjoint}, \quad 0 \le \delta \le \mathbb{1}_{\{\mu\}}(H_*).$$

Proof. Let us show that Π_μ belongs to $\mathcal{K}$. First, Π_μ is a self-adjoint fibered stationary operator and $0 \le \Pi_\mu \le 1$, which is clear since the function $0 \le \mathbb{1}_{(-\infty,\mu)} \le 1$ is measurable and real-valued, and H_* is affiliated with $\mathcal{A}$. We then have to show that $\Pi_\mu \in W_0^{1,1}(\mathcal{A},\tau)$. We claim that, for a.e. $\omega \in \Omega$, we have

$$\rho_{A_\omega} \le Ce^{(\|V_{*,\omega}^-\|_{L^\infty(S_L)} + |v_0|/2)}, \quad \text{where} \quad A_\omega := e^{-H_{*,\omega}} = e^{\frac{1}{2}\Delta_0 - V_{*,\omega} - v_0 g_0}. \tag{II.4.20}$$

This follows from the Feynman-Kac formula for the Dirichlet Laplacian [Sim79], and uses the fact that the heat kernel of $-\Delta_0$ on S_L is pointwise bounded by the heat kernel of $-\Delta$ on $\mathbb{R}^3$. The remainder of the argument is then identical to [CLL13, Prop. 2.11].

The right-hand-side of the inequality in (II.4.20) is finite and bounded uniformly in ω, by (II.4.17). Using the inequality $\mathbb{1}_{(-\infty,\mu)}(x) \leq e^{\mu-x}$, we get that $\varrho_{\Pi_\mu} \in L^\infty(\Omega \times I_L) \subset L^1(\Omega \times I_L)$, so that $\Pi_\mu \in L^1(\mathcal{A}, \tau)$:

$$0 \leq \varrho_{\Pi_\mu} \leq C_{p,\mathfrak{m},L} e^{\mu}, \tag{II.4.21}$$

where the constant $C_{p,\mathfrak{m},L} > 0$ comes from (II.4.17). Furthermore, the inequality $x\mathbb{1}_{(-\infty,\mu)}(x) \leq \mu \mathbb{1}_{\mu \geq 0} e^{\mu - x}$ combined to the fact that $H_* \geq -C$ yields

$$-C\Pi_\mu \leq H_*\Pi_\mu \leq \mu \mathbb{1}_{\mu \geq 0}\, e^{-(H_* - \mu)}.$$

Hence $H_*\Pi_\mu \in L^1(\mathcal{A}, \tau)$, which implies that $\Pi_\mu \in W_0^{1,1}(\mathcal{A}, \tau)$.

Finally, adapting [CLL13, Prop. 2.12], we show that the minimizers of (II.4.19) are of the stated form. Let $\gamma \in \mathcal{K}$. Since $0 \leq \gamma \leq 1$, we get that

$$\Pi_\mu^\perp(\gamma - \Pi_\mu)\Pi_\mu^\perp - \Pi_\mu(\gamma - \Pi_\mu)\Pi_\mu \geq (\gamma - \Pi_\mu)^2,$$

yielding formally that

$$\tau(\Pi_\mu^\perp |H_* - \mu| \Pi_\mu^\perp(\gamma - \Pi_\mu)) - \tau(\Pi_\mu |H_* - \mu| \Pi_\mu(\gamma - \Pi_\mu)) \geq \tau(|H_* - \mu|(\gamma - \Pi_\mu)^2).$$

Now, the left-hand-side, using the definition of Π_μ, is equal to $\tau((H_* - \mu)(\gamma - \Pi_\mu))$, leading to

$$\tau((H_* - \mu)(\gamma - \Pi_\mu)) \geq \tau(|H_* - \mu|(\gamma - \Pi_\mu)^2). \tag{II.4.22}$$

Define the functional

$$\mathcal{F}(\gamma) := \frac{1}{2}\tau(-\Delta_0 \gamma) + \int_{\Omega \times I_L} (\mathcal{V}_* + v_0 g_0)\varrho_\gamma - \mu\tau(\gamma). \tag{II.4.23}$$

As mentioned above, the regularity of $\mathcal{V}_*$ allows to write that

$$\mathcal{F}(\gamma) = \tau((H_* - \mu)\gamma). \tag{II.4.24}$$

Then, (II.4.22) makes sense, and it yields for $\gamma \in \mathcal{K}$, that

$$\mathcal{F}(\gamma) - \mathcal{F}(\Pi_\mu) \geq \tau(|H_* - \mu|^{1/2}(\gamma - \Pi_\mu)^2 |H_* - \mu|^{1/2}) \geq 0. \tag{II.4.25}$$

Hence Π_μ is clearly a minimizer of (II.4.18), and every minimizer γ must verify that $|H_* - \mu|^{1/2}(\gamma - \Pi_\mu) = 0$, so that $\mathrm{Ran}(\gamma - \Pi_\mu) \subset \mathrm{Ker}(H_* - \mu)$. □

To conclude the proof of the theorem, we proceed as in [CLL13, Prop. 4.5]. As mentioned above, γ_* is a minimizer of the linearized problem (II.4.18), that is, for $\gamma \in \mathcal{K}$ such that $\tau(\gamma) = \lambda$,

$$\tau(H_*(\gamma - \gamma_*)) \geq 0.$$

Define, for $\mu \in \mathbb{R}$,

$$E(\mu) := \inf \{\tau(H_*(\gamma - \gamma_*)),\ \gamma \in \mathcal{K},\ \tau(\gamma) = \mu\}.$$

The function $E : \mathbb{R} \to \mathbb{R}$ is convex, hence continuous and left and right differentiable at each $\mu \in \mathbb{R}$. Choose any

$$\varepsilon_F \in [E'(\lambda^-), E'(\lambda^+)],$$

where $E'(\lambda^-)$ and $E'(\lambda^+)$ respectively denote the left and right derivatives of E at λ. Using the fact that $E(\lambda) = 0$, we have that for any $\gamma \in \mathcal{K}$,

$$\varepsilon_F \tau(\gamma - \gamma_*) = \varepsilon_F(\tau(\gamma) - \lambda) \leq E(\tau(\gamma)) - E(\lambda) = E(\tau(\gamma)) \leq \tau(H_*(\gamma - \gamma_*)),$$

which gives $\tau((H_* - \varepsilon_F)(\gamma - \gamma_*)) \geq 0$. It follows that γ_* is also a minimizer of (II.4.19) with $\mu = \varepsilon_F$, so that, by Lemma II.4.8, γ_* is of the form (II.4.14).

Finally, since any minimizer γ_* satisfies $\gamma_* \leq \Pi_{\varepsilon_F+1}$, we have $\varrho_* \leq \varrho_{\Pi_{\varepsilon_F+1}}$, which, in view of the bound (II.4.21), implies that $\varrho_* \in L^\infty(\Omega \times I_L)$. It follows from Theorem II.3.7 that the potential $\mathcal{V}_*$ is bounded as well. This concludes the proof of Theorem II.4.5. □

II.5 Kohn–Sham models for encapsulated periodic 2D materials

For completeness, we provide a proof of existence of a Kohn-Sham LDA ground state for encapsulated 2D periodic materials. Unfortunately, our proof does not extend to the quasi-periodic setting. The most usual and practical way to write down a periodic Kohn–Sham model is to rely on Bloch theory. In order to recycle what has been done in the previous section, we will however follow a different (but equivalent) approach consisting in reformulating the periodic problem as a stationary ergodic problem.

II.5.1 The periodic setting as a stationary one

Consider a 2D material with $\mathbb{L}$-periodic nuclear charge distribution $m : S_L \to \mathbb{R}$, where $\mathbb{L}$ is a periodic lattice of $\mathbb{R}^2$. We introduce the configuration space $\widetilde{\Omega} := \mathbb{R}^2/\mathbb{L}$, and the group action $\mathbb{R}^2 \ni x \mapsto \widetilde{\alpha}_x$, defined by

$$\widetilde{\alpha}_x : \widetilde{\Omega} \to \widetilde{\Omega}, \qquad \widetilde{\alpha}_x(\widetilde{\omega}) = \widetilde{\omega} - x \mod (\mathbb{L}).$$

For a function $f : S_L \to \mathbb{R}$, periodic in the sense $f(x + \ell, z) = f(x, z)$ for all $\ell \in \mathbb{L}$, we introduce the function $\widetilde{f} : \widetilde{\Omega} \times \mathbb{R}^2 \times I_L \to \mathbb{R}$ defined by

$$\widetilde{f}(\widetilde{\omega}, x, z) := \widetilde{f}_{\widetilde{\omega}}(x, z) := f(x - \widetilde{\omega}, z).$$

Then $\widetilde{f}$ is stationary for the group action $\widetilde{\alpha}$. In this context, the configuration $\widetilde{\omega} \in \widetilde{\Omega}$ encodes a global shift of the lattice $\mathbb{L}$. Note that $\widetilde{f}_{\widetilde{\omega}=0} = f$ and that the generating function $\widetilde{\mathfrak{f}}$ of $\widetilde{f}$ can be seen as the original function f, up to inversion:

$$\forall \widetilde{\omega} \in \widetilde{\Omega}, \quad \widetilde{\mathfrak{f}}(\omega, z) = f(-\omega, z).$$

This proves that periodic functions can be identified with stationary function, with this specific group action.

The action $\widetilde{\alpha}$ is ergodic (the only functions invariant by all shifts are the constant functions). Actually, Birkhoff's theorem becomes trivial in this special case: for any configuration $\widetilde{\omega}$, and any locally-integrable $\mathbb{L}$-periodic function f, we have with the above notation

$$\begin{aligned}\lim_{R\to\infty} \frac{1}{(2R)^2} \int_{[-R,R]^2\times I_L} \widetilde{f}_{\widetilde{\omega}}(x,z)\,\mathrm{d}x\,\mathrm{d}z &= \lim_{R\to\infty} \frac{1}{(2R)^2} \int_{[-R,R]^2\times I_L} f(x-\widetilde{\omega},z)\,\mathrm{d}x\,\mathrm{d}z \\ = \lim_{R\to\infty} \frac{1}{(2R)^2} \int_{([-R,R]^2-\widetilde{\omega})\times I_L} f(x,z)\,dx &= \int_{I_L} \fint_{\mathbb{R}^2/\mathbb{L}} f(x,z)\,\mathrm{d}x\,\mathrm{d}z = \int_{I_L} \fint_{\mathbb{R}^2/\mathbb{L}} \widetilde{\mathfrak{f}}(\omega,z)\,\mathrm{d}\omega\mathrm{d}z.\end{aligned}$$

Using this observation, the original periodic Kohn–Sham model for the nuclear charge distribution m, can be transformed into a stationary Kohn–Sham problem with stationary density $\widetilde{m}$.

Before writing down the corresponding encapsulated LDA model, let us emphasize the fundamental difference between the stationary ergodic settings corresponding to periodic and quasi-periodic 2D materials respectively. To illustrate this point, consider the simple case of twisted bilayer graphene (TBG) with twist angle θ. We define the configuration space Ω as $\Omega = \mathbb{R}^2/\mathbb{L}_1 \times \mathbb{R}^2/\mathbb{L}_2$, with $\mathbb{L}_1 := R_{-\theta/2}\mathbb{L}_{\mathrm{mg}}$, $\mathbb{L}_2 := R_{\theta/2}\mathbb{L}_{\mathrm{mg}}$, where $\mathbb{L}_{\mathrm{mg}}$ is the monolayer graphene triangular lattice, and R_φ the 2D rotation matrix of angle φ, and the action α of $\mathbb{R}^2$ on Ω by $\alpha_x(\omega_1,\omega_2) := (\omega_1 - x \mod (\mathbb{L}_1), \omega_2 - x \mod (\mathbb{L}_2))$. Recall that the system is in the configuration $\omega = (\omega_1,\omega_2) \in \Omega$ if the first graphene layer has a carbon atom of type A at $(\omega_1, z_1) \in S_L$ and the second layer a carbon atom of type A at $(\omega_2, z_2) \in S_L$, where z_j is the height of the plane containing the jth monolayer. The action α encodes uniform longitudinal translations of the system. Let

$$\Theta_{\mathrm{per}} := \left\{ \theta \in [0,\pi/3] :\ \cos\theta = \frac{3p^2-q^2}{3p^2+q^2}, \quad p,q \in \mathbb{N}^*, \quad p \wedge q = 1 \right\} \cup \{0\}.$$

If $\theta \in \Theta_{\mathrm{per}}$, then the system is $\mathbb{L}$-periodic, with $\mathbb{L} := \mathbb{L}_1 \cap \mathbb{L}_2$. If $\theta \in [0,\pi/3] \setminus \Theta_{\mathrm{per}}$, it is quasi-periodic (the lattices are incommensurate). If $\theta \in [0,\pi/3] \setminus \Theta_{\mathrm{per}}$, α is an ergodic action and all configurations have the same energy. If $\theta \in \Theta_{\mathrm{per}}$, then α is not ergodic and the energy depends on the configuration ω. Of course, if ω_0 is the configuration corresponding to the periodic charge distribution m and $O_{\omega_0} := \{\alpha_x\omega_0,\, x \in \mathbb{R}^2\}$ the orbit of ω_0, all the configurations in O_{ω_0} have the same energy per unit area. These are precisely the configurations sampled by the action of $\widetilde{\alpha}$ on $\widetilde{\Omega}$. In the periodic setting, all the configurations in $\widetilde{\Omega}$ are strictly identical from a physical viewpoint (they are mere

translations of one another), while in the quasi-periodic setting, all the configurations in Ω are not identical (only the ones in a given orbit O_{ω_0} are identical), although they have the same energy per unit area.

II.5.2 The periodic Kohn–Sham in the stationary framework

That being said, let us now formulate the encapsulated Kohn–Sham LDA model for a 2D material with $\mathbb{L}$-periodic nuclear charge distribution m. We make the assumption that the ground-state one-body density operator (and its corresponding density) is also $\mathbb{L}$-periodic, i.e. that there is no symmetry breaking although this assumption may fail due to the non convexity of the problem, see [GLN21, Section 3] (following [Ric18]).

It is easy to see that the von Neumann algebra $\mathcal{A}$ associated with $(S_L, \widetilde{\Omega}, \widetilde{\alpha})$ is isometric to the von Neumann algebra of $\mathbb{L}$-periodic bounded operator on $L^2(S_L)$, and that the trace $\widetilde{\tau}$ on $\widetilde{\mathcal{A}}$ agrees (for a suitable scaling factor) with the trace per unit area $\underline{\mathrm{Tr}}$ on the space of $\mathbb{L}$-periodic operators on $L^2(S_L)$. More precisely, if A is a locally-trace-class bounded $\mathbb{L}$-periodic operator on $L^2(S_L)$ and $\widetilde{A}$ its counterpart in $\widetilde{\mathcal{A}}$, it holds that $\widetilde{A} \in L^1(\widetilde{\mathcal{A}}, \widetilde{\tau})$ and we have

$$\widetilde{\tau}(\widetilde{A}) = \frac{1}{|\mathbb{R}^2/\mathbb{L}|} \fint_{\mathbb{R}^2/\mathbb{L}^*} \mathrm{Tr}_{L^2(\mathbb{R}^2/\mathbb{L}\times I_L)}\left(A_k\right) \mathrm{d}k = \underline{\mathrm{Tr}}(A),$$

where

$$A = \fint_{\mathbb{R}^2/\mathbb{L}^*}^{\oplus} A_k \,\mathrm{d}k$$

is the Bloch decomposition of A. Likewise, the ergodic Hartree energy per unit area $\widetilde{D}_{\mathrm{erg}}$ of the generating function $\widetilde{\mathfrak{f}}$ is related to the periodic Hartree energy per unit cell of the periodic charge distribution f (see (II.3.18)) by

$$\widetilde{D}_{\mathrm{erg}}(\widetilde{\mathfrak{f}}, \widetilde{\mathfrak{f}}) = \frac{1}{|\mathbb{R}^2/\mathbb{L}|} D_{\mathrm{per}}(f, f).$$

Finally, the LDA exchange-correlation energy per unit area of a periodic electronic density ρ, or equivalently of the generating function $\widetilde{\varrho}$ since $\widetilde{\varrho}(\omega, z) = \rho(-\omega, z)$, is given by

$$E^{\mathrm{xc}}(\rho) := E^{\mathrm{xc}}(\widetilde{\varrho}) = \int_{I_L} \fint_{(\mathbb{R}^2/\mathbb{L})} \mathrm{e}^{\mathrm{xc}}\big(\widetilde{\varrho}(\omega, z)\big) \mathrm{d}\omega \mathrm{d}z,$$

where the function $\mathrm{e}^{\mathrm{xc}} : \mathbb{R}_+ \to \mathbb{R}_-$ is the exchange-correlation energy per unit volume of the homogeneous electron gas (or an approximation of this function). Recall that the Local Density Approximation (LDA) was first introduced by Kohn and Sham [KS65], and was recently justified by rigorous mathematical arguments in some regime [LLS19]. In what follows, we assume that the function $\mathrm{e}^{\mathrm{xc}} : \mathbb{R}^+ \to \mathbb{R}^-$ is continuous and satisfies the bound

$$\forall r \geq 0, \qquad 0 \geq \mathrm{e}^{\mathrm{xc}}(r) \geq -C(r^{\beta_1} + r^{\beta_2}), \qquad 1 \leq \beta_1 \leq \beta_2 < \frac{5}{3}. \tag{II.5.1}$$

This assumption is satisfied for the Dirac exchange term $\mathrm{e}^{\mathrm{xc}}(r) = -c_D r^{4/3}$, as well as for most functionals used in practice [PZ81; PW92].

We finally define the Kohn-Sham energy functional of the $\mathbb{L}$-periodic density matrix γ as (we set $v_0 = 0$ for simplicity)

$$\begin{aligned}\mathcal{E}^{\mathrm{KS}}_{\mathrm{per}}(\gamma) :=& \frac{1}{2|\mathbb{R}^2/\mathbb{L}|} \fint_{\mathbb{R}^2/\mathbb{L}^*} \mathrm{Tr}_{L^2(\mathbb{R}^2/\mathbb{L}\times I_L)} \left(\left((-i\nabla_x + k)^2 - \partial_{zz,0} \right) \gamma_k \right) \, dk \\ &+ \frac{1}{2|\mathbb{R}^2/\mathbb{L}|} \mathcal{D}_{\mathrm{per}}(\rho_\gamma - m, \rho_\gamma - m) + E^{\mathrm{xc}}(\rho_\gamma),\end{aligned}$$

or equivalently, in terms of the stationary operator $\widetilde{\gamma}$ associated with γ, as

$$\boxed{\widetilde{\mathcal{E}}^{\mathrm{KS}}_{\mathrm{per}}(\widetilde{\gamma}) := \frac{1}{2}\widetilde{\tau}(-\Delta_0\widetilde{\gamma}) + \frac{1}{2}\mathcal{D}_{\mathrm{per}}(\widetilde{\varrho}_{\widetilde{\gamma}} - \widetilde{\mathfrak{m}}, \widetilde{\varrho}_{\widetilde{\gamma}} - \widetilde{\mathfrak{m}}) + E^{\mathrm{xc}}(\widetilde{\varrho}_\gamma).}$$

Theorem II.5.1. *For all $\lambda > 0$, the encapsulated periodic model*

$$\inf\left\{\mathcal{E}^{\mathrm{KS}}_{\mathrm{per}}(\gamma), \quad \gamma \in \mathcal{B}(L^2(S_L)), \quad 0 \le \gamma = \gamma^* \le 1, \ \gamma \ \mathbb{L}\text{-periodic}, \ \underline{\mathrm{Tr}}(\gamma) = \lambda, \ \underline{\mathrm{Tr}}(-\Delta_0\gamma) < \infty\right\},$$

or equivalently

$$\inf\left\{\widetilde{\mathcal{E}}^{\mathrm{KS}}_{\mathrm{per}}(\widetilde{\gamma}), \quad \widetilde{\gamma} \in W^{1,1}_0(\widetilde{\mathcal{A}}, \widetilde{\tau}), \quad 0 \le \widetilde{\gamma} = \widetilde{\gamma}^* \le 1, \ \widetilde{\tau}(\widetilde{\gamma}) = \lambda\right\}. \tag{II.5.2}$$

has a ground state.

This time, the energy is no longer convex, due to the LDA term, so do not expect uniqueness in general.

Proof. We only highlight the main differences with the previous proof, focusing on the minimization problem (II.5.2). For the sake of readability, we drop the tilde notations in the proof.

Step 1 (the minimization set is non empty) is similar. We consider γ of the form (II.4.15), with h continuous, then ϱ is continuous on $\Omega \times I_L$, and $E^{\mathrm{xc}}(\rho)$ is bounded.

Step 2 bis. The energy is bounded from below. The LDA term is negative, and we need to control it. Using hypothesis (II.5.1), we get

$$E^{\mathrm{xc}}(\varrho) \ge -C\left(\|\varrho\|^{\beta_1}_{\beta_1} + \|\varrho\|^{\beta_2}_{\beta_2}\right).$$

Since $1 \le \beta_1 < 5/3$, we have the Hölder's inequality

$$\|\varrho\|^{\beta_1}_{\beta_1} \le \|\varrho\|^{\theta_1\beta_1}_1 \|\varrho\|^{(1-\theta_1)\beta_1}_{5/3}, \qquad \text{with } 0 < \theta_1 \le 1 \text{ chosen so that} \quad \theta_1 + \frac{3(1-\theta_1)}{5} = \frac{1}{\beta_1}.$$

Together with the fact that $\|\varrho\|_1 = \lambda$ and Lieb–Thirring inequality, we obtain

$$\|\varrho\|^{\beta_1}_{\beta_1} \le C\lambda^{\theta_1\beta_1}\, \tau(-\Delta_0\gamma)^{\frac{3(1-\theta_1)\beta_1}{5}} = C\lambda^{\theta_1\beta_1}\, \tau(-\Delta_0\gamma)^{1-\theta_1\beta_1}.$$

We have a similar computation for β_2, with some $0 < \theta_2 \leq 1$. This proves that

$$\mathcal{E}^{\mathrm{KS}}(\gamma) \geq \frac{1}{2}\tau(-\Delta_0\gamma) - C_\lambda\left[\tau(-\Delta_0\gamma)^{1-\theta_1\beta_1} + \tau(-\Delta_0\gamma)^{1-\theta_2\beta_2}\right].$$

The function $X \mapsto X - C_\lambda(X^{1-\theta_1\beta_1} + X^{1-\theta_2\beta_2})$ is coercive. We deduce first that the energy is bounded from below, and that any minimizing sequence γ_n has $\tau(-\Delta_0\gamma_n)$ bounded.

Step 3 and Step 4 are similar as before (extraction and convergence of a subsequence). From the periodic Hoffmann–Ostenhof inequality

$$\tau(-\Delta_0\gamma) \geq \int_{I_L} \fint_{\mathbb{R}^2/\mathbb{L}} |\nabla\sqrt{\varrho}|^2 \mathrm{d}\omega\mathrm{d}z, \tag{II.5.3}$$

we also deduce that the sequence $\nabla\sqrt{\varrho_n}$ is bounded in $L^2(\Omega \times I_L)$. Together with Rellich—Kondrachov embedding (compact Sobolev embedding), we deduce that ϱ_n converges **strongly** to ϱ_* in $L^p(\Omega \times I_L)$ for all $1 \leq p < 3$, and almost everywhere up to a subsequence (we used the fact that the set $\Omega \times I_L$ is compact).

Step 5 bis. Convergence of the LDA energy. Reasoning as before, we can prove that

$$\tau(-\Delta_0\gamma_*) \leq \liminf_{n\to\infty} \tau(-\Delta_0\gamma_n) \quad \text{and} \quad \mathcal{D}_{\mathrm{per}}(\varrho_* - \mathfrak{m}, \varrho_* - \mathfrak{m}) \leq \liminf_{n\to\infty} \mathcal{D}_{\mathrm{per}}(\varrho_n - \mathfrak{m}, \varrho_n - \mathfrak{m}).$$

It remains to prove the convergence of the LDA term. We write

$$\mathrm{e}^{\mathrm{xc}}(r) := -C(r^{\beta_1} + r^{\beta_2}) + f(r), \quad \text{with} \quad f(r) \geq 0.$$

Since e^{xc} is continuous, so is f. This gives

$$\mathrm{e}^{\mathrm{xc}}(\varrho_n) = -C(\varrho_n^{\beta_1} + \varrho_n^{\beta_2}) + f(\varrho_n).$$

From the strong convergence of ϱ_n to ϱ_* in $L^p(\Omega \times I_L)$ for all $1 \leq p < 3$, we get

$$\int_{I_L} \fint_\Omega \varrho_n^{\beta_1} + \varrho_n^{\beta_2} \xrightarrow[n\to\infty]{} \int_{I_L} \fint_\Omega \varrho_*^{\beta_1} + \varrho_*^{\beta_2}.$$

On the other hand, ϱ_n converges pointwise to ϱ_* a.e. By continuity of f, we deduce that $f(\varrho_n)$ pointwise to $f(\varrho_*)$ a.e. In addition, f is positive, so Fatou's lemma implies that

$$\int_{I_L} \fint_\Omega f(\varrho_*) \leq \liminf_{n\to\infty} \int_{I_L} \fint_\Omega f(\varrho_n).$$

Altogether, we proved that

$$E^{\mathrm{xc}}(\gamma_*) \leq \liminf_{n\to\infty} E^{\mathrm{xc}}(\gamma_n), \quad \text{hence} \quad \mathcal{E}^{\mathrm{KS}}(\gamma_*) \leq \liminf_{n\to\infty} \mathcal{E}^{\mathrm{KS}}(\gamma_n).$$

This proves as before that γ_* is a minimizer of the Kohn–Sham energy $\mathcal{E}^{\mathrm{KS}}$. □

Remark II.5.2. *The additional property that we have in the periodic setting is the strong convergence of $\widetilde{\varrho}_n$ to $\widetilde{\varrho}_*$ in $L^p(\widetilde{\Omega} \times I_L)$, which allows to prove the convergence of the LDA term. This strong convergence comes from the Hoffmann–Ostenhof inequality* (II.5.3) *involving the ∇ term, which is such that $\nabla^*\nabla$ is coercive. In the quasi-periodic setting, the same inequality involves Λ, and $\Lambda^*\Lambda$ is not coercive. There is no equivalent of the compact Sobolev embedding with the operator Λ.*

Acknowledgements

The authors are grateful to Huajie Chen, Antoine Levitt, Mathieu Lewin, Amine Marrakchi, Basile Morando, and Reinhold Schneider for useful discussions.

III A Mathematical Analysis of IPT-DMFT

This chapter corresponds to a published article [CKP26], written in collaboration with Éric Cancès[1] and Alfred Kirsch[1] ; https://doi.org/10.1007/s00220-026-05647-9 .

III.1 Introduction

The Dynamical Mean-Field Theory (DMFT) is an approximation method for the quantum many-body problem, which can be regarded as a quantum analog of the mean-field solution of Ising models. It was introduced by Georges and Kotliar in 1992 and first applied to the case of the Hubbard model [MV89; GK92; RKZ94; Geo+96; Kot+01] (throughout this paper, we will only consider the fermionic version of the Hubbard model, also known as Fermi-Hubbard model). It has since been extended to other settings [SK02] and coupled with Density Functional Theory (DFT) within the so-called DFT+DMFT method [Hel+01]. The latter is one of the reference methods for first-principle computations of electronic structures of strongly correlated materials. DMFT belongs to the class of quantum embedding methods, and has since been joined by many other methods such as Density-Matrix Embedding Theory (DMET) [KC12], Rotationally-Invariant Slave Boson (RISB) method [Lec+07], Energy-weighted DMET [FB19], Quantum Embedding Theory [Ma+21], and related methods.

At the time of writing, the mathematical analysis of quantum embedding methods is very limited. The rigorous results we are aware of are those on DMFT contained in Lindsey's PhD thesis [Lin19], the ones on DMET recently obtained by the first two authors and their collaborators [Can+23], and a few others with a numerically oriented approach such as [Fau+22; Wu+20].

The purpose of this article is to establish a rigorous mathematical formulation of the DMFT equations and to prove, in particular, the existence of a solution to the DMFT equations for the Hubbard model within the Iterated Perturbative Theory (IPT) approximation [Geo+96]. This relies on the extension of some results from [Lin19, Part VII] and

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is based upon a reformulation of the DMFT equations as a fixed-point problem in the space of probability measures on the real line.

In the language of linear algebra, solving the fermionic quantum many-body problem consists in computing some spectral properties of a Hermitian matrix $\widehat{H} \in \mathbb{C}^{M\times M}$, the Hamiltonian of the system, such as its ground-state energy (i.e. its lowest eigenvalue), or the partition function

$$Z_{\beta,\varepsilon_F} := \exp\left(-\beta(\widehat{H} - \varepsilon_F \hat{N})\right),$$

where $\beta = \frac{1}{k_B T}$ is the inverse temperature, $\varepsilon_F \in \mathbb{R}$ is the chemical potential, and $\hat{N}$ is the number operator, as well as derivatives of Z_{β,ε_F} with respect to β, ε_F or parameters of $\widehat{H}$.

The difficulty is that the size M of the matrix $\widehat{H}$ can be huge (up to 10^{30} or more in some applications). Fortunately, the Hamiltonian $\widehat{H}$ has specific properties, allowing one to use taylored methods. Indeed, in most applications, $\widehat{H}$ is the matrix of a Hamiltonian operator acting on a fermionic Fock space $\mathcal{F}$, containing only one- and two-body terms, and satisfying symmetry properties (particle number conservation, spin, and possibly space, isospin, or time-reversal symmetries). Identifying the one-body state space $\mathcal{H}[]$ with $\mathbb{C}^{2L}$, it holds $M = 2^{2L}$ and

$$\mathcal{F} = \bigoplus_{N=0}^{L} \mathcal{H}[N],$$

where the N-particle sector $\mathcal{H}[N] = \bigwedge^N \mathcal{H}[]$ of the Fock space is of dimension $\binom{2L}{N}$. In this decomposition, $\hat{N}$ is a block diagonal operator, the block corresponding to $\mathcal{H}[N]$ being equal to N times the identity matrix. If $\widehat{H}$ is particle-number conserving, then it is also block-diagonal in this decomposition (equivalently $\widehat{H}$ and $\hat{N}$ commute). Since it only contains one- and two-body terms, then $\widehat{H}$ has a compact representation in the second quantization formalism involving a Hermitian matrix $H^0 \in \mathbb{C}^{2L\times 2L}$ and a fourth-order tensor $V \in \mathbb{C}^{2L\times 2L\times 2L\times 2L}$. Spin, space, isospin, or time-reversal symmetries allow one to further reduce the complexity of the representation of $\widehat{H}$ and refine its block diagonal structure. Still, solving the quantum many-body problem remains extremely challenging.

Quantum embedding methods can be seen as domain decomposition methods in the Fock space, using a partition of the L "sites" (also called orbitals) of the model into P non-overlapping clusters $(\Lambda_p)_{1\leq p\leq P}$ of cardinalities $L_p := |\Lambda_p|$. Without loss of generality, we can assume that the first cluster consists of the first L_1 orbitals, the second cluster of the next L_2 orbitals, and so on. Each cluster is modeled by an impurity consisting in L_p sites. The impurity is coupled with virtual sites called bath orbitals, resulting in a quantum many-body problem known as an impurity model. In DMET, the number of bath orbitals is exactly equal to L_p so that the impurity quantum many-body problem is of size $M_p := 2^{4L_p}$. In practice, DMET impurity problems are solved either by brute-force diagonalization (full CI) if L_p is not too large, or by low-rank tensor methods (e.g. Density Matrix Renormalization Group, DMRG [Whi92]). In DMFT, the impurity problem can be

much larger, but the impurity Hamiltonian has the relatively simple form of an Anderson Impurity Model (AIM): within each of the P impurity models, bath orbitals do not contribute to two-body interactions, and are coupled with impurity orbitals only via one-body interactions. It can be shown that the AIM associated with the p-th cluster can be completely described by the restriction of $\widehat{H}$ to the p-th impurity's orbitals and a hybridization function $\Delta_p : \mathbb{C} \setminus \mathbb{R} \to \mathbb{C}^{L_p \times L_p}$. AIMs are usually solved in practice either by a quantum Monte Carlo method [RSL05], or by an approximate solver such as the IPT (Iterative Perturbation Theory) solver [Geo+96] considered in this article. The IPT solver was introduced in the seminal paper [ZRK93], and is still used to study very challenging systems such as moiré heterobilayers [Tan+23].

Quantum embedding methods are self-consistent theories: the P impurity problems are coupled through a mean-field defined on the whole quantum system with L orbitals.

In DMET, the role of the mean-field is played by an approximation $D \in \mathbb{C}^{2L \times 2L}$ of the ground-state one-body density matrix (1-RDM) of the system. The matrix D allows one to define an impurity problem for each cluster, and the self-consistent condition is that for each cluster p, the diagonal block of D corresponding to this cluster agrees with the restriction of the exact ground-state 1-RDM of the p-th impurity problem to the cluster p. It is expected that at self-consistency the diagonal blocks of D corresponding to the cluster decomposition are good approximations of the diagonal blocks of the exact ground-state 1-RDM of the whole system [Can+23].

In DMFT, the role of the mean-field is played by an approximation G of the exact one-body Green's function [MRC16] associated with some steady state, usually the grand-canonical thermodynamical equilibrium state. One-body Green's functions can be represented by analytic functions $G : \mathbb{C} \setminus \mathbb{R} \to \mathbb{C}^{2L \times 2L}$ and are thus computationally tractable objets for values of L up to a few thousands. The function G is a particular holomorphic extension of the Fourier transform of the time-ordered Green's function. Loosely speaking, the latter is a stationary time-correlation function obtained by creating (resp. annihilating) a particle at time $t_0 = 0$ (resp. at $t < 0$), letting the system evolve from t_0 to t (resp. from t to t_0), and annihilating the extra particle (resp. restoring the missing particle) at time t (resp. at time t_0). The exact one-body Green's function contains a lot of valuable information about the quantum system under investigation. In particular, the 1-RDM of the steady state, hence the expectation value of any one-body observable, can be easily extracted from it. The same holds true for the average energy, thanks to Galitski-Migdal's formula [Gon15; MRC16]. Also, the poles of the analytic continuation of G to the real-axis correspond to the one-particle excitation energies measured in photoemission and inverse-photoemission spectroscopies [Zho+20]. Remarkably, the exact Green's function G^0 of a non-interacting system, i.e. of a many-body Hamiltonian $d\Gamma(H^0)$ which is the second quantization of a one-body Hamiltonian is simply the resolvent of H^0: $G^0(z) = (z - H^0)^{-1}$, whatever the reference steady state. The self-energy of an interacting system with Hamiltonian $\widehat{H} = d\Gamma(H^0) + \widehat{H}^I$, where $\widehat{H}^I$ accounts for the

(two-body) interactions, is the function $\Sigma : \mathbb{C} \setminus \mathbb{R} \to \mathbb{C}^{2L \times 2L}$ defined by

$$\Sigma(z) = G^0(z)^{-1} - G(z)^{-1} \quad \text{or equivalently} \quad G(z) = (z - H^0 - \Sigma(z))^{-1}.$$

DMFT consists in

- approximating the exact self-energy of the whole system by a block diagonal self-energy $\Sigma = \text{block-diag}(\Sigma_1, \dots, \Sigma_P)$, with $\Sigma_p : \mathbb{C} \setminus \mathbb{R} \to \mathbb{C}^{2L_p \times 2L_p}$ compatible with the cluster decomposition. This condition is sometimes called the DMFT approximation;
- imposing the self-consistent conditions that for each cluster
  - the self-energy Σ_p agrees with the restriction to the impurity of the exact self-energy of the associated AIM,
  - the restriction to the cluster of the approximate Green's function of the whole system agrees with the restriction to the impurity of the exact Green's function of the AIM. This condition is often referred to as the self-consistent condition.

In practice, the DMFT equations are solved by fixed-point iterations. The input of iteration n is a collection of P hybridization functions $(\Delta_p^{(n)})_{1 \le p \le P}$. At step 1, the P AIM problems with hybridization functions $\Delta_p^{(n)}$ are solved in parallel, in order to compute P cluster self-energies $\Sigma_p^{(n)}$, yielding an approximation $\Sigma^{(n)} = \text{block-diag}(\Sigma_1^{(n)}, \cdots \Sigma_P^{(n)})$ of the self-energy of the whole system. At step 2, the above two self-consistent conditions are combined yielding a new set $(\Delta_p^{(n+1)})_{1 \le p \le P}$ of hybridization functions. The DMFT iteration scheme can therefore be sketched as

$$\Delta^{(n)} := (\Delta_p^{(n)})_{1 \le p \le P} \longrightarrow \Sigma^{(n)} := (\Sigma_p^{(n)})_{1 \le p \le P} \longrightarrow \Delta^{(n+1)} := (\Delta_p^{(n+1)})_{1 \le p \le P}, \tag{III.1.1}$$

or written in the more compact form

$$\Delta^{(n+1)} = f^{\text{DMFT}}\left(\Delta^{(n)}\right). \tag{III.1.2}$$

Of course, this basic self-consistent loop can be stabilized and accelerated using e.g. damping and Anderson-Pulay extrapolation methods. In this article, we forego an in-depth analysis of the iterative scheme and its convergence, opting instead to direct our attention towards a fundamental inquiry: the existence of solutions of the DMFT equations. Specifically, we address the question of the existence of a fixed-point of the DMFT map f^{DMFT}, a critical aspect which, to our knowledge remains unestablished in the current literature.

This article is organized as follows. In Section III.3, we briefly introduce the DMFT approximation of a Hubbard model, emphasizing the mathematical framework rather than the approximation's derivation (see Appendix III.A for the latter). Of particular importance is the fact that the key mathematical objects involved in DMFT (exact and

approximate one-body Green's functions and self-energies, hybridization functions) are all negatives of Pick functions. Recall that scalar Pick functions are analytic functions from the open upper half-plane to the closed upper-half plane [Nev19], [Pic15]. An interesting property of scalar Pick functions, which we use extensively in our analysis, is that any Pick function admits an integral representation involving a positive Borel measure on $\mathbb{R}$, called its Nevanlinna-Riesz measure [Nev19]. Analogous properties hold true for matrix-valued Pick functions [GT00]. For the Hubbard model with a finite number of sites, the exact Green's function and self-energy can be extended to meromorphic functions on $\mathbb{C}$ with finite numbers of poles (which appear to lie on the real-line), and are therefore represented by finite and finitely supported Nevanlinna-Riesz measures.

We then focus on the single-site, translation-invariant, paramagnetic IPT-DMFT approximation of the Hubbard model, for which $P = L$ and $L_1 = \cdots = L_P = 1$. We show that these equations have no solutions in the class of (negatives of) Pick functions with finite and finitely supported Nevanlinna-Riesz measures, but do have solutions in the set of (negatives of) Pick functions. More precisely, equation (III.1.2) has a translation invariant fixed point $(\Delta, \dots, \Delta)$, Δ being the negative of a Pick function whose Nevanlinna-Riesz measure has the form $c\nu$, where $c \in \mathbb{R}_+$ is a fixed constant only depending on the matrix H^0, and ν a Borel probability measure on $\mathbb{R}$. To obtain the latter result, we show that the IPT-DMFT iteration map f^{DMFT} in (III.1.2) can be rewritten as a map $F^{\mathrm{DMFT}} : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$, which is continuous for the weak topology. We conclude by checking that the Schauder-Singbal's fixed-point theorem [Sha16] can be applied to this setting.

In Appendix III.A, we provide a mathematical introduction to DMFT for the Hubbard model aimed at being accessible to readers unfamiliar with this theory. The Hubbard model provides insights into the behavior of electrons in strongly correlated systems. Its integration within the DMFT framework offers a powerful tool for understanding the interplay between electrons, shedding light on phenomena such as metal-insulator transitions. We recall the basics of second quantization formalism, the formulation of the Hubbard and Anderson impurity models, the definitions of one-body Green's functions, self-energies, and hybridization functions, and the precise formulation of the DMFT equations. We then present Matsubara's formalism, allowing for a justified introduction of the paramagnetic IPT solver we have been interested in. Finally, we highlight how, with such a solver, spin degeneracy can be lifted to reduce the single-site, translation-invariant, paramagnetic IPT-DMFT equations to a set of equations we have been focusing on in the main part of this article.

Finally, the IPT solver being formulated in terms of an Analytic Continuation Problem (ACP), Appendix III.B is devoted to an introduction to these problems. In particular, we introduce a uniqueness result which proved to be very useful for the analysis of the IPT-DMFT equations.

III.2 IPT-DMFT framework

Let us introduce and recall some useful notation. We denote by

$$\mathbb{C}_+ := \{z \in \mathbb{C} \mid \Im(z) > 0\}$$

the complex open upper half-plane, that is the set of complex numbers with positive imaginary part. For a matrix $M \in \mathbb{C}^{n\times n}$, $n \geq 1$, the imaginary part of the Hermitian matrix M is defined by

$$\Im(M) = \frac{M - M^\dagger}{2i}.$$

The set of Hermitian matrices of size n is denoted by $\mathcal{S}_n(\mathbb{C})$, and the set of positive-semidefinite matrices by $\mathcal{S}_n^+(\mathbb{C})$. The notation $M \geq 0$ (resp. $M > 0$) means that the matrix M is positive semidefinite (resp. positive definite). In the following, we will also deal with measure and probability theory. The set of finite signed Borel measures on $\mathbb{R}$ is denoted by $\mathcal{M}(\mathbb{R})$. The subset $\mathcal{M}_+(\mathbb{R}) \subset \mathcal{M}(\mathbb{R})$ is the set of finite positive Borel measures on $\mathbb{R}$, and finally, the subset $\mathcal{P}(\mathbb{R}) \subset \mathcal{M}_+(\mathbb{R})$ denotes the set of Borel probability measures on $\mathbb{R}$.

For a positive Borel measure μ on $\mathbb{R}$, we say that μ has a finite moment of order $k \in \mathbb{N}$ if $\int_{\mathbb{R}} |\varepsilon|^k d\mu(\varepsilon) < \infty$. In this case, we denote by $m_k(\mu) = \int_{\mathbb{R}} \varepsilon^k d\mu(\varepsilon)$ its k-th moment. In particular, μ is finite if and only if it has a finite moment of order 0. In this case, $\mu \in \mathcal{M}_+(\mathbb{R})$ and its 0-th moment is called the *mass* of μ, denoted by $\mu(\mathbb{R}) = m_0(\mu) = \int_{\mathbb{R}} d\mu$. These notation and considerations extend to the case of matrix-valued measures, as will be discussed in the next section.

All graphs considered here are *undirected* and *loopless* (i.e. there is no edge joining a vertex to itself). Apart from the results stated in Section III.3.4, we restrict ourselves to finite graphs. We emphasize that the graphs considered in this article are not assumed to be embedded in some d-dimensional physical space. We will however make use of the expression "translation-invariant setting" to refer to cases when the underlying graph is vertex-transitive. Recall that a graph $\mathcal{G}_H = (\Lambda, E)$ is called *vertex-transitive* if for all $\lambda_1, \lambda_2 \in \Lambda$, there exists a graph automorphism $\tau : \Lambda \to \Lambda$ such that $\tau(\lambda_1) = \lambda_2$ (a graph automorphism on $\mathcal{G}_H$ is a bijection $\tau : \Lambda \to \Lambda$ such that $\{i, j\} \in E$ if and only if $\{\tau(i), \tau(j)\} \in E$). In the particularly important case of a d-dimensional periodic lattice (for instance, a cubic or hypercubic lattice), the vertex-transitivity of the graph is precisely a consequence of the lattice's translation invariance. We hope that this slight abuse of terminology will not cause any confusion.

III.2.1 One-body Green's function and self-energy as Pick functions

Our mathematical framework intersects with the realm of complex analysis pioneered by Pick [Pic15] and Nevanlinna [Nev19], focusing on the study of so-called Pick functions.

A *Pick function* is a map $f : \mathbb{C}_+ \to \overline{\mathbb{C}_+}$ which is analytic. In this article, we use the term Pick functions, but several terminologies coexist in the literature: Nevanlinna functions, Herglotz functions, Riesz functions, or R-functions. A *Pick matrix* is an analytic map $f : \mathbb{C}_+ \to \mathbb{C}^{n\times n}$, $n \geq 1$, such that for all $z \in \mathbb{C}_+, \Im(f(z)) \geq 0$ (i.e. $\Im(f(z)) \in \mathcal{S}_n^+(\mathbb{C})$). Sometimes, it is convenient to extend Pick matrices to the lower-half-plane $\mathbb{C}_-$. In this case, the usual convention is to set for all $z \in \mathbb{C}_-, f(z) = f(\overline{z})^\dagger$ [GT00].

As mentioned in the introduction, DMFT is an approximation method of the quantum one-body Green's function G (which we refer to as Green's function in short). In this section, we recall the main mathematical properties of this object, and refer to Appendix III.A for a short, pedagogical introduction to Green's functions.

Consider a many-body quantum system with finite-dimensional one-body state space $\mathcal{H}$ and Hamiltonian $\widehat{H} \in \mathcal{S}(\mathcal{F})$ on the Fock space $\mathcal{F}$ associated with $\mathcal{H}$. Assume that the system is in an equilibrium state characterized by the density operator $\widehat{\rho} \in \mathcal{S}(\mathcal{F}(\mathcal{H}))$. Recall that an equilibrium state is a stationary solution to the quantum Liouville equation, and thus a state commuting with the Hamiltonian ($[\widehat{H}, \widehat{\rho}] = 0$). The Green's function $G : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H})$ is an operator-valued function defined on the upper half-plane and depending on both $\widehat{H}$ and $\widehat{\rho}$. A crucial mathematical property of G used in our analysis is the following (see Proposition III.A.1 in Appendix III.A):

The negative of the Green's function $-G$ is a Pick matrix.

For a given pair of operators $(\widehat{H}, \widehat{\rho})$, finding the associated Green's function is very challenging. Nevertheless, when the Hamiltonian is the second quantization $\widehat{H} = d\Gamma(H^0)$ of a one-body Hamiltonian $H^0 \in \mathcal{S}(\mathcal{H})$, the Green's function, denoted by G^0 in this special case, is the resolvent of the one-body Hamiltonian, regardless of the density operator $\widehat{\rho}$: for all $z \in \mathbb{C}_+$,

$$G^0(z) = \left(z - H^0\right)^{-1}.$$

For a Hamiltonian of the form $\widehat{H} = d\Gamma(H^0) + \widehat{H}^I$ (these are the only Hamiltonians considered in this article), the self-energy $\Sigma : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H})$ quantifies the discrepancy between the interacting Green's function G associated with the pair $(\widehat{H}, \widehat{\rho})$ and the non-interacting Green's function G^0 associated with $d\Gamma(H^0)$: it is defined, for all $z \in \mathbb{C}_+$, by

$$\Sigma(z) = G^0(z)^{-1} - G(z)^{-1} = z - H^0 - G(z)^{-1}.$$

The second crucial property used in our analysis is the following (see again Proposition III.A.1):

The negative of the self-energy $-\Sigma$ is a Pick matrix.

III.2.2 IPT-DMFT equations

In this article, we are interested in proving the existence of solution(s) to the DMFT equations, formulated within the approximation of the Iterated Perturbation Theory impurity

(IPT) solver. A detailed introduction to IPT can be found in Appendix III.A. In the present section, we only briefly motivate this approach and introduce the corresponding equations.

The purpose of DMFT is to provide an approximation of the Green's function of the Hubbard model in a generic Gibbs state. Let us recall that the Hubbard model considers particles living on a finite, connected, unoriented, loopless graph $\mathcal{G}_H = (\Lambda, E)$, where each vertex can host up to two fermions. The Hubbard Hamiltonian $\hat{H}_H$ models the fact that fermions can jump from one site to another — the amplitude of this effect being entirely defined by the hopping matrix $T : E \to \mathbb{R}$ — and that fermions interact with one another strictly locally, when they sit on the same site— the amplitude of this interaction being described by the on-site interaction $U : \Lambda \to \mathbb{R}$ (see Definition III.A.7 for a precise formulation).

The DMFT approximation is defined through a partition $\mathfrak{P}$ of the Hubbard vertices Λ: to each of the induced subgraphs (Λ_p, E_p) is associated an Anderson Impurity model. The latter describes fermions that live in two subsystems: the impurity on the one-hand, which is prescribed by the induced subgraph in the present case and whose one-particle space is denoted by $\mathcal{H}[\text{imp}]$, and the bath on the other hand, with one-partice space $\mathcal{H}[\text{bath}]$ which is an unknown of the DMFT approximation. The Anderson Impurity Model (AIM) Hamiltonian $\widehat{H}_{\text{AIM}}$ models the fact that fermions can jump between the bath levels and the impurity vertices — the magnitude of this process being described by the coupling $V : \mathcal{H}[\text{imp}] \to \mathcal{H}[\text{bath}]$ — between impurity vertices — the magnitude of this process being described by the restriction of the hopping matrix T to E_p — and that fermions interact with one another only when they sit on the same impurity site — the magnitude of this interaction being described by the restriction of the on-site interaction U to Λ_p (see Appendix III.A.3 for a precise formulation).

In an Anderson Impurity model, interactions between fermions are localized within the impurity subsystem. As a consequence, the self-energy of an AIM satisfies a sparsity pattern (see Theorem III.A.10): using the decomposition $\mathcal{H} = \mathcal{H}[\text{imp}] \oplus \mathcal{H}[\text{bath}]$, we have

$$\Sigma(z) = \begin{pmatrix} \Sigma_{\text{imp}}(z) & 0 \\ 0 & 0 \end{pmatrix}. \tag{III.2.1}$$

Moreover, the impurity self-energy Σ_{imp} only depends on the impurity parameters (the restriction of T and U to (Λ_p, E_p)) and on the so-called hybridization function $\Delta : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H}[\text{imp}])$, defined, for all $z \in \mathbb{C}_+$, by

$$\Delta(z) = V \left(z - H^0_{\text{bath}}\right)^{-1} V^\dagger,$$

where $H^0_{\text{bath}} \in \mathcal{S}(\mathcal{H}[\text{bath}])$ is the restriction of the AIM one-body Hamiltonian to the bath subspace (see Definition III.A.9 for a precise formulation). Note that

the negative of the hybridization function $-\Delta$ is a Pick matrix.

For fixed impurity parameters, the map that associates the hybridization function Δ to the corresponding impurity self-energy Σ_{imp} is called an impurity solver. In this work, we considered an approximate impurity solver, the Iterated Perturbation Theory (IPT), in the single-site, translation-invariant, paramagnetic setting: as explained in Appendix III.A.6, this setting allows one to consider that Δ and Σ_{imp} are scalar-valued. In other words,

> Within the single-site, translation-invariant, paramagnetic IPT-DMFT framework, the negative of the hybridization function and of the self-energy $-\Delta, -\Sigma$ are Pick functions.

The IPT solver provides the self-energy as the solution to an Analytic Continuation Problem (ACP) (see Appendix III.B for an introduction to the Nevanlinna-Pick analytic continuation problem): given a hybridization function $-\Delta : \mathbb{C}_+ \to \overline{\mathbb{C}_+}$, the impurity self-energy Σ_{imp} is a solution to

$$-\Sigma_{\text{imp}} : \mathbb{C}_+ \to \overline{\mathbb{C}_+}, \quad \Sigma_{\text{imp}} \text{ is analytic}, \quad \forall n \in \mathbb{N},\ \Sigma_{\text{imp}}(i\omega_n) = \Sigma_{\text{imp},n}, \tag{III.2.2}$$

with, for all $n \in \mathbb{N}$,

$$\Sigma_{\text{imp},n} = U^2 \int_0^\beta e^{i\omega_n \tau} \left(\frac{1}{\beta} \sum_{n' \in \mathbb{Z}} (i\omega_{n'} - \Delta(i\omega_{n'}))^{-1} e^{-i\omega_{n'}\tau} \right)^3 d\tau, \tag{III.2.3}$$

where $\omega_n = (2n+1)\pi/\beta$ is the n-th Matsubara's frequency at given inverse temperature $\beta \in \mathbb{R}_+^*$ (see Appendix III.A.5 for an introduction to Matsubara's formalism).

This solver, if well-defined, provides one relation between the hybridization function and the impurity self-energy. The second relation between these quantities is provided by the DMFT approximation: as explained in details in Appendix III.A.4, in the single-site, translation-invariant, paramagnetic setting (in which the hopping matrix is constant $T \in \mathbb{R}$), this equation reads, for all $z \in \mathbb{C}_+$,

$$\Delta(z) = W \left(z - H_\perp^0 - \Sigma(z) \right)^{-1} W^\dagger, \tag{III.2.4}$$

where $W \in \mathbb{R}^{L-1}$ and $H_\perp^0 \in \mathcal{S}(\mathbb{R}^{L-1})$ are defined by the following block decomposition of the adjacency matrix $\text{Adj}_{\mathcal{G}_H}$ of the considered Hubbard graph $\mathcal{G}_H$:

$$T\text{Adj}_{\mathcal{G}_H} = \begin{pmatrix} 0 & W \\ W^\dagger & H_\perp^0 \end{pmatrix}. \tag{III.2.5}$$

Finally, restricting ourselves to the single-site, translation-invariant, paramagnetic framework, and denoting by Σ the single-site impurity self-energy Σ_{imp}, we get from (III.2.1)-(III.2.5) the set of coupled equations

$$\Delta(z) = W(z - H_\perp^0 - \Sigma(z))W^\dagger, \tag{III.2.6a}$$

$-\Sigma$ is a Pick function solution to the interpolation problem

$$\forall n \in \mathbb{N}, \Sigma(i\omega_n) = U^2 \int_0^\beta e^{i\omega_n \tau} \left(\frac{1}{\beta} \sum_{n' \in \mathbb{Z}} (i\omega_{n'} - \Delta(i\omega_{n'}))^{-1} e^{-i\omega_{n'}\tau} \right)^3 d\tau. \tag{III.2.6b}$$

III.3 Main results

The main results of this article deal with the existence and properties of solution(s) to the IPT-DMFT equations in the single-site, translation-invariant, paramagnetic setting: we will prove the following.

Main results. *For all connected, undirected, loopless and vertex-transitive graph $\mathcal{G}_H = (\Lambda, E)$, for all $T, U \in \mathbb{R}$, $\beta \in \mathbb{R}_+^*$, there exist Pick functions $-\Delta, -\Sigma$ satisfying the single-site, translation-invariant, paramagnetic IPT-DMFT equations (III.2.6a)-(III.2.6b). More precisely, such solutions have the following form: there exist a probability measure $\nu \in \mathcal{P}(\mathbb{R})$ and a finite, positive, Borel measure $\mu \in \mathcal{M}_+(\mathbb{R})$ such that, for all $z \in \mathbb{C}_+$,*

$$\Delta(z) = |W|^2 \int_{\mathbb{R}} \frac{1}{z - \varepsilon} d\nu(\varepsilon), \quad \Sigma(z) = \int_{\mathbb{R}} \frac{1}{z - \varepsilon} d\mu(\varepsilon). \tag{III.3.1}$$

Apart from the limit cases $U = 0$ or $T = 0$ (for which the solution is unique), any solution of this form is such that ν, μ are not finitely supported. Finally, any solution of this form is such that ν, μ have finite moments up to any order.

To get a deeper insight into this statement, let us remind the reader that Pick functions are tightly related to positive measures, according to the following integral representation theorem, which appears to be particularly useful in our analysis.

Theorem III.3.1 (Nevanlinna-Riesz representation theorem [GT00])**.** *Let $f : \mathbb{C}_+ \to \mathbb{C}^{n \times n}$ be a Pick matrix. There exist $a \in \mathcal{S}_n^+(\mathbb{C})$, $b \in \mathcal{S}_n(\mathbb{C})$ and μ a Borel $\mathcal{S}_n^+(\mathbb{C})$-valued measure on $\mathbb{R}$ such that $(1 + |\varepsilon|)^{-1}$ is μ-integrable and*

$$\forall z \in \mathbb{C}_+, \quad f(z) = az + b + \int_{\mathbb{R}} \left(\frac{1}{\varepsilon - z} - \frac{\varepsilon}{1 + \varepsilon^2} \right) d\mu(\varepsilon). \tag{III.3.2}$$

The measure μ is called the Nevanlinna-Riesz measure of f, and

$$a = \lim_{y \to +\infty} \frac{1}{iy} f(iy), \qquad b = \Re(f(i)) := (f(i) + f(i)^\dagger)/2.$$

In the particular case of Pick functions, i.e. $n = 1$, we have $a \geq 0$, $b \in \mathbb{R}$, and μ is a positive Borel measure on $\mathbb{R}$, with the same integrability condition.

Our result therefore states that there exists a solution (Δ, Σ), such that $-\Delta$ and $-\Sigma$ are not generic Pick functions but Cauchy-Stieltjes transforms [GH03] of a probability measure (up to a fixed multiplicative constant) and a finite, positive, Borel measure respectively.

An important remark is that, given a sequence $(\Sigma_{\mathrm{imp},n})_{n \in \mathbb{N}}$ computed via (III.2.3) from an admissible hybridization function Δ, there is no guarantee that the Analytic Continuation Problem (III.2.2) admits a solution nor, if it does, that this solution is unique.

In Section III.3.1, we study this problem for the functional spaces of hybridization functions/self-energies associated with finite-dimensional Anderson Impurity models. Based on an interpolation result stated in Appendix III.B, we show in Proposition III.3.2 that for such a hybridization function (which is the Cauchy-Stieltjes transform of a *finitely supported* measure ν), the considered ACP admits a *unique* solution $-\Sigma$, which appears to also be the Cauchy-Stieltjes transform of a finitely supported measure μ. We then show in Proposition III.3.3 that, apart from the limit cases $U = 0$ or $T = 0$, there is no solution $(-\Delta, -\Sigma)$ to (III.2.6a)-(III.2.6b) which are Cauchy-Stieltjes transforms of finitely-supported measures. We prove this by reformulating (III.2.6a)-(III.2.6b) as a fixed point problem on the set of finitely-supported probability measures, and by proving that the cardinality of the support of the measure strictly increases at each iteration.

In Section III.3.2, we extend the set of admissible $(-\Delta, -\Sigma)$ to include Cauchy-Stieltjes transforms of measures with arbitrary support. In this setting, we show that the two maps in (III.1.1) whose combination leads to the DMFT map f^{DMFT} are well-defined between appropriate functional spaces. More precisely, we show in Proposition III.3.5 that the Bath Update map (III.2.6a), reformulated in terms of Nevanlinna-Riesz measures, is well-defined, and in Proposition III.3.7 that the IPT solver (solving (III.2.6b)), reformulated in terms of Nevanlinna-Riesz measures, admits a unique continuous extension for the weak topology. This strategy allows us to avoid dealing with ACPs for which we were unable to conclude on the uniqueness, and provides, eventually, solutions to the IPT-DMFT equations.

In Section III.3.3, we state our main result (Theorem III.3.8) on the existence of a solution to the single-site, translation-invariant, paramagnetic IPT-DMFT equations, which is a fixed-point of the DMFT map. We also state Proposition III.3.9 which shows that such solutions have finite moments at any order.

III.3.1 Absence of finite dimensional bath solutions

As recalled previously, the mathematical study of the DMFT equations is very limited. The only results the authors are aware of are those contained in M. Lindsey's PhD Thesis [Lin19, Part VII], and are concerned with the well-posedness of the so-called "DMFT loop". Note that this map is referred to as the "Bath Update map" ($\Sigma \to \Delta$) in this article.

To introduce the results by Lindsey et al., let us recall that, for finite-dimensional Anderson Impurity Models, the self-energy Σ and the hybridization function Δ have the following integral representation ([Lin19], Definition III.A.9): there exist $C \in \mathbb{R}$ and μ, ν

two finite, finitely supported and positive Borel measures on $\mathbb{R}$ such that, for all $z \in \mathbb{C}_+$,

$$\Sigma(z) = C + \int_{\mathbb{R}} \frac{1}{z-\varepsilon} d\mu(\varepsilon) = C + \sum_{\varepsilon \in \mathrm{supp}(\mu)} \frac{\mu(\{\varepsilon\})}{z-\varepsilon}, \tag{III.3.3}$$

$$\Delta(z) = \int_{\mathbb{R}} \frac{1}{z-\varepsilon} d\nu(\varepsilon) = \sum_{\varepsilon \in \mathrm{supp}(\nu)} \frac{\nu(\{\varepsilon\})}{z-\varepsilon}, \tag{III.3.4}$$

where the sums are finite. The main result by Lindsey et al. [Lin19, Corollary 20, Part VII] states that, given a self-energy Σ of the form (III.3.3), the hybridization function Δ defined from Σ by Eq. (III.2.6a) is of the form (III.3.4).

Implementing on their result, we took this specific integral representation as a starting point for our analysis: the first result deals with the Iterated Perturbation Theory impurity solver, which we show below to be well-defined for such hybridization functions and self-energies.

Proposition III.3.2 (IPT solver for finite-dimensional AIMs)**.** *Let $U \in \mathbb{R}$, $\Delta : \mathbb{C}_+ \to \mathbb{C}$ of the form*

$$\forall z \in \mathbb{C}_+, \quad \Delta(z) = \sum_{k=1}^{K} \frac{a_k}{z-\varepsilon_k}, \quad \textit{with} \quad \varepsilon_1 < \varepsilon_2 < \cdots < \varepsilon_K \textit{ and } a_k > 0 \textit{ for all } 1 \leq k \leq K, \tag{III.3.5}$$

and $(\Sigma_n)_{n\in\mathbb{N}}$ defined by

$$\forall n \in \mathbb{N}, \quad \Sigma_n = U^2 \int_0^\beta e^{i\omega_n \tau} \left(\frac{1}{\beta} \sum_{n' \in \mathbb{Z}} \left(i\omega_{n'} - \Delta(i\omega_{n'})\right)^{-1} e^{-i\omega_{n'}\tau} \right)^3 d\tau.$$

Then, the Analytic Continuation Problem (ACP)

$$-\Sigma : \mathbb{C}_+ \to \overline{\mathbb{C}_+}, \quad \Sigma \textit{ is analytic}, \quad \forall n \in \mathbb{N}, \Sigma(i\omega_n) = \Sigma_n \tag{III.3.6}$$

has a unique solution, which is given by

$$\forall z \in \mathbb{C}_+, \quad \Sigma^{\mathrm{IPT}}(z) = U^2 \sum_{k_1,k_2,k_3=1}^{K+1} \frac{a'_{k_1,k_2,k_3}}{z - \left(\varepsilon'_{k_1} + \varepsilon'_{k_2} + \varepsilon'_{k_3}\right)}, \tag{III.3.7}$$

where $\varepsilon'_1 < \varepsilon'_2 < \cdots < \varepsilon'_{K+1}$ are the $(K+1)$ real roots of the equation

$$\varepsilon - \Delta(\varepsilon) = 0,$$

which satisfy the interlacing relation

$$\varepsilon'_1 < \varepsilon_1 < \varepsilon'_2 < \varepsilon_2 < \cdots < \varepsilon_K < \varepsilon'_{K+1},$$

and for all $k_1, k_2, k_3 \in [\![1, K+1]\!]$, a_{k_1,k_2,k_3} is defined by

$$a'_{k_1,k_2,k_3} = \frac{1 + e^{-\beta(\varepsilon'_{k_1} + \varepsilon'_{k_2} + \varepsilon'_{k_3})}}{\left(1 + e^{-\beta\varepsilon'_{k_1}}\right)\left(1 + e^{-\beta\varepsilon'_{k_2}}\right)\left(1 + e^{-\beta\varepsilon'_{k_3}}\right)} \prod_{i=1}^{3} \left(1 - \Delta'(\varepsilon'_{k_i})\right)^{-1} > 0. \tag{III.3.8}$$

Combined with the results by Lindsey et al., this proves that the DMFT iteration loop

$$\Delta^{(n)} \xrightarrow{\text{Eqs. (III.2.2)–(III.2.3)}} \Sigma^{(n)} \xrightarrow{\text{Eq. (III.2.6a)}} \Delta^{(n+1)}$$

is well-defined on the set of Cauchy-Stieltjes transforms of finite, finitely-supported and positive Borel measures. Nevertheless, the following proposition highlights the fact that the single-site, translation-invariant, paramagnetic IPT-DMFT equations have no solution with hybridization functions Δ corresponding to finite-dimensional AIMs.

Proposition III.3.3 (Non-existence of solutions to the finite-dimensional bath single-site, translation-invariant, paramagnetic IPT-DMFT equations)**.** *Apart from the limit cases $U = 0$ or $T = 0$, the single-site, translation-invariant, paramagnetic IPT-DMFT equations* (III.2.6a)-(III.2.6b) *have no finite-dimensional bath solution, that is no solution Δ of the form*

$$\forall z \in \mathbb{C}_+, \quad \Delta(z) = \sum_{k=1}^{K} \frac{a_k}{z - \varepsilon_k}, \quad K \geq 1, \quad a_k > 0, \quad \varepsilon_k \in \mathbb{R}.$$

This result originates from the fact that an iteration of the IPT-DMFT map increases the bath dimension K. It implies that finding a solution to the DMFT equations requires considering infinite-dimensional bath hybridization functions. The appropriate function space can be characterized in terms of Nevanlinna-Riesz measures, as will be shown in the subsequent section.

Remark III.3.4 (Limit cases $U = 0$ or $T = 0$.)**.** *The limit cases considered in the previous proposition are covered in Proposition III.A.14, for the exact impurity solver. In each of these limits, the (unique) solution to the DMFT equations provides the exact solution to the quantum many-body problem, which is indeed the Cauchy-Stieltjes transform of a finite, finitely-supported, positive Borel measure. See also Remark III.3.4 below for a discussion of the limit case $U = 0$ for the extension of the results to infinite graphs of finite degree.*

III.3.2 Functional setting: the Bath Update and IPT maps

The DMFT map is the composition of the IPT map and the Bath Update (BU) map, which we will study separately. First, we formalize in our setting the definition of the BU map F^{SC}. The following proposition extends the result by Lindsey et al. to the case of finite and positive Borel measures of arbitrary support, by using a different approach. These definitions are motivated by the consequences of Proposition III.3.3 and the statements of Propositions III.3.5 and III.3.7.

We introduce the self-energy space

$$\mathfrak{S} = \left\{ \mathbb{C}_+ \ni z \mapsto U^2 \int_{\mathbb{R}} \frac{d\mu(\varepsilon)}{z - \varepsilon} \in \mathbb{C} \ ; \ \mu \in \mathcal{M}_+(\mathbb{R}) \right\}, \tag{III.3.9}$$

which is in one-to-one correspondence with the set $\mathcal{M}_+(\mathbb{R})$ of finite positive Borel measures on $\mathbb{R}$, and the hybridization space

$$\mathfrak{D} = \left\{ \mathbb{C}_+ \ni z \mapsto |W|^2 \int_{\mathbb{R}} \frac{d\nu(\varepsilon)}{z-\varepsilon} \in \mathbb{C} \ ; \ \nu \in \mathcal{P}(\mathbb{R}) \right\}, \tag{III.3.10}$$

which is in one-to-one correspondence with the set $\mathcal{P}(\mathbb{R})$ of Borel probability measures on $\mathbb{R}$. These one-to-one correspondences (which are, up to a multiplicative constant, Cauchy-Stieltjes transforms) allow us to focus on the measure spaces $\mathcal{M}_+(\mathbb{R})$ and $\mathcal{P}(\mathbb{R})$, and study the DMFT iteration scheme in terms of measures. The BU map can then be defined as a function F^{SC} between measure spaces as follows.

Proposition III.3.5 (Bath Update ($F^{\mathrm{SC}} : \Sigma \mapsto \Delta$) map in the single-site, translation-invariant, paramagnetic IPT-DMFT framework)**.** *Let $\Sigma \in \mathfrak{S}$, and let μ be the Nevanlinna-Riesz measure associated with Σ. Then, the hybridization function Δ, given by*

$$\Delta(z) = W \left(z - H_\perp^0 - \Sigma(z) \right)^{-1} W^\dagger, \tag{III.3.11}$$

for $z \in \mathbb{C}_+$ is well-defined. With this definition, $-\Delta$ is a Pick function and there exists a Borel probability measure $\nu \in \mathcal{P}(\mathbb{R})$ such that

$$\forall z \in \mathbb{C}_+, \quad \Delta(z) = |W|^2 \int_{\mathbb{R}} \frac{d\nu(\varepsilon)}{z-\varepsilon}$$

The Bath Update (BU) map in the single-site, translation-invariant, paramagnetic IPT-DMFT framework is defined as the function $F^{\mathrm{SC}} : \mathcal{M}_+(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ such that

$$F^{\mathrm{SC}}(\mu) = \nu,$$

with

$$W_p^\dagger W_p \int_{\mathbb{R}} \frac{d\nu(\varepsilon)}{z-\varepsilon} = W \left(z - H_\perp^0 - U^2 \int_{\mathbb{R}} \frac{d\mu(\varepsilon)}{z-\varepsilon} \right)^{-1} W^\dagger. \tag{III.3.12}$$

Remark III.3.6 (Bath Update map in the general DMFT formalism)**.** *In Section III.4.3, we prove an extension of Proposition III.3.5 to the general case presented in Appendix III.A.4, where we do not make the assumptions (single-site, translation-invariant, paramagnetic) under which the DMFT equations reduce to a set of two equations whose unknowns are two scalar-valued functions.*

In the DMFT loop, the impurity solver is the focus of the second stage. Within our model, we define the IPT map F^{IPT}, which transforms a given hybridization function Δ into a self-energy Σ. As well as the Bath Update map, this mapping operates across measure spaces and maps Borel probability measures to finite positive Borel measures.

Proposition III.3.7 (Definition of the IPT map in the single-site, translation-invariant, paramagnetic framework)**.** *Let $\nu \in \mathcal{P}(\mathbb{R})$ and $\Delta \in \mathfrak{D}$ the hybridization function associated with ν: for all $z \in \mathbb{C}_+$,*

$$\Delta(z) = |W|^2 \int_{\mathbb{R}} \frac{d\nu(\varepsilon)}{z-\varepsilon}. \tag{III.3.13}$$

There exists $\xi \in \mathcal{P}(\mathbb{R})$, such that for all $z \in \mathbb{C}_+$,

$$\int_{\mathbb{R}} \frac{d\xi(\varepsilon)}{z-\varepsilon} = \frac{1}{z-\Delta(z)}. \tag{III.3.14}$$

Then define

$$\tilde{\xi}(d\varepsilon) := \frac{\xi(d\varepsilon)}{1+e^{-\beta\varepsilon}}, \tag{III.3.15}$$

$$\tilde{\mu} := \tilde{\xi} * \tilde{\xi} * \tilde{\xi}, \tag{III.3.16}$$

where $*$ is the convolution product, and

$$\mu(d\varepsilon) := (1+e^{-\beta\varepsilon})\tilde{\mu}(d\varepsilon). \tag{III.3.17}$$

Finally, define the self-energy associated with the measure μ by

$$\Sigma(z) = U^2 \int_{\mathbb{R}} \frac{d\mu(\varepsilon)}{z-\varepsilon}. \tag{III.3.18}$$

The measure μ is a positive finite measure: $\mu \in \mathcal{M}_+(\mathbb{R})$, hence $\Sigma \in \mathfrak{S}$, where $\mathfrak{S}$ is defined in (III.3.9). The IPT map $F^{\mathrm{IPT}} : \mathcal{P}(\mathbb{R}) \to \mathcal{M}_+(\mathbb{R})$ is defined by

$$F^{\mathrm{IPT}}(\nu) = \mu.$$

Moreover, the map $\mathfrak{D} \ni \Delta \mapsto \Sigma \in \mathfrak{S}$ defined by (III.3.14)-(III.3.18) is continuous with respect to the weak topology of measures, and coincides with the IPT solver defined in Proposition III.3.2 on the dense subset of finite-dimensional bath hybridization functions.

Now that we have defined the maps F^{IPT} and F^{SC}, we define the single-site, translation-invariant, paramagnetic IPT-DMFT map as

$$F^{\mathrm{DMFT}} := F^{\mathrm{SC}} \circ F^{\mathrm{IPT}} : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R}). \tag{III.3.19}$$

For the sake of brevity, we will sometimes refer to F^{DMFT} as the IPT-DMFT map and not restate all the simplifying assumptions considered in this analysis: single-site, translation-invariant, paramagnetic framework. In the same way, we refer to fixed points of F^{DMFT} as IPT-DMFT solutions. Such fixed points (more precisely the associated hybridization functions and self-energies) are indeed solutions of the single-site, translation-invariant, paramagnetic IPT-DMFT equations (III.2.6a)-(III.2.6b).

III.3.3 Existence and properties of IPT-DMFT solutions

The main result of this paper is the existence of a solution to the DMFT equations (III.2.6a)-(III.2.6b) in the single-site, translation-invariant, paramagnetic framework, using the IPT impurity solver.

Theorem III.3.8 (Existence of a fixed-point)**.** *The IPT-DMFT map F^{DMFT} has a fixed point $\nu \in \mathcal{P}(\mathbb{R})$.*

Moreover, IPT-DMFT solutions that are fixed-points of F^{DMFT} have finite moments at all orders.

Proposition III.3.9. *Let $\nu^0 \in \mathcal{P}(\mathbb{R})$, $\nu = F^{\mathrm{DMFT}}(\nu^0)$, and $k \in 2\mathbb{N}$. If $\nu^0 \in \mathcal{P}(\mathbb{R})$ has finite k-th moment, then ν has finite $(k+4)$-th moment. In particular, any fixed point of the IPT-DMFT map has finite moments of all orders.*

This property of the DMFT map is actually one of the key elements of the proof of Theorem III.3.8, as it induces compactness. The fact that a fixed point ν of the DMFT map F^{DMFT} has finite moments of any order also raises the following question: are there constants $A, \alpha > 0$ such that, for all $R \in \mathbb{R}$,

$$\nu(\mathbb{R} \setminus [-R, R]) \leq Ae^{-\alpha R} \quad ? \tag{III.3.20}$$

A positive answer to this question would imply that F^{DMFT} has a unique fixed point at high temperature (in the limit case $\beta = 0$). The argument is the following. Using Theorem III.4.1 and some simple algebraic manipulations of equations (III.3.11), (III.3.13)-(III.3.18), it can be shown that there exists a sequence $(m_n^*)_{n\in\mathbb{N}} \in \mathbb{R}^{\mathbb{N}}$, depending on $\mathcal{G}_H, T$ and U, and whose terms can be computed recursively, such that any fixed point ν of F^{DMFT} for $\beta = 0$ satisfies

$$\forall n \in \mathbb{N}, \quad m_n(\nu) = m_n^*, \tag{III.3.21}$$

where we recall that $m_n(\nu)$ is the n-th moment of ν. More precisely, fix $k \geq 0$ and ν, ξ, μ defined as in Proposition III.3.7 for a solution of the IPT-DMFT equations. Given equation (III.3.14) and using Theorem III.4.1, there exists a polynomial $P_k^1 \in \mathbb{R}[X_0, \ldots, X_k]$ such that $m_{k+2}(\xi) = P_k^1(m_0(\nu), \ldots, m_k(\nu))$. Then, using equations (III.3.15)-(III.3.17), there exists another polynomial $P_k^2 \in \mathbb{R}[X_0, \ldots, X_k]$ such that $m_k(\mu) = P_k^2(m_0(\xi), \ldots, m_k(\xi))$. Finally, given equation (III.3.18) and (III.3.11), and using Theorem III.4.1 once again, there exists $P_k^3 \in \mathbb{R}[X_0, \ldots, X_k]$ such that $m_{k+2}(\nu) = P_k^3(m_0(\mu), \ldots, m_k(\mu))$. Putting these considerations together yields the existence of a polynomial $P_k \in \mathbb{R}[X_0, \ldots, X_k]$, such that $m_{k+4}(\nu) = P_k(m_0(\nu), \ldots, m_k(\nu))$. These polynomials can be computed explicitly and are actually independent of ν, ξ, μ, so that P_k is independent of ν (but it depends on the global parameters $\mathcal{G}_H, T$ and U). Additionally, the same reasoning and a direct computation (see Lemma III.4.3) yield that the first moments $m_j(\nu)$ for $0 \leq j \leq 3$ are also independent of ν. In other words, by induction, any fixed point of the IPT-DMFT map satisfies the *Hamburger moment problem* (III.3.21) defined by the sequence $(m_n^*)_{n\in\mathbb{N}}$. Now, if ν satisfies the exponential decay property (III.3.20), it is easy to show that $m_n^* = m_n(\nu)$ satisfies the following Carleman's condition [Akh20]

$$\sum_{n=1}^{\infty} (m_{2n}^*)^{-\frac{1}{2n}} = +\infty, \tag{III.3.22}$$

which implies that the Hamburger moment problem has a unique solution, hence, that ν is unique.

Unfortunately, despite numerical evidence for the simplest case of the Hubbard dimer, we were unable to prove that F^{DMFT} has at least one fixed point ν satisfying the exponential decay (III.3.20), nor that the Hamburger moment problem defined by $(m_n^*)_{n\in\mathbb{N}}$ is well-posed.

III.3.4 Extension of the formalism to countably infinite graphs with finite degree

Many of our results extend to undirected, loopless and countably infinite graphs with finite degree, meaning that $\sup_{\lambda\in\Lambda}\deg(\lambda) = \deg(\mathcal{G}_H) < +\infty$. These graphs are indeed locally finite, and include the nearest neighbor graphs induced by the d-dimensional hypercubic lattices, the triangular lattice, and all the regular trees of finite degree. For such a graph $\mathcal{G}_H$, the formalism, based on the theory of adjacency operators introduced in [Moh82; MW89] for locally finite graphs, can be extended as follows under the assumption that the graph is vertex transitive (a property fulfilled by all the aforementioned examples), in the single-site and paramagnetic setting. Let $l^2(\Lambda)$ be the space of sequences indexed by Λ which are of finite 2-norm (for all $\phi \in l^2(\Lambda)$, $\|\phi\|_2^2 = \sum_{i\in\Lambda}|\phi_i|^2 < +\infty$). The adjacency operator $\mathrm{Adj}_{\mathcal{G}_H} \in \mathcal{B}(l^2(\Lambda))$ is the bounded operator defined on $l^2(\Lambda)$ by linearity with, for all $i \in \Lambda$,

$$\mathrm{Adj}_{\mathcal{G}_H} e_i = \sum_{\{i,j\}\in E} e_j, \quad \text{where } e_i \in l^2(\Lambda) \text{ is defined by } (e_i)_j = \delta_{i,j}.$$

Defined as such, this operator is indeed bounded ($\|\mathrm{Adj}_{\mathcal{G}_H}\| \leq \deg(\mathcal{G}_H)$) and symmetric, hence self-adjoint on $l^2(\Lambda)$. In particular, for any $T \in \mathbb{R}$ and any Pick function f, we have that

$$z \mapsto -W\left(z - H_\perp^0 + f(z)\right)^{-1} W^\dagger$$

defines a Pick function, where $W \in l^2(\Lambda)\setminus\{0\}$ and $H_\perp^0 \in \mathcal{S}(\ell^2(\Lambda\setminus\{0\}))$ are defined similarly as in the finite dimensional case. This fact, in addition with an asymptotic expansion and Theorem III.4.1, shows that the proof of Proposition III.3.5 extends easily to infinite graphs of finite degree.

Note that Proposition III.3.3 also holds, with the slight difference that for the limit case $U = 0$, the solution, being given by $\Sigma = 0$ and

$$\forall z \in \mathbb{C}_+, \quad \Delta(z) = W\left(z - H_\perp^0\right)^{-1} W^\dagger,$$

may not be associated with a finitely supported measure ν, the latter being given by

$$\nu = W\pi_{H_\perp^0} W^\dagger,$$

where $\pi_{H_\perp^0}$ is the spectral measure associated with $H_\perp^0$.

For example, the Nevanlinna-Riesz measure ν of the hybridization function $\Delta : z \mapsto W\left(z - H_\perp^0\right)W^\dagger$ is, for the nearest neighbor graph of the 1d-lattice $\mathbb{Z}$, within the single-site

DMFT approach, absolutely continuous with respect to the Lebesgue measure $d\varepsilon$, and with density ρ given, for all $\varepsilon \in \mathbb{R}$, by

$$\rho(\varepsilon) = \lim_{\eta \to 0^+} \Im \left(\left(\int_{-2}^{2} \frac{1}{\varepsilon + i\eta - T\lambda} \frac{1}{\sqrt{4-\lambda^2}} d\lambda \right)^{-1} \right) \chi_{[-2T,2T]}(\varepsilon).$$

This result follows from a classical spectral graph result concerning the 1d-chain [MW89], in combination with the Stieltjes inversion formula [GT00], after noting that we have, using Schur complement, for all $z \in \mathbb{C}_+$,

$$\Delta(z) = z - \left(\langle e_0, \left(z - T\mathrm{Adj}_{\mathcal{G}_H} \right)^{-1} e_0 \rangle \right)^{-1}.$$

Finally, let us add that the proofs of Theorem III.3.8 and Proposition III.3.9 extend also to infinite graphs with finite degrees.

The reader familiar with the mathematical results regarding thermodynamic limits of fermion lattice systems may be surprised by the way the results stated in this article extend so straightforwardly to infinite systems. All these results hold because of the way DMFT takes interaction between fermions into account: within this approximation, only equation (III.2.6a) depends on the size of the system, which involves only the one-body Hamiltonian and one-body quantities (self-energy, hybridization function).

III.4 Proofs

In this section, we give the proofs of the results stated in Section III.3. Among other things, we will make use of the results already stated about Pick functions, especially Theorem III.3.1 and of some results from measure theory, which will be recalled when needed. As we will discuss continuity of functions defined on measure spaces, we must specify the topology we are considering. Recall that a sequence $(\mu_n)_{n\in\mathbb{N}}$ of finite Borel measures on $\mathbb{R}$ is said to converge weakly to μ if for all bounded continuous function $f \in \mathcal{C}_b^0(\mathbb{R})$,

$$\int_{\mathbb{R}} f \, d\mu_n \to \int_{\mathbb{R}} f \, d\mu. \tag{III.4.1}$$

It converges vaguely to μ if (III.4.1) holds for all $f \in \mathcal{C}_0^0(\mathbb{R})$, the space of continuous functions from $\mathbb{R}$ to $\mathbb{R}$ vanishing at infinity. Weak convergence clearly implies vague convergence, and the converse is also true in $\mathcal{P}(\mathbb{R})$, since $\mathbb{R}$ is locally compact [Vil21, Remark 7.2]. We will also make use of the notions of Wasserstein distance and optimal transportation on $\mathbb{R}$. The Wasserstein 2-distance between two Borel probability measures μ and ν on $\mathbb{R}$ is defined by

$$W_2(\mu,\nu) := \left(\inf_{\pi \in \Pi(\mu,\nu)} \int_{\mathbb{R}^2} |x-y|^2 d\pi(x,y) \right)^{\frac{1}{2}}, \tag{III.4.2}$$

where $\Pi(\mu,\nu)$ is the set of all couplings of μ and ν, i.e. of Borel probability measures on $\mathbb{R}\times\mathbb{R}$ whose marginals with respect to the first and second variables are respectively μ

and ν. If μ and ν have finite moments of order 2, the infimum in the definition (III.4.2) is actually a minimum, and there exists in fact a unique $\pi_{\mu,\nu} \in \Pi(\mu,\nu)$ such that $W_2(\mu,\nu)^2 = \int |x-y|^2 d\pi_{\mu,\nu}(x,y)$ [San15].

Moreover, we will make an extensive use of the following theorem, which extends to Pick matrices a result on Pick functions, which can be found in [Eck22] and [Akh20, Theorem 3.2.1]. It states that the moments of the Nevanlinna-Riesz measure of a Pick function or matrix are related to its expansion on the imaginary axis at $+\infty$.

Theorem III.4.1. *Let $f : \mathbb{C}_+ \to \mathbb{C}^{n\times n}$ be a Pick matrix and μ its Nevanlinna-Riesz measure. Let $n \in \mathbb{N}$. The function f satisfies:*

$$-f(iy) = m_{-2}(iy) + m_{-1} + \frac{m_0}{iy} + \frac{m_1}{(iy)^2} + \frac{m_2}{(iy)^3} + \cdots + \frac{m_{2n}}{(iy)^{2n+1}} + o_{y\to+\infty}\left(\frac{1}{y^{2n+1}}\right) \quad \text{(III.4.3)}$$

if and only if μ has finite moments of order less than or equal to $2n$, i.e. for all $x \in \mathbb{C}^n$, $\langle x, \int_{\mathbb{R}} |\varepsilon|^k d\mu(\varepsilon) x\rangle < \infty$ for $0 \le k \le 2n$. For $0 \le k \le 2n$, the coefficient m_k is then the k-th moment of μ, i.e. $m_k = \int_{\mathbb{R}} \varepsilon^k d\mu(\varepsilon) \in \mathcal{S}_n(\mathbb{C})$.

Proof. The result for Pick functions can be found in [Eck22] and [Akh20, Theorem 3.2.1]. To extend the result to Pick matrices, it suffices to notice f is a Pick matrix if and only if for all $x \in \mathbb{C}^n$, the map $f_x : z \in \mathbb{C}_+ \mapsto \langle x, f(z)x\rangle$ is a Pick function. □

III.4.1 Proof of Proposition III.3.2 (IPT map for finite-dimensional bath)

We have for all $z \in \mathbb{C}_+$,

$$(z - \Delta(z))^{-1} = \left(z - H^0_{\mathrm{AIM}}\right)^{-1}_{1,1}, \qquad \text{where } H^0_{\mathrm{AIM}} = \begin{pmatrix} 0 & \sqrt{a_1} & \sqrt{a_2} & \cdots & \sqrt{a_K} \\ \sqrt{a_1} & \varepsilon_1 & 0 & \cdots & 0 \\ \sqrt{a_2} & 0 & \varepsilon_2 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ \sqrt{a}_K & 0 & \dots & 0 & \varepsilon_K \end{pmatrix},$$

which holds true in particular for $z = i\omega_n$. Note that H^0_{AIM} is self-adjoint and that for all $n \in \mathbb{Z}$, using functional calculus,

$$\int_0^\beta e^{i\omega_n \tau} \frac{-e^{-\tau H^0_{\mathrm{AIM}}}}{1 + e^{-\beta H^0_{\mathrm{AIM}}}} d\tau = \left(i\omega_n - H^0_{\mathrm{AIM}}\right)^{-1},$$

so that we can perform explicitly the Fourier summation : for all $\tau \in (0,\beta)$,

$$\frac{1}{\beta}\sum_{n'\in\mathbb{Z}} e^{-i\omega_{n'}\tau}(i\omega_{n'} - \Delta(i\omega_{n'}))^{-1} = \left(\frac{-e^{-\tau H^0_{\mathrm{AIM}}}}{1 + e^{-\beta H^0_{\mathrm{AIM}}}}\right)_{1,1} = -\sum_{k=1}^{K+1} \frac{|P_{1,k}|^2 e^{-\tau\varepsilon'_k}}{1 + e^{-\beta\varepsilon'_k}}, \quad \text{(III.4.4)}$$

where $P \in \mathbb{C}^{(K+1)\times(K+1)}$ is a unitary matrix such that $H^0_{\mathrm{AIM}} = P\mathrm{diag}(\varepsilon'_1, \cdots, \varepsilon'_{K+1})P^\dagger$. The right-hand side of (III.4.4) is a continuous function on $[0, \beta)$ and the following integral is well-defined and reads

$$\int_0^\beta e^{i\omega_n\tau}\left(\frac{1}{\beta}\sum_{n'\in\mathbb{Z}}(i\omega_{n'} - \Delta(i\omega_{n'}))^{-1}e^{-i\omega_{n'}\tau}\right)^3 d\tau = \sum_{k_1,k_2,k_3=1}^{K+1}\frac{1 + e^{-\beta(\varepsilon'_{k_1}+\varepsilon'_{k_2}+\varepsilon_{k_3})}}{(1+e^{-\beta\varepsilon'_{k_1}})(1+e^{-\beta\varepsilon'_{k_2}})(1+e^{-\beta\varepsilon'_{k_3}})}\frac{|P_{1,k_1}|^2|P_{1,k_2}|^2|P_{1,k_3}|^2}{i\omega_n - (\varepsilon'_{k_1}+\varepsilon'_{k_2}+\varepsilon'_{k_3})}.$$

Let us now compute the spectrum of H^0_{AIM}: a simple calculation shows that its characteristic polynomial $\chi_{H^0_{\mathrm{AIM}}}$ reads

$$\chi_{H^0_{\mathrm{AIM}}}(\varepsilon) = \left(\prod_{k=1}^K(\varepsilon - \varepsilon_k)\right)\varepsilon - \sum_{k=1}^K a_k \prod_{l=1,l\neq k}^K(\varepsilon - \varepsilon_l).$$

By assumption on the a_k's and ε_k's, we have $\chi_{H^0_{\mathrm{AIM}}}(\varepsilon_k) \neq 0$, so that

$$\chi_{H^0_{\mathrm{AIM}}}(\varepsilon) = 0 \iff \varepsilon - \Delta(\varepsilon) = 0.$$

This in fact straightforwardly follows from the Schur complement approach. Moreover, one can compute explicitly $|P_{1,k}|^2$: by definition, we have for all $k \in [\![1, K+1]\!]$ and $l \in [\![1, K]\!]$,

$$\sqrt{a_l}P_{1,k} + \varepsilon_l P_{l+1,k} = \varepsilon'_k P_{l+1,k} \implies \frac{a_l}{(\varepsilon'_k - \varepsilon_l)^2}|P_{1,k}|^2 = |P_{l+1,k}|^2,$$

which gives after summation on l and using the fact that $PP^\dagger = 1$,

$$|P_{1,k}|^2 = (1 - \Delta'(\varepsilon'_k))^{-1}.$$

This shows that $\Sigma^{\mathrm{IPT}}(i\omega_n) = \Sigma_n$, hence Σ^{IPT} is a solution to the Analytic Continuation Problem (ACP) defined in (III.3.6). Then, Theorem III.B.8 ensures that there is no other solution, which concludes the proof.

III.4.2 Proof of Proposition III.3.3 (no finite-dimensional bath solution)

In this section, we prove the statement of Proposition III.3.3, which states that there is no solution to the IPT-DMFT equations considering only hybridization functions with a finite-dimensional bath. The idea is simply that the DMFT loop imposes an increase in the number of poles of a hybridization function corresponding to a finite-dimensional bath, which leads to a contradiction if it is supposed to be a solution to the equations.

Suppose Δ is a hybridization function associated with a bath of finite dimension and a solution to the IPT-DMFT equations. In particular, there exist $K \geq 1$, $a_1, \ldots, a_K > 0$ and $\epsilon_1 < \cdots < \epsilon_K$ such that

$$\Delta(z) = \sum_{k=1}^K \frac{a_k}{z - \epsilon_k}.$$

Let Σ be the self-energy given by the IPT impurity solver, see Proposition III.3.2. We have

$$\Sigma(z) = U^2 \sum_{1\leq k_1,k_2,k_3\leq K'} \frac{a'_{k_1,k_2,k_3}}{z-\varepsilon'_{k_1}-\varepsilon'_{k_2}-\varepsilon'_{k_3}} \text{ and}$$

$$a'_{k_1,k_2,k_3} = \left(1+e^{-\beta(\varepsilon'_{k_1}+\varepsilon'_{k_2}+\varepsilon'_{k_3})}\right)\prod_{i=1}^{3}\frac{1-\Delta'(\varepsilon'_{k_i})}{1+e^{-\beta\varepsilon'_{k_i}}} > 0,$$

where $\varepsilon'_1 < \cdots < \varepsilon'_{K'}$ are the poles of the rational function $(z-\Delta(z))^{-1}$. Since Δ has exactly K real poles, he number of poles of $(z-\Delta(z))^{-1}$ is exactly $K+1$ and the latter are real numbers, so that $K' = K+1$. Indeed, the K' poles are the fixed points of the real-valued function $\varepsilon \in \mathbb{R} \mapsto \Delta(\varepsilon)$. This function is strictly decreasing on its domain of continuity: $\Delta'(\varepsilon) = -\sum_k \frac{a_k}{(\varepsilon-\varepsilon_k)^2} < 0$; it goes to zero at infinity, and it goes to $\pm\infty$ at $\varepsilon_k^{\pm}$, so that there is exactly one fixed point in each interval $(-\infty,\varepsilon_1),(\varepsilon_1,\varepsilon_2),\ldots,(\varepsilon_K,+\infty)$. Now, since $U \neq 0$ by assumption, Σ has more than $K+1$ poles (we do not need a better estimation of the number of poles). Hence we can write

$$\Sigma(z) = U^2 \sum_{k=1}^{K''} \frac{a''_k}{z-\varepsilon''_k},$$

with $K'' \geq K+1$, $a''_k > 0$ and $\varepsilon_1 < \cdots < \varepsilon_{K''}$. As Δ is assumed to be a solution to the IPT-DMFT equations, it reads

$$\Delta(z) = W\left(z - H_\perp^0 - \Sigma(z)\right)^{-1} W^\dagger.$$

This expression can be written using the spectral decomposition of the matrix $H_\perp^0$:

$$\Delta(z) = \sum_{\lambda\in \mathrm{Sp}(H_\perp^0)} \frac{|\Pi_\lambda W^\dagger|^2}{z-\lambda-\Sigma(z)}.$$

Since $W \neq 0$ because $T \neq 0$ and due to the hypotheses on the Hubbard graph, there is at least one $\lambda \in \mathrm{Sp}(H_\perp^0)$ such that $|\Pi_\lambda W^\dagger| > 0$. The same argument as above proves that the associated term in the previous expression is a rational function with exactly $K''+1$ real poles. There is no possible compensation between two same poles in two different terms of the sum because the associated weights are all of the same sign, hence Δ has more than $K+2$ poles. As K is by definition the number of poles of Δ, it is impossible and Δ cannot be a solution to the IPT-DMFT equations.

III.4.3 Proof of Proposition III.3.5 (Bath Update map)

As already mentioned in Remark III.3.6, we will actually prove an extension of Proposition III.3.5 in which we do not restrict ourselves to the single-site, translation-invariant, paramagnetic framework (see Remark III.3.6). We consider a finite graph Λ of cardinality $L := |\Lambda| \geq 2$, partitioned into a set of $P \geq 2$ non-overlapping fragments $(\Lambda_p)_{1\leq p\leq P}$ with

$L_p := |\Lambda_p| \geq 1$. This leads to an orthogonal decomposition of the one-electron state $\mathcal{H} \cong \mathbb{C}^{2L}$ into

$$\mathcal{H} = \mathcal{H}_1 \oplus \cdots \oplus \mathcal{H}_P \quad \text{with} \quad \mathcal{H}_p \cong \mathbb{C}^{2L_p} \text{ for all } 1 \leq p \leq P.$$

Identifying $\mathcal{H}_p$ with $\mathbb{C}^{2L_p}$, the non-interacting Hamiltonian matrix in the decomposition $\mathcal{H}_p \oplus \mathcal{H}_p^\perp$ reads

$$H_0 \cong \begin{pmatrix} H_p^0 & W_p \\ W_p^\dagger & H_{\bar{p}}^0 \end{pmatrix} \quad \text{with} \quad H_p^0 \in \mathbb{C}^{2L_p \times 2L_p}, \quad W_p \in \mathbb{C}^{2(L-L_p)\times 2L_p}, \quad H_{\bar{p}}^0 \in \mathbb{C}^{2(L-L_p)\times 2(L-L_p)},$$

and the self-energy and hybridization spaces are then respectively given by

$$\mathfrak{S}_p = \left\{ z \in \mathbb{C}_+ \mapsto C + \int_{\mathbb{R}} \frac{d\mu(\varepsilon)}{z-\varepsilon} \; ; \; C \in \mathcal{S}_{2L_p}(\mathbb{C}), \mu \in \mathcal{M}(\mathbb{R}, \mathcal{S}^+_{2L_p}(\mathbb{C})) \right\}, \tag{III.4.5}$$

where $\mathcal{M}(\mathbb{R}, \mathcal{S}_n^+(\mathbb{C}))$ is the set of finite $\mathcal{S}_n^+(\mathbb{C})$-valued Borel measures on $\mathbb{R}$, and

$$\mathfrak{D}_p = \left\{ z \in \mathbb{C}_+ \mapsto \int_{\mathbb{R}} \frac{d\nu(\varepsilon)}{z-\varepsilon} \; ; \; \nu \in \mathcal{M}(\mathbb{R}, \mathcal{S}^+_{2L_p}(\mathbb{C})), \; \nu(\mathbb{R}) = W_p^\dagger W_p \right\}.$$

We refer to Appendix III.A.4 for more details on this framework.

Proposition III.4.2. *Let $\Sigma = (\Sigma_1, \cdots, \Sigma_P) \in \mathfrak{S}_1 \times \cdots \times \mathfrak{S}_P$. For each $1 \leq p \leq P$, the hybridization function Δ_p given by*

$$\forall z \in \mathbb{C}_+, \quad \Delta_p(z) = W_p \left(z - H_{\bar{p}}^0 - \bigoplus_{q \neq p} \Sigma_q(z) \right)^{-1} W_p^\dagger \tag{III.4.6}$$

is well-defined, $-\Delta_p$ is a Pick matrix and there exists a finite measure $\nu_p \in \mathcal{M}(\mathbb{R}, \mathcal{S}^+_{2L_p}(\mathbb{C}))$ such that

$$\forall z \in \mathbb{C}_+, \quad \Delta_p(z) = \int_{\mathbb{R}} \frac{d\nu_p(\varepsilon)}{z-\varepsilon} \quad \text{and} \quad \nu_p(\mathbb{R}) = W_p W_p^\dagger. \tag{III.4.7}$$

In other words, $\Delta_p \in \mathfrak{D}_p$.

Proof. Let $C_p \in \mathcal{S}_{2L_p}(\mathbb{C})$ and $\mu_p \in \mathcal{M}(\mathbb{R}, \mathcal{S}^+_{2L_p}(\mathbb{C}))$ be such that for all $z \in \mathbb{C}_+$, $\Sigma_p(z) = C_p + \int_{\mathbb{R}} \frac{d\mu_p(\varepsilon)}{z-\varepsilon}$. Since for all $1 \leq p \leq P$, $-\Sigma_p : \mathbb{C}_+ \to \mathbb{C}^{2L_p \times 2L_p}$ is a Pick matrix,

$$-\bigoplus_{q \neq p} \Sigma_q : \mathbb{C}_+ \to \mathbb{C}^{2(L-L_p)\times 2(L-L_p)}$$

is a Pick matrix. As $H_{\bar{p}}^0$ is Hermitian, we have for all $z \in \mathbb{C}_+$,

$$\Im \left(z - H_{\bar{p}}^0 - \bigoplus_{q \neq p} \Sigma_q(z) \right) = \Im(z) - \bigoplus_{q \neq p} \Im(\Sigma_q(z)) \geq \Im(z) > 0,$$

where $M_1 \geq M_2$ means that $M_1 - M_2$ is a positive semidefinite matrix. Moreover, if $\Im(M) > 0$, then M is invertible. Thus $z - H_{\bar{p}}^0 - \bigoplus_{q \neq j} \Sigma_q(z)$ is invertible and $\Delta_p(z)$ is well

defined. As $\Im(M) > 0$ if and only if $\Im(M^{-1}) < 0$, and as the congruence preserves the sign of the imaginary part, we have $\Im(\Delta_p(z)) \leq 0$ for all $z \in \mathbb{C}_+$, so that $-\Delta_p$ is a Pick matrix. To show formula (III.4.7), we will make use of the results on Pick functions stated in Section III.2.1. As μ_q, the Nevanlinna-Riesz measure of Σ_q, is finite for $1 \leq q \leq P$, we have for $x \in \mathbb{C}^{2L_q}$, that the positive Borel measure μ_q^x defined as the Nevanlinna-Riesz measure of the Pick function $z \mapsto -\langle x, \Sigma_q(z)x\rangle$, is also a finite positive measure. Then, for $x \in \mathbb{C}^{2L_q}$ and $y \geq 1$,

$$\left|\left\langle x, \int_{\mathbb{R}} \frac{d\mu_q(\varepsilon)}{iy-\varepsilon} x\right\rangle\right| = \left|\int_{\mathbb{R}} \frac{d\mu_q^x(\varepsilon)}{iy-\varepsilon}\right| \leq \int_{\mathbb{R}} \frac{d\mu_q^x(\varepsilon)}{|iy-\varepsilon|} \leq \int_{\mathbb{R}} d\mu_q^x < \infty.$$

This coarse upper bound is enough to ensure that

$$\begin{aligned} iy\Delta_p(iy) &= iyW_p\left(iy - H_{\overline{p}}^0 - \bigoplus_{q\neq p}\Sigma_q(iy)\right)^{-1} W_p^\dagger \\ &= W_p\left(1 - \frac{1}{iy}\left(H_{\overline{p}}^0 + \bigoplus_{q\neq p}\left(C_q + \int_{\mathbb{R}}\frac{d\mu_q(\varepsilon)}{iy-\varepsilon}\right)\right)\right)^{-1} W_p^\dagger \\ &\underset{y\to+\infty}{\longrightarrow} W_pW_p^\dagger. \end{aligned}$$

This gives the expansion $\Delta_p(iy) = \dfrac{W_pW_p^\dagger}{iy} + o\left(\dfrac{1}{y}\right)$, as y goes to $+\infty$. By Theorem III.4.1, it follows that the Nevanlinna-Riesz measure of $-\Delta_p$, denoted by ν_p, is finite, and its mass is precisely the quantity $W_p^\dagger W_p$. Thus the Nevanlinna-Riesz representation of $-\Delta_p$ reads

$$-\Delta_p(z) = az + b - \int_{\mathbb{R}} \frac{d\nu_p(\varepsilon)}{z-\varepsilon},$$

for some $a \in \mathcal{S}_{2L_p}^+(\mathbb{C})$ and $b \in \mathcal{S}_{2L_p}(\mathbb{C})$. Now, because of the aforementioned expansion, we must have $a = b = 0$, which concludes the proof. □

III.4.4 Proof of Proposition III.3.7 (IPT map for infinite-dimensional bath)

Let $\nu \in \mathcal{P}(\mathbb{R})$, and define the associated hybridization function $\Delta(z) = W_p^\dagger W_p \int_{\mathbb{R}} \frac{d\nu(\varepsilon)}{z-\varepsilon}$. We first establish the following lemma, which we will use several times in our analysis.

Lemma III.4.3. *Let $c \in \mathbb{R}$, $\mu_0 \in \mathcal{M}_+(\mathbb{R})$. For $z \in \mathbb{C}_+$, we set*

$$f(z) = \int_{\mathbb{R}} \frac{d\mu_0(\varepsilon)}{z-\varepsilon} \quad \text{and} \quad g(z) = \frac{1}{z-c-f(z)}.$$

Then $-g$ is a Pick function and its Nevanlinna-Riesz representation reads $g(z) = \int_{\mathbb{R}} \frac{d\mu(\varepsilon)}{z-\varepsilon}$ with $\mu \in \mathcal{P}(\mathbb{R})$. In particular, μ has finite moments of order less than or equal to 2, given by

$$m_0(\mu) = 1 \text{ (μ is a probability measure)}, \quad m_1(\mu) = c \quad \text{and} \quad m_2(\mu) = \mu_0(\mathbb{R}) + c^2.$$

Proof. The result follows from the expansion of the function $g(z)$ when $z = iy$, $y \to +\infty$ and Theorem III.4.1. One has

$$g(iy) = \frac{1}{iy} \frac{1}{1 - \frac{c}{iy} - \frac{f(iy)}{iy}}.$$

Since $f(iy) = \dfrac{\mu_0(\mathbb{R})}{iy} + o\left(\dfrac{1}{y}\right)$, we have

$$\begin{aligned} g(iy) = \frac{1}{iy} \frac{1}{1 - \frac{c}{iy} - \frac{\mu_0(\mathbb{R})}{(iy)^2} + o\left(\frac{1}{y^2}\right)} &= \frac{1}{iy}\left(1 + \frac{c}{iy} + \frac{\mu_0(\mathbb{R}) + c^2}{(iy)^2} + o\left(\frac{1}{y^2}\right)\right) \\ &= \frac{1}{iy} + \frac{c}{(iy)^2} + \frac{\mu_0(\mathbb{R}) + c^2}{(iy)^3} + o\left(\frac{1}{y^3}\right). \end{aligned}$$

Using again Theorem III.4.1, we obtain the desired result. □

We apply Lemma III.4.3 with $c = 0$ and $f = \Delta$. It follows that there exists $\xi \in \mathcal{P}(\mathbb{R})$ satisfying (III.3.14). Then, following equations (III.3.15)-(III.3.17), set $\tilde{\xi}(d\varepsilon) = \dfrac{\xi(d\varepsilon)}{1 + e^{-\beta\varepsilon}}$, and $\tilde{\mu} = \tilde{\xi} * \tilde{\xi} * \tilde{\xi}$. We must verify that, setting $\mu(d\varepsilon) = (1 + e^{-\beta\varepsilon})\tilde{\mu}(d\varepsilon)$, μ is indeed a positive finite measure, so that the self-energy Σ associated with the measure μ belongs to $\mathfrak{S}^{\mathrm{IPT}}$.

The multiplication by positive functions and the convolution preserves positivity, so μ is a positive measure. Moreover, we have

$$\mu(\mathbb{R}) = \int_{\mathbb{R}^3} \frac{\left(1 + e^{-\beta(\varepsilon_1+\varepsilon_2+\varepsilon_3)}\right) d\xi(\varepsilon_1)d\xi(\varepsilon_2)d\xi(\varepsilon_3)}{(1 + e^{-\beta\varepsilon_1})(1 + e^{-\beta\varepsilon_2})(1 + e^{-\beta\varepsilon_3})} \leq \int_{\mathbb{R}^3} d\xi(\varepsilon_1)d\xi(\varepsilon_2)d\xi(\varepsilon_3) = 1. \tag{III.4.8}$$

Thus μ is finite, and the map F^{IPT} is well defined. To prove that this map is actually continuous with respect to the weak topology of measures, we need to establish Lemma III.4.4. This result states the continuity of the map $\mathcal{M}_+(\mathbb{R}) \ni \mu_0 \mapsto \mu \in \mathcal{P}(\mathbb{R})$ defined in Lemma III.4.3, which is central in the IPT-DMFT equations. Note that it holds

$$\forall z \in \mathbb{C}_+, \quad \int_{\mathbb{R}} \frac{d\mu(\varepsilon)}{z - \varepsilon} = \frac{1}{z - c - \int_{\mathbb{R}} \frac{d\mu_0(\varepsilon)}{z - \varepsilon}}. \tag{III.4.9}$$

Lemma III.4.4. *The map $\Phi : \mathcal{M}_+(\mathbb{R}) \ni \mu_0 \mapsto \mu \in \mathcal{P}(\mathbb{R})$ defined in Lemma III.4.3 is weakly continuous. More precisely, the following stronger result holds true: if $(\mu_0^n)_{n\in\mathbb{N}}$ converges weakly to μ_0 in $\mathcal{M}_+(\mathbb{R})$, then $W_2(\Phi(\mu_0^n), \Phi(\mu)) \underset{n\to\infty}{\longrightarrow} 0$, where W_2 is the Wasserstein distance of order 2.*

Proof. Let $(\mu_0^n)_{n\in\mathbb{N}} \subset \mathcal{M}_+(\mathbb{R})$ converging weakly to μ_0 in $\mathcal{M}_+(\mathbb{R})$, $\mu^n := \Phi(\mu_0^n)$ and $\mu := \Phi(\mu_0)$. In view of Lemma III.4.3, μ^n has finite moments of orders 1 and 2, given by $m_1(\mu^n) = c$ and $m_2(\mu^n) = \mu_0^n(\mathbb{R}) + c^2$.

As μ_0^n converges weakly to μ_0, we have $\int_{\mathbb{R}} f d\mu_0^n \to \int_{\mathbb{R}} f d\mu_0$ for all $f \in \mathcal{C}_b^0(\mathbb{R})$. Taking $f \equiv 1$, we get $\mu_0^n(\mathbb{R}) = \int_{\mathbb{R}} d\mu_0^n \underset{n\to\infty}{\longrightarrow} \int_{\mathbb{R}} d\mu_0 = \mu_0(\mathbb{R})$. Hence, $m_2(\mu^n) = \mu_0^n(\mathbb{R}) + c^2 \underset{n\to\infty}{\longrightarrow} \mu_0(\mathbb{R}) + c^2 = m_2(\mu)$.

Now, for all $z \in \mathbb{C}_+$, the function $\mathbb{R} \ni \varepsilon \mapsto \dfrac{1}{z-\varepsilon} \in \mathbb{C}$ is also bounded and continuous. Thus, we can pass to the limit in formula (III.4.9):

$$\int_{\mathbb{R}} \frac{d\mu^n(\varepsilon)}{z-\varepsilon} = \frac{1}{z - c - \int_{\mathbb{R}} \frac{d\mu_0^n(\varepsilon)}{z-\varepsilon}} \underset{n\to\infty}{\longrightarrow} \frac{1}{z - c - \int_{\mathbb{R}} \frac{d\mu_0(\varepsilon)}{z-\varepsilon}} = \int_{\mathbb{R}} \frac{d\mu(\varepsilon)}{z-\varepsilon}. \tag{III.4.10}$$

This can be extended to complex numbers $z \in \mathbb{C} \setminus \mathbb{R}$ by taking the complex conjugate of the limit (III.4.10). Let $\mathcal{A}$ be the algebra generated by the functions $\mathbb{R} \ni \varepsilon \mapsto \dfrac{1}{z-\varepsilon} \in \mathbb{C}$ for $z \in \mathbb{C} \setminus \mathbb{R}$. It is known that $\mathcal{A}$ is dense in $\mathcal{C}_0^0(\mathbb{R})$ (this can be shown using Helffer-Sjöstrand formula [HS05]). Together with (III.4.10), this implies that for all $f \in \mathcal{C}_0^0(\mathbb{R})$, $\int_{\mathbb{R}} f d\mu^n \underset{n\to\infty}{\longrightarrow} \int_{\mathbb{R}} f d\mu$, i.e. that $(\mu^n)_{n\in\mathbb{N}}$ vaguely converges to μ. Since $\mathbb{R}$ is locally compact, and because μ and the μ^n's are probability measures, the vague convergence is equivalent in this case to the weak convergence. The map Φ is therefore weakly continuous. Since on the space of Borel probability measures on $\mathbb{R}$, W_2-convergence is equivalent to weak convergence and the convergence of the second moment [AGS05, Section 7.1], the proof is complete. □

Let us now prove that the impurity solver $F^{\mathrm{IPT}} : \mathcal{P}(\mathbb{R}) \to \mathcal{M}_+(\mathbb{R})$ is weakly continuous. Let $(\nu^n)_{n\in\mathbb{N}} \subset \mathcal{P}(\mathbb{R})$ converging weakly to $\nu \in \mathcal{P}(\mathbb{R})$. Define $\xi^n \in \mathcal{P}(\mathbb{R})$ and $\mu^n := F^{\mathrm{IPT}}(\nu^n)$ as in Proposition III.3.7, see equation (III.3.14), as well as $\xi \in \mathcal{P}(\mathbb{R})$ and $\mu := F^{\mathrm{IPT}}(\nu)$. We want to show that μ^n converges weakly to μ. First, because of the definition of ξ through equation (III.3.14), and thanks to Lemma III.4.4, we know that $W_2(\xi^n, \xi) \underset{n\to\infty}{\longrightarrow} 0$. Moreover, using the same density argument as in the proof of Lemma III.4.4, it is sufficient to show that for all $z \in \mathbb{C}_+$, the following convergence holds:

$$\int_{\mathbb{R}} \frac{d\mu^n(\varepsilon)}{z-\varepsilon} \underset{n\to\infty}{\longrightarrow} \int_{\mathbb{R}} \frac{d\mu(\varepsilon)}{z-\varepsilon}.$$

We have for $z \in \mathbb{C}_+$,

$$\int_{\mathbb{R}} \frac{d\mu^n(\varepsilon)}{z-\varepsilon} = \int_{\mathbb{R}^3} \frac{1}{z-(\varepsilon_1+\varepsilon_2+\varepsilon_3)} \frac{1+e^{-\beta(\varepsilon_1+\varepsilon_2+\varepsilon_3)}}{(1+e^{-\beta\varepsilon_1})(1+e^{-\beta\varepsilon_2})(1+e^{-\beta\varepsilon_3})} d\xi^n(\varepsilon_1) d\xi^n(\varepsilon_2) d\xi^n(\varepsilon_3).$$

Let $\psi(\varepsilon_1, \varepsilon_2, \varepsilon_3) = \dfrac{1}{z-(\varepsilon_1+\varepsilon_2+\varepsilon_3)} \dfrac{1+e^{-\beta(\varepsilon_1+\varepsilon_2+\varepsilon_3)}}{(1+e^{-\beta\varepsilon_1})(1+e^{-\beta\varepsilon_2})(1+e^{-\beta\varepsilon_3})}$, for $(\varepsilon_1, \varepsilon_2, \varepsilon_3) \in \mathbb{R}^3$. Then,

$$\begin{aligned}
\left| \int_{\mathbb{R}} \frac{d\mu^n(\varepsilon)}{z-\varepsilon} - \int_{\mathbb{R}} \frac{d\mu(\varepsilon)}{z-\varepsilon} \right| &= \left| \int_{\mathbb{R}^3} \psi d\xi^n d\xi^n d\xi^n - \int_{\mathbb{R}^3} \psi d\xi d\xi d\xi \right| \\
&\leq \left| \int_{\mathbb{R}^3} \psi d\xi^n d\xi^n d\xi^n - \int_{\mathbb{R}^3} \psi d\xi^n d\xi^n d\xi \right| && \text{(III.4.11)} \\
&+ \left| \int_{\mathbb{R}^3} \psi d\xi^n d\xi^n d\xi - \int_{\mathbb{R}^3} \psi d\xi^n d\xi d\xi \right| && \text{(III.4.12)} \\
&+ \left| \int_{\mathbb{R}^3} \psi d\xi^n d\xi d\xi - \int_{\mathbb{R}^3} \psi d\xi d\xi d\xi \right|. && \text{(III.4.13)}
\end{aligned}$$

We now prove that the last term (III.4.13) goes to zero when n goes to ∞. The same arguments apply to the other two terms, (III.4.11) and (III.4.12). The function ψ is smooth and a simple calculation shows that its partial derivative with respect to ε_1 is bounded by

$$|\partial_1\psi(\varepsilon_1,\varepsilon_2,\varepsilon_3)| \leq \frac{1}{|\Im(z)|^2} + \frac{2\beta}{|\Im(z)|} =: \kappa_{z,\beta}. \tag{III.4.14}$$

Let $\varepsilon_2, \varepsilon_3 \in \mathbb{R}$. Using the W_2 convergence of ξ^n towards ξ, set π_n the optimal coupling between these two measures [San15]. We have that

$$\int_{\mathbb{R}} \psi(\varepsilon_1,\varepsilon_2,\varepsilon_3) d\xi^n(\varepsilon_1) = \int_{\mathbb{R}^2} \psi(\varepsilon_1,\varepsilon_2,\varepsilon_3) d\pi_n(\varepsilon_1,\varepsilon_1').$$

By Taylor's Theorem and since ψ is continuously derivable, we have

$$\int_{\mathbb{R}} \psi(\varepsilon_1,\varepsilon_2,\varepsilon_3) d\xi^n(\varepsilon_1) = \int_{\mathbb{R}^2} \psi(\varepsilon_1',\varepsilon_2,\varepsilon_3) d\pi_n(\varepsilon_1,\varepsilon_1') + \int_{\mathbb{R}^2}\int_{\varepsilon_1'}^{\varepsilon_1} (\varepsilon_1 - t)\partial_1\psi(t,\varepsilon_2,\varepsilon_3) dt d\pi_n(\varepsilon_1,\varepsilon_1').$$

On the one hand, the first term is exactly $\int_{\mathbb{R}} \psi(\varepsilon_1,\varepsilon_2,\varepsilon_3) d\xi(\varepsilon_1)$ by definition of π_n, and on the other hand, using (III.4.14), we can bound the second term by

$$\begin{aligned}
\left| \int_{\mathbb{R}^2}\int_{\varepsilon_1'}^{\varepsilon_1} (\varepsilon_1 - t)\partial_1\psi(t,\varepsilon_2,\varepsilon_3) dt d\pi_n(\varepsilon_1,\varepsilon_1') \right| &\leq \frac{1}{2}\int_{\mathbb{R}^2} |\varepsilon_1 - \varepsilon_1'|^2 \|\partial_1\psi\|_\infty d\pi_n(\varepsilon_1,\varepsilon_1') \\
&\leq \frac{\kappa_{z,\beta}}{2}\int_{\mathbb{R}^2} |\varepsilon_1 - \varepsilon_1'|^2 d\pi_n(\varepsilon_1,\varepsilon_1') \\
&= \frac{\kappa_{z,\beta}}{2} W_2(\xi^n,\xi)^2.
\end{aligned}$$

Finally, the term (III.4.13) can be bounded by

$$\begin{aligned}
\left| \int_{\mathbb{R}^3} \psi d\xi^n d\xi d\xi - \int_{\mathbb{R}^3} \psi d\xi d\xi d\xi \right| &\leq \int_{\mathbb{R}^2} \left| \int_{\mathbb{R}} \psi(\varepsilon_1,\cdot,\cdot) d\xi^n(\varepsilon_1) - \int_{\mathbb{R}} \psi(\varepsilon_1,\cdot,\cdot) d\xi(\varepsilon_1) \right| d\xi d\xi \\
&= \int_{\mathbb{R}^2} \left| \int_{\mathbb{R}^2}\int_{\varepsilon_1'}^{\varepsilon_1} (\varepsilon_1 - t)\partial_1\psi(t,\varepsilon_2,\varepsilon_3) dt d\pi_n(\varepsilon_1,\varepsilon_1') \right| d\xi(\varepsilon_2) d\xi(\varepsilon_3) \\
&\leq \int_{\mathbb{R}^2} \frac{\kappa_{z,\beta}}{2} W_2(\xi^n,\xi)^2 d\xi(\varepsilon_2) d\xi(\varepsilon_3) = \frac{\kappa_{z,\beta}}{2} W_2(\xi^n,\xi)^2 \underset{n\to\infty}{\longrightarrow} 0.
\end{aligned}$$

This shows that the map F^{IPT} is weakly continuous. It remains to prove that the map $\mathfrak{D}^{\mathrm{IPT}} \ni \Delta \mapsto \Sigma \in \mathfrak{S}^{\mathrm{IPT}}$ defined by (III.3.14)-(III.3.18) coincides on the set of discrete probability measures with finite support with the map defined in Proposition III.3.2.

Let $\Delta \in \mathfrak{D}^{\mathrm{IPT}}$ with a Nevanlinna-Riesz measure of the form $\nu = \sum_{k=1}^K a_k \delta_{\varepsilon_k}$, where $a_k > 0$, $\sum a_k = W_p^\dagger W_p$ and $\varepsilon_1 < \cdots < \varepsilon_K$. It follows that the rational function $(z - \Delta(z))^{-1}$ is of the form $\sum_{k=1}^{K+1} \frac{a_k'}{z - \varepsilon_k'}$ (see the proof of Proposition III.3.3 in Section III.4.2). This means by (III.3.14) that $\xi = \sum_{k=1}^{K+1} a_k' \delta_{\varepsilon_k'}$, and the ε_k''s are the zeros of the rational function $z - \Delta(z)$, so that the residues are given by $a_k' = (1 - \Delta'(\varepsilon_k'))^{-1}$. The self-energy Σ given

by (III.3.18) then reads for all $z \in \mathbb{C}_+$,

$$\begin{aligned}\Sigma(z) &= U^2 \int_{\mathbb{R}^3} \frac{1+e^{-\beta(\varepsilon_1+\varepsilon_2+\varepsilon_3)}}{(1+e^{-\beta\varepsilon_1})(1+e^{-\beta\varepsilon_2})(1+e^{-\beta\varepsilon_3})} \frac{d\xi(\varepsilon_1)d\xi(\varepsilon_2)d\xi(\varepsilon_3)}{z-(\varepsilon_1+\varepsilon_2+\varepsilon_3)} \\ &= U^2 \sum_{k_1,k_2,k_3=1}^{K+1} \frac{1+e^{-\beta(\varepsilon'_{k_1}+\varepsilon'_{k_2}+\varepsilon_{k_3})}}{(1+e^{-\beta\varepsilon'_{k_1}})(1+e^{-\beta\varepsilon'_{k_2}})(1+e^{-\beta\varepsilon'_{k_3}})} \frac{a'_{k_1}a'_{k_2}a'_{k_3}}{z-(\varepsilon'_{k_1}+\varepsilon'_{k_2}+\varepsilon'_{k_3})} \\ &= U^2 \sum_{k_1,k_2,k_3=1}^{K+1} \frac{a'_{k_1,k_2,k_3}}{z-(\varepsilon'_{k_1}+\varepsilon'_{k_2}+\varepsilon'_{k_3})},\end{aligned}$$

where a'_{k_1,k_2,k_3} is given by (III.3.8). This complies with the result stated in Proposition III.3.2.

Finally, since the set of discrete probability measures with finite support is dense in the set of probability measures $\mathcal{P}(\mathbb{R})$ for the weak topology and since F^{IPT} is weakly continuous, F^{IPT} is the only continuous extension of the IPT map defined in Proposition III.3.2 for a finite-dimensional bath.

III.4.5 Continuity of the IPT-DMFT map

The following result is central to prove the existence of a fixed point to the DMFT equations.

Theorem III.4.5 (Continuity of the IPT-DMFT map)**.** *The IPT-DMFT map F^{DMFT} is weakly continuous on $\mathcal{P}(\mathbb{R})$. More precisely, the following stronger results holds true: if $(\nu^n)_{n\in\mathbb{N}}$ converges weakly to ν, then*

$$W_2\left(F^{\mathrm{DMFT}}(\nu^n), F^{\mathrm{DMFT}}(\nu)\right) \underset{n\to\infty}{\longrightarrow} 0,$$

where W_2 is the Wasserstein 2-distance.

We have proven in the previous section the continuity of the map F^{IPT}. In order to prove the continuity of F^{SC}, we need to adapt Lemma III.4.4 to equation (III.3.12). Let $\mu \in \mathcal{M}_+(\mathbb{R})$. Applying Theorem III.4.1 to the measure μ, and using the definition (III.3.12) of the hybridization function Δ associated with $F^{\mathrm{SC}}(\mu)$, we obtain

$$\begin{aligned}\Delta(iy) &= \frac{1}{iy} W \left(1 - \frac{H_\perp^0}{iy} - \frac{1}{iy} U^2 \int_{\mathbb{R}} \frac{d\mu(\varepsilon)}{iy-\varepsilon}\right)^{-1} W^\dagger \\ &= \frac{1}{iy} W \left(1 - \frac{H_\perp^0}{iy} - \frac{1}{iy}\left(\frac{U^2\mu(\mathbb{R})}{iy} + o\left(\frac{1}{y}\right)\right)\right)^{-1} W^\dagger \\ &= W\left[\frac{1}{iy} + \frac{H_\perp^0}{(iy)^2} + \frac{(H_\perp^0)^2 + U^2\mu(\mathbb{R})}{(iy)^3}\right] W^\dagger + o\left(\frac{1}{y^3}\right).\end{aligned}$$

It follows that, when $y \to +\infty$, we have the expansion

$$\int_{\mathbb{R}} \frac{dF^{\mathrm{SC}}(\mu)(\varepsilon)}{iy-\varepsilon} = \frac{1}{W_p^\dagger W_p}\Delta(iy) = \frac{1}{iy} + \frac{s^1}{(iy)^2} + \frac{s^2(\mu)}{(iy)^3} + o\left(\frac{1}{y^3}\right), \tag{III.4.15}$$

with s^1 and $s^2(\mu)$ given by

$$s^1 = \frac{W H_\perp^0 W^\dagger}{W_p^\dagger W_p}, \tag{III.4.16}$$

$$s^2(\mu) = \frac{W\left((H_\perp^0)^2 + U^2\mu(\mathbb{R})\right)W^\dagger}{W_p^\dagger W_p}. \tag{III.4.17}$$

In view of Theorem III.4.1, this implies that $\nu := F^{\mathrm{SC}}(\mu)$ has finite moments of orders 1 and 2, respectively given by $m_1(\nu) = s^1$ and $m_2(\nu) = s^2(\mu)$. The arguments in the proof of Lemma III.4.4 can then be used to show that the following result holds true.

Proposition III.4.6. *$F^{\mathrm{SC}} : \mathcal{M}_+(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ is weakly continuous. More precisely, if $(\mu^n)_{n\in\mathbb{N}}$ converges weakly to μ in $\mathcal{M}_+(\mathbb{R})$, then $W_2(F^{\mathrm{SC}}(\mu^n), F^{\mathrm{SC}}(\mu)) \underset{n\to\infty}{\longrightarrow} 0$.*

We are now in position to complete the proof of Theorem III.4.5. Let $(\nu^n)_n \subset \mathcal{P}(\mathbb{R})$ converging weakly to ν in $\mathcal{P}(\mathbb{R})$. Denoting by $\mu^n := F^{\mathrm{IPT}}(\nu^n)$ and $\mu := F^{\mathrm{IPT}}(\nu)$, we have shown in the proof of Proposition III.3.7 that μ^n converges weakly to μ in $\mathcal{M}_+(\mathbb{R})$, see Section III.4.4. Proposition III.4.6 then shows that $W_2(F^{\mathrm{DMFT}}(\nu^n), F^{\mathrm{DMFT}}(\nu)) = W_2(F^{\mathrm{SC}}(\mu^n), F^{\mathrm{SC}}(\mu)) \underset{n\to\infty}{\longrightarrow} 0$. In particular, $F^{\mathrm{DMFT}}(\nu^n)$ converges weakly to $F^{\mathrm{DMFT}}(\nu)$.

III.4.6 Proof of Theorem III.3.8 (existence of a fixed point)

The existence of a fixed point to the IPT-DMFT map, that is a solution to the IPT-DMFT equations, is a consequence of the following fixed point theorem [Sha16].

Theorem III.4.7 (Schauder-Singbal)**.** *Let E be a locally convex Hausdorff linear topological space, C a nonempty closed convex subset of E, and F a continuous map from C into itself, such that $F(C)$ is contained in a compact subset of C. Then F has a fixed point.*

Let us consider the vector space $E = \mathcal{M}(\mathbb{R})$ endowed with the Kantorovitch-Rubinstein norm $\|\cdot\|_0$. Recall that the latter is defined as

$$\|\mu\|_0 := \sup\left\{\int_{\mathbb{R}} f d\mu \;;\; f \in \mathrm{Lip}_1, \|f\|_\infty \leq 1\right\},$$

where Lip_1 is the set of continuous functions on $\mathbb{R}$ with Lipschitz constant less than or equal to 1.

Let us then set $C := \mathcal{P}(\mathbb{R}) = \{\mu \in E \mid \mu \geq 0,\ \int_{\mathbb{R}} d\mu = 1\}$. Since E is a normed vector space on $\mathbb{R}$, it is a locally convex Hausdorff linear topological space, and C is obviously a non-empty convex subset of E.

Besides, on the set of finite positive measures, weak convergence is equivalent to convergence for the Kantorovitch-Rubinstein norm [Bog07, Chapter 8.3]. We can thus work with the weak topology on C.

The fact that C is weakly closed means that $\mathcal{P}(\mathbb{R})$ is a weakly closed subset of $\mathcal{M}(\mathbb{R})$. This result can be found in [Bog18, Section 3.2]. Moreover, we already proved in Theorem III.4.5 that $F^{\mathrm{DMFT}} : C \to C$ was weakly continuous.

To apply Schauder-Singbal's theorem to our setting, it thus remains to show that $F^{\mathrm{DMFT}}(C)$ is relatively compact for the weak topology. This is in fact a consequence of Prokhorov's Theorem [Bog18, Theorem 2.3.4].

Indeed, let $\nu \in F^{\mathrm{DMFT}}(C)$ and $\nu_0 \in \mathcal{P}(\mathbb{R})$, $\mu = F^{\mathrm{IPT}}(\nu_0) \in \mathcal{M}_+(\mathbb{R})$ such that $\nu = F^{\mathrm{DMFT}}(\nu_0) = F^{\mathrm{SC}}(\mu)$. As we have seen in the proof of Proposition III.4.6, ν has finite moments of order 1 and 2, given respectively by $m_1(\nu) = s^1$ and $m_2(\nu) = s^2(\mu)$, where s^1 is defined by (III.4.16) and $s^2(\mu)$ by (III.4.17). The inequality (III.4.8) states that the mass of the measure μ is bounded by 1. Hence, $m_2(\nu) = s^2(\mu)$ is bounded independently on ν:

$$m_2(\nu) = s^2(\mu) \leq c := \frac{W\left((H_\perp^0)^2 + U^2\right)W^\dagger}{W_p^\dagger W_p}. \tag{III.4.18}$$

This allows us to show that $F^{\mathrm{DMFT}}(C)$ is tight. For $\eta > 0$, take $K = [-a, a]$, with $a \geq 1$ large enough so that $c/a^2 \leq \eta$. Then for $\nu \in F^{\mathrm{DMFT}}(C)$, (III.4.18) holds and

$$\nu(\mathbb{R} \setminus K) = \int_{\mathbb{R}\setminus K} \frac{\varepsilon^2}{\varepsilon^2} d\nu(\varepsilon) \leq \frac{1}{a^2}\int_{\mathbb{R}\setminus K} \varepsilon^2 d\nu(\varepsilon) \leq \frac{1}{a^2}\int_{\mathbb{R}} \varepsilon^2 d\nu(\varepsilon) = \frac{m_2(\nu)}{a^2} \leq \frac{c}{a^2} \leq \eta.$$

Hence $F^{\mathrm{DMFT}}(C)$ is tight. By Prokhorov's Theorem, it is thus weakly relatively compact.

This concludes the proof of our main result.

III.4.7 Proof of Proposition III.3.9 (finiteness of the moments of a solution)

Let $\nu^0 \in \mathcal{P}(\mathbb{R})$ and $k \in 2\mathbb{N}$, and assume that ν^0 has finite k-th moment, i.e. $\int_{\mathbb{R}} |\varepsilon|^k d\nu^0(\varepsilon) < \infty$. By Theorem III.4.1, the following expansion holds:

$$\int_{\mathbb{R}} \frac{d\nu^0(\varepsilon)}{iy - \varepsilon} = \frac{1}{iy} + \cdots + \frac{m_k(\nu^0)}{(iy)^{k+1}} + o\left(\frac{1}{y^{k+1}}\right). \tag{III.4.19}$$

Then define $\Delta^0(z) = W_p^\dagger W_p \displaystyle\int_{\mathbb{R}} \frac{d\nu^0(\varepsilon)}{z - \varepsilon}$ and ξ by (III.3.14):

$$\int_{\mathbb{R}} \frac{d\xi(\varepsilon)}{z - \varepsilon} = \frac{1}{z - \Delta^0(z)}.$$

This function can be asymptotically expanded to order $k+3$ using (III.4.19).

$$\begin{aligned}
\int_{\mathbb{R}} \frac{d\xi(\varepsilon)}{iy - \varepsilon} &= \frac{1}{iy - W_p^\dagger W_p\left(\frac{1}{iy} + \cdots + \frac{m_k(\nu^0)}{(iy)^{k+1}} + o\left(\frac{1}{y^{k+1}}\right)\right)} \\
&= \frac{1}{iy}\left(1 - \frac{W_p^\dagger W_p}{iy}\left(\frac{1}{iy} + \cdots + \frac{m_k(\nu^0)}{(iy)^{k+1}} + o\left(\frac{1}{(iy)^{k+1}}\right)\right)\right)^{-1} \\
&= \frac{1}{iy} + \cdots + \frac{m_{k+2}(\xi)}{(iy)^{k+3}} + o\left(\frac{1}{y^{k+3}}\right).
\end{aligned}$$

By Theorem III.4.1, ξ has finite moments up to order $k+2$, which is even, denoted by $m_0(\xi) = 1, \dots, m_{k+2}(\xi)$. Now let $\mu := F^{\mathrm{IPT}}(\nu^0)$, so that $\mu \in \mathcal{M}_+(\mathbb{R})$ is given by (III.3.17) in the statement of Proposition III.3.7. In particular, there exists $C_k \in \mathbb{R}_+$ such that

$$\begin{aligned}
\int_{\mathbb{R}} |\varepsilon|^{k+2} d\mu(\varepsilon) &= \int_{\mathbb{R}^3} |\varepsilon_1+\varepsilon_2+\varepsilon_3|^{k+2} \frac{1+e^{-\beta(\varepsilon_1+\varepsilon_2+\varepsilon_3)}}{(1+e^{-\beta\varepsilon_1})(1+e^{-\beta\varepsilon_2})(1+e^{-\beta\varepsilon_3})} d\xi(\varepsilon_1)d\xi(\varepsilon_2)d\xi(\varepsilon_3) \\
&\leq \int_{\mathbb{R}^3} |\varepsilon_1+\varepsilon_2+\varepsilon_3|^{k+2} d\xi(\varepsilon_1)d\xi(\varepsilon_2)d\xi(\varepsilon_3) \\
&\leq \int_{\mathbb{R}^3} C_k \sum_{\substack{1\leq i_1,i_2,i_3\leq k+2 \\ i_1+i_2+i_3=k+2}} |\varepsilon_1|^{i_1}|\varepsilon_2|^{i_2}|\varepsilon_3|^{i_3} d\xi(\varepsilon_1)d\xi(\varepsilon_2)d\xi(\varepsilon_3) \\
&= C_k \sum_{\substack{1\leq i_1,i_2,i_3\leq k+2 \\ i_1+i_2+i_3=k+2}} \int_{\mathbb{R}} |\varepsilon_1|^{i_1} d\xi(\varepsilon_1) \int_{\mathbb{R}} |\varepsilon_2|^{i_2} d\xi(\varepsilon_2) \int_{\mathbb{R}} |\varepsilon_3|^{i_3} d\xi(\varepsilon_3) < \infty,
\end{aligned}$$

since for $l \leq k+2$, $\int_{\mathbb{R}} |\varepsilon|^l d\xi(\varepsilon) < \infty$. Thus μ has finite moments of order less than or equal to $k+2$. Finally, let $\nu := F^{\mathrm{SC}}(\mu) = F^{\mathrm{DMFT}}(\nu^0)$. Equations (III.4.6) and (III.4.7) read

$$\Delta(z) = W_p^\dagger W_p \int_{\mathbb{R}} \frac{d\nu(\varepsilon)}{z-\varepsilon} = W \left(z - H_\perp^0 - \Sigma(z)\right)^{-1} W^\dagger,$$

with $\Sigma(z) = U^2 \int_{\mathbb{R}} \frac{d\mu(\varepsilon)}{z-\varepsilon}$. By Theorem III.4.1, we can expand $\int_{\mathbb{R}} \frac{d\mu(\varepsilon)}{iy-\varepsilon}$ as y goes to $+\infty$ and get

$$\begin{aligned}
\Delta(iy) &= W \left(iy - H_\perp^0 - U^2 \left(\frac{m_0(\mu)}{iy} + \cdots + \frac{m_{k+2}(\mu)}{(iy)^{k+3}} + o\left(\frac{1}{y^{k+3}}\right)\right)\right)^{-1} W^\dagger \\
&= W_p^\dagger W_p \left(\frac{1}{iy} + \cdots + \frac{m_{k+4}(\nu)}{(iy)^{k+5}} \right) + o\left(\frac{1}{y^{k+5}}\right).
\end{aligned}$$

This means that

$$\int_{\mathbb{R}} \frac{d\nu(\varepsilon)}{iy-\varepsilon} = \frac{1}{iy} + \cdots + \frac{m_{k+4}(\nu)}{(iy)^{k+5}} + o\left(\frac{1}{y^{k+5}}\right),$$

which proves, by Theorem III.4.1, that ν has finite moments up to order $k+4$.

Acknowledgements

The authors thank Michel Ferrero, David Gontier and Mathias Dus for useful discussions.

In the appendices, we present two topics in a pedagogical manner to ease the reader's understanding of the article.

The first section is devoted to a mathematical derivation of the DMFT equations, which departs from usual presentations in two main respects:

- First, we introduce a mathematically rigorous Green's functions formalism. To do so, we stick to the setting in which the one-particle Hilbert space is *finite dimensional*. This choice is motivated by the fact that the statements of the main results and the corresponding proofs are greatly simplified, and the reader is referred to [CGS16] and [Kir24] for an introduction to Green's function in the infinite dimensional setting.

Note also that, instead to considering the (real-frequency) Fourier transform of the Green's function, as usually done [MRC16; Gon15], we reinvest the notion of Generalized Fourier Transform, already introduced by Titchmarsh [Tit48], which is defined for *any* complex frequency. This enables us to establish a fruitful connection between Green's functions and (matrix-valued) Pick functions, which forms the basis of our analysis. Adapting our presentation to the infinite dimensional setting would require, in the light of this connection, handling with operator-valued Pick functions [Ges+01]. Such an approach, if possible, would involve technical complications that are unnecessary for a pedagogical introduction to DMFT.

A last justification for the proposed finite-dimensional one-particle Hilbert space approach is that the Hubbard and Anderson Impurity Models as introduced in this article are indeed defined within the frame of finite-dimensional Hilbert spaces. It is only when the DMFT equations are derived that we extend the formalism to infinite systems, as detailed in Section III.3.4.

- Second, we introduce the Hubbard and Anderson Impurity models using graph theory formalism. This way of defining the models is not new (see e.g. [BLS94] for the fermionic and [AFP24] for the bosonic setting), and allows us to consider various applications. For instance, the notion of "translation invariance" is captured by vertex-transitive graphs, and avoids any spatial embedding. Moreover, we believe that the DMFT formalism is very concisely defined when using notions such as induced subgraphs.

The second section consists of a basic introduction to an Analytic Continuation Problem (ACP) formulated in the set of (matrix-valued) Pick functions. Such a problem appears in the DMFT equations considered in this work because of the deep connection between the formalism of Green's functions as introduced in Section III.A.1 and the Matsubara formalism introduced in Section III.A.5. The most important result in this section is Theorem III.B.8: it allows to prove the unambiguous connection between these two formalisms (see Theorem III.A.17) and more importantly that the IPT solver is well-defined in the framework of a finite-dimensional AIM (leading to Proposition III.3.2).

Finally, we hope that the presentation of this section will convince the reader that, by solving analytic continuation problem for finitely supported measure and then extending the IPT solver by continuity, we circumvent the theoretical and computational technicality related to ACPs.

III.A DMFT of the Hubbard model

We provide in this section a mathematical description of the models and quantities of interest involved in Dynamical Mean-Field Theory (DMFT) for the Hubbard model. We

first recall the definitions of one-body Green's functions and the self-energy. We then introduce the Hubbard model and the Anderson Impurity Model (AIM). Next, we derive the DMFT equations and finally present the Iterated Perturbative Theory (IPT) solver, which is the approximate impurity solver considered in this work, together with a derivation of the Matsubara formalism of Green's functions. We then detail the spin independence of the equations within the paramagnetic framework, and the following simplification of the equations. Finally, in section III.A.7, we give the proofs of the results stated in this appendix.

The following result, which can also be found in [Lin19], underlines the relation between Pick functions and Green's function methods. It is central in our mathematical analysis of the IPT-DMFT equations, and justifies the framework of Section III.2. The mathematical objects involved, namely the Green's function, the hybridization function and the self-energy are introduced later in this section. Recall that a Pick matrix M is a matrix-valued analytic function defined on the upper-half plane, such that for all $z \in \mathbb{C}_+$, the matrix $(M(z) - M(z)^\dagger)/2i$ is positive semi-definite.

Proposition III.A.1 ($-G, -\Delta, -\Sigma$ are Pick)**.**

1. *Let $\widehat{H} \in \mathcal{S}(\mathcal{F})$ be a Hamiltonian and $G : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H}) \simeq \mathbb{C}^{n\times n}$ the one-body Green's function of $\widehat{H}$ in a steady state. Then $-G$ is a Pick matrix.*

2. *Let $\widehat{H} \in \mathcal{S}(\mathcal{F})$ be a Hamiltonian, with non-interacting component $d\Gamma(H^0) \in \mathcal{S}(\mathcal{F})$, G^0 the one-body Green's function of $d\Gamma(H^0)$, G the one-body Green's function of $\widehat{H}$ in a steady state of $\widehat{H}$, and $\Sigma : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H}) \simeq \mathbb{C}^{n\times n}$ the self-energy defined by*

$$\Sigma(z) := G^0(z)^{-1} - G(z)^{-1}.$$

   Then $-\Sigma$ is a Pick matrix.

3. *Let $\Delta : \mathbb{C}_+ \to \mathbb{C}^{n\times n}$ be the hybridization function of some Anderson Impurity Model (AIM) with impurity one-particle state space $\mathbb{C}^n$. Then $-\Delta$ is a Pick matrix.*

III.A.1 One-body Green's functions and the self-energy

One-body Green's functions are key objects in DMFT. To avoid technicalities, we will define Green's functions in a finite-dimensional setting and assume that the one-body state space is a finite-dimensional Hilbert space $(\mathcal{H}, \langle\cdot,\cdot\rangle)$, $\dim(\mathcal{H}) = 2L \in \mathbb{N}^*$. The associated Fock space

$$\mathcal{F} = \bigoplus_{n=0}^{2L} \bigwedge^n \mathcal{H},$$

where the n-particle sector $\bigwedge^n \mathcal{H}$ is the anti-symmetrized tensor product of n copies of $\mathcal{H}$, is then of dimension 2^{2L}. Given a one-particle state $\phi \in \mathcal{H}$, we denote by $\widehat{a}_\phi$ (resp.

$\widehat{a}_\phi^\dagger$) the usual annihilation (resp. creation) operator defined on $\mathcal{F}$ (see e.g. [BR97]), which satisfy the Canonical Anti-commutation Relations (CAR):

$$\forall \phi, \phi' \in \mathcal{H}, \quad \{\widehat{a}_\phi, \widehat{a}_{\phi'}\} = \{\widehat{a}_\phi^\dagger, \widehat{a}_\phi^\dagger\} = 0, \quad \{\widehat{a}_\phi, \widehat{a}_{\phi'}^\dagger\} = \langle \phi, \phi' \rangle$$

where $\{\widehat{O}, \widehat{O}'\} = \widehat{O}\widehat{O}' + \widehat{O}'\widehat{O}$ is the anti-commutator of the two operators $\widehat{O}, \widehat{O}' \in \mathcal{L}(\mathcal{F})$.

Steady states. A *state* Ω is a linear form on the set of operators $\mathcal{L}(\mathcal{F})$, which is positive ($\Omega(\widehat{O}^\dagger \widehat{O}) \geq 0$) and normalized (i.e. $\sup\{|\Omega(\widehat{O})|, \|\widehat{O}\| = 1\} = 1$). In the finite-dimensional case, any state Ω can be represented by a unique self-adjoint operator $\widehat{\rho} \in \mathcal{S}(\mathcal{F})$ such that for all $\widehat{O} \in \mathcal{L}(\mathcal{F})$, $\Omega(\widehat{O}) = \mathrm{Tr}(\widehat{\rho}\widehat{O})$. The operator $\widehat{\rho}$ is positive and satisfies $\mathrm{Tr}(\widehat{\rho}) = 1$. It is called the *density operator* associated with the state Ω. For an isolated quantum system described by a time-independent Hamiltonian $\widehat{H} \in \mathcal{S}(\mathcal{F})$, a *steady state* corresponds to a stationary solution to the quantum Liouville equation

$$i\frac{d\widehat{\rho}}{dt}(t) = [\widehat{H}, \widehat{\rho}(t)],$$

where $[\widehat{O}, \widehat{O}'] = \widehat{O}\widehat{O}' - \widehat{O}'\widehat{O}$ is the commutator of $\widehat{O}, \widehat{O}' \in \mathcal{L}(\mathcal{F})$. It follows that a state is a steady state if and only if its density $\widehat{\rho}$ commutes with the Hamiltonian $\widehat{H}$, namely $[\widehat{H}, \widehat{\rho}] = 0$. Important examples of steady states are thermal and osmotic equilibrium states known as Gibbs states, as well as ground and excited states of $\widehat{H}$ with a prescribed number of particles (for particle-number conserving Hamiltonians).

One-body Green's functions. The *one-body time-ordered Green's functions* are then defined as follows:

Definition III.A.2 (One-body time-ordered Green's function)**.** *Given a Hamiltonian $\widehat{H} \in \mathcal{S}(\mathcal{F})$ and an associated steady state Ω, one defines the $\mathcal{L}(\mathcal{H})$-valued function $\tilde{G} : \mathbb{R} \to \mathcal{L}(\mathcal{H})$, known as* one-body time-ordered Green's function*, by requiring that $i\tilde{G}(t)$ is the operator represented by the sesquilinear form defined, for all $\phi, \phi' \in \mathcal{H}$, by*

$$\langle \phi, (i\tilde{G}(t))\phi' \rangle = \chi_{\mathbb{R}_+}(t)\, \Omega(\mathbb{H}(\widehat{a}_\phi)(t)\widehat{a}_{\phi'}^\dagger) - \chi_{\mathbb{R}_-^*}(t)\, \Omega(\widehat{a}_{\phi'}^\dagger \mathbb{H}(\widehat{a}_\phi)(t)), \tag{III.A.1}$$

where for all $\widehat{O} \in \mathcal{L}(\mathcal{F})$, $\mathbb{H}(\widehat{O}) : \mathbb{R} \ni t \mapsto e^{it\widehat{H}}\widehat{O}e^{-it\widehat{H}}$ denotes the Heisenberg picture *of $\widehat{O}$ and χ_A is the characteristic function of the set A.*

Let us comment on the terminology. First, the term "body" encompasses cases of a "particle" or a "hole": the first term of the r.h.s. of (III.A.1) can be interpreted as describing the propagation from $t_0 = 0$ to $t > 0$ of a particle added to the system at $t_0 = 0$, while the second term can be interpreted as the propagation from $t < 0$ to $t_0 = 0$ of a hole created at $t < 0$. Second, it is "time-ordered": the r.h.s. of (III.A.1) can be rewritten as

$$\Omega\left(\mathcal{T}\left(\mathbb{H}(\widehat{a}_\phi), \mathbb{H}(\widehat{a}_{\phi'}^\dagger)\right)(t, 0)\right)$$

where for all operators-valued functions $\mathbb{R} \ni t \mapsto \widehat{O}(t) \in \mathcal{L}(\mathcal{F}), \mathbb{R} \ni t \mapsto \widehat{O}'(t) \in \mathcal{L}(\mathcal{F})$, the fermionic *time-ordered product* $\mathcal{T}(\widehat{O}, \widehat{O}')$ is the operator-valued function $\mathbb{R}^2 \to \mathcal{L}(\mathcal{F})$ defined as

$$\mathcal{T}(\widehat{O}, \widehat{O}')(t, t') = \begin{cases} \widehat{O}(t)\widehat{O}'(t') \text{ if t} \geq \text{t'} \\ -\widehat{O}'(t')\widehat{O}(t) \text{ otherwise,} \end{cases}$$

where the minus sign is specific to the fermionic case. Up to a sign, it is the product of the operators applied in the order of increasing time.

The i prefactor in the l.h.s. of (III.A.1) is a convention that facilitates the expression of the results to come, especially Proposition III.A.4.

Finally, note that $\tilde{G}$ is real-analytic on $(-\infty, 0) \cup (0, +\infty)$ with a $-i\mathbf{1}$ jump at $t = 0$ due to the CAR.

As $\mathcal{F}$ is finite-dimensional, the Green's function can be expanded in a joint orthonormal eigenbasis $\mathcal{B}$ of $\widehat{\rho}$ and $\widehat{H}$, leading to the Källén-Lehmann (KL) representation [Käl52; Leh54]

$$\forall \phi, \phi' \in \mathcal{H}, \quad \langle \phi, i\tilde{G}(t)\phi' \rangle = \sum_{\psi, \psi' \in \mathcal{B}} e^{it(E_\psi - E_{\psi'})} \langle \psi, \widehat{a}_\phi \psi' \rangle \langle \psi', \widehat{a}^\dagger_{\phi'} \psi \rangle \left(\rho_\psi \chi_{\mathbb{R}_+}(t) - \rho_{\psi'} \chi_{\mathbb{R}_-^*}(t) \right), \tag{III.A.2}$$

where $\forall \psi \in \mathcal{B}, \widehat{H}\psi = E_\psi \psi$ (with $E_\psi \in \mathbb{R}$) and $\widehat{\rho}\psi = \rho_\psi \psi$ (with $\rho_\psi \in \mathbb{R}_+$, $\sum_{\psi \in \mathcal{B}} \rho_\psi = 1$).

Other types of Green's functions are encountered in the physics literature, notably retarded and advanced Green's functions. These objects encode the same information on the spectral properties of $\widehat{H}$ as the time-ordered Green's function, but this information is stored in a different way. A suitable way to highlight this information is to consider specific holomorphic extensions to the complex plane of the time-Fourier transform of these Green's functions [MRC16; CGS16]. In the case of the time-ordered one-body Green's function, the suitable holomorphic extension is provided by the generalized Fourier transform introduced by Titchmarsh [Tit48].

Definition III.A.3 (Generalized Fourier Transform (GFT))**.** *The Generalized Fourier Transform (GFT) of the one-body time-ordered Green's function $\tilde{G}$ is the analytic function on the upper half-plane $G : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H})$, also called (one-body) Green's function, defined by*

$$\forall z \in \mathbb{C}_+, G(z) = G_+(z) + G_-(\overline{z})^\dagger \tag{III.A.3}$$

with

$$\forall z \in \mathbb{C}_+ := \{z \in \mathbb{C} \mid \Im(z) > 0\}, \quad G_+(z) = \int_{\mathbb{R}_+} e^{izt} \tilde{G}(t) dt, \tag{III.A.4}$$

$$\forall z \in \mathbb{C}_- := \{z \in \mathbb{C} \mid \Im(z) < 0\}, \quad G_-(z) = \int_{\mathbb{R}_-} e^{izt} \tilde{G}(t) dt. \tag{III.A.5}$$

Note that the Green's function G can be extended to $\mathbb{C} \setminus \mathbb{R}$ by reflection, namely by setting

$$\forall z \in \mathbb{C}_-, \quad G(z) = G(\overline{z})^\dagger. \tag{III.A.6}$$

By construction, $G(z)$ is analytic on $\mathbb{C}_+$. In addition, it follows from the KL representation (III.A.2) that

$$\forall z \in \mathbb{C}_+, \quad \forall \phi, \phi' \in \mathcal{H}, \quad \langle \phi, G(z)\phi' \rangle = \sum_{\psi,\psi' \in \mathcal{B}} \frac{\rho_\psi + \rho_{\psi'}}{z + (E_\psi - E_{\psi'})} \langle \psi, \widehat{a}_\phi \psi' \rangle \langle \psi', \widehat{a}^\dagger_{\phi'} \psi \rangle. \tag{III.A.7}$$

An important observation is that

$$\forall z \in \mathbb{C}_+, \quad \forall \phi \in \mathcal{H} \backslash \{0\}, \quad \Im\left(\langle \phi, G(z)\phi \rangle\right) = -\Im(z) \sum_{\psi,\psi' \in \mathcal{B}} \frac{\rho_{\psi'} + \rho_\psi}{|z + (E_\psi - E_{\psi'})|^2} |\langle \psi, \widehat{a}_\phi \psi' \rangle|^2 < 0, \tag{III.A.8}$$

which shows in particular that $G(z)$ is invertible for all $z \in \mathbb{C}_+$.

Up to now, we have not specified the Hamiltonian $\widehat{H}$; in the sequel, we will assume that it is of the form

$$\widehat{H} = d\Gamma(H^0) + \widehat{H}^I, \quad H^0 \in \mathcal{S}(\mathcal{H}), \quad \widehat{H}^I \in \mathcal{S}(\mathcal{F}) \tag{III.A.9}$$

where $d\Gamma(H^0)$ is the second quantization of the one-particle Hamiltonian $H^0 \in \mathcal{S}(\mathcal{H})$ (see e.g. [BR97]) and $\widehat{H}^I \in \mathcal{S}(\mathcal{F})$ some interaction Hamiltonian. We say that $\widehat{H}$ is *non-interacting* if $\widehat{H}^I = 0$.

Depending on the formalism of interest, one can consider the grand canonical Hamiltonian $\widehat{H}' = \widehat{H} - \varepsilon_F \hat{N}$ without loss of generality.

The following is an essential property of the Green's function, to which it owes its name: the Green's function of a non-interacting system in a steady state is the resolvent of the one-particle Hamiltonian.

Proposition III.A.4 (Non-interacting Green's function)**.** *Let $H^0 \in \mathcal{S}(\mathcal{H})$ be a one-particle Hamiltonian and Ω a steady state of the non-interacting Hamiltonian $d\Gamma(H^0)$. The Green's function G^0 of $d\Gamma(H^0)$ associated with Ω is the resolvent of H^0:*

$$\forall z \in \mathbb{C}_+, \quad G^0(z) = (z - H^0)^{-1}. \tag{III.A.10}$$

In particular, G^0 is independent of Ω.

The proof of this classical result is recalled in Section III.A.7. It is a consequence of the fact that the one-body Green's function $\tilde{G}^0$ of the non-interacting Hamiltonian $d\Gamma(H^0)$ is solution to the equation

$$\left(i\frac{d}{dt} - H^0\right) \tilde{G} = \delta \mathbf{1} \quad \text{in } \mathcal{D}'(\mathbb{R}; \mathcal{L}(\mathcal{H})) \tag{III.A.11}$$

so that the time-ordered Green's function $\tilde{G}^0$ is indeed a Green's function of the linear differential operator $i\frac{d}{dt} - H^0$. The various avatars of the Green's function (retarded/advanced) also satisfy this equation in $\mathcal{D}'(\mathbb{R}^*; \mathcal{L}(\mathcal{H}))$, but with different asymptotic conditions at infinity and jumps at $t = 0$. The properties (III.A.10)-(III.A.11) hold *only* for non-interacting Hamiltonians.

Self-energy. For interacting systems, the general relation between the Hamiltonian and the Green's function is more involved: the discrepancy with the non-interacting case is characterized by the *self-energy*.

Definition III.A.5 (Self-energy)**.** *The* self-energy $\Sigma : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H})$ *of an Hamiltonian* $\widehat{H}$ *of the form* (III.A.9) *associated with a steady state* Ω *of* $\widehat{H}$ *is defined by*

$$\forall z \in \mathbb{C}_+, \quad \Sigma(z) = G^0(z)^{-1} - G(z)^{-1}, \tag{III.A.12}$$

where the non-interacting Green's function G^0 *is the Green's function of any steady state of* $d\Gamma(H^0)$.

Recall that $G^0(z)^{-1} = z - H^0$ (III.A.10) and that $G(z)$ is invertible in view of (III.A.8). Let us emphasize that the definition of the self-energy only depends on $\widehat{H}$, H^0, and the considered steady state Ω of $\widehat{H}$, since G^0 is independent of the steady state associated with $d\Gamma(H^0)$.

Using again (III.A.10) and the definition of the self-energy, the Green's function G can be rewritten as

$$\forall z \in \mathbb{C}_+, \quad G(z) = \big(z - (H^0 + \Sigma(z))\big)^{-1}$$

so that, for a given complex frequency z, $H^0 + \Sigma(z)$ can be considered as an *effective one-particle Hamiltonian* (compare with (III.A.10)): the self-energy is thus the extra term to be added to H^0 to obtain a representation of the interacting system of particles in terms of a system of non-interacting ones. The frequency dependence of Σ comes from the fact that particles do interact in the original system.

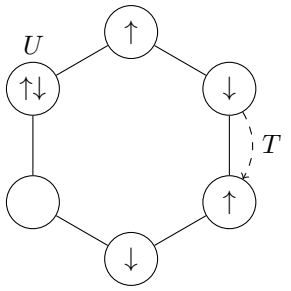
Remark III.A.6. *In condensed matter physics, the Nevanlinna-Riesz measure of* $-G$ *is the so-called spectral function [MRC16].*

III.A.2 Hubbard model

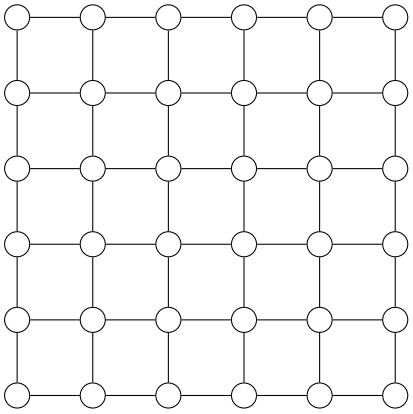
Originally introduced in quantum chemistry [PP53; Pop53], the Hubbard model [Kan63; HF63; Gut63] is an idealized model that minimally describes *interacting electrons* in a molecule or a crystalline material. From the mathematical physics perspective, it is a prototypical example of short-range fermionic lattice system, and its mathematical study has been pioneered soon after its introduction in [LW68; LW03], triggering an extensive study of its ground-states properties [Lie04]. More recently, the discovery of cuprate-based high-temperature superconductors [BM86; Yu+19], exhibiting a *layered structure*, shed new light on the *square lattice* Hubbard model, for which much remains to be discovered [Lie04]. Since then, *approximation methods* have been derived for this model such as *generalized Hartree-Fock* [BLS94], and Dynamical Mean-Field Theory [Geo+96].

In this article, we restrict ourselves to *finite-dimensional* Hubbard models, i.e. to Hubbard models defined on *finite graphs*. The reason for this is threefold. First, we stick to the case of finite-dimensional state spaces, for which all the objects involved in the

mathematical formulation of DMFT are well-defined. Second, *graph theory* provides a suitable formalism to describe on the same footing different physical settings, ranging from molecular systems to truncated lattices, and with arbitrary hopping parameters (nearest neighbours, next nearest neighbors,...). Third, this is precisely the language in which DMFT can be formulated concisely, as described in Section III.A.4.



(a) The Pariser-Parr-Pople model of benzene C_6H_6: the cyclic graph C_6.



(b) A truncated lattice $\mathbb{Z}^2/6\mathbb{Z}$ with nearest neighbors : the square grid graph $P_6^{\square 2}$.

Figure III.1: The Hubbard model for different graphs

Consider now a finite undirected graph $\mathcal{G}_H = (\Lambda, E)$ that describes the "sites" Λ hosting electrons, as depicted in Figure III.1. The state space of a site is the vector space spanned by the orthonormal vectors $|\emptyset\rangle$ (empty site), $|\uparrow\rangle$ (site occupied by one spin-up electron), $|\downarrow\rangle$ (site occupied by one spin-down electron), and $|\uparrow\downarrow\rangle$ (doubly-occupied site). Note that $|\uparrow\downarrow\rangle$ is a shorthand notation for the two-electron singlet state $2^{-1/2}(|\uparrow\rangle\otimes|\downarrow\rangle - |\downarrow\rangle\otimes|\uparrow\rangle)$. In chemistry language, to each site is associated a single orbital (and thus two spin-orbitals); in physics, this setting is referred to as the *one-band* Hubbard model. Since the sites are distinguishable, the state space of the full system is the tensor product of the state space of each site. It is therefore of dimension $4^{|\Lambda|} = 2^{2L}$ where $L = |\Lambda|$ is the number of sites.

Let us now specify the Hamiltonian. In the Hubbard model, electrons can jump from one site to any other neighboring site. This models the tunnel effect, whose intensity is described by the *hopping matrix* T, as for tight-binding Hamiltonians. Electrons repel each other due to (screened) Coulomb interaction. The simplicity of the Hubbard model lies in the *range* of this interaction, which is the shortest possible: it is only effective for two electrons occupying the same site, and if the site i is doubly occupied, the energy of the system is increased by an *on-site repulsion* U_i.

More precisely, a finite-dimensional Hubbard model can be defined as follow.

Definition III.A.7 (Hubbard model of a finite graph)**.** *Given a finite graph $\mathcal{G}_H = (\Lambda, E)$, a hopping matrix $T : E \to \mathbb{R}$, and an on-site repulsion $U : \Lambda \to \mathbb{R}$, the Hubbard Fock*

Space $\mathcal{F}_H$ is defined as

$$\mathcal{F}_H = \bigotimes_{i\in\Lambda} \mathcal{F}_1, \quad \mathcal{F}_1 = \mathrm{Span}(|\emptyset\rangle, |\uparrow\rangle, |\downarrow\rangle, |\uparrow\downarrow\rangle)$$

and the Hubbard Hamiltonian $\hat{H}_H \in \mathcal{S}(\mathcal{F}_H)$ as

$$\hat{H}_H = \sum_{\{i,j\}\in E, \sigma\in\{\uparrow,\downarrow\}} T_{i,j}\left(\widehat{a}^\dagger_{i,\sigma}\widehat{a}_{j,\sigma} + \widehat{a}^\dagger_{j,\sigma}\widehat{a}_{i,\sigma}\right) + \sum_{i\in\Lambda} U_i \hat{n}_{i,\uparrow}\hat{n}_{i,\downarrow}$$

where the usual annihilation and creation operators $\widehat{a}_{i,\sigma}$ and $\widehat{a}^\dagger_{i,\sigma}$ of an electron on site j with spin σ satisfy the CAR

$$\forall i,j\in\Lambda, \quad \sigma\in\{\uparrow,\downarrow\}, \quad \{\widehat{a}_{i,\sigma}, \widehat{a}_{j,\sigma'}\} = \{\widehat{a}^\dagger_{i,\sigma}, \widehat{a}^\dagger_{j,\sigma'}\} = 0, \quad \{\widehat{a}_{i,\sigma}, \widehat{a}^\dagger_{j,\sigma'}\} = \delta_{i,j}\delta_{\sigma,\sigma'}, \tag{III.A.13}$$

and the site number operators $\widehat{n}_{i,\sigma}$ are defined by $\widehat{n}_{i,\sigma} = \widehat{a}^\dagger_{i,\sigma}\widehat{a}_{i,\sigma}$.

In this paper, we do not consider external magnetic field, so that we can assume without loss of generality that the hopping matrix T is real-valued [Lie04]. For standard Coulomb interaction, the on-site repulsion U is positive, but we do not need to make this assumption here. Let us finally mention that this model falls within the class of quantum lattice systems (see [BR97],[Bd+23],[You23] for a discussion about the mathematical and physical properties of these systems).

Remark III.A.8. *The Hubbard Hamiltonian $\hat{H}_H$ is particle-number and spin conserving, i.e. it commutes with the number operator $\hat{N}_H = \sum_{i\in\Lambda}(\hat{n}_{i,\uparrow} + \hat{n}_{i,\downarrow})$ and the spin operators. In particular, it commutes with the z-component $\hat{S}^z_H = \frac{1}{2}\sum_{i\in\Lambda}(\hat{n}_{i,\uparrow} - \hat{n}_{i,\downarrow})$ of the spin operator. This property will be used later, in Appendix III.A.6, to make the IPT-DMFT model spin-independent.*

III.A.3 Anderson Impurity Model (AIM)

Impurity models Impurity models are models in which an "impurity" is coupled to a "bath" in such a way that particles in the bath do not interact, and the coupling Hamiltonian between the impurity and the bath only involves one-body terms. Otherwise stated, the one-particle state space and the Fock space of the total system can be decomposed as

$$\mathcal{H}[\mathrm{IM}] = \mathcal{H}_{\mathrm{imp}} \oplus \mathcal{H}_{\mathrm{bath}} \quad \text{and} \quad \mathcal{F}_{\mathrm{IM}} = \mathcal{F}_{\mathrm{imp}} \otimes \mathcal{F}_{\mathrm{bath}}, \tag{III.A.14}$$

and its Hamiltonian as

$$\widehat{H}_{\mathrm{IM}} = \widehat{H}_{\mathrm{imp}} \otimes 1_{\mathrm{bath}} + 1_{\mathrm{imp}} \otimes \widehat{H}_{\mathrm{bath}} + \widehat{H}_{\mathrm{coupling}}, \tag{III.A.15}$$

with

$$\widehat{H}_{\mathrm{imp}} = d\Gamma(H^0_{\mathrm{imp}}) + \widehat{H}^1_{\mathrm{imp}}, \quad \widehat{H}_{\mathrm{bath}} = d\Gamma(H^0_{\mathrm{bath}}), \quad \widehat{H}_{\mathrm{coupling}} = d\Gamma(H^0_{\mathrm{coupling}}), \tag{III.A.16}$$

and H^0_{coupling} can be decomposed according to (III.A.14) as

$$H^0_{\text{coupling}} = \begin{pmatrix} 0 & V \\ V^\dagger & 0 \end{pmatrix}.$$

Reshuffling the terms, we also have

$$\widehat{H}_{\text{IM}} = d\Gamma(H^0_{\text{IM}}) + \widehat{H}^{\text{I}}_{\text{imp}} \otimes 1_{\text{bath}}, \quad \text{with} \quad H^0_{\text{IM}} = \begin{pmatrix} H^0_{\text{imp}} & V \\ V^\dagger & H^0_{\text{bath}} \end{pmatrix}. \tag{III.A.17}$$

As will be seen below, a key step of the DMFT iteration loop is to compute the restriction of the Green's function G_{IM} of an impurity model to the impurity space $\mathcal{H}[\text{imp}]$. This is in general a difficult task. On the other hand, computing the restriction of the *non-interacting* Green's function can be done straightforwardly using a Schur complement technique. This leads to the concept of hybridization function, which, as announced in the introduction, is of paramount importance in the mathematical formulation of DMFT.

Definition III.A.9 (Hybridization function Δ of an impurity model)**.** *Given an impurity model of the form* (III.A.14)-(III.A.17)*, its* hybridization function $\Delta : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H}[\text{imp}])$ *is defined by*

$$\forall z \in \mathbb{C}_+, \quad \Delta(z) = V \left(z - H^0_{\text{bath}}\right)^{-1} V^\dagger. \tag{III.A.18}$$

Using this definition and Proposition III.A.4, the non-interacting Green's function of the impurity model is then given in the decomposition (III.A.14) by

$$G^0(z) = (z - H^0_{\text{IM}})^{-1} = \begin{pmatrix} (z - H^0_{\text{imp}} - \Delta(z))^{-1} & * \\ * & * \end{pmatrix}.$$

The hybridization function Δ thus plays a role analogous to the self-energy Σ. The equation

$$\left(G^0(z)\right)_{\text{imp}} = \left(z - (H^0_{\text{imp}} + \Delta(z))\right)^{-1}$$

means that $H^0_{\text{imp}} + \Delta(z)$ can be considered as an *effective one-particle impurity Hamiltonian*: the hybridization function is the extra term to be added to H^0_{imp} so that, in the non-interacting case, a particle in the whole system can be regarded as a particle localized on the impurity. In the time domain, the hybridization function describes the phenomenon of a particle localized on the impurity, jumping into the bath, and jumping back to the impurity, hence the name *hybridization*.

The most important result in this section is the following.

Theorem III.A.10 (Σ sparsity pattern of an impurity model)**.** *Given an impurity model of the form* (III.A.14)-(III.A.17)*, the self-energy* $\Sigma_{\text{IM}} : \mathbb{C}_+ \to \mathcal{H}[\text{IM}]$ *associated with any steady state of* $\widehat{H}_{\text{IM}}$ *reads in the decomposition* (III.A.14)

$$\forall z \in \mathbb{C}_+, \quad \Sigma_{\text{IM}}(z) = \begin{pmatrix} \Sigma_{\text{imp}}(z) & 0 \\ 0 & 0 \end{pmatrix}. \tag{III.A.19}$$

In practical DMFT computations and as mentioned already in [FV63], the self-energy Σ of an impurity problem depends solely on the hybridization function Δ and on the impurity Hamiltonian defined by H^0_{imp} and $\widehat{H}^I$, via a quantum action defined by path integrals [Gul+11, eq. 9]. In this article, we focus on the IPT approximation (see section III.A.5), which makes this dependence almost explicit, and postpone the study of the exact impurity solver in a more general framework to future works.

Anderson Impurity Model The original Anderson impurity model (AIM) [And61] is a specific impurity model in which the impurity consists of a single-site Hubbard model. It was introduced by Anderson back in 1961 to explain the low-temperature behavior of the conductivity of metallic alloys with dilute magnetic impurities, later called the *Kondo effect* [Kon64; Sar68]. The AIM was then extended to multiple-site Hubbard impurities. Figure III.2 provides a graphical illustration of an AIM model with a 4-site Hubbard impurity and 5 bath orbitals ($\dim(\mathcal{H}_{\text{imp}}) = 8$, $\dim(\mathcal{H}_{\text{bath}}) = 10$). Without loss of generality, we can indeed identify the bath space $\mathcal{H}_{\text{bath}}$ with $\mathbb{C}^{2B}$ and assume that $H^0_{\text{bath}} = \text{diag}(\epsilon_1, \epsilon_1, \cdots, \epsilon_B, \epsilon_B)$. In this picture, the canonical basis of $\mathbb{C}^{2B}$ corresponds to an orthonormal basis of bath spin-orbitals with energies $\epsilon_1, \epsilon_1, \ldots, \epsilon_B, \epsilon_B$. The matrix V models jumps from the bath to the impurity, while the matrix $V^\dagger$ models jumps from the impurity to the bath. The coupling terms $V_{i,j}$ are also assumed to be real-valued due to the absence of magnetic field.

In the seminal paper [And61], the *electronic bath* was thought as the set of *conducting electrons*, for instance described by a tight-binding model on a (truncated) lattice.

Remark III.A.11. *As for the Hubbard model, the z-component of the total spin operator $\hat{S}^z_{AI} = \hat{S}^z_H + \sum_{k=1}^B \hat{n}_{k,\uparrow} - \hat{n}_{k,\downarrow}$, the total number operator $\hat{N}_{AI} = \hat{N}_H + \sum_{k=1}^B \hat{n}_{k,\uparrow} + \hat{n}_{k,\downarrow}$ and the Anderson Impurity Hamiltonian $\widehat{H}_{\text{AIM}}$ pairwise commute. This property will also be used to make the IPT-DMFT model spin-independent.*

III.A.4 Dynamical Mean-Field Theory (DMFT)

Consider a Hubbard model defined by $(\mathcal{G}_H, T, U)$ in one of its steady states Ω. The purpose of DMFT is to provide an *approximation* of the corresponding Green's function G, and more precisely of some blocks of this Green's function.

To do so, DMFT uses a self-consistent mapping between the Hubbard model $(\mathcal{G}_H, T, U)$ and a collection of Anderson Impurity Models associated with a partition $\mathfrak{P}$ of the vertices of the Hubbard graph $\mathcal{G}_H$. The process, illustrated in Figure III.3, is the following.

1. Choose a partition $\mathfrak{P} = \{\Lambda_p, p \in [\![1, P]\!]\}$ of the L sites of the Hubbard model into P impurities $\Lambda_1, \cdots \Lambda_P$, , $\sqcup_{p=1}^P \Lambda_p = \Lambda$, and consider for all $p \in [\![1, P]\!]$ the *induced subgraphs* $\mathcal{G}_p = (\Lambda_p, E_p)$ with $E_p = \{\{i, j\} \in E;\ i, j \in \Lambda_p\}$. This partition leads to

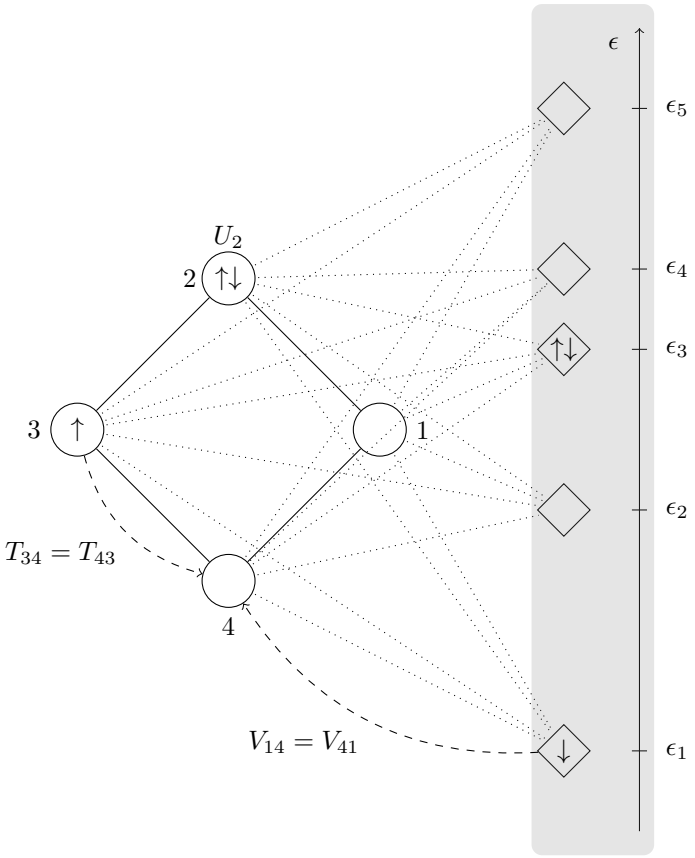


Figure III.2: The Anderson Impurity Model of an impurity ($\mathcal{G}_H = C_4, T, U$) and a *bath* ($B = 5, \epsilon, V$) in a 6 electrons state.

the canonical orthogonal decomposition

$$\mathcal{H}[H] = \bigoplus_{p=1}^{P} \mathcal{H}[p], \quad \dim(\mathcal{H}[p]) = 2|\Lambda_p| = 2L_p, \tag{III.A.20}$$

from which follows the decomposition of the Fock space

$$\mathcal{F}_H = \bigotimes_{p=1}^{P} \mathcal{F}_p, \quad \dim \mathcal{F}_p = 4^{L_p}. \tag{III.A.21}$$

The decomposition (III.A.20) of the one-particle state space of the original Hubbard model gives rise to block-operator representations of the exact Green's function and self-energy (for the state Ω) of the original Hubbard model

$$G(z) = \begin{pmatrix} G_1(z) & * & \cdots & * \\ * & G_2(z) & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \cdots & G_P(z) \end{pmatrix}, \qquad \Sigma(z) = \begin{pmatrix} \Sigma_1(z) & * & \cdots & * \\ * & \Sigma_2(z) & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \cdots & \Sigma_P(z) \end{pmatrix};$$

2. The decomposition (III.A.20) also leads to a block-operator representation of the one-body Hamiltonian corresponding to the non-interacting part of the Hubbard

Hamiltonian

$$H^0_{\mathrm{H}} = \begin{pmatrix} H^0_1 & * & \cdots & * \\ * & H^0_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \cdots & H^0_P \end{pmatrix},$$

where the diagonal block H^0_p is constructed from the *induced hopping matrices* $T_p = T_{|E_p}$, $1 \leq p \leq P$. Due to the locality of the interactions in the Hubbard model, the interaction term is "block-diagonal" in the decomposition (III.A.21). It is represented by the *induced on-site repulsion* $U_p = U_{|\Lambda_p}$, and it reads $\widehat{H}^I = \oplus_{p=1}^P \widehat{H}^I_p$, with $\widehat{H}^I_p = \sum_{i\in\Lambda_p} U_i \hat{n}_{i,\uparrow}\hat{n}_{i,\downarrow}$;

3. The AIM associated with the p-th impurity is defined by (i) the impurity state space $\mathcal{H}[\mathrm{imp}, p] := \mathcal{H}[p]$, (ii) the impurity Hamiltonian defined by the induced Hubbard model $(\mathcal{G}_p, T_p, U_p)$, (iii) some bath state space $\mathcal{H}[\mathrm{bath}, p]$, bath one-body Hamiltonian $H^0_{\mathrm{bath},p}$ and coupling one-body Hamiltonian $H^0_{\mathrm{coupling},p} = \begin{pmatrix} 0 & V_p^\dagger \\ V_p & 0 \end{pmatrix}$ to be specified. From each of these P AIMs, one can compute the Green's functions and self-energies

$$G_{\mathrm{AI},p}(z) = \begin{pmatrix} G_{\mathrm{imp},p}(z) & * \\ * & * \end{pmatrix}, \qquad \Sigma_{\mathrm{AI},p}(z) = \begin{pmatrix} \Sigma_{\mathrm{imp},p}(z) & 0 \\ 0 & 0 \end{pmatrix},$$

for AIM steady states Ω_p of the same nature as Ω (see Remark III.A.12 below for a comment on this important point).

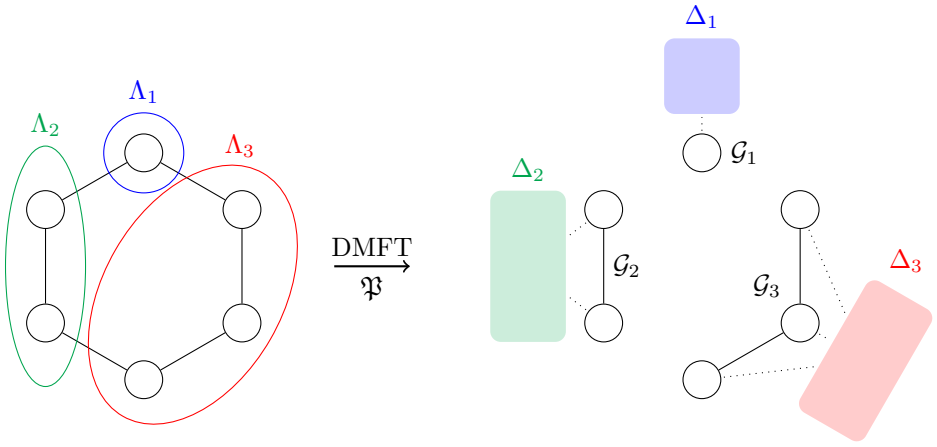


Figure III.3: A partition $\mathfrak{P} = \Lambda_1 \sqcup \Lambda_2 \sqcup \Lambda_3$ of the 6 vertices of the C_6 graph, and the 3 induced graphs $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$: the impurities of the corresponding AIMs are defined using the original Hubbard model, while the bath and the impurity-bath coupling are specified by the DMFT approximation.

DMFT aims at computing approximations of the diagonal-blocks $G_1(z), \cdots, G_P(z)$ of the exact Green's function $G(z)$. Ideally, the baths should be ajusted in such a way that

$G_{\mathrm{imp},p} = G_p$, but of course this is not possible since the functions G_p are unknown. To make DMFT a practical tool, the idea is to introduce approximate Green's functions and self-energies of the form

$$G_{\mathrm{DMFT}}(z) = \begin{pmatrix} G_{\mathrm{DMFT},1}(z) & * & \cdots & * \\ * & G_{\mathrm{DMFT},2}(z) & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \cdots & G_{\mathrm{DMFT},P}(z) \end{pmatrix}, \tag{III.A.22}$$

$$\Sigma_{\mathrm{DMFT}}(z) = \begin{pmatrix} \Sigma_{\mathrm{DMFT},1}(z) & 0 & \cdots & 0 \\ 0 & \Sigma_{\mathrm{DMFT},1}(z) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \Sigma_{\mathrm{DMFT},P}(z) \end{pmatrix}, \tag{III.A.23}$$

related by

$$\begin{aligned} G_{\mathrm{DMFT}}(z) &= ((G^0_{\mathrm{DMFT}})^{-1}(z) - \Sigma_{\mathrm{DMFT}}(z))^{-1}, \\ G^0_{\mathrm{DMFT}}(z) &= \left(z - H^0_{\mathrm{H}}\right)^{-1}, \end{aligned}$$

and to choose the baths in such a way that

$$\forall 1 \leq p \leq P, \quad G_{\mathrm{DMFT},p}(z) = G_{\mathrm{imp},p}(z), \tag{III.A.24}$$

$$\Sigma_{\mathrm{DMFT},p}(z) = \Sigma_{\mathrm{imp},p}(z). \tag{III.A.25}$$

Of course, it is not clear whether a collection of baths that satisfies the above conditions exists, and this article partially answers this question.

Note that the DMFT Green's function $G_{\mathrm{DMFT}}(z)$ is *not*, a priori, the Green's function of some interacting quantum many-body problem which could be defined as in Section III.A.1. Instead, it is defined from an approximate self-energy $\Sigma_{\mathrm{DMFT}}(z)$. Equations (III.A.23) and (III.A.25) indicate that, in the DMFT approximation, each impurity interacts with its neighborhood as if the former were an impurity and the latter a bath, in accordance with Theorem III.A.10, hence the name of *mean-field*. This mean-field is *dynamical* because it is frequency dependent, in contrast with static mean-field theory such as in Hartree-Fock or Density-Matrix Embedding Theory (DMET).

Remark III.A.12. *In the above sketch of the DMFT framework, we did not specify the states* $(\Omega_p)_{p\in[\![1,P]\!]}$ *of the associated AIMs from which the self-energies* $(\Sigma_{\mathrm{imp},p})_{p\in[\![1,P]\!]}$ *are computed. In [Geo+96], it is implicitly stated that, for a Hubbard model in the* Gibbs *state* $\Omega_{\mathrm{H},\beta,\varepsilon_F}$ *as defined below in Section III.A.5, the AIM's steady states* Ω_p *are defined to be* Gibbs *states as well* $\Omega_p = \Omega_{\mathrm{AIM},\beta,\varepsilon'_F}$*, with the same temperature as that of the* Gibbs *state of the whole system, but with* a priori *different chemical potential* ε'_F*. The latter is to be chosen to satisfy appropriate* filling *conditions. We will address this question in a future work and simply assume here that the* $\varepsilon'_F = \varepsilon_F$*. Moreover, when working with the IPT solver, the chemical potential is fixed by the on-site interaction, as detailed in Section III.A.5.*

Our analysis is based on a reformulation of conditions (III.A.24)-(III.A.25) using the hybridization functions $(\Delta_p(z))_{1\leq p\leq P}$ of the P impurity problems as the main variables. As mentionned previously, knowing Δ_p as well as T_p, U_p and Ω_p, suffices in principle to compute $\Sigma_{\mathrm{imp},p}$. In practice, this is done by using an approximate impurity solver. A particular example of such a solver will be presented in Section III.A.5. Conversely, knowing $(\Sigma_{\mathrm{imp},p})_{1\leq p\leq P}$ and T suffices to characterize the unique set $(\Delta_p(z))_{1\leq p\leq P}$ for which (III.A.24)-(III.A.25) hold true. Indeed, by denoting

$$\mathcal{H}[\overline{p}] := \bigoplus_{p\neq q=1}^{P} \mathcal{H}[q],$$

$$\Sigma_{\mathrm{DMFT},\overline{p}}(z) = \begin{pmatrix} \Sigma_{\mathrm{DMFT},1}(z) & \cdots & 0 & \cdots & 0 & \\ \vdots & \ddots & \vdots & & & \\ 0 & \cdots & \Sigma_{\mathrm{DMFT},p-1}(z) & 0 & & \\ 0 & \cdots & 0 & \Sigma_{\mathrm{DMFT},p+1}(z) & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & & \\ 0 & \cdots & 0 & 0 & \cdots & \Sigma_{\mathrm{DMFT},P}(z) \end{pmatrix},$$

and using the Schur complement formula, we have on the one hand

$$\begin{aligned} G_{\mathrm{DMFT},p}(z) &= \left(\left(z-(H^0_{\mathrm{H}}+\Sigma_{\mathrm{DMFT}}(z))\right)^{-1}\right)_p \\ &= \left(z-\left(H^0_p+\Sigma_{\mathrm{DMFT},p}(z)\right)-W_p\left(z-\left(H^0_{\overline{p}}+\Sigma_{\mathrm{DMFT},\overline{p}}(z)\right)\right)^{-1}W_p^\dagger\right)^{-1}, \end{aligned} \tag{III.A.26}$$

where

$$\begin{pmatrix} H^0_p & W_p \\ W_p^\dagger & H^0_{\overline{p}} \end{pmatrix} \text{ and } \begin{pmatrix} z-(H^0_p+\Sigma_{\mathrm{DFMT,p}}(z)) & -W_p \\ -W_p^\dagger & z-(H^0_{\overline{p}}+\Sigma_{\mathrm{DFMT},\overline{\mathrm{p}}}(z)) \end{pmatrix} \tag{III.A.27}$$

are the block-representations of H^0_{H} and $(z-(H^0_{\mathrm{H}}+\Sigma_{\mathrm{DMFT}}(z)))$, respectively, in the decomposition $\mathcal{H}=\mathcal{H}[p]\oplus\mathcal{H}[\overline{p}]$. Note that $H^0_p\in\mathcal{L}(\mathcal{H}[p])$, $W_p\in\mathcal{L}(\mathcal{H}[\overline{p}];\mathcal{H}[p])$, and $H^0_{\overline{p}},\Sigma_{\mathrm{DFMT},\overline{\mathrm{p}}}(z)\in\mathcal{L}(\mathcal{H}[\overline{p}])$. On the other hand, using again the Schur complement formula, we have

$$\begin{aligned} G_{\mathrm{AI},p}(z) &= \left(G^0_{\mathrm{AI},p}(z)^{-1}-\Sigma_{\mathrm{AI},p}(z)\right)^{-1} = \left(z-H^0_{\mathrm{AI},p}-\Sigma_{\mathrm{AI},p}(z)\right)^{-1} \\ &= \begin{pmatrix} z-H^0_{\mathrm{p}}-\Sigma_{\mathrm{imp},p}(z) & -V_p \\ -V_p^\dagger & z-H^0_{\mathrm{bath},p} \end{pmatrix}^{-1} = \begin{pmatrix} (z-H^0_{\mathrm{p}}-\Sigma_{\mathrm{imp},p}(z)-\Delta_p(z))^{-1} & * \\ * & * \end{pmatrix}, \end{aligned}$$

so that

$$G_{\mathrm{imp},p}(z) = (z-H^0_{\mathrm{p}}-\Sigma_{\mathrm{imp},p}(z)-\Delta_p(z))^{-1}. \tag{III.A.28}$$

Conditions (III.A.24)-(III.A.25), together with (III.A.26)-(III.A.28) yield the self-consistent condition

$$\forall 1 \leq p \leq P, \quad \Delta_p(z) = W_p \left(z - H^0_{\bar{p}} - \bigoplus_{q=1, q\neq p}^{P} \Sigma_{\mathrm{imp},q}(z) \right)^{-1} W_p^\dagger. \tag{III.A.29}$$

Note that the matrices W_p, $H^0_{\bar{p}}$ depend only on the hopping matrix T through

$$[W_p]_{i\sigma,j\sigma'} = \begin{cases} T_{i,j} \text{ if } i \in \Lambda_p, j \in \Lambda \setminus \Lambda_p, \{i,j\} \in E, \ \sigma = \sigma', \\ 0 \qquad\qquad\qquad \text{otherwise,} \end{cases}$$

$$[H^0_{\bar{p}}]_{i\sigma,j\sigma'} = \begin{cases} T_{i,j} \text{ if } i,j \in \Lambda \setminus \Lambda_p, \{i,j\} \in E, \ \sigma = \sigma', \\ 0 \qquad\qquad\qquad \text{otherwise,} \end{cases}$$

where $i, j \in \Lambda$ denote site indices and $\sigma, \sigma' \in \{\uparrow, \downarrow\}$ spin indices.

Remark III.A.13. *This formulation of DMFT agrees with the original one [Geo18; Geo+96] in the translation-invariant setting where*

$$\forall p \in [\![1, P]\!], \quad \mathcal{H}[p] = \mathcal{H}[1], \quad H^0_p = H^0_1, \quad W_p = W_1, \quad H^0_{\bar{p}} = H^0_{\bar{1}}, \quad U_p = U_1. \tag{III.A.30}$$

In this setting, we can consider translation-invariant solutions to the DMFT equations, for which

$$\forall p \in [\![1, P]\!], \quad \forall z \in \mathbb{C}_+, \quad \Sigma_{\mathrm{imp},p}(z) = \Sigma_{\mathrm{imp}}(z), \quad \Delta_p(z) = \Delta(z),$$

where $\Sigma_{\mathrm{imp}}(z)$ and $\Delta(z)$ are related by the translation-invariant self-consistent condition

$$\Delta(z) = W_1 \left(z - H^0_{\bar{1}} - \bigoplus_{q=2}^{P} \Sigma_{\mathrm{imp}}(z) \right)^{-1} W_1^\dagger. \tag{III.A.31}$$

This setting is sketched in Figure III.4. When $\mathfrak{P}$ is a partition into singletons (single-site DMFT), condition (III.A.30) is equivalent to the fact that U is constant and $(\mathcal{G}_H, T)$ is vertex-transitive*, namely that for all $\lambda_1, \lambda_2 \in \Lambda$, there exists a graph isomorphism $\tau : \Lambda \to \Lambda$ such that*

$$\tau(\lambda_1) = \lambda_2, \quad \forall \lambda, \lambda' \in \Lambda, \quad T_{\tau(\lambda),\tau(\lambda')} = T_{\lambda,\lambda'}$$

In particular, the graph Cartesian product $C_N^{\square d}$ of d copies of the N-cycle, which is the nearest neighbor graph of a truncation of the d-dimensional square lattice, with constant hopping and on-site repulsion, and artificial periodic boundary conditions (supercell method), is vertex-transitive due to the translation invariance of the corresponding lattice. This setting was the one considered in the first DMFT computations [Geo+96]. Due to translation invariance, a single impurity problem has to be solved at each iteration. Recall that the impurity solver is the computational bottleneck in DMFT.

The DMFT self-consistent equations (III.A.29), combined with an exact impurity solver, give the exact Green's function of the original Hubbard model in the following trivial limits.

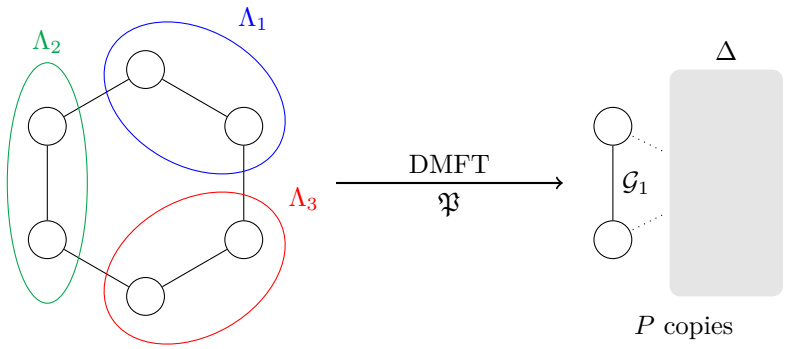


Figure III.4: Translation-invariant DMFT formalism.

Proposition III.A.14 (Exactness of DMFT in some trivial limits)**.** *Consider a Hubbard model $(\mathcal{G}_H, T, U)$ in a Gibbs state $\Omega_{\beta,\varepsilon_F}$. The self-consistent DMFT equations (III.A.29), combined with an exact impurity solver, admit a unique solution in each of the following two settings:*

1. *non-interacting particles. If the on-site repulsion term is equal to zero ($U_i = 0$ for all $i \in \Lambda$), then this solution is given by*

$$\forall p \in [\![1, P]\!], \quad \forall z \in \mathbb{C}_+, \quad \Sigma_{\mathrm{imp},p}(z) = 0, \qquad \Delta_p(z) = W_p \left(z - H_{\overline{p}}^0\right)^{-1} W_p^\dagger; \tag{III.A.32}$$

2. *disconnected graphs and atomic limit. If the partition $\mathfrak{P}$ of $\mathcal{G}_H$ is such that $\mathcal{G}_H = \bigoplus_{p=1}^P \mathcal{G}_p$ (meaning $E_H = \sqcup_{p=1}^P E_p$), that is if the partition decomposes the original Hubbard model $(\mathcal{G}_H, T, U)$ into P independent Hubbard models $(\mathcal{G}_p, T_p, U_p)$, or if the hopping matrix T vanishes, then this solution is given by*

$$\forall p \in [\![1, P]\!], \quad \forall z \in \mathbb{C}_+, \quad \Sigma_{\mathrm{imp},p}(z) = \Sigma_p(z), \qquad \Delta_p(z) = 0, \tag{III.A.33}$$

   where Σ_p is the exact self-energy associated with the p-th Hubbard model in the associated Gibbs state $\Omega^p_{\beta,\varepsilon_F}$.

In both settings, DMFT is exact in the sense that

$$\forall z \in \mathbb{C}_+, \quad G_{\mathrm{DMFT}}(z) = G(z).$$

The purpose of DMFT [Geo18] is to build an appoximation that complies with these two limiting cases. Another limiting case in which DMFT is claimed to be exact is in the "infinite dimension" limit [MV89; Geo+96]. We leave the mathematical investigation of this limit to future works.

We deduce from (III.A.32)-(III.A.33) that in the trivial limits considered in Proposition III.A.14, the hybridization functions $\Delta_p(z)$ are either identically zero, or have a finite number of poles, so that the corresponding baths can be chosen finite dimensional.

III.A.5 A specific impurity solver : the Iterated Perturbative Theory (IPT) solver.

To define properly the IPT solver, we need first to introduce *Matsubara*'s formalism, and more precisely *Matsubara's Green's function* $\tilde{G}^M$.

Matsubara's Green's functions

Matsubara's Green's functions are defined only for *Gibbs states* $\Omega_{\beta,\varepsilon_F}$ at a given inverse temperature β and chemical potential ε_F. These functions have been more extensively studied mathematically than time-ordered Green's functions [BR87; BR97]. In this section, we recall their definition and state an analytic continuation result based on Theorem III.B.8, the latter being useful for our analysis. The setup remains the same as the one introduced in Section III.A.1.

Definition III.A.15 (Gibbs's state and Matsubara's time-ordered Green's function)**.** *Given a particle-number conserving Hamiltonian* $\widehat{H} \in \mathcal{S}(\mathcal{F})$*, that is* $[\widehat{H}, \hat{N}] = 0$*, the* Gibbs state $\Omega_{\beta,\varepsilon_F}$ *at inverse temperature* $\beta \in \mathbb{R}_+^*$ *and chemical potential* $\varepsilon_F \in \mathbb{R}$ *is defined through its density operator by*

$$\widehat{\rho} = \frac{1}{Z_{\beta,\varepsilon_F}} e^{-\beta\left(\widehat{H}-\varepsilon_F\hat{N}\right)}, \quad Z_{\beta,\varepsilon_F} = \mathrm{Tr}\left(e^{-\beta\left(\widehat{H}-\varepsilon_F\hat{N}\right)}\right).$$

The Matsubara's Green's function $\tilde{G}^M$ *is defined as the* $\mathcal{L}(\mathcal{H})$*-valued map* $\tilde{G}^M : [-\beta, \beta) \to \mathcal{L}(\mathcal{H})$ *represented by the sesquilinear form verifying for all* $\phi, \phi' \in \mathcal{H}$*,*

$$\forall \tau \in [-\beta, \beta), \quad \langle \phi, -\tilde{G}^M(\tau)\phi' \rangle = \chi_{\mathbb{R}_+}(\tau)\, \Omega_{\beta,\varepsilon_F}\left(\mathbb{M}(\widehat{a}_\phi)(\tau)\widehat{a}^\dagger_{\phi'}\right) - \chi_{\mathbb{R}_-^*}(\tau)\, \Omega_{\beta,\varepsilon_F}\left(\widehat{a}^\dagger_{\phi'}\mathbb{M}(\widehat{a}_\phi)(\tau)\right), \tag{III.A.34}$$

where for all $\widehat{O} \in \mathcal{L}(\mathcal{F})$*,*

$$\mathbb{M}(\widehat{O}) : [-\beta, \beta] \ni \tau \mapsto e^{\tau\left(\widehat{H}-\varepsilon_F\hat{N}\right)}\widehat{O}e^{-\tau\left(\widehat{H}-\varepsilon_F\hat{N}\right)}$$

is the Matsubara picture *of* $\widehat{O}$*.*

Recall that in our setting, any operator is trace-class since $\mathcal{F}$ is finite dimensional, hence Z_{β,ε_F} is always finite.

As it is the case for the time-ordered Green's function $\tilde{G}$, one can recast equation (III.A.34) using the time-ordered product :

$$\forall \phi, \phi' \in \mathcal{H},\ \forall \tau \in [-\beta, \beta), \quad \langle \phi, -\tilde{G}^M(\tau)\phi' \rangle = \Omega_{\beta,\varepsilon_F}\left(\mathcal{T}\left(\mathbb{M}(\widehat{a}_\phi), \mathbb{M}(\widehat{a}^\dagger_{\phi'})\right)(\tau, 0)\right).$$

Note that the negative sign in the definition of $\tilde{G}^M$ must be consistent with the i prefactor in the definition of $\tilde{G}$ for Theorem III.A.17 below to hold. Considering the grand canonical Hamiltonian $\widehat{H}' = \widehat{H} - \varepsilon_F\hat{N}$ as the Hamiltonian from which the Green's function $\tilde{G}$ is

defined, the Matsubara formalism involves the following formal connection (known as *Wick rotation* [Wic54]):

$$\tau \leftrightarrow it$$

or, in other words, working with $t \leftrightarrow -i\tau$ where τ is real. Hence, the term *imaginary-time* Green's function [MRC16].

Contrary to the Heisenberg picture $\widehat{O} \mapsto \mathbb{H}(\widehat{O})$, the Matsubara picture does not consist in a family of C^*-morphisms: one has for all $\tau \in [-\beta, \beta]$,

$$\Big(\mathbb{M}(\widehat{O})(\tau)\Big)^\dagger = \mathbb{M}(\widehat{O}^\dagger)(-\tau).$$

A consequence of this property is that the Matsubara's Green's function is Hermitian:

$$\Big(\tilde{G}^M(\tau)\Big)^\dagger = \tilde{G}^M(\tau).$$

Note moreover that Gibbs states are *Kubo-Martin-Schwinger (KMS)* states [BR97], meaning that they satisfy the following property:

$$\forall\ \widehat{O}, \widehat{O}' \in \mathcal{L}(\mathcal{F}),\ \forall \tau \in [-\beta, 0], \quad \Omega_{\beta,\varepsilon_F}(\mathbb{M}(\widehat{O})(\tau+\beta)\widehat{O}') = \Omega_{\beta,\varepsilon_F}(\widehat{O}'\mathbb{M}(\widehat{O})(\tau)).$$

This implies that $\tilde{G}^M$ is *β-anti-periodic*, i.e.

$$\forall \tau \in [-\beta, 0), \quad \tilde{G}^M(\tau+\beta) = -\tilde{G}^M(\tau).$$

As for the time-ordered Green's function, the Matsubara's Green's function has a Källén-Lehmann's representation: given an orthonormal basis $\mathcal{B}$ of $\mathcal{F}$ which jointly diagonalizes $\widehat{H}$ and $\hat{N}$ ($\forall \psi \in \mathcal{B}, \widehat{H}\psi = E_\psi \psi,\ \hat{N}\psi = N_\psi \psi$), we have for all $\tau \in [0, \beta)$, for all $\phi, \phi' \in \mathcal{H}$,

$$\begin{aligned}\langle \phi, -\tilde{G}^M(\tau)\phi' \rangle &= \sum_{\psi,\psi' \in \mathcal{B}} e^{\tau((E_\psi - \varepsilon_F N_\psi) - (E_{\psi'} - \varepsilon_F N_{\psi'}))} \langle \psi, \widehat{a}_\phi \psi' \rangle \langle \psi', \widehat{a}^\dagger_{\phi'} \psi \rangle e^{-\beta(E_\psi - \varepsilon_F N_\psi)} \\ &= \sum_{\psi,\psi' \in \mathcal{B}} e^{\tau(E_\psi - E_{\psi'} + \varepsilon_F)} \langle \psi, \widehat{a}_\phi \psi' \rangle \langle \psi', \widehat{a}^\dagger_{\phi'} \psi \rangle e^{-\beta(E_\psi - \varepsilon_F N_\psi)},\end{aligned}$$

as $N_{\psi'} = N_\psi + 1$ whenever $\langle \psi', \widehat{a}^\dagger_{\phi'} \psi \rangle \neq 0$.

Similarly to the time-ordered Green's function $\tilde{G}$, it is convenient to work with a Fourier representation of the Matsubara Green's function $\tilde{G}^M$. Since the latter is defined only on $[-\beta, \beta)$, it is quite natural to extend $\tilde{G}^M$ to the real-line by periodicity and introduce the associated Fourier series with coefficients defined by

$$\forall n \in \mathbb{Z}, \quad \frac{1}{2}\int_{-\beta}^{\beta} \tilde{G}^M(\tau) e^{in\frac{\pi}{\beta}\tau} d\tau.$$

Due to the β-anti-periodicity of $\tilde{G}^M$, the even Fourier coefficients vanish and it holds

$$\frac{1}{2}\int_{-\beta}^{\beta} \tilde{G}^M(\tau) e^{in\frac{\pi}{\beta}\tau} d\tau = \begin{cases} \int_0^\beta \tilde{G}^M(\tau) e^{in\frac{\pi}{\beta}\tau} d\tau & \text{if } n \text{ is odd,} \\ 0 & \text{otherwise.} \end{cases}$$

Definition III.A.16 (Matsubara's Fourier series and frequencies)**.** *The* Matsubara's Fourier coefficients $\left(G_n^M\right)_{n\in\mathbb{Z}}$ *are defined by*

$$\forall n \in \mathbb{Z}, \quad G_n^M = \int_0^\beta \tilde{G}^M(\tau) e^{i\omega_n \tau} d\tau$$

where $\omega_n = (2n+1)\frac{\pi}{\beta}$ *is the* n*-th* Matsubara's frequency.

We thus have

$$\forall \tau \in [-\beta, \beta), \quad \tilde{G}^M(\tau) = \frac{1}{\beta} \sum_{n\in\mathbb{Z}} e^{-i\omega_n \tau} G_n^M.$$

The very reason these coefficients are useful in Green's functions methods lies in the following theorem.

Theorem III.A.17 (Matsubara's Fourier coefficients analytic continuation)**.** *Let* $\widehat{H} \in \mathcal{S}(\mathcal{F})$ *be a particle-number-conserving Hamiltonian,* $\Omega_{\beta,\varepsilon_F}$ *the associated Gibbs state, and* $G : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H})$ *the Generalized Fourier Transform of the associated time-ordered Green's function* $\tilde{G}$ *defined from the grand canonical Hamiltonian* $\widehat{H}' = \widehat{H} - \varepsilon_F \hat{N}$*. Then* G *is the only analytic matrix-valued function such that*

$$\forall z \in \mathbb{C}_+, \ \Im(G(z)) := \frac{G(z) - G(z)^\dagger}{2i} \leq 0, \ \textit{and} \tag{III.A.35}$$

$$\forall n \in \mathbb{N}, \quad G(i\omega_n) = G_n^M. \tag{III.A.36}$$

Note that, since $\omega_{-n} = -\omega_{n-1}$ and $\tilde{G}^M$ is Hermitian, it holds

$$\forall n \in \mathbb{N}^*, \quad \left(G_{n-1}^M\right)^\dagger = G_{-n}^M,$$

so that (III.A.36) actually holds for all $n \in \mathbb{Z}$, with the extension of G to $\mathbb{C}_-$ defined as in (III.A.6).

Remark III.A.18. *The requirement that* G *is analytic and verifies* (III.A.35) *is crucial for uniqueness: for instance, for each* $m \in 2\mathbb{Z}+1$*, the function*

$$\mathbb{C}_+ \ni z \mapsto \frac{1 - e^{m\beta z}}{2} G(z) \in \mathcal{L}(\mathcal{H})$$

also satisfies (III.A.36) *and is analytic, but its imaginary part is not negative semi-definite.*

Theorem III.A.17 is extensively used for practical computations: it is sufficient to run the computations for the Matsubara frequencies and then perform an analytic continuation [Fei+21].

Note that Theorem III.A.17 works in particular for non-interacting Hamiltonians, which leads to the following definition.

Definition III.A.19 (Matsubara's self-energy)**.** *The* Matsubara's self-energy Fourier coefficients $\left(\Sigma_n^M\right)_{n\in\mathbb{Z}}$ *are defined by*

$$\forall n \in \mathbb{Z}, \quad \Sigma_n^M = i\omega_n + \varepsilon_F - H^0 - \left(G_n^M\right)^{-1}.$$

In fact, the self-energy $\Sigma : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H})$ defined in (III.A.12) is the only analytic function with negative imaginary part such that

$$\forall n \in \mathbb{N}, \quad \Sigma(i\omega_n) = \Sigma_n^M.$$

This follows from Proposition III.A.1 and Theorem III.B.8. One can define the *Matsubara's self-energy* $\tilde{\Sigma}^M(\tau)$ by Fourier summation formula as in Definition III.A.16, but this function will not play any role in what follows.

With all these definitions in place, we can now introduce the final ingredient of the model under investigation, namely the single-site, translation-invariant, paramagnetic Iterated Perturbative Theory (IPT).

IPT approximation

In this article, we will not discuss the derivation nor the validity of the Iterative Perturbation Theory (IPT) approximation and refer the interested reader to [AST12]. As in the review paper [Geo+96], we only consider *single-site* and *translation-invariant* DMFT. This seems to be a setting in which the usual IPT approximation is generally considered as valid in the physics literature, while constructing an IPT-like approximation for multi-site DMFT is still an active field of research [AST12]. In the former case, $\mathfrak{P}$ is a partition of the L sites into $P = L$ singletons and the on-site repulsion U is constant as exposed in remark III.A.13. Also, we will focus on the paramagnetic case [RKZ94]. In other words, we will assume that there is no spin-symmetry breaking, so that the spin components can be factored out as detailed in Appendix III.A.6. In the single-site, translation-invariant, paramagnetic IPT-DMFT approximation considered in the sequel, we thus have $P = L$ and for $z \in \mathbb{C}_+$,

$$H^0_{\mathrm{H}},\ G^{DMFT}(z),\ \Sigma^{DMFT}(z) \in \mathbb{C}^{P\times P}.$$

Recall that in translation invariant DMFT, we restrict ourselves to translation invariant solutions to the DMFT equations, so that we consider only one hybridization function Δ and self-energy Σ, with for all $z \in \mathbb{C}_+$,

$$\Delta(z),\ \Sigma(z) \in \mathbb{C},$$

and we use the following notations

$$W^\dagger := W_p^\dagger \in \mathbb{C}^{P-1}, \quad H^0_\perp := H^0_{\overline{p}} \in \mathbb{C}^{(P-1)\times(P-1)}.$$

Moreover, we stick to the *half-filled* setting [Geo+96], for which the chemical potential ε_F of the Anderson Impurity Model is set to $U/2$. The Hamiltonians of interest are then on the one-hand the grand canonical Hubbard Hamiltonian $\widehat{H}_H - \varepsilon_F \hat{N}_H$, and on the other hand the "impurity" grand canonical Hamiltonian $\widehat{H}_{\mathrm{AIM}} - \varepsilon_F \hat{N}_{\mathrm{imp}}$ where $\hat{N}_{\mathrm{imp}}$ is the impurity number operator, complying with [Geo+96, eq. 14].

The IPT solver is based on a second order *perturbation expansion* of the Matsubara's self-energy Fourier coefficients $\Sigma^M_{\mathrm{imp},n}$ of the single-site impurity problem in the parameter U : the Matsubara's self-energy Fourier coefficients of the impurity problem is approximated by

$$\forall n \in \mathbb{N}, \quad \Sigma^M_{\mathrm{imp},n} \approx \Sigma^{M,\mathrm{IPT}}_{\mathrm{imp},n} := \frac{U}{2} + U^2 \int_0^\beta e^{i\omega_n \tau} \left(\tilde{G}^{M,0}_{\mathrm{imp}}(\tau) \right)^3 d\tau,$$

where $\tilde{G}^{M,0}_{\mathrm{imp}}$ is the restriction to $\mathcal{H}[\mathrm{imp}]$ of the Matsubara's Green's function of the non-interacting Hamiltonian H^0_{AI}. The Fourier coefficients of $\tilde{G}^{M,0}_{\mathrm{imp}}$ are given by

$$G^{M,0}_{\mathrm{imp},n} = \left(\left(G^0_{\mathrm{imp}}(i\omega_n) \right)^{-1} - \varepsilon_F \right)^{-1} = \left(i\omega_n - H^0_{\mathrm{imp}} - \Delta(i\omega_n) \right)^{-1} = \left(i\omega_n - \Delta(i\omega_n) \right)^{-1},$$

since $H^0_{\mathrm{imp}} = 0$ and where the first equality is a consequence of a shift to enforce particle-hole symmetry [Geo+96, p.50]. Finally, noticing that

$$\Delta(z) = W \left(z + \varepsilon_F - H^0_\perp + \Sigma(z) \right)^{-1} W^\dagger = W \left(z - H^0_\perp - (\Sigma(z) - \varepsilon_F) \right)^{-1} W^\dagger, \quad \text{(III.A.37)}$$

we make the change of variable $\Sigma \leftarrow \Sigma - \varepsilon_F$ and we thus have, due to the filling condition,

$$\Sigma^{M,IPT}_{\mathrm{imp},n} = U^2 \int_0^\beta e^{i\omega_n \tau} \left(\frac{1}{\beta} \sum_{n' \in \mathbb{Z}} \left(i\omega_{n'} - \Delta(i\omega_{n'}) \right)^{-1} e^{-i\omega_{n'}\tau} \right)^3 d\tau. \quad \text{(III.A.38)}$$

The IPT approximation therefore provides an explicit expression of the Matsubara Fourier coefficients of the impurity self-energy Σ_{imp} as a function of U and Δ. To reconstruct $\Sigma^{\mathrm{IPT}}_{\mathrm{imp}}(z)$ from these Fourier coefficients, we have to solve an Analytic Continuation Problem. The following result shows that, in the case of finite-dimensional baths, this problem has a unique solution, which can be computed (almost) explicitly. Our results are similar to computations already obtained in [KK96, eq. 4].

We denote by

$$\Sigma^{\mathrm{IPT}}_{\mathrm{imp}} = \mathrm{IPT}_\beta(U, \Delta)$$

the output of this solver. At this time, this solver is defined only for finite baths, that is for hybridization functions that are rational functions of the form (III.3.5). We will see later (in Proposition III.3.7) that this map can be extended by (weak) continuity to the space of all physically admissible hybridization functions. We thus finally obtain the system of single-site, translation-invariant, paramagnetic IPT-DMFT equations

$$\forall z \in \mathbb{C}_+, \quad \Delta(z) = W \left(z - H^0_\perp - \Sigma(z) \right)^{-1} W^\dagger \quad \text{(III.A.39)}$$

$$\Sigma = \mathrm{IPT}_\beta(U, \Delta), \quad \text{(III.A.40)}$$

where the on-site interaction $U \in \mathbb{R}$, the inverse temperature $\beta > 0$, the vector $W^\dagger \in \mathbb{C}^{P-1}$ and the matrix $H^0_\perp \in \mathbb{C}^{(P-1)\times(P-1)}$ obtained from the hopping matrix T are the parameters of the model, and where $\Delta : \mathbb{C}_+ \to \mathbb{C}$ and $\Sigma : \mathbb{C}_+ \to \mathbb{C}$ are the unknowns.

In the remainder of this article, our main focus will be the existence of solutions to the above equations.

III.A.6 Paramagnetic IPT-DMFT equations

In this appendix, we detail the spin independence of the paramagnetic IPT-DMFT equations.

As detailed in Remark III.A.11, the Hamiltonian $\widehat{H}_{\mathrm{AIM}}$ commutes with the total spin operator $\hat{S}_{AI}$. More precisely, $\mathcal{H}[AI]$ is $\hat{S}_{AI}$-invariant, and in the decomposition

$$\mathcal{H}[AI] = \mathcal{H}[\uparrow] \oplus \mathcal{H}[\downarrow],$$

$$\mathcal{H}[\sigma] = \mathrm{Span}\left(|\emptyset\rangle \otimes \cdots \otimes |\emptyset\rangle \otimes \underset{m\text{-th}}{|\sigma\rangle} \otimes |\emptyset\rangle \otimes \cdots \otimes |\emptyset\rangle, m \in \Lambda \sqcup [\![1, B]\!] \right),$$

the total spin operators reads $\hat{S}_{AI} = \mathbf{1}_{\uparrow} \oplus (-\mathbf{1}_{\downarrow})$:

$$\hat{S}_{AI} = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}.$$

Writing in this decomposition $H^0 = H^0_{\uparrow} \oplus H^0_{\downarrow}$, we then have

$$G^0(z) = (z - H^0_{\uparrow})^{-1} \oplus (z - H^0_{\downarrow})^{-1}.$$

Since no magnetic field is included in the model, $H^0_{\uparrow}$ and $H^0_{\downarrow}$ act the same way on their respective domains: denoting by $\hat{F} \in \mathcal{L}(\mathcal{H}[AI])$ the *spin flip* isomorphism defined by linearity with $\forall m \in \Lambda \sqcup [\![1, B]\!], \forall \sigma \in \{\uparrow, \downarrow\}$

$$\hat{F}\left(|\emptyset\rangle \otimes \cdots \otimes |\emptyset\rangle \otimes \underset{m\text{-th}}{|\sigma\rangle} \otimes |\emptyset\rangle \otimes \cdots \otimes |\emptyset\rangle \right) = |\emptyset\rangle \otimes \cdots \otimes |\emptyset\rangle \otimes \underset{m\text{-th}}{|\bar{\sigma}\rangle} \otimes |\emptyset\rangle \otimes \cdots \otimes |\emptyset\rangle,$$

we have

$$H^0_{\uparrow} = \hat{F}_{\uparrow,\downarrow} H^0_{\downarrow} \hat{F}_{\downarrow,\uparrow}.$$

Now in the impurity-spin decomposition

$$\mathcal{H} = \mathcal{H}[\uparrow, \mathrm{imp}] \oplus \mathcal{H}[\uparrow, \mathrm{bath}] \oplus \mathcal{H}[\downarrow, \mathrm{imp}] \oplus \mathcal{H}[\downarrow, \mathrm{bath}]$$

$$\mathcal{H}[\sigma, imp] = \mathrm{Span}\left(|\emptyset\rangle \otimes \cdots \otimes |\emptyset\rangle \otimes \underset{i\text{-th}}{|\sigma\rangle} \otimes |\emptyset\rangle \otimes \cdots \otimes |\emptyset\rangle, i \in \Lambda \right)$$

$$\mathcal{H}[\sigma, bath] = \mathrm{Span}\left(|\emptyset\rangle \otimes \cdots \otimes |\emptyset\rangle \otimes \underset{k\text{-th}}{|\sigma\rangle} \otimes |\emptyset\rangle \otimes \cdots \otimes |\emptyset\rangle, k \in [\![1, B]\!] \right),$$

we have $\hat{S}_{AI} = \mathbf{1} \oplus \mathbf{1} \oplus (-\mathbf{1}) \oplus (-\mathbf{1})$ and $\hat{N}_{\mathrm{imp}} = \mathbf{1} \oplus 0 \oplus \mathbf{1} \oplus 0$:

$$\hat{S}_{AI} = \begin{pmatrix} \mathbf{1} & 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & -\mathbf{1} & 0 \\ 0 & 0 & 0 & -\mathbf{1} \end{pmatrix}, \quad \hat{N}_{\mathrm{imp}} = \begin{pmatrix} \mathbf{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \mathbf{1} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

and we have for the impurity orthogonal restriction of the non-interacting Green's function G^0_{imp}:

$$G^0_{\mathrm{imp}}(z) = \left(z - H^0_{\uparrow,\mathrm{imp}} - \Delta_{\uparrow}(z)\right) \oplus \left(z - H^0_{\downarrow,\mathrm{imp}} - \Delta_{\downarrow}(z)\right).$$

Indeed, as it is the case for the non-interacting Hamiltonian and the non-interacting Green's function, $\Delta_\uparrow$ acts similarly as $\Delta_\downarrow$ on their respective domain, and are related by conjugation with the spin flip operator

$$\Delta_\uparrow = \hat{F}_{\uparrow,\downarrow}\Delta_\downarrow\hat{F}_{\downarrow,\uparrow}$$

With that being said, in the IPT approximation outlined in this document, we have for $n \in \mathbb{Z}$,

$$\Sigma^M_{\mathrm{imp},n} = \Sigma^M_{\uparrow,\mathrm{imp},n} \oplus \Sigma^M_{\downarrow,\mathrm{imp},n},$$

with for all $\sigma \in \{\uparrow,\downarrow\}$,

$$\Sigma^M_{\sigma,\mathrm{imp},n} = U^2 \int_0^\beta e^{i\omega_n\tau}\left(\frac{1}{\beta}\sum_{n'\in\mathbb{Z}}\left(i\omega_{n'} - \Delta_\sigma(i\omega_{n'})^{-1}\right)\right)^3 d\tau.$$

And indeed, we have the conjugation relation

$$\Sigma^M_{\uparrow,\mathrm{imp},n} = \hat{F}_{\uparrow,\downarrow}\Sigma^M_{\downarrow,\mathrm{imp},n}\hat{F}_{\downarrow,\uparrow}.$$

These last equations show that the impurity-spin orthogonal restrictions of the Matsubara self-energy Fourier coefficients are copies one of another. Since we assume that the partition $\mathfrak{P}$ is in singletons, it follows that $|\Lambda| = 1$ and $\dim(\mathcal{H}[\sigma,\mathrm{imp}]) = 1$, so that using the bases $\mathcal{B}_{\sigma,\mathrm{imp}}$ given by

$$\forall\sigma \in \{\uparrow,\downarrow\}, \quad \mathcal{B}_{\sigma,\mathrm{imp}} = \{|\sigma\rangle \otimes |\emptyset\rangle \otimes \cdots \otimes |\emptyset\rangle\},$$

we have for all $n \in \mathbb{Z}$,

$$\mathrm{mat}_{\mathcal{B}_{\uparrow,\mathrm{imp}}}(\Sigma^M_{\uparrow,\mathrm{imp},n}) = \mathrm{mat}_{\mathcal{B}_{\downarrow,\mathrm{imp}}}(\Sigma^M_{\downarrow,\mathrm{imp},n}) := \Sigma_n.$$

Finally, we have for $z \in \mathbb{C}_+$,

$$\mathrm{mat}_{\mathcal{B}_{\uparrow,\mathrm{imp}}}(\Delta_\uparrow(z)) = \mathrm{mat}_{\mathcal{B}_{\downarrow,\mathrm{imp}}}(\Delta_\downarrow(z)) = \sum_{k=1}^B \frac{V_k V_k^\dagger}{z-\epsilon_k},$$

so that we indifferently refer to $\Delta(z)$ by abuse of notation.

III.A.7 Proofs of the results in Appendix III.A

Most of the results presented in Appendix III.A are known in settings similar to ours. However, the proofs of Propositions III.A.4 and III.A.10 available in the literature typically apply only to specific classes of states (such as ground states or Gibbs states) and do not emphasize the importance of the notion of *steady* state. Our proofs remove this artificial distinction. Additionally, Proposition III.A.14 is often treated as obvious in the physics literature, although it is usually proven only in translation-invariant settings. Our

proof facilitates the understanding of the DMFT equations, and we hope it will help the reader grasp the machinery introduced in this section.

Furthermore, Theorem III.A.17 has long been considered proven in [BM61] within the physics community. However, the proof given in [BM61] does not make use of the classical analytic continuation techniques already available at the time. Our proof is entirely different and relies on Nevanlinna-Pick analytic continuation theory (especially on Theorem III.B.8, see Appendix III.B for an introduction to this subject). We hope it sheds light on certain aspects of [BM61] in the specific *finite*-dimensional Hilbert space setting considered here. In particular, our proof of the uniqueness of the analytic continuation of the Matsubara's Fourier coefficients does not rely on the asymptotic behavior of the Green's function.

Proof of Proposition III.A.4

Our proof is based on the time evolution formula of annihilation/creation propagators for ideal Fermi gases: one has, as detailed in [BR97, p.46],

$$\mathbb{H}(\widehat{a}_\phi)(t) = \widehat{a}_{e^{itH^0}\phi}. \tag{III.A.41}$$

For all $z \in \mathbb{C}_+$, we have to show that $(z - H^0)G(z) = (z - H^0)(G_+(z) + G_-(\bar{z})^\dagger) = \mathbf{1}$. Integrating by parts, one has

$$(z - H^0)G_+(z) = i\tilde{G}(0^+) + \int_{\mathbb{R}_+^*} e^{izt}\left(i\frac{d}{dt}\tilde{G}(t) - H^0\tilde{G}(t)\right)dt,$$

and for $t > 0$, $i\tilde{G}(t)$ represents the sesquilinear form defined by

$$\forall \phi, \phi' \in \mathcal{H}, \quad \langle \phi, (i\tilde{G})(t)\phi'\rangle = \Omega(\mathbb{H}(\widehat{a}_\phi)(t)\widehat{a}^\dagger_{\phi'}),$$

so that

$$\begin{aligned}\langle \phi, \left(i\frac{d}{dt}\tilde{G}\right)(t)\phi'\rangle = \Omega\left(\frac{d}{dt}(\mathbb{H}(\widehat{a}_\phi))(t)\widehat{a}^\dagger_{\phi'}\right) &= -i\Omega\left(\mathbb{H}(\widehat{a}_{H^0\phi})(t)\widehat{a}^\dagger_{\phi'}\right)\\ &= \langle H^0\phi, \tilde{G}(t)\phi'\rangle\\ &= \langle \phi, H^0\tilde{G}(t)\phi'\rangle.\end{aligned}$$

Similarly, another integration by parts leads to

$$(z - H^0)G_-(\bar{z})^\dagger = i\tilde{G}(0^-)^\dagger + \int_{\mathbb{R}_-} e^{-izt}\left(i\frac{d}{dt}\tilde{G}(t) - \tilde{G}(t)H^0\right)^\dagger dt.$$

For $t < 0$, $i\tilde{G}(t)$ represents the sesquilinear form defined by

$$\forall \phi, \phi' \in \mathcal{H}, \langle \phi, (i\tilde{G})(t)\phi'\rangle = -\Omega(\widehat{a}^\dagger_{\phi'}\mathbb{H}(\widehat{a}_\phi)(t)).$$

Note that, because Ω is a steady state and by the cyclic property of the trace, one has

$$\langle \phi, (i\tilde{G})(t)\phi'\rangle = -\Omega(\mathbb{H}(\widehat{a}^\dagger_{\phi'})(-t)\widehat{a}_\phi), \tag{III.A.42}$$

(steady state propagators are time-translation-invariant), so that

$$\langle \phi, (i\frac{d}{dt}\tilde{G})(t)\phi' \rangle = i\Omega(\mathbb{H}(\widehat{a}^\dagger_{H^0\phi'})(-t)\widehat{a}_\phi) = \langle \phi, \tilde{G}(t)H^0\phi' \rangle.$$

Finally, one has for $\phi, \phi' \in \mathcal{H}$,

$$\langle \phi, (z - H^0)G(z)\phi' \rangle = \langle \phi, i\tilde{G}(0^+)\phi' \rangle + \langle \phi, i\tilde{G}(0^-)^\dagger \phi' \rangle = \Omega(\widehat{a}_\phi \widehat{a}^\dagger_{\phi'} + \widehat{a}^\dagger_{\phi'}\widehat{a}_\phi) = \langle \phi, \phi' \rangle,$$

because Ω is a state and the annihilation/creation operators satisfy the CAR.

Proof of Proposition III.A.1 ($-G, -\Sigma, -\Delta$ are Pick matrices)

The fact that the Green's function G is a Pick matrix readily follows from the KL representation (III.A.7) and inequality (III.A.8).

Combining (III.A.10) and (III.A.12), the self-energy can be written as

$$\forall z \in \mathbb{C}_+, \quad \Sigma(z) = z - H^0 - G(z)^{-1}.$$

Recall that we know from (III.A.8) that $G(z)$ is invertible for all $z \in \mathbb{C}_+$. Since $-G$ is Pick, G^{-1} is Pick. This readily implies that Σ is analytic. Let us now prove that $-\Sigma$ is Pick. First, we infer from the KL representation (III.A.7) that for all $k \in \mathbb{N}$, there exists $m_0, \ldots, m_{2k} \in \mathbb{C}^{n\times n}$ such that it holds

$$G(iy) = \frac{m_0}{iy} + \frac{m_1}{(iy)^2} + \frac{m_2}{(iy)^3} + \cdots + \frac{m_{2k}}{(iy)^{2k+1}} + o_{y\to+\infty}\left(\frac{1}{y^{2k+1}}\right).$$

Using the anti-commutation relation $\widehat{a}_\phi \widehat{a}^\dagger_{\phi'} + \widehat{a}^\dagger_{\phi'}\widehat{a}_\phi = \langle \phi, \phi' \rangle$ and the normalization condition $\sum_{\psi\in\mathcal{B}} \rho_\psi = 1$, we obtain that $m_0 = I_n$. As a consequence, we have

$$G(iy)^{-1} = (iy)I_n - m_1 + \frac{1}{iy}(m_1^2 - m_2) + o_{y\to+\infty}\left(\frac{1}{y}\right).$$

In view of Theorem III.4.1, the Pick matrix G^{-1} has a Nevanlinna-Riesz representation of the form

$$G(z)^{-1} = z - H^0 - \Sigma_\infty + \int_{\mathbb{R}} \frac{d\mu(\epsilon)}{\epsilon - z},$$

with $\Sigma_\infty \in \mathcal{S}_n(\mathbb{C})$ and μ a finite Borel $\mathcal{S}_n^+(\mathbb{C})$-valued measure on $\mathbb{R}$. We thus obtain that

$$\forall z \in \mathbb{C}_+, \quad \Sigma(z) = \Sigma_\infty + \int_{\mathbb{R}} \frac{d\mu(\epsilon)}{z - \epsilon},$$

from which we deduce that $-\Sigma$ is Pick. As a matter of fact, Σ is a matrix-valued rational function, that is μ is a weighted sum of finitely many Dirac measures.

Let Δ be a hybridization function of an AIM defined as (III.A.18). Since H^0_{bath} is self-adjoint, its spectrum is real and (III.A.18) thus defines an analytic function on $\mathbb{C}_+$. In addition, for all $z \in \mathbb{C}_+$, $\Im(z - H^0_{\text{bath}}) = \Im(z) > 0$ hence $\Im((z - H^0_{\text{bath}})^{-1}) < 0$. Since the congruence preserves the sign of the imaginary part we have $\Im(\Delta(z)) \leq 0$. This shows that $-\Delta$ is a Pick matrix.

Proof of Theorem III.A.10

For the simplicity of the proof, note first that an equivalent definition of an impurity problem is that there exists an *impurity space* $\mathcal{H}[\mathrm{imp}] \subset \mathcal{H}$ such that the interacting part $\widehat{H}^I$ of the Hamiltonian $\widehat{H}$ as introduced in (III.A.9) belongs to the following subalgebra

$$\widehat{H}^I \in \mathcal{A}\{\widehat{a}^\dagger_\phi \widehat{a}_{\phi'}, \phi, \phi' \in \mathcal{H}[\mathrm{imp}]\}. \tag{III.A.43}$$

In other words, the interacting part of the Hamiltonian is an element of the Gauge Invariant Canonical Anti-commutation Relations (GICAR) algebra generated by $\mathcal{H}[\mathrm{imp}]$ (see [Sat24] for a concise introduction to GICAR algebras).

From that, the proof is a generalization of [LL20]. To prove the sparsity pattern of Σ, we have to prove that for all $z \in \mathbb{C}_+$, for all $\phi \in \mathcal{H}, \phi' \in \mathcal{H}[\mathrm{imp}]^\perp$,

$$\langle \phi, G(z)\Sigma(z)\phi' \rangle = 0 \text{ and } \langle \phi', \Sigma(z)G(z)\phi \rangle = 0. \tag{III.A.44}$$

The first equality is equivalent to

$$\langle \phi, G(z)(z - H^0)\phi' \rangle = \langle \phi, \phi' \rangle,$$

and, similarly as in the previous proof, we have

$$\begin{aligned}\langle \phi, G(z)(z - H^0)\phi' \rangle = \langle \phi, \phi' \rangle &+ \int_{\mathbb{R}_+} e^{izt} \left(\frac{d}{dt} \langle \phi, i\tilde{G}(t)\phi' \rangle - \langle \phi, \tilde{G}(t)H^0\phi' \rangle \right) dt \\ &+ \int_{\mathbb{R}_-^*} e^{-izt} \left(\frac{d}{dt} \langle i\tilde{G}(t)\phi, \phi' \rangle - \langle \tilde{G}(t)H^0\phi, \phi' \rangle \right) dt.\end{aligned}$$

Now, for all $t \in \mathbb{R}_+$, we have using the cyclicity of the trace, similarly as in (III.A.42),

$$\frac{d}{dt} \langle \phi, i\tilde{G}(t)\phi' \rangle = -i\Omega \left(\widehat{a}_\phi \mathbb{H}([\widehat{H}, \widehat{a}^\dagger_{\phi'}])(-t) \right).$$

To compute the commutator, note first that for all $\phi_1, \phi_2 \in \mathcal{H}[\mathrm{imp}]$, we have using the CAR

$$[\widehat{a}^\dagger_{\phi_1} \widehat{a}_{\phi_2}, \widehat{a}^\dagger_{\phi'}] = \langle \phi_2, \phi' \rangle \widehat{a}^\dagger_{\phi_1} = 0$$

since $\phi' \in \mathcal{H}[\mathrm{imp}]^\perp$, so that $\widehat{a}^\dagger_{\phi'}$ commutes with the generators of the algebra to which $\widehat{H}^I$ belongs, hence with $\widehat{H}^I$. Moreover, we have using (III.A.41),

$$[d\Gamma(H^0), \widehat{a}^\dagger_{\phi'}] = -i \frac{d}{dt} \left(t \mapsto \mathbb{H}(\widehat{a}^\dagger_{\phi'})_{H^0} \right)(0) = \widehat{a}^\dagger_{H^0\phi'},$$

where $\mathbb{H}(\cdot)_{H^0}$ denotes the Heisenberg picture associated with the non-interacting Hamiltonian $d\Gamma(H^0)$, so that we end up with

$$\frac{d}{dt} \langle \phi, i\tilde{G}(t)\phi' \rangle = -i\Omega \left(\widehat{a}_\phi \mathbb{H}(\widehat{a}^\dagger_{H^0\phi'})(-t) \right) = \langle \phi, \tilde{G}(t)H^0\phi' \rangle.$$

One shows, using the same techniques, that for all $t \in \mathbb{R}_-^*$,

$$\frac{d}{dt} \langle i\tilde{G}(t)\phi, \phi' \rangle = \langle \tilde{G}(t)H^0\phi, \phi' \rangle$$

hence the first equality of (III.A.44) is proven. The second equality can be proved similarly.

Proof of Proposition III.A.14

We stick to the case in Remark III.A.12, where the AIM states are Gibbs states at inverse temperature β and chemical potential ε_F. Now for the first case, if $U = 0$, the AIM are non-interacting, hence the Green's functions are the non-interacting Green's functions, the self-energies $\Sigma_{\mathrm{imp},p}$ are identically zero, and the hybridization functions are given by, for all $z \in \mathbb{C}_+$,

$$\Delta_p(z) = W_p \left(z - H_{\overline{p}}^0\right)^{-1} W_p^\dagger.$$

Moreover, $\Sigma_{\mathrm{DMFT}} = 0$ so that $G_{\mathrm{DMFT}} = G^0 = G$ since the original Hubbard model is not interacting in this setting.

Now if the partition $\mathfrak{P}$ is such that $\mathcal{G}_H = \bigoplus_{p=1}^P \mathcal{G}_p$, or if $T = 0$, we have

$$W_p = 0,$$

so that the hybridization functions Δ_p are identically zero. This is equivalent to zero-dimensional baths, and all the AIMs reduce to Hubbard models defined by $(\mathcal{G}_p, T_p, U_p)$. Hence the self-energies Σ_p are the self-energies associated with the Gibbs states of the corresponding Hubbard models.

Moreover, the clusters do not interact in this setting: the Hamiltonian $\hat{H}_H$ of the original Hubbard model reads

$$\hat{H}_H = \sum_{p=1}^P \widehat{H}_p, \tag{III.A.45}$$

where $\widehat{H}_p$ is the Hamiltonians of the p-th Hubbard model defined on the induced subgraph $(\mathcal{G}_p, T_p, U_p)$. By assumption, these Hamiltonians pairwise commute, so that the grand-canonical Gibbs state factorizes, and the associated Green's functions G and G^0 read

$$G = \bigoplus_{p=1}^P G_p, \quad G^0 = \bigoplus_{p=1}^P G_p^0, \tag{III.A.46}$$

where G_p (resp. G_p^0) is the Green's functions of the Gibbs state of the p-th Hubbard model (resp. p-th non-interacting Hubbard model). Therefore $\Sigma = \bigoplus_{p=1}^P \Sigma_p = \Sigma_{\mathrm{DMFT}}$, which proves the claim.

Proof of Theorem III.A.17

We start by proving the equality. On the one hand, the Källén-Lehmann representation (III.A.7) of $\tilde{G}$ associated with the grand canonical Hamiltonian $\widehat{H}' = \widehat{H} - \varepsilon_F \hat{N}$ reads for $\phi, \phi' \in \mathcal{H}$ and $z \in \mathbb{C}_+$,

$$\langle \phi, G(z)\phi' \rangle = \sum_{\psi,\psi' \in \mathcal{B}} (\rho_\psi + \rho_{\psi'}) \langle \psi, \widehat{a}_\phi \psi' \rangle \langle \psi', \widehat{a}_{\phi'}^\dagger \psi \rangle \frac{1}{z + \varepsilon_F + (E_\psi - E_{\psi'})}.$$

On the other hand, the Källén-Lehmann representation of $\tilde{G}^M$ reads for $\phi, \phi' \in \mathcal{H}$ and $n \in \mathbb{N}$,

$$\langle \phi, G_n^M \phi' \rangle = \sum_{\psi,\psi' \in \mathcal{B}} \left(\rho_\psi + \rho_\psi e^{\beta(E_\psi - E_{\psi'} + \varepsilon_F)}\right) \langle \psi, \widehat{a}_\phi \psi' \rangle \langle \psi', \widehat{a}^\dagger_{\phi'} \psi \rangle \frac{1}{i\omega_n + \varepsilon_F + (E_\psi - E_{\psi'})}.$$

Note now that whenever $\langle \psi, \widehat{a}_\phi \psi' \rangle \neq 0$, we have $N_{\psi'} = N_\psi + 1$ and then

$$\rho_\psi e^{\beta(E_\psi - E_{\psi'} + \varepsilon_F)} = e^{-\beta(E_{\psi'} - \varepsilon_F(N_\psi + 1))} = \rho_{\psi'},$$

yielding

$$\forall n \in \mathbb{N},\ G(i\omega_n) = G_n^M.$$

To prove uniqueness, we use the fact that $-G$ is a Pick matrix (see Proposition III.A.1), and its Nevanlinna-Riesz measure is a weighted sum of finitely many Dirac measures. It follows that for all $\phi \in \mathcal{H}$, the Analytic Continuation Problem defined by $\left(i\omega_n, -\langle \phi, G_n^M \phi \rangle\right)_{n \in \mathbb{N}}$ has no other solution thanks to Theorem III.B.8, which concludes.

III.B Uniqueness theorem for an interpolation problem

The Nevanlinna-Pick Analytic Continuation Problem (ACP), which we will also call *interpolation problem*, has been studied in the beginning of the 20th century independently by Nevanlinna ([Nev19], 1919) and Pick ([Pic15], 1915). The results presented in Section III.B.1 are gathered in the book [Wal35]. Many other references address this problem, such as [Gar06], [Abr79], [Sar67]. In Section III.B.1, we set up the problem and then give some results that are important to our analysis. Other important results on the ACP and characterizations of the solutions are detailed in the references. For example, we will not discuss the question of extremal solutions ([Nic16], [Str91]), nor the use of Blaschke products ([Wal35], [Gar06], [Nic94]). The approaches of Nevanlinna and Pick are different, we chose here to focus on Nevanlinna's approach.

III.B.1 Introduction and some general properties

The Nevanlinna-Pick Analytic Continuation Problem (ACP) can be stated as follows. Let $(z_n)_{n \in I}$ be a sequence of distinct points in the open upper half-plane $\mathbb{C}_+$ and let $(w_n)_{n \in I}$ be a sequence in $\overline{\mathbb{C}_+}$. We want to answer the following question.

Question III.B.1. *Is there an analytic function $f : \mathbb{C}_+ \to \overline{\mathbb{C}_+}$ interpolating the given values at the prescribed points? In other words, we look for a Pick function f such that*

$$\forall n \in I,\ f(z_n) = w_n, \tag{III.B.1}$$

where I is a (at most) countable set.

Without loss of generality, we can assume that $I = \{1, 2, \ldots\}$. We denote by $AC_{\mathbb{C}_+}(z_n, w_n)_I$ the set of solutions to this problem (we will omit the dependence on I, unless when needed). Both sets $\mathbb{C}_+$ and $\overline{\mathbb{C}_+}$ are invariant under the action of the subset $\mathcal{T}$ of affine transformations of $\overline{\mathbb{C}_+}$ given by $\mathcal{T} = \{\tau : \overline{\mathbb{C}_+} \to \overline{\mathbb{C}_+}, z \mapsto az + b,\ b \in \mathbb{R},\ a > 0\}$, and we have

$$\forall \tau_1, \tau_2 \in \mathcal{T}, \quad AC_{\mathbb{C}_+}(\tau_1(z_n), \tau_2(w_n)) = \tau_2 \circ AC_{\mathbb{C}_+}(z_n, w_n) \circ \tau_1^{-1}. \tag{III.B.2}$$

Moreover, question III.B.1 can equivalently be stated in the unit disc instead of the upper half-plane:

Question III.B.2. *Let (z_n) and (w_n) be sequences in the unit disc $\mathbb{D} = \{z \in \mathbb{C}, |z| < 1\}$ and the closed disc $\overline{\mathbb{D}}$ respectively. Is there an analytic function $f : \mathbb{D} \to \overline{\mathbb{D}}$ such that $\forall n \in I, \quad f(z_n) = w_n$?*

The reason for the equivalence between both formulations is that the upper half-plane can be mapped to the unit disc through the Cayley transform, which is biholomorphic between these sets. The Cayley transform $\mathcal{W}$ and its reciprocal $\mathcal{W}^{-1}$ are given by:

$$\begin{array}{rccc} \mathcal{W} : \mathbb{C}\backslash\{-i\} & \longrightarrow & \mathbb{C}\backslash\{1\} \\ z & \mapsto & \dfrac{z-i}{z+i} \end{array} \qquad \begin{array}{rccc} \mathcal{W}^{-1} : \mathbb{C}\backslash\{1\} & \longrightarrow & \mathbb{C}\backslash\{-i\} \\ z & \mapsto & i\dfrac{1+z}{1-z}. \end{array}$$

As the Cayley transform $\mathcal{W}$ is biholomorphic from $\mathbb{C}_+$ to $\mathbb{D}$ and maps the real line $\mathbb{R}$ to the unit circle deprived of the point 1, we have the equivalence of the following statements, with $F : \mathbb{C}_+ \to \overline{\mathbb{C}_+}$, $(z_n) \subset \mathbb{C}_+$, $(w_n) \subset \overline{\mathbb{C}_+}$, $f = \mathcal{W} \circ F \circ \mathcal{W}^{-1} : \mathbb{D} \to \overline{\mathbb{D}}$, $\tilde{z_n} = \mathcal{W}(z_n) \in \mathbb{D}$ and $\tilde{w_n} = \mathcal{W}(w_n) \in \overline{\mathbb{D}}$.

$$\forall n \geq 1,\ F(z_n) = w_n \iff \forall n \geq 1,\ f(\tilde{z_n}) = \tilde{w_n}.$$

In other words, denoting by $AC_{\mathbb{D}}(z_n, w_n)$ the set of solutions to Question III.B.2, we have

$$AC_{\mathbb{D}}(\mathcal{W}(z_n), \mathcal{W}(w_n)) = \mathcal{W} \circ AC_{\mathbb{C}_+}(z_n, w_n) \circ \mathcal{W}^{-1}. \tag{III.B.3}$$

As a matter of fact, we can answer Question III.B.2 when the sequence (w_n) is constant and of modulus 1, as a direct consequence of the maximum modulus principle.

Lemma III.B.3. *Given $C \in \overline{\mathbb{D}} \setminus \mathbb{D}$, $AC_{\mathbb{D}}(z_n, C) = \{C\}$.*

Combined with equation (III.B.3), we have a first statement about the set of solutions $AC_{\mathbb{C}_+}$ to the interpolation problem (III.B.1).

Corollary III.B.4. *Given $C \in \mathbb{R}$, $AC_{\mathbb{C}_+}(z_n, C) = \{C\}$.*

The next definition introduces the functions b_a (the notation suggests Blaschke products, see [Nic16] and [Gar06]), which will be useful for our analysis of $AC_{\mathbb{D}}(z_n, w_n)$.

Definition III.B.5. *Let $a \in \mathbb{D}$. Define for $z \in \mathbb{D}$, $b_a(z) = \frac{z-a}{1-\bar{a}z} \in \mathbb{D}$.*

These functions are in fact biholomorphic from $\mathbb{D}$ to itself and b_a only vanishes in $a \in \mathbb{D}$. This zero is of order 1. The function b_a can actually be extended to the closed disc $\overline{\mathbb{D}}$, and we will also denote this extension b_a. Considering our interpolation problem, we notice the following. If f is a solution to the interpolation problem stated in question III.B.2, we have $f(z_1) = w_1$, with $|w_1| \leq 1$. If the modulus of w_1 is 1, then f is constant because of the maximum modulus principle. Suppose this is not the case and define the function $f^{(1)}$ by

$$f^{(1)}(z) = \frac{b_{w_1}(f(z))}{b_{z_1}(z)}. \tag{III.B.4}$$

$f^{(1)}$ is then well-defined on $\mathbb{D}$, because the only zero of b_{z_1} is z_1 and is of order 1, and z_1 is also a zero of order at least 1 of $b_{w_1} \circ f$. One can check that $f^{(1)}$ takes values in $\overline{\mathbb{D}}$, so that $f^{(1)}$ is analytic from $\mathbb{D}$ to $\overline{\mathbb{D}}$ if and only if $f = b_{w_1}^{-1} \circ (b_{z_1} f^{(1)})$ is analytic from $\mathbb{D}$ to $\overline{\mathbb{D}}$.

Now suppose $|I| \geq 2$ and define for all $n \in I \setminus \{1\}$, $w_n^{(1)} := \frac{b_{w_1}(w_n)}{b_{z_1}(z_n)}$. We notice that $f^{(1)}(z_n) = \frac{b_{w_1}(f(z_n))}{b_{z_1}(z_n)} = \frac{b_{w_1}(w_n)}{b_{z_1}(z_n)} = w_n^{(1)}$. This means that $f^{(1)}$ is the solution to the Nevanlinna-Pick interpolation problem

$$g(z_n) = w_n^{(1)}, \ \forall n \in I \setminus \{1\}. \tag{III.B.5}$$

We have proven the following lemma, which is the main element of the Schur interpolation algorithm [Gul+11].

Lemma III.B.6 (Schur iteration)**.** *Assume $|I| \geq 2$ and $w_1 \in \mathbb{D}$. We then have the following equivalence:*

$$f \in AC_{\mathbb{D}}(z_n, w_n)_I \iff f^{(1)} : z \mapsto \frac{b_{w_1}(f(z))}{b_{z_1}(z)} \in AC_{\mathbb{D}}(z_n, w_n^{(1)})_{I\setminus\{1\}}.$$

In other words,

$$AC_{\mathbb{D}}(z_n, w_n)_I = b_{w_1}^{-1} \circ \left(b_{z_1} \cdot AC_{\mathbb{D}}(z_n, w_n^{(1)})_{I\setminus\{1\}}\right).$$

Both previous lemmas give the intuition about the following theorem, which can be found in any reference dealing with the issue of Nevanlinna-Pick interpolation, e.g. [Wal35], [Gar06], [Nic94]. Refinements of this result and characterizations of the solutions are detailed in the references given at the beginning of this section.

Theorem III.B.7. *Let $(z_n)_{n\geq 1}$ and $(w_n)_{n\geq 1}$ be sequences in the unit disc $\mathbb{D}$ and the closed disc $\overline{\mathbb{D}}$ respectively. Define by induction, for $1 \leq l$ and $k > l$,*

$$w_k^{(l)} := \frac{w_k^{(l-1)} - w_l^{(l-1)}}{1 - \overline{w_l^{(l-1)}} w_k^{(l-1)}} \frac{1 - \overline{z_l} z_k}{z_k - z_l} = \frac{b_{w_l^{(l-1)}}(w_k^{(l-1)})}{b_{z_l}(z_k)}, \tag{III.B.6}$$

with $w_k^{(0)} = w_k$ for $k \geq 1$.

There are three distinct cases.

1. *If there exists $k \geq 1$ such that $|w_k^{(k-1)}| > 1$, then there is no solution to the interpolation problem (III.B.1).*

2. *Else, if there exists $K \geq 1$ such that for all $1 \leq k < K$, $|w_k^{(k-1)}| < 1$, $|w_K^{(K-1)}| = 1$ and for all $l \geq K$, $w_l^{(K-1)} = w_K^{(K-1)}$, then there exists a unique solution to the interpolation problem (III.B.1).*

3. *Else, we have for all $k \geq 1$, $|w_k^{(k-1)}| < 1$ and there is either 1 or infinitely many solutions to the interpolation problem.*

III.B.2 A uniqueness result for ACPs with a rational solution

In our mathematical framework, we need the following uniqueness theorem (Theorem III.B.8). In this section, we are tackling the uniqueness of the interpolation problem (III.B.1), in the case where we have already found one solution of some specific form. We assume that we have found a solution f such that its Nevanlinna-Riesz measure in the integral representation (III.3.2) is a finite sum of weighted Dirac measures, that is f is a rational function. This means that we have $f(z) = \alpha z + C + \sum_{k=1}^{K} \frac{a_k}{z-\varepsilon_k}$, with $\alpha \geq 0$, $C \in \mathbb{R}$, $K \in \mathbb{N}$, the a_k's are negative numbers and the ε_k's are distinct real numbers. The following theorem states that it is then the only solution to the interpolation problem (III.B.1). The theorem is actually a corollary of a result by Donoghue [Don74b], stated within the Pick formalism of Pick functions. The approach and the proof presented here are however different, and may therefore be of interest to the reader.

Theorem III.B.8. *Let $f : \mathbb{C}_+ \to \overline{\mathbb{C}_+}$ be a Pick function such that its Nevanlinna-Riesz measure μ is a finite sum of Dirac measures: there exist $K \in \mathbb{N}$, $a_1, \ldots, a_K < 0$ and distinct real numbers $\varepsilon_1, \ldots, \varepsilon_K$ such that*

$$\mu = \sum_{k=1}^{K} a_k \delta_{\varepsilon_k}.$$

Then, if f is a solution to an interpolation problem $AC_{\mathbb{C}_+}(z_n, w_n)_I$ with $|I| \geq K+2$, this problem has no other solution.

Proof. We prove this result by strong induction on K, for $K \in \mathbb{N}$. If $K = 0$, the expression of f reads $f(z) = \alpha z + C$, with $\alpha \geq 0$ and $C \in \mathbb{R}$. Now suppose $f \in AC_{\mathbb{C}_+}(z_n, w_n)$. Then for all $n \in I$, we have $\alpha z_n + C = w_n$, so that

$$AC_{\mathbb{C}_+}(z_n, w_n) = AC_{\mathbb{C}_+}(z_n, \alpha z_n + C).$$

If $\alpha = 0$, then $AC_{\mathbb{C}_+}(z_n, C) = \{C\}$ by corollary III.B.4. Else, we extend continuously f to $\overline{\mathbb{C}_+}$ into the affine transformation $\widetilde{f} \in \mathcal{T}$ and we have

$$AC_{\mathbb{C}_+}(z_n, w_n) = \widetilde{f} \circ \mathcal{W}^{-1} \circ AC_{\mathbb{D}}(\mathcal{W}(z_n), \mathcal{W}(z_n)) \circ \mathcal{W}.$$

Denoting by $\widetilde{z}_n = \widetilde{w}_n = \mathcal{W}(z_n)$, we find that for all $n \in I \setminus \{1\}, \widetilde{w}_n^{(1)} = 1$. Hence by Theorem III.B.7, $AC_{\mathbb{D}}(\mathcal{W}(z_n), \mathcal{W}(z_n)) = \{\mathrm{id}_{\mathbb{D}}\}$ and $AC_{\mathbb{C}_+}(z_n, w_n) = \{f\}$.

Now assume the result holds for $l \leq K$ and take f such that its Nevanlinna-Riesz measure is a sum of $K+1$ Dirac measures. Now consider an ACP $AC_{\mathbb{C}}(z_n, w_n)_I$ with $|I| \geq K+3$ to which f is a solution, namely $f(z_n) = w_n$ and there exist $\alpha \geq 0$, $C \in \mathbb{R}$, $a_1, \dots, a_{K+1} < 0$ and real numbers $\varepsilon_1 < \cdots < \varepsilon_{K+1}$, such that for all $z \in \mathbb{C}_+$,

$$f(z) = \alpha z + C + \sum_{k=1}^{K+1} \frac{a_k}{z - \varepsilon_k}.$$

We start by making use of the property (III.B.2) to simplify the problem. It is possible to chose two affine transformations τ_{z_1} and $\check{\tau}_{z_1}$ in $\mathcal{T}$ such that, setting $\widetilde{f} = \check{\tau}_{z_1} \circ f \circ \tau_{z_1}^{-1}$, we have

$$\widetilde{f}(z) = \widetilde{C} + \widetilde{\alpha} z + \sum_{k=1}^{K+1} \frac{\widetilde{a}_k}{z - \widetilde{\varepsilon}_k},$$

where the parameters $\widetilde{C}$, $\widetilde{\alpha}$, $\widetilde{a}_k$, and $\widetilde{\varepsilon}_k$ satisfy the same properties as their counterparts without the tildes, and such that

$$\tau_{z_1}(z_1) = i, \quad \Re(\widetilde{f}(\widetilde{z}_1)) = 0 \quad \text{and} \quad \sum_{k=1}^{K+1} \frac{-\widetilde{a}_k}{1 + \widetilde{\varepsilon}_k^2} = 1. \tag{III.B.7}$$

Using (III.B.2), we have

$$AC_{\mathbb{C}_+}(z_n, w_n) = \check{\tau}_{z_1}^{-1} \circ AC_{\mathbb{C}_+}(\underbrace{\tau_{z_1}(z_n)}_{\widetilde{z}_n}, \underbrace{(\check{\tau}_{z_1} \circ f \circ \tau_{z_1}^{-1})}_{\widetilde{f}}(\tau_{z_1}(z_n))) \circ \tau_{z_1}.$$

For the remaining of the proof, we will then focus on $AC_{\mathbb{C}_+}(\widetilde{z}_n, \widetilde{f}(\widetilde{z}_n))$. We will omit the tildes for the sake of simplicity, and assume that $z_1 = i$ and that (III.B.7) holds. With that being said, notice that $\Im(f(z_1)) = \Im(f(i)) = \alpha + \sum_{k=1}^{K+1} \frac{-a_k}{1+\varepsilon_k^2} = \alpha + 1 > 0$, hence $\mathcal{W}(f(z_1)) \in \mathbb{D}$. Therefore, using Lemma III.B.6, we have

$$AC_{\mathbb{D}}(\mathcal{W}(z_n), \mathcal{W}(f(z_n)))_I = b_{\mathcal{W}(f(z_1))}^{-1} \circ \left(b_{z_1} \cdot AC_{\mathbb{D}}(\mathcal{W}(z_n), \mathcal{W}(f(z_n))^{(1)})_{I \setminus \{1\}}\right),$$

and, using equation (III.B.3),

$$AC_{\mathbb{D}}(\mathcal{W}(z_n), \mathcal{W}(f(z_n))^{(1)})_{I\setminus\{1\}} = \mathcal{W} \circ AC_{\mathbb{C}_+}\left(z_n, \mathcal{W}^{-1}([\mathcal{W}(f(z_n))]^{(1)})\right)_{I\setminus\{1\}} \circ \mathcal{W}^{-1}.$$

We now compute $g(z) = \mathcal{W}^{-1} \circ [\mathcal{W} \circ f]^{(1)}(z)$, for $z \in \mathbb{C}_+ \setminus \{i\}$. Since $\mathcal{W}(z_1) = \mathcal{W}(i) = 0$ and $z \neq i$, we have $b_{\mathcal{W}(z_1)}(\mathcal{W}(z)) = \mathcal{W}(z) \neq 0$ and $g(z)$ is well defined. The computation reads

$$g(z) = i \frac{(f(z_1) + i)\overline{\frac{f(z_1) - f(z)}{z+i}} + \overline{(f(z_1) + i)}\frac{f(z) - f(z_1)}{z-i}}{(f(z_1) + i)\overline{\frac{f(z_1) - f(z)}{z+i}} - \overline{(f(z_1) + i)}\frac{f(z) - f(z_1)}{z-i}}.$$

We used the fact that since $\overline{\mathcal{W}(f(z_1))}\mathcal{W}(f(z)) \in \mathbb{D}$, it is not equal to 1 and that $\frac{1}{2i}(f(z)+i)|(f(z_1)+i)|^2 \neq 0$. Now, realize that, since $z_1 = i$, we have

$$\frac{f(z)-f(z_1)}{z-i} = \alpha + \sum_{k=1}^{K+1} \frac{1}{\varepsilon_k - i}\frac{a_k}{z-\varepsilon_k},$$

and similarly

$$\frac{\overline{f(z_1)}-f(z)}{z+i} = -\alpha - \sum_{k=1}^{K+1} \frac{1}{\varepsilon_k + i}\frac{a_k}{z-\varepsilon_k},$$

so that

$$g(z) = -\frac{\alpha(\Im(f(z_1))+1) + \sum_{k=1}^{K+1} \Im\left(\frac{f(z_1)+i}{\varepsilon_k+i}\right)\frac{a_k}{z-\varepsilon_k}}{\alpha\Re(f(z_1)) + \sum_{k=1}^{K+1} \Re\left(\frac{f(z_1)+i}{\varepsilon_k+i}\right)\frac{a_k}{z-\varepsilon_k}}, \tag{III.B.8}$$

which holds for $z = i$ as well. As we set $\Re(f(z_1)) = 0$, we have

$$\Re\left(\frac{f(z_1)+i}{\varepsilon_k+i}\right) = \frac{\Im(f(z_1))+1}{1+\varepsilon_k^2} > 0 \quad \text{and} \quad \Im\left(\frac{f(z_1)+i}{\varepsilon_k+i}\right) = \varepsilon_k\frac{\Im(f(z_1))+1}{1+\varepsilon_k^2}.$$

Multiplying the numerator and denominator in (III.B.8) by $\frac{1}{\Im(f(z_1))+1}\prod_{k=1}^{K+1}(z-\varepsilon_k)$, we end up with $g(z) = \dfrac{P(z)}{Q(z)}$, with

$$P(z) := \alpha\prod_{k=1}^{K+1}(z-\varepsilon_k) + \sum_{k=1}^{K+1}\varepsilon_k\frac{a_k}{1+\varepsilon_k^2}\prod_{l=1,l\neq k}^{K+1}(z-\varepsilon_l),$$

and

$$Q(z) := \sum_{k=1}^{K+1}\frac{-a_k}{1+\varepsilon_k^2}\prod_{l=1,l\neq k}^{K+1}(z-\varepsilon_l).$$

We have $Q(\varepsilon_k) = \frac{-a_k}{1+\varepsilon_k^2}\prod_{l=1,l\neq k}^{K+1}(\varepsilon_k-\varepsilon_l) \neq 0$ by hypothesis on the ε_k's, so that

$$Q(z) = 0 \iff \sum_{k=1}^{K+1}\frac{\frac{-a_k}{1+\varepsilon_k^2}}{z-\varepsilon_k} = 0,$$

hence Q admits exactly K distinct roots $\varepsilon'_k \in (\varepsilon_k, \varepsilon_{k+1}) \subset \mathbb{R}$ by the intermediate value theorem (as in the proof of Proposition III.3.3 in Section III.4.2), and it is unitary due to (III.B.7). The partial fraction decomposition of g finally reads

$$g(z) = \alpha' z + C' + \sum_{k=1}^{K}\frac{a'_k}{z-\varepsilon'_k},$$

where

$$\begin{aligned}
\alpha' &= \alpha, \\
C' &= \lim_{y\to\infty} g(iy) - i\alpha' y = (1-\alpha)\sum_{k=1}^{L+1}\varepsilon_k\frac{a_k}{1+\varepsilon_k^2} \in \mathbb{R}, \\
a'_k &= \lim_{y\to 0^+} iyg(\varepsilon'_k+iy) \\
&= \left(\prod_{l=1,l\neq k}^{K}\underbrace{\left(\frac{\varepsilon'_k-\varepsilon_l}{\varepsilon'_k-\varepsilon'_l}\right)}_{>0}\right)\underbrace{(\varepsilon'_k-\varepsilon_k)(\varepsilon'_k-\varepsilon_{L+1})}_{<0}\underbrace{(\alpha+1)}_{>0} < 0.
\end{aligned}$$

This computation shows that the Nevanlinna-Riesz measure of g is a sum of K Dirac measures as described in the statement of the theorem. It is a solution to the Analytic Continuation Problem

$$AC_{\mathbb{C}_+}\left(z_n, \mathcal{W}^{-1}([\mathcal{W}(f(z_n))]^{(1)})\right)_{I\setminus\{1\}},$$

where $|I \setminus \{1\}| \geq K+2$ by assumption. By the induction hypothesis, g is the only solution to this ACP, and therefore f is the only solution to the ACP $AC_{\mathbb{C}_+}(w_n, z_n)_I$. □

IV A mathematical analysis of the discretized IPT-DMFT equations

This chapter corresponds to a preprinted article [CKP25], written in collaboration with Éric Cancès[1] and Alfred Kirsch[1] ; https://doi.org/10.48550/arXiv.2505.21287.

IV.1 Introduction

Dynamical Mean-Field Theory (DMFT) [GK92; Geo+96; MV89; ZRK93] is a numerical method for computing approximate solutions to quantum many-body problems in condensed-matter physics. It is particularly efficient for studying strong correlation effects in very large or infinite systems, such as high critical temperature superconductivity or magnetism. This method was first applied to the celebrated Hubbard model [Lie04], and then extended to more realistic models of materials, in combination with first-principle methods such as Density Functional Theory (leading to the DFT+DMFT method [Hel+01]) or the GW approximation (leading to the GW+DMFT method [SK02]).

Like the GW approximation [Hed65; AG98; CGS16], DMFT belongs to the class fo Green's function methods, in the sense that it provides an approximation to the one-body Green's function of the system, a dynamical quantity from which many physical quantities of interest can be computed. In this sense, it differs from two other methods that have been used study Hubbard models and analyzed mathematically, namely the Generalized Hartree-Fock [BLS94] method and the Density Matrix Embedding Theory (DMET) [KC12; Can+25b], which only provide an approximation of the one-body reduced density matrix, a static quantity, which can be computed from the one-body Green's function, but contains much less information than the latter.

DMFT also belongs to the class of quantum embedding methods. It is in some sense a quantum analog to the mean-field approach for the Ising model [Geo+96]. In fact, DMFT is the prototypical example of quantum embedding methods, the first one in chronological order, and the most widely used.

Like other self-consistent methods in physics, the DMFT formalism essentially consists

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of a fixed-point equation, which is solved numerically by an iterative algorithm. To fix the ideas, consider a Hubbard model with L sites. In the *single-site* DMFT setting, the set of sites is partitioned into L singletons. A DMFT iteration then goes in two steps:

Step 1. For each site of the original Hubbard model, an auxiliary Anderson Impurity Model (AIM) is solved via a so-called *impurity solver*.

Step 2. The collection of solutions from Step 1 are combined together to provide the parameters of the AIMs to solve at the next iteration.

From a numerical point of view, the main bottleneck of a DMFT loop is the first step, i.e. the resolution of the L impurity problems [BG17]. In the *translation-invariant* setting considered here, the L AIMs to be solved at each iteration, are identical copies of one another. Accurate impurity solvers are available, among which Continuous Time Quantum Monte Carlo (CTQMC) solvers [Gul+11; Wer+06] or Inchworm Monte Carlo methods [EGC20], both being supposedly exact up to statistical noise. However, their computational costs can be too high for specific applications, such as moiré heterostructures [Tan+23], so that computations in this setting must be performed using *approximate solvers*. One of the computationally cheapest (and least accurate) impurity solvers is the Iterated Perturbative Theory (IPT) solver [GK92; ZRK93; Geo+96; Van22]. We have studied its analytical properties in [CKP26], in the *single-site, translation-invariant, paramagnetic* setting, to which we will also stick in this article. For this setting, the IPT-DMFT equations read as:

$$\forall z \in \mathbb{C}_+, \quad \Delta(z) = W^T \left(z - H_\perp^0 - \Sigma(z)\right)^{-1} W, \tag{IV.1.1}$$

$$\Sigma = U^2 F_\beta^{\mathrm{IPT}}(\Delta). \tag{IV.1.2}$$

The parameters $W \in \mathbb{R}^{L-1}$ and $H_\perp^0 \in \mathbb{R}_{\mathrm{sym}}^{(L-1)\times(L-1)}$ can be computed from the physical parameters $\mathcal{G}_H$ (the Hubbard graph) and $T \in \mathbb{R}$ (the hoping parameter) according to simple formulae recalled in the following section. The parameters $U \in \mathbb{R}$ and $\beta > 0$ are respectively the on-site interaction energy and the inverse temperature. The functional F_β^{IPT} is the IPT solver. Its explicit form is given in the next section. The unknown Δ and Σ are respectively called the hybridization function and the local self-energy. Because of physical constraints related to causality, the functions $-\Delta$ and $-\Sigma$ must be *Pick functions*, i.e. analytic maps defined on the upper half-plane $\mathbb{C}_+ = \{z \in \mathbb{C}, \Im(z) > 0\}$ and valued in its closure $\overline{\mathbb{C}_+}$. This property plays a crucial role in the mathematical analysis of the IPT-DMFT equations we have performed in [CKP26]. On the one hand, it offers a mathematically rigorous and powerful setting to analyze the Analytic Continuation Problem involved in the definition of the IPT solver. On the other hand, it allows for a reformulation of the IPT-DMFT equations in terms of the Nevanlinna-Riesz measure of the $-\Delta, -\Sigma$, from which we are able to prove that Eqs (IV.1.1)–(IV.1.2) have a solution (Δ, Σ) for *any* set of parameters $\mathcal{G}_H$, T, U, β.

The purpose of this article is to analyze the finite-dimensional equations obtained by discretizing (IV.1.1)–(IV.1.2) by a collocation method on the set $(i\omega_n)_{n\in[\![0,N_\omega]\!]} \subset \mathbb{C}_+$, where $\omega_n = (2n+1)\pi/\beta$, $n \in \mathbb{Z}$, are the Matsubara frequencies, and N_ω the discretization parameter. This discretized scheme provides an approximation $\mathbf{\Delta} = (\Delta_n)_{n\in[\![0,N_\omega]\!]} \in -\overline{\mathbb{C}_+}^{N_\omega+1}$ and $\mathbf{\Sigma} = (\Sigma_n)_{n\in[\![0,N_\omega]\!]} \in -\overline{\mathbb{C}_+}^{N_\omega+1}$ of the values $(\Delta(i\omega_n))_{n\in[\![0,N_\omega]\!]}$ and $(\Sigma(i\omega_n))_{n\in[\![0,N_\omega]\!]}$ of the hybridization function and self-energy at the first $(N_\omega + 1)$ imaginary Matsubara frequencies of the upper half-plane. The unknown of the Matsubara frequency (MaF) discretization of the IPT-DMFT equations is then the pair $(\mathbf{\Delta}, \mathbf{\Sigma}) \in -\overline{\mathbb{C}_+}^{N_\omega+1} \times -\overline{\mathbb{C}_+}^{N_\omega+1}$. The derivation of these equations, which is done in detail in Section IV.2, can be sketched as follows. As Equation (IV.1.1) is local (in the sense that, for all $z \in \mathbb{C}_+$, $\Delta(z)$ depends on $\Sigma(z)$), its discretized counterpart is given by the evaluation of (IV.1.2) at the first N_ω imaginary Matsubara frequencies:

$$\forall n \in [\![0, N_\omega]\!], \quad \Delta_n = W^T \left(i\omega_n - H_\perp^0 - \Sigma_n\right)^{-1} W. \tag{IV.1.3}$$

The discrete counterpart

$$\mathbf{\Sigma} = U^2 F^{\mathrm{IPT}}_{\beta,N_\omega}(\mathbf{\Delta}) \tag{IV.1.4}$$

of the nonlocal equation (IV.1.2), is a bit more involved and will be discussed below. To fix the ideas, let us only mention at this point that $F^{\mathrm{IPT}}_{\beta,N_\omega}(\mathbf{\Delta})$ is a rational fraction in the Δ_n's and $\overline{\Delta_n}$'s, so that (IV.1.3)–(IV.1.4) can be reformulated as a set of real algebraic equations in the variables $(\Re(\Delta_n), \Im(\Delta_n))_{0\le n\le N_\omega}$. In fact, the discretized equations (IV.1.3)–(IV.1.4) are the only existing in the physics literature. The functional equations (IV.1.1)–(IV.1.2) were introduced in [CKP26] as the natural limit of the finite-dimensional equations (IV.1.3)–(IV.1.4) when N_ω goes to infinity.

In practice, Eqs (IV.1.3)–(IV.1.4) are solved numerically with iterative algorithms, typically a simple fixed-point algorithm stabilized and accelerated by possible Anderson-Pulay [Pul80; Pul82; Chu+21] or Broyden's [Bro65; Žit09] methods. Recall that the simple fixed-point algorithm is defined by an initial guess $\mathbf{\Delta}^{(0)}$ and the recurrence relation $\mathbf{\Delta}^{(n+1)} = W^T(i\omega_n - H_\perp^0 - U^2 F^{\mathrm{IPT}}_{\beta,N_\omega}(\mathbf{\Delta}^{(n)}))W$, defined according to Eqs (IV.1.3)-(IV.1.4). If this algorithm converges to a solution $\mathbf{\Delta}^{(\infty)}$ (this is observed in practice for a large set of parameters [Geo+96]), the values $(\Delta_n^{(\infty)})_{n\in[\![0,N_\omega]\!]}$ at the imaginary Matsubara frequencies $(i\omega_n)_{n\in[\![0,N_\omega]\!]}$ can be analytically interpolated to the whole upper half-plane $\mathbb{C}_+$ via several approaches such as Padé approximants, Maximum Entropy method [Geo+96] or the more recent Schur algorithm [FYG21]. In particular, based on the Stieltjes inversion formula [GT00], these interpolation methods allow for a numerical computation of (a regularization of) the *spectral function* A, which is nothing but the Nevanlinna-Riesz measure of the (negative of) the Green's function G [CKP26]. The latter quantity is fundamental in many-body physics. For instance, its behavior around the origin encodes the conductivity properties of the material. The demonstration that DMFT was able to correctly reproduce the interaction-driven metal-to-insulator transition known as Mott transition [RKZ94; Geo+96] was a major advance in computational physics theory, and

made DMFT very popular in the condensed-matter physics community. In spite of the practical importance of this method, it has not been much studied from a mathematical viewpoint. The only mathematical results we are aware of are those in our previous contribution [CKP26], which extend results by Lin, Lindsey and Schneider that can be found in [Lin19].

Main results. In this article, we study the set of discretized equations (IV.1.3)–(IV.1.4) from a theoretical and numerical viewpoints. We start in Section IV.2 by introducing and motivating the Matsubara frequency discretization of the IPT-DMFT equations under consideration. For the clarity of the mathematical analysis, we also introduce a dimensionless version of the discretized equations (IV.1.3)–(IV.1.4), in which all the quantities are expressed in units of the inverse temperature β. In Section IV.3, we then present our main theoretical results:

- **Theorem IV.3.1** is concerned with the existence of physically admissible solutions. Note that our criterion of a physically admissible solution is less stringent for the discretized equations than for the continuous equations: we impose $(-\mathbf{\Delta}, -\mathbf{\Sigma}) \in \overline{\mathbb{C}_+}^{N_\omega+1}$, which is a necessary but not sufficient condition for the Analytic Continuation Problems $\mathrm{ACP}(i\omega_n, -\Delta_n)_{n\in[\![0,N_\omega]\!]}$ and $\mathrm{ACP}(i\omega_n, -\Sigma_n)_{n\in[\![0,N_\omega]\!]}$ to have a solution [Nic16]. Although the original IPT functional F_β^{IPT} preserves the set of negative of Pick functions, its discretized version $F_{\beta,N_\omega}^{\mathrm{IPT}}$ does not map $-\overline{\mathbb{C}_+}^{N_\omega+1}$ into itself for $N_\omega > 0$. Due to this possible limitation of the MaF discretization scheme of (IV.1.1)–(IV.1.2), we were only able to prove the existence of physically admissible solutions to (IV.1.3)–(IV.1.4) in some parameter range. Namely, we prove that a solution to the discretized equations exists only for $\beta T\sqrt{\deg(\mathcal{G}_H)}$ smaller than a critical value depending on N_ω (recall that $\deg(\mathcal{G}_H) \in \mathbb{N}$ denotes the degree of the vertex-transitive Hubbard graph $\mathcal{G}_H$). This condition on the parameters ensures that the DMFT fixed-point map preserves a specific compact set, from which Brouwer fixed-point theorem permits to conclude.

- **Theorem IV.3.5**, which does not have a counterpart in our study [CKP26] of the continuous IPT-DMFT equations, states that the solution to (IV.1.3)–(IV.1.4) is unique under the additional constraint on the parameters that $\beta^2 TU\sqrt{\deg(\mathcal{G}_H)}$ is smaller than a critical value depending on N_ω. This uniqueness theorem is based on Picard fixed-point theorem, from which we also prove that the aforementioned simple iterative algorithm converges linearly toward the solution at a given rate, provided that the initial guess $\mathbf{\Delta}^{(0)}$ lies within the compact set left invariant by the discretized DMFT map.

We next observe that, much like in the continuous setting, these two results extend without major technical difficulties to infinite graphs of finite degrees and, to a certain extent, even to the celebrated "$d = \infty$" limit [MV89; GK92]. We also discuss a slightly

different discretization strategy in which a so-called "tail-fitting" procedure is carried out at each DMFT iteration. The remainder of the paper is devoted to complementing and further developing these general theoretical results.

In Section IV.4, we first focus on the special case of bipartite systems at half filling. Such systems exhibit particle-hole symmetry, allowing us to focus on purely imaginary solutions. Under this restriction, Equations (IV.1.3)–(IV.1.4) can then be reformulated as a problem of real algebraic geometry: find the solutions in $(0,1]^{N_\omega+1}$ of a set of $(N_\omega+1)$ real, sparse, polynomial equations in $(N_\omega+1)$ real variables with degree $(6M+4)$, where M is an integer depending on the adjacency matrix of the Hubbard graph. We consider in particular the Hubbard dimer (for which $L=2$ and $M=0$), and study the corresponding algebraic equations. More precisely, we characterize all solutions analytically in the low temperature regime in **Theorem** IV.4.2 for $N_\omega = 0$, and we compute numerically the number of purely imaginary solutions for $N_\omega = 1$. These two results show that, in general, the discretized IPT-DMFT equations have multiple solutions. This fact has been known in the physics community since the early days of DMFT [Geo+96]; more recently, a bifurcation analysis was carried out in [Van22].

To proceed further, we provide in Section IV.5 some numerical simulations on the Hubbard dimer for arbitrary values of N_ω. The purpose of this section is by no means to obtain new physical insights on the Hubbard model, as the Hubbard dimer can be solved analytically, but to illustrate on a simple example the properties of the MaF discretization method and the associated DMFT iteration scheme. We start by showing how DMFT predicts a metal-to-insulator-like transition in this simple system using the `TRIQS` library. In fact, these simulations reveal that the Hubbard dimer (which is the less coordinated graph, hence a system for which DMFT is not expected to be accurate) shares qualitative properties with larger graphs within this approximation. More importantly, we also show that numerical analytic continuation methods can lead to negative values of the spectral function A. To make sure that this problem is not a numerical artifact, nor due to the "tail-fitting" procedure, we developed our own implementation of the simple fixed point algorithm in `Julia`, supporting arbitrary precision calculations. The results of the simulations carried out with this homemade code are presented in Section IV.5.2. They highlight the fact that, for a large range of parameters (T,U,β) of the Hubbard model, the numerically obtained values of the hybridization function, self-energy and Green's function at the lowest $(N_\omega+1)$ positive Matsubara frequencies do not satisfy the Pick criterion [Nic16]. In other words, there exists a solution $\mathbf{\Delta},\mathbf{\Sigma}$ to Eqs. (IV.1.3)–(IV.1.4) such that the Analytic Continuation Problems $\mathrm{ACP}(i\omega_n,-\Delta_n)_{n\in[\![0,N_\omega]\!]}$, $\mathrm{ACP}(i\omega_n,-\Sigma_n)_{n\in[\![0,N_\omega]\!]}$ have no solution. This numerical finding demonstrates that the discretization studied in this article does not preserve the structure of the functional equations. The last part of this section focuses on the iterative algorithm for $N_\omega = 0$, for which we show that our result on the convergence rate holds for a certain range of parameters, and we provide a complete analysis of the convergence rate.

An appendix, containing a brief introduction to the derivation of the IPT-DMFT functional equations and a result on Green's functions of particle-hole symmetric systems, completes this article.

IV.2 Matsubara frequency discretization of IPT-DMFT equations

Let us first briefly recall the IPT-DMFT functional equations. As previously mentioned, these equations involve two Pick functions $-\Delta, -\Sigma : \mathbb{C}_+ \to \overline{\mathbb{C}_+}$. The four parameters of the IPT-DMFT equations are

- a finite, undirected, loopless and vertex-transitive graph $\mathcal{G}_H = (\Lambda, E)$. This graph is one of the three parameters of the translation-invariant Hubbard model from which the DMFT approximation is derived (Λ is the set of sites, identified with the vertices of the graph, E the set of edges connecting sites between which hopping is allowed, and $L = |\Lambda|$ is the number of sites);
- a hopping parameter $T \in \mathbb{R}$ and an on-site interaction energy $U \in \mathbb{R}$, which are the other two parameters required to define the underlying Hubbard model (in particular the Hubbard Hamiltonian);
- an inverse temperature $\beta > 0$.

The vector $W \in \mathbb{R}^{L-1}$ and the real symmetric matrix $H_\perp^0 \in \mathbb{R}_{\mathrm{sym}}^{(L-1)\times(L-1)}$ are defined from $\mathcal{G}_H$ and T by the relation

$$T \operatorname{Adj}_{\mathcal{G}_H} = \begin{pmatrix} 0 & W^T \\ W & H_\perp^0 \end{pmatrix},$$

where $\operatorname{Adj}_{\mathcal{G}_H}$ denotes the adjacency matrix of $\mathcal{G}_H$. Note here that we slightly changed notations with respect to [CKP26]: W is now a column vector. With these parameters, the IPT-DMFT functional equations are given by (IV.1.1)–(IV.1.2), where F_β^{IPT} is the so-called IPT solver, parameterized by β. For a Pick function $-\Delta$, $F_\beta^{\mathrm{IPT}}(\Delta)$ is formally defined as the (negative of) the solution to the following Analytic Continuation Problem:

$$\text{Find a Pick function } f \text{ s.t., } \forall n \in \mathbb{N},$$

$$f(i\omega_n) = \int_0^\beta e^{i\omega_n \tau} \left(\frac{1}{\beta} \sum_{n' \in \mathbb{Z}} e^{-i\omega_{n'}\tau} \left(i\omega_{n'} - \Delta(i\omega_{n'}) \right)^{-1} \right)^3 d\tau, \tag{IV.2.1}$$

where $\omega_n = (2n+1)\pi/\beta$ is the n-th Matsubara frequency for $n \in \mathbb{Z}$, and Δ is extended to negative imaginary Matsubara frequencies in accordance with the conventional extension of a Pick function to the lower half-plane [GT00]

$$\forall z \in \mathbb{C}_- := -\mathbb{C}_+, \quad \Delta(z) := \overline{\Delta(\overline{z})}. \tag{IV.2.2}$$

It is well-known that an analytic function defined on an open subset Ω of the complex plane is fully characterized by its values on a countable number of points of Ω with an accumulation point in Ω. Yet, this argument does not allow one to conclude in the present case, as the sequence of imaginary Matsubara frequencies $(i\omega_n)_{n\in\mathbb{N}}$ has no accumulation point in $\mathbb{C}_+$. Analytic Continuation Problems formulated in the set of Pick functions, also known as Nevanlinna-Pick interpolation problems [Nic16], have either no, one, or infinitely many solutions, so that Eq. (IV.2.1) does not automatically sdefine a unique Pick function. However, this is the case if the function Δ in the RHS of (IV.2.1) is the negative of a meromorphic Pick function with a finite number of poles. In addition, the so-defined map $\Delta \mapsto \Sigma$ can be uniquely extended by weak-continuity to a function F_β^{IPT} from the set of physically-admissible hybridization function to the set of physically-admissible self-energies. We refer to [CKP26] for more details on these points, as well as for a proof that Eqs (IV.1.1)–(IV.1.2) have a solution for any set of parameters.

IV.2.1 Matsubara frequency (MaF) discretization

The Matsubara frequency (MaF) discretization of the IPT-DMFT equations provides an approximation $(\mathbf{\Delta}, \mathbf{\Sigma}) \in -\overline{\mathbb{C}_+}^{N_\omega+1} \times -\overline{\mathbb{C}_+}^{N_\omega+1}$ *of the values of the hybridization function* Δ *and self-energy* Σ *at the first* $(N_\omega+1)$ *imaginary Matsubara frequencies*:

$$\mathbf{\Delta} = (\Delta_n)_{n\in[\![0,N_\omega]\!]}, \quad \Delta_n \simeq \Delta(i\omega_n), \qquad \mathbf{\Sigma} = (\Sigma_n)_{n\in[\![0,N_\omega]\!]}, \quad \Sigma_n \simeq \Sigma(i\omega_n), \tag{IV.2.3}$$

for some Matsubara frequency cutoff $N_\omega \in \mathbb{N}$, an additional parameter specific to this discretization.

As already mentioned in the introduction, Eq. (IV.1.1) is local (diagonal in the Matsubara representation in the terminology used in condensed-matter physics [Van22]), in the sense that the value of Δ at a given collocation point $z \in \mathbb{C}_+$ depends only (up to the other parameters) on the value of Σ at the same point z. This holds in particular for the first $(N_\omega+1)$ imaginary Matsubara frequencies $(i\omega_n)_{n\in[\![0,N_\omega]\!]}$, so that the discretized counterpart of the first equation is

$$\forall n \in [\![0, N_\omega]\!], \quad \Delta_n = W^T \left(i\omega_n - H_\perp^0 - \Sigma_n\right)^{-1} W. \tag{IV.2.4}$$

The particular choice of the collocation points $(i\omega_n)_{n\in[\![0,N_\omega]\!]}$ in $\mathbb{C}_+$, represented in Figure IV.1, is best motivated by the discretization of Eq. (IV.1.2). Let us detail this point. First, it is convenient to set, for all $n \in [\![0, N_\omega]\!]$,

$$\Delta_{-(n+1)} = \overline{\Delta_n} \in \overline{\mathbb{C}_+}, \quad \Sigma_{-(n+1)} = \overline{\Sigma_n} \in \overline{\mathbb{C}_+}, \tag{IV.2.5}$$

in accordance with the extension of Δ, Σ defined in Eq. (IV.2.2) and the fact that $\omega_{-(n+1)} = -\omega_n$. By choosing a subset of imaginary Matsubara frequencies, the need for analytic continuation in the IPT solver is by-passed, as the latter is precisely formulated by an Analytic Continuation Problem from the values of the solver at these points (see Eq.

(IV.2.1)). However, the values $[F_\beta^{\mathrm{IPT}}(\Delta)](i\omega_n)$, as defined in Eq. (IV.2.1), depends in principle on the values of Δ at *all Matsubara frequencies*. For this reason, we consider the following discretization, in which the Fourier series in Eq. (IV.2.1) is truncated (see Section IV.3.3 for a slightly different approach):

$$\boldsymbol{\Sigma} = U^2 F^{\mathrm{IPT}}_{\beta,N_\omega}(\boldsymbol{\Delta}), \tag{IV.2.6}$$

with

$$\forall n \in [\![0, N_\omega]\!], \quad [F^{\mathrm{IPT}}_{\beta,N_\omega}(\boldsymbol{\Delta})]_n := \int_0^\beta e^{i\omega_n \tau} \left(\frac{1}{\beta} \sum_{n'=-(N_\omega+1)}^{N_\omega} e^{-i\omega_{n'}\tau} \left(i\omega_{n'} - \Delta_{n'}\right)^{-1} \right)^3 d\tau. \tag{IV.2.7}$$

Note that the discretization of (IV.1.2) is nonlocal, in the sense that Σ_n depends on $(\Delta_{n'})_{n' \in [\![0,N_\omega]\!]}$ and not only on Δ_n. More generally, imaginary Matsubara frequencies coincide with the poles of the Fermi–Dirac function at inverse temperature β when extended to the complex plane, and therefore are natural collocation points when evaluating physical observables at positive temperature using complex-plane contour integral techniques. The following definition consolidates the content of this section.

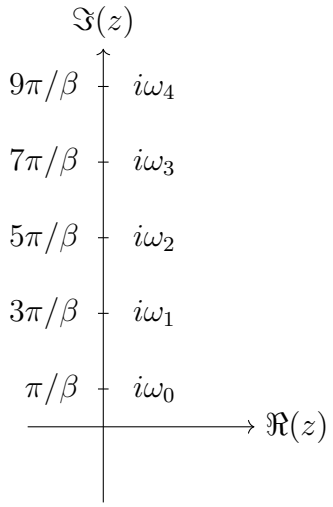


Figure IV.1: The set $(i\omega_n)_{n \in [\![0,N_\omega]\!]}$ of points in $\mathbb{C}_+$ used for the discretization of Δ and Σ with $N_\omega = 4$.

Definition IV.2.1 (MaF-discretized IPT-DMFT equations)**.** *Consider a finite, undirected, loopless, vertex-transitive graph $\mathcal{G}_H = (\Lambda, E)$ with $|\Lambda| = L \in \mathbb{N}$ and associated adjacency matrix $\mathrm{Adj}_{\mathcal{G}_H} \in \mathbb{R}^{L\times L}_{\mathrm{sym}}$. Given a hopping amplitude $T \in \mathbb{R}$, introduce $H^0_\perp \in \mathbb{R}^{(L-1)\times(L-1)}_{\mathrm{sym}}$ and $W \in \mathbb{R}^{L-1}$ defined by*

$$T\,\mathrm{Adj}_{\mathcal{G}_H} = \begin{pmatrix} 0 & W^T \\ W & H^0_\perp \end{pmatrix}.$$

Given an on-site interaction energy $U \in \mathbb{R}$, an inverse temperature $\beta \in \mathbb{R}^*_+$ and a Matsubara frequency cutoff $N_\omega \in \mathbb{N}$, the MaF-discretized IPT-DMFT equations *consist of the*

following set of equations: find $\mathbf{\Delta} = (\Delta_n)_{n\in[\![0,N_\omega]\!]} \in -\overline{\mathbb{C}_+}^{N_\omega+1}$ *and* $\mathbf{\Sigma} = (\Sigma_n)_{n\in[\![0,N_\omega]\!]} \in -\overline{\mathbb{C}_+}^{N_\omega+1}$ *such that*

$$\forall n \in [\![0, N_\omega]\!], \quad \Delta_n = W^T \left(i\omega_n - H_\perp^0 - \Sigma_n\right)^{-1} W, \tag{IV.2.8}$$

$$\mathbf{\Sigma} = U^2 F^{\mathrm{IPT}}_{\beta,N_\omega}(\mathbf{\Delta}), \tag{IV.2.9}$$

where $\omega_n = (2n+1)\pi/\beta$ *is the* n*-th Matsubara frequency and* $F^{\mathrm{IPT}}_{\beta,N_\omega}$ *is defined by* (IV.2.7).

Remark IV.2.2 (Admissible solutions to the MaF-discretized IPT-DMFT equations)**.** *Note that we restrict ourselves to solutions satisfying the condition* $\mathbf{\Delta}, \mathbf{\Sigma} \in -\overline{\mathbb{C}_+}^{N_\omega+1}$*. This is motivated by the fact that the unknown* Δ *and* Σ *of the IPT-DMFT functional equations are negatives of Pick functions. Nevertheless, for* $N_\omega > 0$*, not every element of* $-\overline{\mathbb{C}_+}^{N_\omega+1}$ *can be seen as the set of values of some negative of a Pick function at the first* $(N_\omega + 1)$ *imaginary Matsubara frequencies. As already mentioned, the problem of finding a Pick function that interpolates these values is known as the Nevanlinna–Pick interpolation problem, and the reader is referred to [CKP26; Nic16] for brief overviews of this topic. In particular, nothing ensures* a priori *that the Analytic Continuation Problems* $\mathrm{ACP}(i\omega_n, -\Delta_n)_{n\in[\![0,N_\omega]\!]}$ *and* $\mathrm{ACP}(i\omega_n, -\Sigma_n)_{n\in[\![0,N_\omega]\!]}$*, with* $\mathbf{\Delta}$ *and* $\mathbf{\Sigma}$ *solutions to* (IV.2.8)–(IV.2.9)*, have a solution. The numerical simulations reported in Section IV.5 reveal that, for a large set of parameters, this is indeed not the case. This issue is well-identified when using approximate impurity solvers in DMFT computations, as already pointed out in e.g. [FYG21; Fei+21].*

IV.2.2 Dimensionless formulation

Let us now provide a dimensionless formulation of the MaF-discretized IPT-DMFT equations, which is more suited for the present mathematical analysis. To do so, consider the following dimensionless quantities (recall that Δ and Σ have energy dimensions [CKP26], while β has the dimension of the inverse of an energy)

$$\mathbf{\Delta}' = \beta\mathbf{\Delta}, \quad \mathbf{\Sigma}' = \beta\mathbf{\Sigma}. \tag{IV.2.10}$$

Indeed, $(\mathbf{\Delta}, \mathbf{\Sigma})$ satisfies equations (IV.2.8)–(IV.2.9) if and only if $(\mathbf{\Delta}', \mathbf{\Sigma}')$ satisfies the following equations for all $n \in [\![0, N_\omega]\!]$,

$$\Delta'_n = t^2 w^T \left(i(2n+1)\pi - t h_\perp^0 - \Sigma'_n\right)^{-1} w, \tag{IV.2.11}$$

$$\Sigma'_n = u^2 \int_0^1 e^{i(2n+1)\pi\tau'} \left(\sum_{n'=-(N_\omega+1)}^{N_\omega} e^{-i(2n'+1)\pi\tau'} \left(i(2n'+1)\pi - \Delta'_{n'}\right)^{-1} \right)^3 d\tau', \tag{IV.2.12}$$

with

$$t = \beta T,\ u = \beta U, \quad w \in \{0,1\}^{L-1},\ h_\perp^0 \in \{0,1\}_{(L-1)\times(L-1)} \text{ such that } \mathrm{Adj}_{\mathcal{G}_H} = \begin{pmatrix} 0 & w^T \\ w & h_\perp^0 \end{pmatrix}.$$

Since for all $n \in [\![0, N_\omega]\!]$,

$$\int_0^1 e^{i(2n+1)\pi\tau'} \left(\sum_{n'=-(N_\omega+1)}^{N_\omega} e^{-i(2n'+1)\pi\tau'} \left(i(2n'+1)\pi - \Delta'_{n'}\right)^{-1} \right)^3 d\tau' = \tag{IV.2.13}$$

$$\sum_{n_1,n_2,n_3=-(N_\omega+1)}^{N_\omega} \prod_{j=1}^{3} \left(i(2n_j+1)\pi - \Delta'_{n_j}\right)^{-1} \int_0^1 e^{i2\pi(n-1-n_1-n_2-n_3)\tau'} d\tau' =$$

$$\sum_{\substack{n_1,n_2,n_3=-(N_\omega+1) \\ n_1+n_2+n_3=n-1}}^{N_\omega} \prod_{j=1}^{3} \left(i(2n_j+1)\pi - \Delta'_{n_j}\right)^{-1}, \tag{IV.2.14}$$

(IV.2.12) can be rewritten as

$$\boldsymbol{\Sigma}' = u^2 F_{N_\omega}(\boldsymbol{\Delta}'), \tag{IV.2.15}$$

where $F_{N_\omega} : -\overline{\mathbb{C}_+}^{N_\omega+1} \to \mathbb{C}^{N_\omega+1}$ is the rational fraction defined, for all $\mathbf{z} \in \mathbb{C}_+^{N_\omega+1}$, by

$$F_{N_\omega}(\mathbf{z})_n = \sum_{\substack{n_1,n_2,n_3=-(N_\omega+1) \\ n_1+n_2+n_3=n-1}}^{N_\omega} \prod_{j=1}^{3} \left(i(2n_j+1)\pi - z_{n_j}\right)^{-1}, \tag{IV.2.16}$$

where for all $n \in \{-(N_\omega+1), \dots, -1\}$, $z_n := \overline{z_{-(n+1)}}$ (as in (IV.2.5)).

Changing Δ', Σ' by Δ, Σ in order to simplify the notation, we end up with the following dimensionless formulation of the MaF-discretization of the IPT-DMFT equations.

Definition IV.2.3 (dimensionless MaF-discretized IPT-DMFT equations)**.** *Consider an undirected, loopless, vertex transitive graph $\mathcal{G}_H = (\Lambda, E)$ with $|\Lambda| = L \in \mathbb{N}^*$ vertices and adjacency matrix $\mathrm{Adj}_{\mathcal{G}_H}$, a hopping parameter $T \in \mathbb{R}$, an on-site interaction energy $U \in \mathbb{R}$, and an inverse temperature $\beta \in \mathbb{R}_+^*$. The dimensionless MaF-discretized IPT-DMFT equations reads as follows: find $\boldsymbol{\Delta} = (\Delta_n)_{n\in[\![0,N_\omega]\!]} \in -\overline{\mathbb{C}_+}^{N_\omega+1}$ and $\boldsymbol{\Sigma} = (\Sigma_n)_{n\in[\![0,N_\omega]\!]} \in -\overline{\mathbb{C}_+}^{N_\omega+1}$ satisfying*

$$\forall n \in [\![0, N_\omega]\!], \qquad \Delta_n = t^2 w^T \left(i(2n+1)\pi - t h_\perp^0 - \Sigma_n\right)^{-1} w, \tag{IV.2.17}$$

$$\Sigma_n = u^2 F_{N_\omega}(\boldsymbol{\Delta})_n, \tag{IV.2.18}$$

with $t = \beta T$, $u = \beta U$, F_{N_ω} given by (IV.2.16) and $w \in \{0,1\}^{L-1}$ and $h_\perp^0 \in \{0,1\}_{\mathrm{sym}}^{(L-1)\times(L-1)}$ such that

$$\mathrm{Adj}_{\mathcal{G}_H} = \begin{pmatrix} 0 & w^T \\ w & h_\perp^0 \end{pmatrix}. \tag{IV.2.19}$$

Remark IV.2.4 (comparison with other dimensionless versions of the IPT-DMFT equations)**.** *In the existing literature [Geo+96], the dimensionless quantities are rather expressed in units of the hopping parameter T, or, more precisely, in units of the half-band width*

$$D := T \,\mathrm{diam}(\sigma(\mathrm{Adj}_{\mathcal{G}_H}))/2, \tag{IV.2.20}$$

where $\mathrm{diam}(\sigma(\mathrm{Adj}_{\mathcal{G}_H}))$ *is the diameter of the spectrum* $\sigma(\mathrm{Adj}_{\mathcal{G}_H})$ *of the real-symmetric adjacency matrix. This choice is made to facilitate the comparison with experimental data, such as temperature-pressure phase diagram of transition metal oxides [Geo+96], but is less suited to the mathematical analysis provided in this article.*

IV.3 Main results

In this section, we present our main theoretical results regarding the solution(s) of the dimensionless MaF-discretized IPT-DMFT equations (IV.2.17)–(IV.2.18).

In **Theorem IV.3.1**, we are primarily concerned with the existence of solutions to the discretized equations. The main difference with the result about the continuous case presented in [CKP26] is that, in the case of the discretized equations, the function $F_{N_\omega} : \mathbb{C}^{N_\omega+1} \to \mathbb{C}^{N_\omega+1}$ does not, in general, map the set $-\overline{\mathbb{C}_+}^{N_\omega+1}$ onto itself. For this reason, our existence result holds only for a certain range of parameters. On the other hand, and this is another difference with the continuous case, we are able to prove in **Theorem IV.3.5** that the solution is unique in a further restricted range of parameters. In this case, the solution can be obtained using a simple fixed-point algorithm.

In the last part of the section, we address the extension of the results to the case of infinite graphs of finite degrees, which, as in the continuous case, does not add any technical difficulties. We also discuss a slightly different discretization scheme, incorporating "tail-fitting".

IV.3.1 Existence of solutions

To analyze the existence of solutions to the MaF discretized IPT-DMFT equations (IV.2.8)–(IV.2.9), our strategy relies essentially on Brouwer's fixed-point theorem [Sha16], which is the finite-dimensional counterpart of the Schauder fixed-point theorem used in the continuous case [CKP26].

Nevertheless, the situation differs in that the map F_{N_ω} (or equivalently $F^{\mathrm{IPT}}_{\beta,N_\omega}$) arising from the discretization of the IPT equation under study does not leave invariant the set of physically admissible solutions in general. For example, taking $N_\omega = 1$ and letting $\boldsymbol{\Delta} = (-i\pi x_0, -3i\pi x_1) \in -\overline{\mathbb{C}_+}^2$ with $x_0, x_1 \in \mathbb{R}_+$, we have

$$F_1(\boldsymbol{\Delta})_1 = -\frac{i}{((1+x_0)\pi)^3}\left(\frac{1}{9}\left(\frac{1+x_0}{1+x_1}\right)^3 + 2\frac{1+x_0}{1+x_1} - 1\right) \xrightarrow{x_1\to+\infty} \frac{i}{((1+x_0)\pi)^3} \notin -\overline{\mathbb{C}_+}. \tag{IV.3.1}$$

This type of limitation does not arise for $N_\omega = 0$, since

$$\forall \Delta \in -\mathbb{C}_+, \quad F_0(\Delta) = \frac{3}{|i\pi - \Delta|^2(i\pi - \Delta)} \in -\overline{\mathbb{C}_+}, \tag{IV.3.2}$$

and we will see that (IV.2.8)–(IV.2.9) always have a solution in this special case. Yet, for the general case $N_\omega > 0$, we need to narrow the range of parameters, particularly of the

dimensionless hopping amplitude t, as we state below.

Theorem IV.3.1 (Existence of solution to the discretized IPT-DMFT equations.)**.** *Let* $N_\omega \in \mathbb{N}^*$ *and*

$$C_{N_\omega} := \left\{ \mathbf{z} = (z_n)_{n\in[\![0,N_\omega]\!]} \in -\overline{\mathbb{C}_+}^{N_\omega+1} \;\middle|\; |z_n| \leq \frac{1}{(2n+1)\pi} \right\}. \tag{IV.3.3}$$

There exists $t_{N_\omega} > 0$ *depending only on* N_ω *such that, for all* $0 \leq t \leq t_{N_\omega}/\sqrt{\deg(\mathcal{G}_H)}$*, the MaF-discretized IPT-DMFT equations* (IV.2.17)–(IV.2.18) *have a solution*

$$(\mathbf{\Delta}, \mathbf{\Sigma}) \in \mathfrak{D}_t^{N_\omega} \times u^2 F_{N_\omega}\left(\mathfrak{D}_t^{N_\omega}\right) \subset -\overline{\mathbb{C}_+}^{N_\omega+1} \times -\overline{\mathbb{C}_+}^{N_\omega+1}, \tag{IV.3.4}$$

where $\mathfrak{D}_t^{N_\omega}$ *is given by*

$$\mathfrak{D}_t^{N_\omega} = t^2 \deg(\mathcal{G}_H) C_{N_\omega} = \left\{ \mathbf{z} = (z_n)_{n\in[\![0,N_\omega]\!]} \in -\overline{\mathbb{C}_+}^{N_\omega+1} \;\middle|\; |z_n| \leq \frac{t^2 \deg(\mathcal{G}_H)}{(2n+1)\pi} \right\}. \tag{IV.3.5}$$

Remark IV.3.2 (non-interacting setting $u = 0$)**.** *The MaF-discretized IPT-DMFT equations admit a unique solution when* $u = 0$*, given by*

$$\forall n \in [\![0, N_\omega]\!], \quad \Sigma_n = 0, \quad \Delta_n = t^2 w^T \left(i(2n+1)\pi - t h_\perp^0\right) w, \tag{IV.3.6}$$

which satisfies the condition $\mathbf{\Delta}, \mathbf{\Sigma} \in -\overline{\mathbb{C}_+}^{N_\omega+1}$*, independently of the value of* t*. This existence and uniqueness result does not extend automatically to the small* u *regime by perturbation, as* $\mathbf{\Sigma}$ *is on the boundary of the admissible domain* $-\overline{\mathbb{C}_+}^{N_\omega+1}$ *for* $u = 0$*.*

Proof of Theorem IV.3.1. Let BU_{N_ω} be defined, for all $n \in [\![0, N_\omega]\!]$, by

$$\mathrm{BU}_{N_\omega}(\mathbf{\Sigma})_n = t^2 w^T \left(i(2n+1)\pi - t h_\perp^0 - \Sigma_n\right)^{-1} w,$$

so that (IV.2.17) $\iff$ $\mathbf{\Delta} = \mathrm{BU}_{N_\omega}(\mathbf{\Sigma})$. We start with the following lemma.

Lemma IV.3.3. *For* $\mathbf{\Sigma} \in -\overline{\mathbb{C}_+}^{N_\omega+1}$*,* $\mathrm{BU}_{N_\omega}(\mathbf{\Sigma})$ *is well defined and belongs to* $-\overline{\mathbb{C}_+}^{N_\omega+1}$*. Moreover, for* $n \in [\![0, N_\omega]\!]$*,*

$$|\mathrm{BU}_{N_\omega}(\mathbf{\Sigma})_n| \leq \frac{t^2 \deg(\mathcal{G}_H)}{(2n+1)\pi}.$$

In other words, $\mathrm{BU}_{N_\omega}(\mathbf{\Sigma}) \in \mathfrak{D}_t^{N_\omega}$*, where* $\mathfrak{D}_t^{N_\omega}$ *is defined in equation* (IV.3.5)*.*

Proof. The sign of the imaginary part is a direct consequence of the definition of the imaginary part of matrices and of the fact that if f is a nonzero Pick matrix, that is $(f - f^\dagger)/(2i)$ is definite positive, then so is $-f^{-1}$. Recall that $h_\perp^0 \in \mathcal{S}_{L-1}(\mathbb{R})$, so that there exists $P \in \mathcal{M}_{L-1}(\mathbb{C})$ unitary with $h_\perp^0 = P\mathrm{diag}(\varepsilon_1, \ldots, \varepsilon_{L-1})P^\dagger$. From that, we have for all $n \in [\![0, N_\omega]\!]$,

$$t^{-2}|\mathrm{BU}_{N_\omega}(\mathbf{\Sigma})_n| \leq \sum_{k=1}^{L-1} \frac{|(wP)_k|^2}{|i(2n+1)\pi - \Sigma_n - t\varepsilon_k|} \leq \frac{\|w\|_2^2}{|(2n+1)\pi - \Im(\Sigma_n)|} \leq \frac{\deg(\mathcal{G}_H)}{(2n+1)\pi}, \tag{IV.3.7}$$

where we have used that $\|w\|_2^2 = \deg(\mathcal{G}_H)$. □

The analogue of this lemma for F_{N_ω} is more restrictive (see the counterexample given in equation (IV.3.1)), as we state now.

Lemma IV.3.4. *For all Matsubara frequency cutoff $N_\omega \in \mathbb{N}^*$, there exists $t_{N_\omega} > 0$ such that for all $0 \leq t \leq t_{N_\omega}$, $F_{N_\omega}(\mathfrak{D}_t^{N_\omega}) \subset -\overline{\mathbb{C}_+}^{N_\omega+1}$.*

Proof. Let us first prove that $\Im(F_{N_\omega}(0)_n) < 0$. For all $n \in [\![0, N_\omega]\!]$, we define

$$\varphi(n, N_\omega) = \pi^3 \Im(F_{N_\omega}(0)_n) = \sum_{\substack{n_1,n_2,n_3=-(N_\omega+1) \\ n_1+n_2+n_3=n-1}}^{N_\omega} \prod_{i=1}^{3} \frac{1}{2n_i+1}$$

Let us show that the rational number $\varphi(n, N_\omega)$ increases with N_ω: by enumerating, we have for all $n \in [\![0, N_\omega]\!]$,

$$\begin{aligned}\varphi(n, N_\omega+1) - \varphi(n, N_\omega) = {} & \frac{3}{2N_\omega+3} \sum_{\substack{n_1,n_2=-(N_\omega+1) \\ n_1+n_2+N_\omega+1=n-1}}^{N_\omega} \frac{1}{(2n_1+1)(2n_2+1)} \\ & - \frac{3}{2N_\omega+3} \sum_{\substack{n_1,n_2=-(N_\omega+1) \\ n_1+n_2-(N_\omega+2)=n-1}}^{N_\omega} \frac{1}{(2n_1+1)(2n_2+1)} \\ & - \frac{6}{(2N_\omega+3)^2(2n+1)}. \end{aligned} \tag{IV.3.8}$$

Now for the first term in the r.h.s. of (IV.3.8), we have for all $n_1, n_2 \in [\![-(N_\omega+1), N_\omega]\!]$ satisfying $n_1 + n_2 + N_\omega + 1 = n - 1$,

$$\frac{1}{(2n_1+1)(2n_2+1)} = \frac{1}{2(n-(N_\omega+1))}\left(\frac{1}{2n_1+1} + \frac{1}{2n_2+1}\right)$$

so that

$$\begin{aligned}\sum_{\substack{n_1,n_2=-(N_\omega+1) \\ n_1+n_2+N_\omega+1=n-1}}^{N_\omega} \frac{1}{(2n_1+1)(2n_2+1)} &= \frac{1}{n-(N_\omega+1)} \sum_{n_1=-(N_\omega+1)}^{n-1} \frac{1}{2n_1+1} \\ &= \frac{1}{N_\omega+1-n} \sum_{n_1=n}^{N_\omega} \frac{1}{2n_1+1},\end{aligned}$$

where the last equality comes from the fact that $\sum_{n_1=-(N_\omega+1)}^{N_\omega} \frac{1}{2n_1+1} = 0$. For the second term, we also have for all $n_1, n_2 \in [\![-(N_\omega+1), N_\omega]\!]$ satisfying $n_1 + n_2 - (N_\omega+2) = n - 1$,

$$\frac{1}{(2n_1+1)(2n_2+1)} = \frac{1}{2(n+N_\omega+2)}\left(\frac{1}{2n_1+1} + \frac{1}{2n_2+1}\right),$$

so that

$$\sum_{\substack{n_1,n_2=-(N_\omega+1) \\ n_1+n_2-(N_\omega+2)=n-1}}^{N_\omega} \frac{1}{(2n_1+1)(2n_2+1)} = \frac{1}{n+N_\omega+2} \sum_{n_1=n+1}^{N_\omega} \frac{1}{2n_1+1}.$$

We end up with

$$\varphi(n, N_\omega+1) - \varphi(n, N_\omega) = \frac{3}{2N_\omega+3}\left(\frac{1}{N_\omega+1-n} - \frac{1}{n+N_\omega+2}\right)\sum_{n_1=n+1}^{N_\omega}\frac{1}{2n_1+1} + \frac{3}{(2N_\omega+3)(2n+1)}\left(\frac{1}{(N_\omega+1-n)} - \frac{2}{2N_\omega+3}\right),$$

which is positive, hence $\varphi(n, N_\omega)$ increases with N_ω and so is $\Im(F_{N_\omega}(0)_n)$. From equations (IV.2.1) and (IV.2.7), we obtain, using (IV.2.1), (IV.2.7), [CKP26, Proposition 2.18], and the dominated convergence theorem that

$$\lim_{N_\omega\to\infty} \Im\left(F_{N_\omega}(0)_n\right) = \frac{1}{\beta}\Im\left(F_\beta^{\mathrm{IPT}}(0)(i\omega_n)\right) < 0.$$

Therefore,

$$\Im(F_{N_\omega}(0)_n) \leq \frac{1}{\beta}\Im(F_\beta^{\mathrm{IPT}}(0)(i\omega_n)) < 0.$$

This shows that $F_{N_\omega}(0) \in -\overline{\mathbb{C}_+}^{N_\omega+1}$. This set is open and F_{N_ω} is continuous on $-\overline{\mathbb{C}_+}^{N_\omega+1}$. Since $\mathfrak{D}_t^{N_\omega}$ is a compact convex set containing $z = 0$, we conclude by setting $\alpha_{N_\omega} := \sup\left\{\alpha > 0 : F_{N_\omega}(\alpha C_{N_\omega}) \subset -\overline{\mathbb{C}_+}^{N_\omega+1}\right\}$, and $t_{N_\omega} := \sqrt{\alpha_{N_\omega}}$. □

We know that $0 < t_{N_\omega}$, and for $N_\omega \geq 1$, $t_{N_\omega} < \infty$, as shown in the counter-example (IV.3.1). We conclude from the two previous lemmata, that for $t < t_{N_\omega}/\sqrt{\deg(\mathcal{G}_H)}$, the map $F_{N_\omega}^{\mathrm{DMFT}} : \mathbf{\Delta} \mapsto \mathrm{BU}_{N_\omega}(u^2 F_{N_\omega}(\mathbf{\Delta}))$ is well-defined and maps $\mathfrak{D}_t^{N_\omega}$ onto itself: indeed, for all $\mathbf{\Delta} \in \mathfrak{D}_t^{N_\omega}$, $u^2 F_{N_\omega}(\mathbf{\Delta}) \in -\overline{\mathbb{C}_+}^{N_\omega+1}$ by lemma IV.3.4, hence $\mathrm{BU}_{N_\omega}(u^2 F_{N_\omega}(\mathbf{\Delta})) \in \mathfrak{D}_t^{N_\omega} \subset -\overline{\mathbb{C}_+}^{N_\omega+1}$ by Lemma IV.3.3.

Since $F_{N_\omega}^{\mathrm{DMFT}}$ is continuous and $\mathfrak{D}_t^{N_\omega}$ is compact, $F_{N_\omega}^{\mathrm{DMFT}}$ has a fixed-point in $\mathfrak{D}_t^{N_\omega}$ by Brouwer's fixed-point theorem, which concludes the proof. □

IV.3.2 Uniqueness

In this section, we state a result which guarantees the uniqueness of the solution. We first note that, similarly to the continuous IPT-DMFT equations (IV.1.1)–(IV.1.2) (see [CKP26, Proposition 2.12]), the MaF-discretized IPT-DMFT equations (IV.2.17)–(IV.2.18) admit a unique solution in the following trivial limiting cases:

- for $u = 0$, the unique solution is given by

$$\forall n \in [\![0, N_\omega]\!], \quad \Sigma_n = 0, \quad \Delta_n = t^2 w^T \left(i(2n+1)\pi - t h_\perp^0\right) w;$$

- for $t = 0$, the unique solution is given by

$$\forall n \in [\![0, N_\omega]\!], \quad \Sigma_n = u^2 F_{N_\omega}(0)_n, \quad \Delta_n = 0.$$

In both cases, these solutions satisfy the condition $\mathbf{\Delta}, \mathbf{\Sigma} \in -\overline{\mathbb{C}_+}^{N_\omega+1}$ (see the proof of Lemma IV.3.4 for the second case). The following result covers the perturbative regime around these two limiting cases: it proves the existence and uniqueness of a solution satisfying the bounds (IV.3.4), as well as the linear convergence of the discretized IPT-DMFT simple fixed-point iteration scheme observed in numerical simulations in both perturbation regimes (see Section IV.5).

Theorem IV.3.5 (Uniqueness of the solution to the discretized IPT-DMFT equations)**.** *Let $N_\omega \in \mathbb{N}$. There exist constants $\eta_{N_\omega} > 0$ and $t_{N_\omega} > 0$ depending only on N_ω, such that if the Hubbard graph $\mathcal{G}_H$ and the parameters $t, u \in \mathbb{R}$ satisfy*

$$0 \leq t < t_{N_\omega}/\sqrt{\deg(\mathcal{G}_H)}, \tag{IV.3.9}$$

$$t^2 u^2 \deg(\mathcal{G}_H) < \eta_{N_\omega}, \tag{IV.3.10}$$

then the dimensionless MaF-discretized IPT-DMFT equations (IV.2.17)–(IV.2.18) have a unique solution $(\mathbf{\Delta}_\star, \mathbf{\Sigma}_\star)$ in $\mathfrak{D}_t^{N_\omega} \times u^2 F_{N_\omega}\left(\mathfrak{D}_t^{N_\omega}\right)$, where $\mathfrak{D}_t^{N_\omega}$ is defined in (IV.3.5). Moreover, the sequence $\left(\mathbf{\Delta}^{(k)}\right)_{k\in\mathbb{N}}$ generated by the simple fixed-point iteration scheme

$$\mathbf{\Delta}^{(0)} \in \mathfrak{D}_t^{N_\omega}, \quad \forall k \in \mathbb{N}, \quad \mathbf{\Delta}^{(k+1)} = F_{N_\omega}^{\mathrm{DMFT}}(\mathbf{\Delta}^{(k)}), \tag{IV.3.11}$$

where $F_{N_\omega}^{\mathrm{DMFT}}$ is defined by

$$\forall n \in [\![0, N_\omega]\!], \quad [F_{N_\omega}^{\mathrm{DMFT}}(\mathbf{\Delta})]_n = t^2 w^T \left(i(2n+1)\pi - t h_\perp^0 - u^2 [F_{N_\omega}(\mathbf{\Delta})]_n\right)^{-1} w, \tag{IV.3.12}$$

is well-defined and converges linearly to $\mathbf{\Delta}_\star$.

The proof is based on Picard fixed-point theorem. We endow here $\mathfrak{D}_t^{N_\omega}$ with the distance induced by the maximum norm $\|\cdot\|_\infty$. This is an arbitrary choice that suffices to make $F_{N_\omega}^{\mathrm{DMFT}}$ a contracting map on $\mathfrak{D}_t^{N_\omega}$ for the range of parameters (t, u) defined by (IV.3.9)-(IV.3.10).

Proof of Theorem IV.3.5. The first ingredient of the proof is an estimate on the bath-update map BU_{N_ω}.

Lemma IV.3.6. *Given $\mathbf{\Sigma}^1, \mathbf{\Sigma}^2 \in -\overline{\mathbb{C}_+}^{N_\omega+1}$, the following estimate holds:*

$$|\mathrm{BU}_{N_\omega}(\mathbf{\Sigma}^1)_n - \mathrm{BU}_{N_\omega}(\mathbf{\Sigma}^2)_n| \leq \frac{t^2 \deg(\mathcal{G}_H)}{((2n+1)\pi)^2} |\Sigma_n^1 - \Sigma_n^2|.$$

Proof. The resolvent identity yields, for all $n \in [\![0, N_\omega]\!]$,

$$\mathrm{BU}_{N_\omega}(\mathbf{\Sigma}^1)_n - \mathrm{BU}_{N_\omega}(\mathbf{\Sigma}^2)_n =$$

$$t^2 \left(\Sigma_n^2 - \Sigma_n^1\right) w^T \left(i(2n+1)\pi - \Sigma_n^1 - t h_\perp^0\right)^{-1} \left(i(2n+1)\pi - \Sigma_n^2 - t h_\perp^0\right)^{-1} w,$$

and we have, diagonalizing $h_\perp^0$ similarly as in the proof of Lemma IV.3.3,

$$|w^T \left(i(2n+1)\pi - \Sigma_n^1 - t h_\perp^0\right)^{-1} \left(i(2n+1)\pi - \Sigma_n^2 - t h_\perp^0\right)^{-1} w| \leq \frac{\deg(\mathcal{G}_H)}{\left((2n+1)\pi\right)^2},$$

which concludes the proof. □

In particular, Lemma IV.3.6 shows that BU_{N_ω} is Lipschitz continuous with Lipschitz constant $t^2 \deg(\mathcal{G}_H)/\pi^2$ for the distance induced by the maximum norm. Now let $t_{N_\omega} > 0$ given by Theorem IV.3.1. We know that under Assumption (IV.3.9), that is $t^2 \deg(\mathcal{G}_H) < t_{N_\omega}^2$, the compact set $\mathfrak{D}_t^{N_\omega}$ is invariant by the map $F_{N_\omega}^{\mathrm{DMFT}}$. Moreover, the function F_{N_ω} is smooth on $\mathfrak{D}_t^{N_\omega} \subset -\overline{\mathbb{C}_+}^{N_\omega+1}$(it is even holomorphic from Hartog's theorem, see Equation (IV.2.16)). Define $L_{N_\omega} = \sup\{\|\nabla F_{N_\omega}(z)\|_\infty \ ; \ z \in \mathfrak{D}_t^{N_\omega}\}$, which is then finite. For all $\Delta_1, \Delta_2 \in \mathfrak{D}_t^{N_\omega}$, we have

$$\begin{aligned}\|F_{N_\omega}^{\mathrm{DMFT}}(\Delta_1) - F_{N_\omega}^{\mathrm{DMFT}}(\Delta_2)\|_\infty &= \|\mathrm{BU}_{N_\omega}(u^2 F_{N_\omega}(\Delta_1)) - \mathrm{BU}_{N_\omega}(u^2 F_{N_\omega}(\Delta_2))\|_\infty \\ &\leq \left(\frac{tu}{\pi}\right)^2 \deg(\mathcal{G}_H)\|F_{N_\omega}(\Delta_1) - F_{N_\omega}(\Delta_2)\|_\infty \\ &\leq \left(\frac{tu}{\pi}\right)^2 \deg(\mathcal{G}_H) L_{N_\omega}\|\Delta_1 - \Delta_2\|_\infty,\end{aligned}$$

and we conclude using Picard fixed-point theorem, with $\eta_{N_\omega} = \pi^2/L_{N_\omega}$. This constant is of course dependent on the choice of the distance. □

IV.3.3 Some remarks

In this section, we elaborate on the prior main results with two major remarks. The first of them explains how both Theorems extend to the case of infinite graphs of finite degrees, similarly as with our previous contribution. Along this extension, the constraints on the parameters of our existence theorem allows to take the so-called "$d = \infty$" limit, from which DMFT has been derived [MV89; GK92]. In the second remark, we mention a slightly different set of discretized equations, which essentially differs in the discretization of the IPT equation by adding a post-processing operation known as tail-fitting, and which we introduce at this point.

Extension of the results to infinite graphs of finite degrees and beyond

In [CKP26], we extend the results on the continuous IPT-DFMT equations obtained on finite Hubbard graphs to infinite graphs of finite degrees. In this work, both **Theorems** IV.3.1 and IV.3.5 can also be stated for these graphs without special technicalities, as we detail below.

First, recall that an infinite graph of finite degree is a graph $\mathcal{G}_H = (\Lambda, E)$ such that Λ is countable, and $\deg(\mathcal{G}_H) = \sup_{\lambda\in\Lambda} \deg(\lambda)$ is finite, where by definition $\deg(\lambda) = |\{\lambda' \in \Lambda, \{\lambda, \lambda'\} \in E\}|$. In this article, we stick to the translation-invariant setting, meaning that the graph under consideration is *vertex-transitive*. Under this extra assumption, the degree is constant along the graph ($\forall \lambda \in \mathcal{G}_H, \deg(\lambda) = \deg(\mathcal{G}_H)$), and the adjacency operator $\mathrm{Adj}_{\mathcal{G}_H}$ is the self-adjoint operator on $l^2(\Lambda)$ defined by

$$\forall \phi \in l^2(\Lambda), \quad \forall \lambda \in \Lambda, \quad (\mathrm{Adj}_{\mathcal{G}_H}\phi)(\lambda) = \sum_{\lambda'\in\Lambda, \{\lambda,\lambda'\}\in E} \phi(\lambda'). \tag{IV.3.13}$$

This operator is bounded (and such that $\|\mathrm{Adj}_{\mathcal{G}_H}\| \leq \deg(\mathcal{G}_H)$), and allows one to define $w \in \{0,1\}^{\Lambda^*}$ and $h_\perp^0 = \mathrm{Adj}_{\mathcal{G}_H^*} \in \mathcal{S}(l^2(\Lambda^*))$ as in Eq. (IV.2.19), where $\mathcal{G}_H^*$ denotes the graph induced by $\mathcal{G}_H$ on Λ^*, the set of vertices deprived from one vertex (this vertex can be chosen arbitrarily under the vertex-transitivity assumption). In particular, Eq (IV.2.17) can be extended with this definition of $w, h_\perp^0$ and constitutes, together with the unmodified Eq (IV.2.18), the non-dimensional MaF-discretized IPT-DMFT equations in the setting of infinite graphs of finite degrees.

With this definition, our results extend with only minor technical refinements. More precisely, the statement of Theorems IV.3.1 and IV.3.5 remain unchanged, *with the same critical values* t_{N_ω} and η_{N_ω}. Only the proof of Lemmata IV.3.3 and IV.3.6, which are concerned with the bath-update map, need to be adapted to this setting. This can be done using the spectral decomposition of the self-adjoint operator $h_\perp^0$ for the first lemma, and the fact that the resolvent identity also holds for self-adjoint operators for the second.

This extension covers the important physical examples of d-regular trees (also known as Bethe lattices in condensed physics) and the nearest-neighbor graph of the d-dimensional hypercubic lattice $\mathbb{Z}^d$.

In addition, our results also extend to the so-called “infinite dimensional" Hubbard model. To define this model, one considers a sequence of Hubbard models defined on the d-dimensional hypercubic lattice (or d-regular tree), and considers the limit $d \to \infty$. The Green's function of this model at thermal equilibrium (which is the main quantity of interest in DMFT computations) has a non-trivial limit when d goes to infinity with the parameters t, u, and β scaling as $t(d) = t'/\sqrt{d}$, $u(d) = u$, and $\beta(d) = \beta$. This limit has been identified [GK92; MV89] and DMFT is claimed to be exact in this limit. To our knowledge, proving this claim remains an open mathematical problem under investigation by some of the authors of this article. However, noting that

- $\deg(\mathcal{G}_H) = 2d$ for the d-dimensional hypercubic lattice and $\deg(\mathcal{G}_H) = d$ for the d-regular tree,
- the compact set $\mathfrak{D}_t^{N_\omega}$ is independent of d for this scaling on these two graphs,

a direct implication of our results is the existence of a limit $\mathbf{\Delta}^{d=\infty}, \mathbf{\Sigma}^{d=\infty}$, provided $t' < t_{N_\omega}/\sqrt{2}$ for the hypercubic lattice and $t' < t_{N_\omega}$ for the d-regular tree.

Tail-fitting in the MaF-discretized IPT-DMFT equations

In this article, we focus on the discretization of the IPT-DMFT equation considered in the seminal DMFT paper [Geo+96]. As will be discussed in Section IV.5, numerical schemes associated with these equations allow to explicit physical phenomena such as interaction-driven metal-to-insulator phase transition. Yet, motivated by the enhancement of numerical simulations, other discretized equations can be considered and we present below one of them.

In the discretization of the IPT equation, we ascribed the Fourier series in Eq. (IV.2.7) to be truncated, as the discretized hybridization $\mathbf{\Delta}$ only contains the first $N_\omega + 1$ values of the hybridization function at the Matsubara frequencies. In other words, the Fourier series is approximated by a discrete Fourier transform, allowing in practice to use the Fast Fourier Transform algorithm (recall that the bottleneck of the DMFT loop is precisely the impurity solver). Still, this discretization can be slightly modified, by adding a post-processing step known as *tail-fitting*. To best motivate this additional procedure, let us recall that the hybridization function Δ and self-energy Σ of an Anderson Impurity Model, as defined rigorously in our previous contribution, read, for all $z \in \mathbb{C}_+$,

$$\Delta(z) = \sum_{k=1}^{N_\Delta} \frac{a_k}{z - \varepsilon_k}, \quad \Sigma(z) = C_\Sigma + \sum_{k=1}^{N_\Sigma} \frac{a'_k}{z - \varepsilon'_k}, \tag{IV.3.14}$$

where $a_k, a'_k > 0$ and $\varepsilon_k, \varepsilon'_k, C_\Sigma \in \mathbb{R}$. Note also that $C_\Sigma = 0$ within the IPT approximation at half-filling (see [CKP26]). If the vector $\mathbf{\Delta}$ (resp. $\mathbf{\Sigma}$) correctly approximates the values of a hybridization function (resp. self-energy) at the prescribed Matsubara frequencies, it is then expected that the high components are small, leading to potential numerical instabilities. In order to alleviate this issue, a tail-fitting procedure makes advantage of the asymptotic expansion of these two functions on the imaginary Matsubara frequencies: one has, for large Matsubara frequencies

$$\Delta(i\omega_n) \approx \frac{\sum_k a_k}{i\omega_n}, \quad \Sigma(i\omega_n) \approx \frac{\sum_k a'_k}{i\omega_n}. \tag{IV.3.15}$$

These expansions allow to fit the high components (say $N_{\text{fit}} \leq n \leq N_\omega$) of each vector with a sequence of the form $C/i\omega_n$, and to replace the high components with the values of this fit. This tail-fitting is implemented by default on the `TRIQS` version we have used for the simulation in Section IV.5, where the fit is performed with a least-square algorithm. It is particularly relevant for more realistic DMFT computations, when e.g. Continuous Time Quantum Monte Carlo solvers [Gul+11] are used, as these solvers produce noisy datas. However, as we will see in the numerical section, its use within the IPT solver is not of great importance, and we did not include it in our mathematical analysis.

IV.4 The case of bipartite systems

The purpose of this section is to investigate the special case of bipartite graphs $\mathcal{G}_H$. Examples of bipartite graphs considered in condensed matter physics include regular trees (also known as Bethe lattices [Geo+96]), the Lieb lattice, and hypercubic lattices of various dimensionalities [Geo+96]. This setting also includes periodic supercell models built from Lieb or hypercubic lattices with an even number of sites in each dimension.

The reason for this focus originates from the quantity DMFT aims at approximating, namely the Green's function of the Hubbard model in its Gibbs state. When defined

on a bipartite graph, the Hubbard model at half-filling satisfies a particle-hole symmetry [Lie04]. The latter implies is a special symmetry of the Green's function: we have, as detailed in Appendix IV.A.2, for all $z \in \mathbb{C}_+$,

$$G(z) = -G(-\overline{z})^\dagger. \tag{IV.4.1}$$

In particular, the values of the Green's function on the first imaginary Matsubara frequencies are anti-Hermitian matrices.

The following corollary ensures that the MaF discretization of the IPT-DMFT equations preserves this property in some sense as will be seen in Section IV.5.

Proposition IV.4.1 (MaF-discretization preserves particle-hole symmetry)**.** *Assume that the Hubbard graph $\mathcal{G}_H$ is* bipartite*, and that the hopping parameter t satisfies assumption* (IV.3.9)*. Then the MaF-discretized IPT-DMFT map $F_{N_\omega}^{\mathrm{DMFT}}$ defined in* (IV.3.12) *preserves purely imaginary vectors:*

$$\boldsymbol{\Delta} \in -i\mathbb{R}_+^{N_\omega+1} \cap \mathfrak{D}_t^{N_\omega} \implies F_{N_\omega}^{\mathrm{DMFT}}(\boldsymbol{\Delta}) \in -i\mathbb{R}_+^{N_\omega+1} \cap \mathfrak{D}_t^{N_\omega}. \tag{IV.4.2}$$

In particular, if the assumptions (IV.3.9)-(IV.3.10) *of Theorem IV.3.5 are satisfied, the solution $(\boldsymbol{\Delta}_\star, \boldsymbol{\Sigma}_\star)$ is purely imaginary, i.e. $\boldsymbol{\Delta}_\star, \boldsymbol{\Sigma}_\star \in -i\mathbb{R}_+^{N_\omega+1}$.*

It follows that the approximation $\mathbf{G} = (G_n)_{n\in[\![0,N_\omega]\!]}$ of the local Green's function $G_{1,1}$ at the lowest $(N_\omega+1)$ Matsubara frequencies defined by

$$G_n \simeq G(i\omega_n) = (i\omega_n - (\Delta(i\omega_n) + \Sigma(i\omega_n))^{-1} \simeq (i\omega_n - (\Delta_n + \Sigma_n)^{-1},$$

is a vector of purely imaginary numbers.

Before providing the proof of this statement, let us remark that this result is not inherent to this discretization, as a similar result can be derived for the original IPT-DMFT equations. Indeed, the IPT-DMFT map introduced in [CKP26, Section 3] preserves symmetric measures when the Hubbard graph is bipartite.

In the next section, we make use of this property to focus on purely imaginary solutions.

Proof. Let $\boldsymbol{\Delta} \in -i\mathbb{R}_+^{N_\omega+1}$. One can easily check that $\boldsymbol{\Sigma} = u^2 F_{N_\omega}(\boldsymbol{\Delta})$ is purely imaginary. If, furthermore, $\boldsymbol{\Delta} \in \mathfrak{D}_t^{N_\omega}$, with t small enough to satisfy the assumption (IV.3.9), then it has been shown in the proof of Theorem IV.3.1 that $\boldsymbol{\Sigma} \in -\overline{\mathbb{C}_+}^{N_\omega+1}$, so that $\boldsymbol{\Sigma} \in -i\mathbb{R}_+^{N_\omega+1}$. Suppose now that $\boldsymbol{\Delta} \in -i\mathbb{R}_+^{N_\omega+1} \cap \mathfrak{D}_t^{N_\omega}$.

Let $X_n \in \mathbb{R}$ be given by $iX_n = i(2n+1)\pi - \Sigma_n$ for $n \in [\![0, N_\omega]\!]$. Using the Schur complement methods, we obtain

$$F_{N_\omega}^{\mathrm{DMFT}}(\boldsymbol{\Delta})_n = \mathrm{BU}_{N_\omega}(\boldsymbol{\Sigma})_n = iX_n - \left(\left(\left(iX_n - t\mathrm{Adj}_{\mathcal{G}_H}\right)^{-1}\right)_{1,1}\right)^{-1}. \tag{IV.4.3}$$

Now, since $\mathcal{G}_H$ is bipartite [BH11], the associated adjacency matrix $\mathrm{Adj}_{\mathcal{G}_H}$ reads

$$\mathrm{Adj}_{\mathcal{G}_H} = \begin{pmatrix} 0 & a^T \\ a & 0 \end{pmatrix} \tag{IV.4.4}$$

and letting $J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, we have $J\mathrm{Adj}_{\mathcal{G}_H}J = -\mathrm{Adj}_{\mathcal{G}_H}$. This implies that x is an eigenvector of $\mathrm{Adj}_{\mathcal{G}_H}$ with eigenvalue λ if and only if Jx is also an eigenvector with eigenvalue $-\lambda$. The matrix $\mathrm{Adj}_{\mathcal{G}_H}$ being symmetric real, it is therefore diagonalizable as follow:

$$\mathrm{Adj}_{\mathcal{G}_H} = PDP^t \text{ with } \begin{cases} D = \mathrm{diag}(\lambda_1, \dots \lambda_p, -\lambda_1, \dots, -\lambda_p, 0, \dots, 0), \\ P = (x_1 | \dots | x_p | Jx_1 | \dots | Jx_p | \dots). \end{cases} \tag{IV.4.5}$$

Using this diagonalization, we have

$$\left(\left(iX_n - \mathrm{Adj}_{\mathcal{G}_H} \right)^{-1} \right)_{1,1} = \sum_{j=1}^{p} \frac{P_{1,j}^2}{iX_n - \lambda_j} + \sum_{j=1}^{p} \frac{P_{1,j+p}^2}{iX_n + \lambda_j} + \sum_{j=2p+1}^{L} \frac{P_{1,j}^2}{iX_n} \tag{IV.4.6}$$

$$= i \left(\sum_{j=1}^{p} \frac{2P_{1,j}^2 X_n}{X_n^2 + \lambda_j^2} - \sum_{j=2p+1}^{L} \frac{P_{1,j}^2}{X_n} \right) \in i\mathbb{R}, \tag{IV.4.7}$$

making use of the fact for all $j \in [\![1, p]\!]$, $P_{1,j} = x_{j,1} = (JxS_j)_1 = P_{1,j+p}$, proving thereby the first claim. Now, if the assumptions (IV.3.9)-(IV.3.10) of Theorem IV.3.5 are satisfied, the sequence $(\boldsymbol{\Delta}^{(k)})_{k\in\mathbb{N}} \in -i\mathbb{R}_+^{N_\omega+1}$ defined by

$$\boldsymbol{\Delta}^{(0)} \in -i\mathbb{R}_+^{N_\omega+1} \cap \mathfrak{D}_t^{N_\omega}, \quad \forall k \in \mathbb{N},\ \boldsymbol{\Delta}^{(k+1)} = F_{N_\omega}^{\mathrm{DMFT}}(\boldsymbol{\Delta}^{(k)}) \tag{IV.4.8}$$

converges toward the unique solution $\boldsymbol{\Delta}_*$, therefore $\boldsymbol{\Delta}_* \in -i\mathbb{R}_+^{N_\omega+1}$, and the same holds for the self-energy: $\boldsymbol{\Sigma}_* \in -i\mathbb{R}_+^{N_\omega+1}$. □

IV.4.1 Algebraic structure of MaF-discretized IPT-DMFT for generic bipartite systems

In this section, we characterize the purely imaginary and physically admissible solutions to the dimensionless MaF-discretized IPT-DMFT equations (IV.2.17)–(IV.2.18) for bipartite systems. After showing that these equations boil down to a set of sparse algebraic equations for generic bipartite systems, we investigate more in depth the case of the Hubbard dimer.

Seeking the purely imaginary, physically admissible solutions to (IV.2.17)–(IV.2.18) amounts to seeking solutions of the form $\boldsymbol{\Delta} = (-i\pi y_0, -i3\pi y_1, \cdots, -i(2N_\omega+1)\pi y_{N_\omega})$, with $\mathbf{y} = (y_0, \cdots, y_{N_\omega}) \in \mathbb{R}_+^{N_\omega+1}$. It is convenient to rewrite this problem in terms of the auxiliary variables

$$\mathbf{x} = (x_0, \cdots, x_{N_\omega}) \quad \text{where} \quad x_n := \frac{1}{1+y_n} \quad \text{for all } 0 \le n \le N_\omega.$$

For bipartite systems, the spectrum of $h_\perp^0$ has a special structure. Moreover, removing a vertex from a bipartite graph does not change the bipartite character. The diagonalization is therefore the same as for the total adjacency matrix (IV.4.5): if L is even, the

dimension of $h_\perp^0$ is odd, 0 is an eigenvalue with odd multiplicity, and the nonzero eigenvalues come in pairs of real eigenvalues with opposite signs. If L is odd, the dimension of $h_\perp^0$ is even and the eigenvalues come in pairs of real eigenvalues with opposite signs (0 can be an eigenvalue, but if it is, it has an even multiplicity). In addition if ν is a positive eigenvalue of $h_\perp^0$, then the spectral projectors Π_ν and $\Pi_{-\nu}$ on $\ker(h_\perp^0 - \nu)$ and $\ker(h_\perp^0 + \nu)$ respectively verify $|\Pi_\nu v| = |\Pi_{-\nu} v|$, with $v = w/|w|$, that is $w = v\sqrt{\deg(\mathcal{G}_H)}$. Denoting Π_0 the orthogonal projector on $\ker h_\perp^0$, which is equal to zero if $0 \notin \sigma(h_\perp^0)$, it follows that (IV.2.17)–(IV.2.18) can be rewritten as

$$\frac{1 - x_n}{x_n a \deg(\mathcal{G}_H)} = \sum_{\substack{\nu \in \sigma(h_\perp^0) \\ \nu > 0}} \frac{2|\Pi_\nu v|^2 \left((2n+1)^2 - bP_{n,N_\omega}(\mathbf{x})\right)}{\left((2n+1)^2 - bP_{n,N_\omega}(\mathbf{x})\right)^2 + (2n+1)^2 a\nu^2} + \frac{|\Pi_0 v|^2}{(2n+1)^2 - bP_{n,N_\omega}(\mathbf{x})},$$

where the parameters a and b are related to t and u by

$$a = \frac{t^2}{\pi^2} \quad \text{and} \quad b = \frac{u^2}{\pi^4},$$

and where the homogeneous cubic polynomials P_{n,N_ω}, $N_\omega \in \mathbb{N}$, $0 \le n \le N_\omega$, are given by

$$\begin{aligned}
\frac{P_{n,N_\omega}(\mathbf{x})}{(2n+1)} = {} & \mathbb{1}_{n \ge 1} \sum_{\substack{n_1,n_2,n_3 = 0 \\ n_1+n_2+n_3 = n-1}}^{N_\omega} \prod_{j=1}^{3} \frac{1}{2n_j+1} x_{n_j} \\
& - 3 \sum_{\substack{n_1,n_2 = 0 \\ n \le n_1+n_2 \le N_\omega + n}}^{N_\omega} \frac{1}{2n_1+1} \frac{1}{2n_2+1} \frac{1}{2n_1+2n_2-2n+1} x_{n_1} x_{n_2} x_{n_1+n_2-n} \\
& + 3\mathbb{1}_{n \le N_\omega - 1} \sum_{\substack{n_1,n_2 = 0 \\ n_1+n_2 \le N_\omega - (n+1)}}^{N_\omega} \frac{1}{2n_1+1} \frac{1}{2n_2+1} \frac{1}{2n_1+2n_2+2n+3} x_{n_1} x_{n_2} x_{n_1+n_2+n+1}.
\end{aligned}$$

The above equations read as a system of $(N_\omega + 1)$ scalar algebraic equations in the $(N_\omega + 1)$ scalar variables $(x_0, \ldots, x_{N_\omega})$. Each of these equations can be written as

$$\sum_{j=0}^{2M+1} (c_{j,n,N_\omega} x_n + d_{j,n,N_\omega}) P_{n,N_\omega}(\mathbf{x})^j = 0, \tag{IV.4.9}$$

where M is the number of distinct positive eigenvalues of $h_\perp^0$, and c_{j,n,N_ω} and d_{j,n,N_ω} are real coefficients, which are themselves polynomials in a and b (or equivalently t and u). Note that M is bounded by $\lfloor (L-1)/2 \rfloor$. As the polynomials P_{n,N_ω} are cubic, homogeneous, the total degree in $\mathbf{x}$ of each of these scalar equations is exactly $6M+4$ for any positive values of a and b. As the number of monomials in each P_{n,N_ω} is of the order of N_ω^2, the number of monomials in (IV.4.9) is of order at most N_ω^{4M+2}. For large values of N_ω, this number is small compared with the total number $\binom{N_\omega + 6M + 5}{6M+4} \sim N_\omega^{6M+4}$ of monomials in a generic polynomial equation of degree $(6M+4)$ with $(N_\omega + 1)$ variables.

A bound on the number of solutions for fixed parameters t, u can therefore be obtained using results from symbolic algebraic geometry, while the set of solutions can be explored numerically by homotopy methods from numerical algebraic geometry, although these methods are limited to relatively small values of N_ω.

IV.4.2 Hubbard dimer

The Hubbard dimer is the simplest non trivial case. Here $L = 2$ and the graph consists of two vertices and one edge. It is clearly bipartite, and we have $\mathrm{Adj}_{\mathcal{G}_H} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, so that $w = 1$ and $h^0_\perp = 0$.

Low-temperature regime for $N_\omega = 0$

In this section, we are interested in the simplest case $N_\omega = 0$, which corresponds to an approximation with only one Matsubara frequency: the unknown is a pair $(\Delta, \Sigma) \in -\overline{\mathbb{C}_+}^2$. Combining (IV.2.17)–(IV.2.18) with (IV.3.2), the problem reduces to finding a solution $\Delta \in -\overline{\mathbb{C}_+}$ to the scalar equation

$$\Delta = t^2 \left(i\pi - \frac{3u^2}{|i\pi - \Delta|^2 (i\pi - \Delta)} \right)^{-1}, \tag{IV.4.10}$$

As such, equation (IV.4.10) can be seen as a polynomial equation in the real variables $(\Re(\Delta), \Im(\Delta))$ of degree 4.

We would like to investigate the solutions to this equation in the physical setting when the hopping parameter T and the on-site interaction parameter U are fixed, and the temperature β^{-1} goes to zero. In our dimensionless setting, this corresponds to

$$t, u \to +\infty \text{ with } \frac{u}{t} = \frac{U}{T} =: \alpha > 0 \text{ fixed},$$

and it is therefore convenient to rewrite (IV.4.10) as

$$\Delta = t^2 \left(i\pi - \frac{3\alpha^2 t^2}{|i\pi - \Delta|^2 (i\pi - \Delta)} \right)^{-1}. \tag{IV.4.11}$$

Theorem IV.4.2. *The solutions to equation (IV.4.11) are purely imaginary. The number of solutions that verify $\Delta \in -\overline{\mathbb{C}_+}$ depends on the parameter α. In the low-temperature ($\alpha > 0$ fixed, t sufficiently large), we have that*

- *if $0 < \alpha < \frac{3\pi}{2}$, then (IV.4.11) has a unique solution $\Delta^\infty_{\alpha,t}$ in $-\overline{\mathbb{C}_+}$ and it holds $i\Delta^\infty_{\alpha,t} \underset{t\to+\infty}{\longrightarrow} +\infty$;*
- *if $\alpha > \frac{3\pi}{2}$, then (IV.4.11) has exactly three solutions in $-\overline{\mathbb{C}_+}$. One solution goes to infinity with t, and the other two solutions converge to distinct nonzero values $\Delta^1_\alpha, \Delta^2_\alpha \in i\mathbb{R}_-$.*

In each case, the branch of solutions escaping to infinity has the following asymptotic behavior:

$$\Delta^{\infty}_{\alpha,t} = -i\frac{t^2}{\pi} + i\frac{3\pi\alpha^2}{t^2} + o\left(\frac{1}{t^2}\right). \tag{IV.4.12}$$

Proof. Setting

$$\epsilon := t^{-1}, \quad z_\epsilon := -i\pi^{-1}\Delta, \quad \lambda := \frac{3\alpha^2}{\pi^2} > 0,$$

the problem consists in finding the solutions $z_\epsilon \in \mathbb{C}$ to the equation

$$z_\epsilon = \frac{(z_\epsilon - 1)|z_\epsilon - 1|^2}{\lambda - \epsilon^2\pi^2(z_\epsilon - 1)|z_\epsilon - 1|^2}, \quad \text{with} \quad \mathrm{Re}(z_\epsilon) \leq 0, \tag{IV.4.13}$$

in the limit $\epsilon \to 0^+$. Let us first study the limiting equation obtained by setting $\epsilon = 0$, namely

$$z_0 = \frac{(z_0 - 1)|z_0 - 1|^2}{\lambda}. \tag{IV.4.14}$$

This equation can be rewritten as

$$z_0 = \frac{|z_0 - 1|^2}{|z_0 - 1|^2 - \lambda} \in \mathbb{R}. \tag{IV.4.15}$$

This shows that any solution z_0 to (IV.4.14) is real and that any solution to (IV.4.14) is therefore a solution to the polynomial equation

$$(z_0 - 1)^3 = \lambda z_0. \tag{IV.4.16}$$

Thus, $x := z_0 - 1$ is a real solution to $x^3 + px + q = 0$ with $p = q = -\lambda < 0$. The discriminant of this equation is $-\lambda^2(27 - 4\lambda)$. It follows that

- if $0 < \lambda < \frac{27}{4}$, then (IV.4.16) has one simple real root and two non real conjugated complex roots. For $\lambda = 0$, the three roots are equal to 1. When λ increases, there are two branches of complex conjugated roots, and the third root is real and increases. In particular, it is positive;
- if $\lambda = \frac{27}{4}$, then the roots of (IV.4.16) are $z_0 = 4$ (simple root) and $z_0 = -1/2$ (double root);
- if $\lambda > \frac{27}{4}$, then (IV.4.16) has three distinct real roots, one positive greater than 1 as above, and the two other solutions z^1_λ, z^2_λ are negative, because they behave continuously with λ, they converge to $-1/2$ when $\lambda \to 27/4$ and they do not cross zero, which is not a solution to (IV.4.16).

Consider now the case when $\epsilon > 0$. Recall that we are looking for solutions to (IV.4.13), which writes

$$z_\epsilon = \frac{(z_\epsilon - 1)|z_\epsilon - 1|^2}{\lambda - \epsilon^2\pi^2(z_\epsilon - 1)|z_\epsilon - 1|^2}, \quad \text{with} \quad \mathrm{Re}(z_\epsilon) < 0. \tag{IV.4.17}$$

Suppose z_ϵ is a solution to (IV.4.17). Let $b_\epsilon = \frac{\lambda}{|z_\epsilon - 1|^2} + \epsilon^2\pi^2 - 1 \in \mathbb{R}$. The discriminant of the quadratic polynomial equation $\epsilon^2\pi^2 z^2 - b_\epsilon z - 1 = 0$ is positive and z_ϵ is a root of this polynomial. Hence z_ϵ is real. We are then looking for real solutions to the quartic polynomial equation

$$\left(1 + \epsilon^2\pi^2 z_\epsilon\right)(z_\epsilon - 1)^3 = \lambda z_\epsilon. \tag{IV.4.18}$$

The above equation has four complex roots (counting multiplicities) which depend continuously on the parameter $\epsilon > 0$ (for fixed λ). As the product of the roots of (IV.4.18) is equal to $-\frac{1}{\pi^2\epsilon^2}$ and as the four roots of this polynomial are bounded away from zero uniformly for $0 < \epsilon \leq 1$, there is a branch of solutions $\epsilon \mapsto z^\infty_{\lambda,\epsilon}$ that escapes to infinity when $\epsilon \to 0^+$, namely $|z^\infty_\epsilon| \underset{\epsilon\to 0^+}{\longrightarrow} \infty$. Then we necessarily have $(1 + \epsilon^2\pi^2 z^\infty_{\lambda,\epsilon}) \to 0$ when $\epsilon \to 0^+$ because of (IV.4.18). We can thus write $z^\infty_{\lambda,\epsilon} = -\frac{y_\epsilon}{\epsilon^2\pi^2}$ with y_ϵ satisfying $y_\epsilon \to 1$ when $\epsilon \to 0^+$ and solution to the quartic polynomial equation

$$1 - y_\epsilon = (\epsilon^2\pi^2)^2 \frac{\lambda y_\epsilon}{(y_\epsilon + \epsilon^2\pi^2)^3} \underset{\epsilon\to 0^+}{\sim} \lambda\pi^4\epsilon^4.$$

We finally obtain

$$z^\infty_{\lambda,\epsilon} = -\frac{1}{\pi^2\epsilon^2} + \lambda\pi^2\epsilon^2 + o(\epsilon^2). \tag{IV.4.19}$$

The four branches of solutions therefore behave as follows: one of them, namely z^∞_ϵ, goes to infinity when $\epsilon \to 0^+$ and behaves as (IV.4.19), while the other three remain in a compact set at positive distance from the origin, and converge when $\epsilon \to 0^+$ to the three roots to the limiting equation (IV.4.16). We then have to discuss whether these three bounded roots are solutions to (IV.4.17) or not, depending on the value of λ.

- For $\lambda < 27/4$, the roots of (IV.4.16) are either non real or positive. Hence, for ϵ small enough, the bounded roots of (IV.4.18) are either non real or positive as well, so that they are not real negative. Therefore, they are not solution to (IV.4.17). The only solution to (IV.4.17) is then $z^\infty_{\lambda,\epsilon}$ for ϵ small enough, which is real, because else its complex conjugate would also be a solution to (IV.4.18).

- For $\lambda > 27/4$, the roots of the limiting equation (IV.4.16) are three distinct real roots: $z^1_\lambda < z^2_\lambda < 0$ and a third root greater than 1. Since the bounded roots of (IV.4.18) are either real or complex conjugated and they converge to the roots of the limiting equation, for ϵ small enough, they must all be real. Moreover, two of them are real negative and bounded, one of them is positive bounded and the last one is $z^\infty_{\lambda,\epsilon}$, which is real for the same reason as above.

□

Purely imaginary solutions for the Hubbard dimer

Let us conclude this section with the research of purely imaginary solutions in the general case $N_\omega \geq 1$, still focusing on the simple case of the Hubbard dimer ($L = 2$, $h^0_\perp = 0$), for

which (IV.4.9) boils down to solving the system of quartic polynomial equations

$$(x_n - 1)\left((2n+1)^2 - bP_{n,N_\omega}(\mathbf{x})\right) + ax_n = 0, \quad \forall\, 0 \leq n \leq N_\omega. \tag{IV.4.20}$$

Whenever $b > 0$, that is $u > 0$, each of these polynomial equations contains a number of monomials exactly equal to $(M_{n,N_\omega} + 1)$, where M_{n,N_ω} is the number of monomials in P_{n,N_ω}, so it scales in $O(N_\omega^2)$. The system of quartic algebraic equations (IV.4.20) has thus a sparsity of order N_ω^{-2} for large values of N_ω.

As a matter of example of symbolic and numerical algebraic geometry methods, let us consider the simple case when $N_\omega = 1$. In this case, (IV.4.20) reads: find the solutions (x_0, x_1) in $(0,1]^2$ to the system of quartic equations

$$(x_0 - 1)(1 - bP_{0,1}(x_0, x_1)) + ax_0 = 0, \tag{IV.4.21}$$
$$(x_1 - 1)(9 - bP_{1,1}(x_0, x_1)) + ax_1 = 0, \tag{IV.4.22}$$

with

$$P_{0,1}(x_0, x_1) = -3x_0^3 + x_0^2 x_1 - \frac{2}{3}x_0 x_1^2,$$
$$P_{1,1}(x_0, x_1) = 3x_0^3 - 6x_0^2 x_1 - \frac{1}{3}x_1^3.$$

The dimension of the algebraic variety defined as the set of solutions of (IV.4.21)-(IV.4.22) for a given set of parameters $(a, b) \in \mathbb{R}_+ \times \mathbb{R}_+$ can be computed using methods from symbolic algebraic geometry. The dimension returned by the `Macaulay2` software [GS] is 0 showing that (IV.4.21)-(IV.4.22) has discrete complex solutions for fixed parameters (a, b). These solutions can be approximated numerically using tools from numerical algebraic geometry. The homotopy methods implemented in the `breiding2018homotopycontinuationjl` Julia package [BT18] indicate that in the range of parameters $(a, b) \in [0, 10] \times [0, 25]$, (IV.4.21)-(IV.4.22) have 16 complex solutions and that the number of admissible solutions satisfying $(x_0, x_1) \in [0, 1]^2$ varies from 1 to 5 (Figure IV.2). The separation line $b = 3(1 + a/10)^3$ corresponds to solutions crossing the bottom boundary of the square $(0, 1] \times (0, 1]$. Indeed, the system (IV.4.21)-(IV.4.22) admits a solution of the form $(x_0, 0)$ if and only if $b = 3(1 + a/10)^3$. The other separation lines correspond to pairs of conjugated solutions merging on the real line in both variables, so that the number of solutions on both sides differs by 2.

IV.5 Iterative algorithm for the Hubbard dimer

In this section, we report numerical simulations of the MaF-discretized IPT-DMFT equations for the Hubbard dimer (the graph contains two vertices linked by one edge: $\Lambda = \{1, 2\}, E = \{\{1, 2\}\}$), which is the simplest non trivial bipartite and vertex-transitive graph.

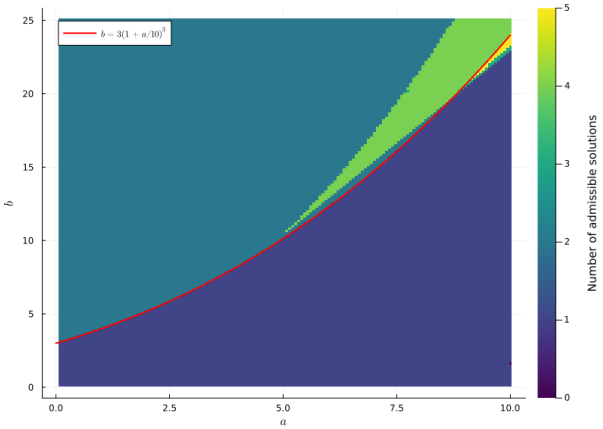


Figure IV.2: Number of admissible solutions $(x_0, x_1) \in (0, 1]^2$ to (IV.4.21)-(IV.4.22) in the range of parameters $(a, b) \in [0, 10] \times [0, 25]$. This gives the number of purely imaginary solutions to the dimensionless MaF-discretized IPT-DMFT equations (IV.2.17)–(IV.2.18) for the Hubbard dimer in the range of parameters $(t, u) \in [0, \sqrt{10}\pi] \times [0, 5\pi^2]$.

In spite of its simplicity, this setting shares qualitative properties with other (potentially infinite) graphs. In particular, we provide numerical results, based on a criterion in terms of the support of the spectral measure of the system, that can be interpreted as an interaction-driven conductor-to-insulator transition for infinite systems. This is conducted in the next section, in which we use the Python/C++ library `TRIQS` 3.1.0 [Par+15], and take advantage of its analytic continuation functionalities.

Unfortunately, this library is not suited to perform simulations with a small number of Matsubara's frequencies (N_ω needs to be larger than 50 for the tail-fitting procedure, described in Section IV.3.3, to work properly). In addition to this, the discretized equations are slightly modified by the addition of the tail-fitting procedure, for reasons mentioned previously (see also Remark IV.5.2). In order to exemplify the results of Section IV.4.2 and to remain as close as possible to the analysis provided in Section IV.3, the results reported below have been obtained with our in-house arbitrary-precision Julia implementation.

With this algorithm, we first perform a series of simulations revealing that the simple fixed point algorithm may converge to a solution that does not fulfill the Pick criterion, meaning that there is no Pick function exactly interpolating the computed values at the $(N_\omega + 1)$ lowest Matsubara frequencies. Because of Remark IV.2.2, this observation indicates that the MaF-discretization do not preserve the fact that $-\Delta, -\Sigma$ are Pick functions [CKP26, Section 3].

In the last part of this section, we focus on the iterative algorithm for the aforementioned studied case $N_\omega = 0$, still on the Hubbard dimer. For this case, we test the validity of the rate of convergence estimated by Theorem IV.3.5, and provide an explanation of the gap, for the parameter range where there is one, in this special case.

IV.5.1 Conductor-to-insulator-like transition using the `TRIQS` library

In this section, we show that the IPT-DMFT iterative algorithm, when applied to the Hubbard dimer, provides results that share qualitative properties with the observed conductor-to-insulator transition in more extended graphs. The simulations are performed using the `TRIQS` library [Par+15].

Before going any further, let us specify the meaning of the terms conductor and insulators in Green's function computations. The system is called a *conductor* if the Fermi level (zero in this setting) is in the support of the spectral function [MRC16] , i.e. of the Nevanlinna-Riesz measure of $-G$, and an *insulator* otherwise. Even if no phase transition is expected to occur for finite systems, we will see that the results obtained with the IPT-DMFT algorithm show such a transition according to this criterion (see Remark IV.5.1 below).

To study the conduction properties of a system with the MaF-discretized IPT-DMFT, a strategy is as follows:

1. **Solve the MaF-discretized IPT-DMFT equations iteratively**. As a self-consistent method, the DMFT equations are usually solved using *an iterative scheme*: for the MaF-discretized IPT-DMFT setting, we implement the simple iterative scheme described in Theorem IV.3.5, namely, for a given $N_{\text{iter}} \in \mathbb{N}^*$, we compute sequences $\left(\mathbf{\Delta}^{(k)}\right)_{k\in[\![0,N_{\text{iter}}]\!]}$ defined by

$$\mathbf{\Delta}^{(0)} \in \mathfrak{D}_t^{N_\omega}, \quad \forall k \in \mathbb{N}, \quad \mathbf{\Delta}^{(k+1)} = F_{N_\omega}^{\text{DMFT}}(\mathbf{\Delta}^{(k)}),$$

   for various initial guesses $\mathbf{\Delta}^{(0)}$. The number of iterations N_{iter} is chosen to ensure that for the considered set of parameters T, U, β, $\mathbf{\Delta}^{(N_{\text{iter}})} \approx \mathbf{\Delta}^{(\infty)}$ reaches a stationary value (see Remark IV.5.2 below). We observe that the algorithm converges linearly toward a stationary solution for the two extreme initial guesses considered: $\mathbf{\Delta}^{(0)} = 0$ and $\mathbf{\Delta}_n^{(0)} = 1/(i\omega_n)$ (Figure IV.4). These initial values are chosen so that $\mathbf{\Delta}^{(0)} \in \mathfrak{D}_t^{N_\omega}$ in both cases and are extreme in the sense that they belong to the boundary of $\mathfrak{D}_t^{N_\omega}$. Note that after this linear convergence, the solution reaches an almost stationary value, with oscillations of amplitude between 10^{-16} and 10^{-19} in the 2-norm.

2. **Compute the MaF-discretized IPT-DMFT local Green's function G**. With the converged solution in the variable $\mathbf{\Delta}$, we extract the associated self-energy $\mathbf{\Sigma}^{(\infty)} = U^2 F_{\beta,N_\omega}^{\text{IPT}}(\mathbf{\Delta}^{(\infty)})$, and an approximation $\mathbf{G} \in -\overline{\mathbb{C}_+}^{N_\omega+1}$ of the discretization of the local Green's function $G_{1,1}$, defined by

$$\forall n \in [\![0, N_\omega]\!], \quad G_n = \left(i\omega_n - \Delta_n^{(\infty)} - \Sigma_n^{(\infty)}\right)^{-1}.$$

3. **Find an interpolation G of $\mathbf{G}$ and extract the spectral function A.** We seek an analytic interpolation G of $\mathbf{G}$ to the whole upper half-plane $\mathbb{C}_+$, namely we solve

the following Nevanlinna-Pick interpolation problem [Nev19; Nic94; Nic16]

$$\text{Find} - G : \mathbb{C}_+ \to \overline{\mathbb{C}_+} \text{ analytic, such that, } \forall n \in [\![0, N_\omega]\!], \quad G(i\omega_n) = G_n. \quad \text{(IV.5.1)}$$

Once G is found (provided a solution to this problem exists, see the following section), we can get the values of the spectral function A by taking the limit of G on the real line, using the Stieljes inversion formula [GT00]: recall that for f a Pick matrix given, for all $z \in \mathbb{C}_+$, by

$$f(z) = \int_{\mathbb{R}} \frac{1}{\varepsilon - z} d\mu(\varepsilon),$$

with μ a positive matrix-valued Borel finite measure, we have for all $x, y \in \mathbb{R}$,

$$\frac{1}{2}\mu(\{x\}) + \frac{1}{2}\mu(\{y\}) + \mu((x,y)) = \frac{1}{\pi} \lim_{\eta \to 0^+} \int_x^y \Im(f(\varepsilon + i\eta)) d\varepsilon.$$

In particular, assuming the spectral function A is absolutely continuous with respect to the Lebesgue measure (which seems to hold in the examples below), and denoting by ρ its density, we have

$$\rho(\varepsilon) = -\frac{1}{\pi} \lim_{\eta \to 0^+} \Im(G(\varepsilon + i\eta)).$$

In practice, several algorithms exist to perform this analytic continuation [Gub+91; JG96; FYG21; Fei+21; HGL23], which, although theoretically well founded, suffer from ill-conditioning issues [Gub+91; JG96].

For the present purpose, we used the `set_from_pade` algorithm, which is based on Pade approximants, for the reason that it is the most widely used interpolation method in this application field. Given the MaF-discretized Green's function $\mathbf{G}$, it returns the values $\rho_{\mathbb{R}}$ of an interpolation of the above points on a given set of points on the real line. The latter set is a regular mesh of $[\varepsilon_{\min}, \varepsilon_{\max}]$ consisting in N_{mesh} points.

Let us mention that, contrary to other methods developed recently in [FYG21], `set_from_pade` does not ensure *a priori* that the analytic continuation is indeed valued in $-\overline{\mathbb{C}_+}$, and may lead to results with negative values of $\rho_{\mathbb{R}}$. This issue is investigated in the next section.

Discretization parameters Discretization parameters are chosen such that the discretization error due to the integral performed in (IV.2.7) can be considered as negligible: we always take a number of points N_τ associated to the discretization of $(0, \beta)$ large compared to N_ω. Another reason for neglecting numerical quadrature errors is that this integral is actually performed using the `Fourier` method with tail-fitting. For the simulations to come, we took the following parameters:

$$N_\tau = 1000, \quad N_\omega = 100 \quad \left(\frac{N_\tau}{N_\omega} = 10\right).$$

We plot in Figure IV.3 the results of this algorithm for the following parameters:

$$\beta = 1, \quad T = 1, \quad N_{\text{iter}} = 50, \quad \boldsymbol{\Delta}^{(0)} = (i\omega_n)^{-1}_{n \in [\![0, N_\omega]\!]}, \quad -\varepsilon_{\min} = \varepsilon_{\max} = 10, \quad N_{\text{mesh}} = 1000, \tag{IV.5.2}$$

while varying U from 2 to 10. As annouced, the main features are similar to what is obtained for the so-called Bethe lattice [Geo+96]:

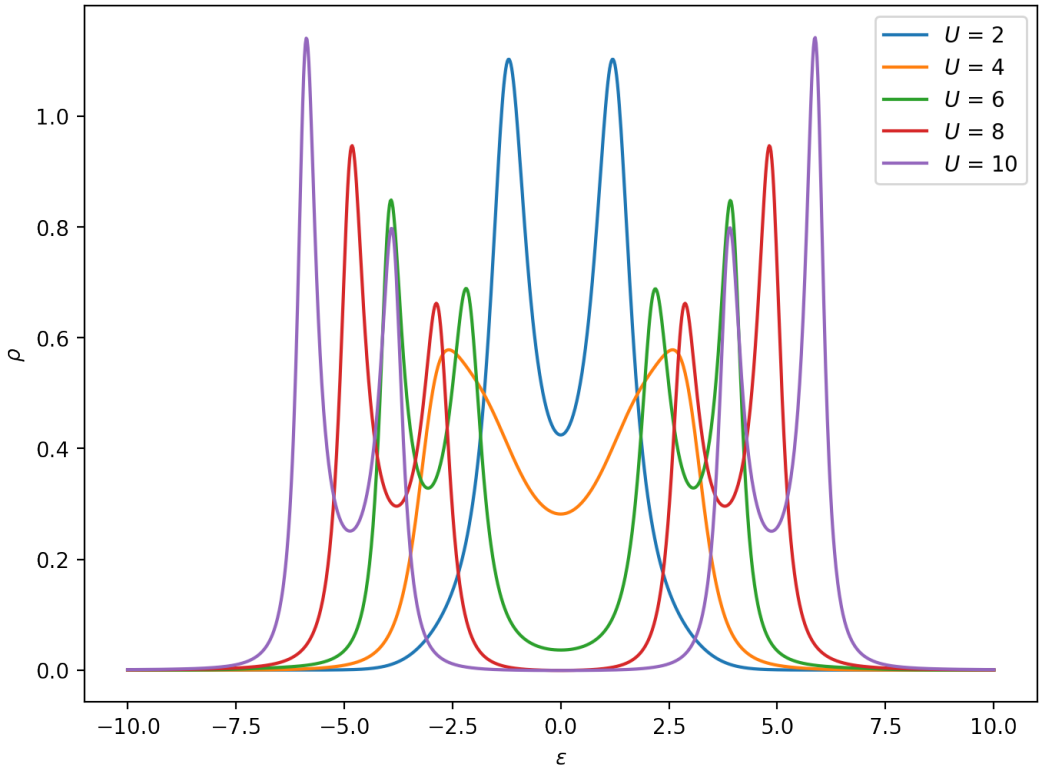
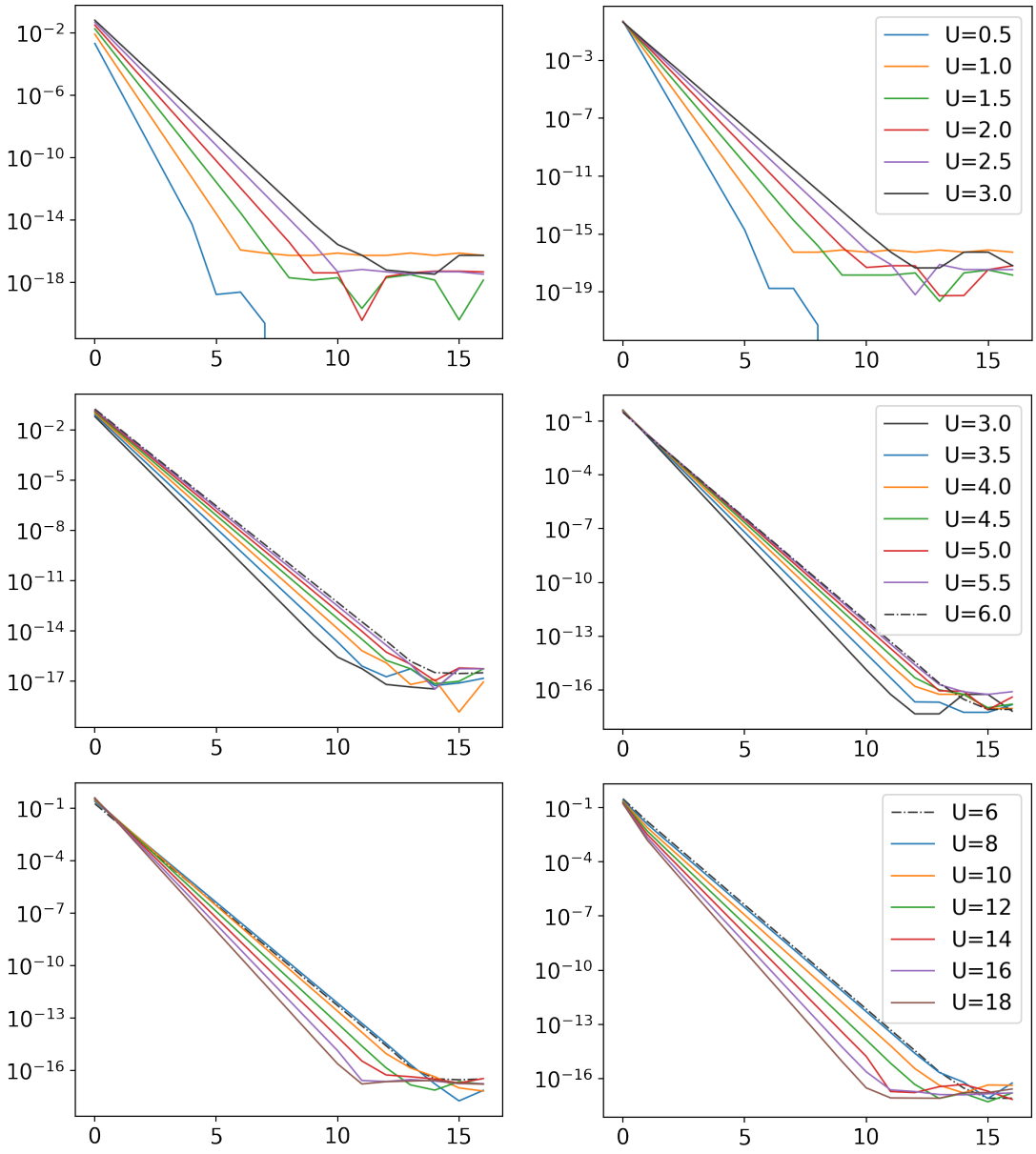


Figure IV.3: Density ρ of the spectral function A obtained by analytic continuation with Pade approximants, for different values of the on-site repulsion U. Other parameters are fixed as in (IV.5.2).

- As U increases, $\rho(0)$ decreases to reach approximately 0 between $U = 6$ and 8: for $U < 6$, the system is a conductor according to the conductivity criterion provided in the introduction of Section IV.5, and an insulator for $U > 8$.
- The density ρ is even, which is consistent with Corollary IV.4.1: as the initialization $\boldsymbol{\Delta}^{(0)} \in -i\mathbb{R}_+^{N_\omega+1}$, the iterates are purely imaginary as well: $\forall k \in \mathbb{N}, \boldsymbol{\Delta}^{(k)} \in -i\mathbb{R}_+^{N_\omega+1}$. Note that we observe that this property also holds true at convergence, for non-purely imaginary initializations.

Further simulations reported in Figure IV.5 indicate that, for some values of U (e.g. $U = 7$), `set_from_pade` returns negative values of $\rho(0)$. This observation holds true for larger values of N_ω, and is not satisfactory for an approximate solution to (IV.5.1). Such an issue may originate from the analytic continuation method we consider, but also from



(a) Initial guess $\Delta^{(0)} = (i\omega_n)^{-1}_{n\in[\![0,N_\omega]\!]}$ (b) Initial guess $\Delta^{(0)} = 0$

Figure IV.4: Linear convergence of the fixed-point algorithm. The residual $\|\Delta^{(n+1)} - \Delta^{(n)}\|_2$ is plotted in log scale, as a function of the iteration $n \in [\![0, N_{\text{iter}}]\!]$. Left and right sides differ in the initial guess $\Delta^{(0)}$.

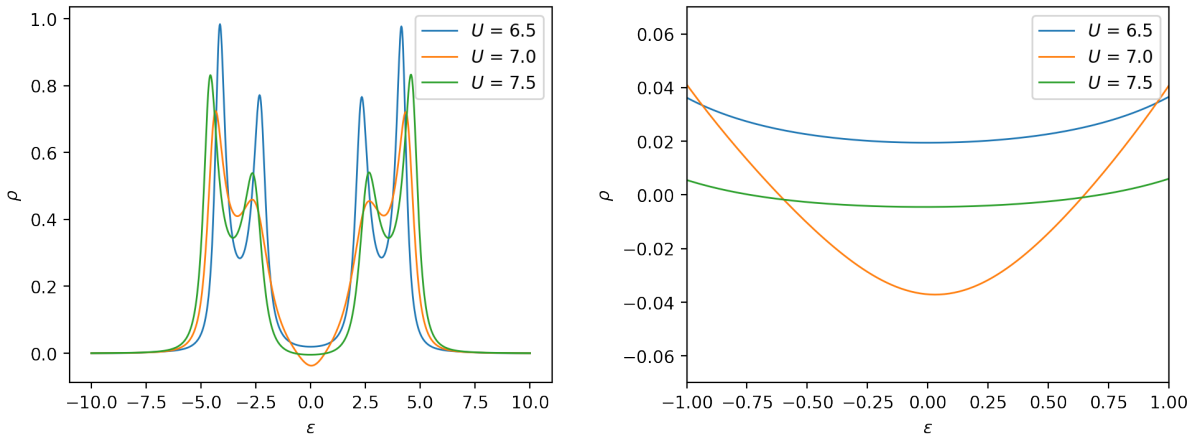


Figure IV.5: Density ρ of the spectral function A, for different values of the on-site repulsion U (zoom around 0 on the right). For $U = 7$ and $U = 7.5$, the density ρ takes negatives values around 0.

the proper analytic continuation problem defined by the discretized Green's function $\mathbf{G}$. In order to find the origin of this problem, we test numerically the existence of a solution to the Nevanlinna-Pick problem (IV.5.1) in the next section.

Remark IV.5.1 (Mott transition on more realistic systems)**.** *The conductivity criterion used here is to be taken with a pinch of salt, as the Hubbard dimer we consider is finite-dimensional and does not model the thermodynamic limit. However, this is the criterion adopted in [Geo+96] for the study of extended lattices, and we stick to it for the purpose of this numerical study, as the results are similar, even though the considered graphs largely differ. A more physical analysis of the conductivity properties would involve simulations for the bipartite cyclic graph C_{2n} of size $2n$ for increasing values of n. We do not repeat it in this article, as it is already done in [Geo+96] (recall that the exact solution is known for the infinite 1d Hubbard chain [Lie04]). The main purpose of this section is actually to introduce the reader to DMFT practice and analytic continuation issues in this simple setting, rather than providing a numerical analysis of DMFT on the 1d Hubbard chain.*

Remark IV.5.2 (Oscillations around the stationnary solution)**.** *In Figure IV.4, we indeed observe linear convergence for the first iterations. After this first phase, the algorithm reaches an almost stationary value, with oscillations that vary in amplitude from 10^{-16} to 10^{-19}. We suspect that these oscillations are due to rounding errors due to machine precision, but also to the tail-fitting procedure. As we shall see in the following, such an artifact is not present in the Julia implementation we developed.*

IV.5.2 Pick criterion on the converged solution

In this section, we are interested in the existence of a solution to the Nevanlinna-Pick interpolation problem set in (IV.5.1). Two criteria are available for determining whether

such a problem has a solution, and whether this solution is unique (see [CKP26, Appendix A] and references therein). In this article, we focus on the Pick criterion, which appears to be more robust that the Nevanlinna criterion with respect to small numerical errors.

As explained in details in [Nic16], the Nevanlinna-Pick interpolation problem (IV.5.1) admits a solution if and only if the Pick matrix $P \in \mathcal{S}_{N_\omega+1}(\mathbb{C})$ given by

$$P_{i,j} = \frac{1 - \overline{\mathcal{W}(-G_i)}\mathcal{W}(-G_j)}{1 - \overline{\mathcal{W}(i\omega_i)}\mathcal{W}(i\omega_j)}, \tag{IV.5.3}$$

where

$$\mathcal{W} : \begin{array}{ccc} \overline{\mathbb{C}_+} & \to & \mathbb{D} = \{z \in \mathbb{C}, |z| \leq 1\} \\ z & \mapsto & (z-i)/(z+i) \end{array} \quad \text{(Cayley transform)}, \tag{IV.5.4}$$

is positive semi-definite. The solution is unique if zero is an eigenvalue of this matrix, in which case the solution is rational with real poles [Don74a; CKP26]. Testing numerically the existence (and uniqueness) of a solution using the Pick criterion therefore amounts to diagonalize the Pick matrix and to check the sign of its lowest eigenvalue.

As briefly mentioned in the previous section, the `TRIQS` library has been developed to run simulations for a large number of Matsubara frequencies. We find in practice that, for $N_\omega \leq 50$, the library does not manage to perform the `Fourier` method, which is used to compute (IV.2.7). Moreover, the latter method uses tail-fitting, which makes the `TRIQS` implementation of the method slighty different from the one analyzed in Theorem IV.3.5. In addition, the `TRIQS` library is limited to double-precision computations. To make sure that numerical errors in the diagonalization do not jeopardize the quality of the results, we use arbitrary-precision finite-arithmetic computations.

For all these reasons, we have used for the rest of Section IV.5 our own implementation in `Julia` of the simple fixed-point algorithm described in (IV.2.7). We use the `BigFloat` package to handle arbitrary precision. In comparison with the `TRIQS` implementation, we do not compute numerically the integral defined in (IV.2.7): instead, we make use of the alternative expression given in (IV.2.15), and compute $F_{N_\omega}(\boldsymbol{\Delta})$ directly. Doing so, our implementation is as close as possible to the simple fixed-point algorithm studied in this article, at the price of greatly reduced performance in terms of computing times.

We proceed as follows:

1. **Solve the MaF-discretized IPT-DMFT equation**. The first step is the same as previously: we implement the simple iterative scheme described in Theorem IV.3.5, namely, for a given $N_{\text{iter}} \in \mathbb{N}^*$, we compute the sequence $\left(\boldsymbol{\Delta}^{(k)}\right)_{k \in [\![0, N_{\text{iter}}]\!]}$ by

$$\boldsymbol{\Delta}^{(0)} \in \mathfrak{D}_t^{N_\omega}, \quad \forall k \in \mathbb{N}, \quad \boldsymbol{\Delta}^{(k+1)} = F_{N_\omega}^{\text{DMFT}}(\boldsymbol{\Delta}^{(k)}). \tag{IV.5.5}$$

   This time, the number of iterations N_{iter} has to be chosen to ensure that for the considered set of parameters T, U, β, $\boldsymbol{\Delta}^{(N_{\text{iter}})} \approx \boldsymbol{\Delta}^{(\infty)}$ up to a precision that allows us to compute the sign of the lowest eigenvalue of the associated Pick matrix. Our Julia implementation is set to tackle this issue, that is, the precision is chosen so

that the numerical error due to the diagonalization of the Pick matrix does not prevent the sign of its smallest eigenvalue from being computed exactly.

2. **Compute the MaF-discretized IPT-DMFT local Green's function $\mathbf{G}$.** As in the previous section, we extract the associated self-energy $\mathbf{\Sigma}^{(\infty)} = U^2 F^{\mathrm{IPT}}_{\beta,N_\omega}(\mathbf{\Delta}^{(\infty)})$ and an approximation $\mathbf{G} \in -\overline{\mathbb{C}_+}^{N_\omega+1}$ of the discretization of the local Green's function $G_{1,1}$, defined by

$$\forall n \in [\![0, N_\omega]\!], \quad G_n = \left(i\omega_n - \Delta_n^{(\infty)} - \Sigma_n^{(\infty)}\right)^{-1}. \tag{IV.5.6}$$

3. **Check the Pick criterion.** Define the Pick matrices $P(\Delta), P(\Sigma), P(G)$ respectively associated to $-\mathbf{\Delta}, -\mathbf{\Sigma}$ and $-\mathbf{G}$, according to formula (IV.5.3). Compute the lowest eigenvalue $\min(\sigma(P(\cdot)))$ of each of these matrices. If it is negative, then there is no solution to the associated Nevanlinna-Pick interpolation problem.

We report the results of these simulations in Figure IV.6. The plots correspond to two different values of N_ω. We see that $N_\omega = 2$ is the lowest value such that there is a range of parameters (T, U) for which the Pick criterion is not satisfied. Note that, because of the structure of (IV.5.6), if the interpolation problems associated to $\mathbf{\Delta}$ and $\mathbf{\Sigma}$ have solutions, then so does the problem for $\mathbf{G}$, namely (IV.5.1). On the other hand, the converse is not true: as highlighted in Figure IV.6, it is possible to have a solution for $\mathbf{G}$ and $\mathbf{\Delta}$, but not for $\mathbf{\Sigma}$.

The plots cover the range or parameters $(T, U) \in [0, 10]^2$. The colormap corresponds to the value of $\min(\sigma(P(\cdot)))$, the lowest eigenvalue of the Pick matrix. For a range of parameters (T around 3 for $\mathbf{\Delta}$ and $\mathbf{G}$, T around 5 for $\mathbf{\Sigma}$, and U around 10), it is clear that the lowest eigenvalue of the Pick matrix is negative. The Pick matrix is not positive semidefinite in this case, hence there is no solution to the Nevanlinna-Pick interpolation problem.

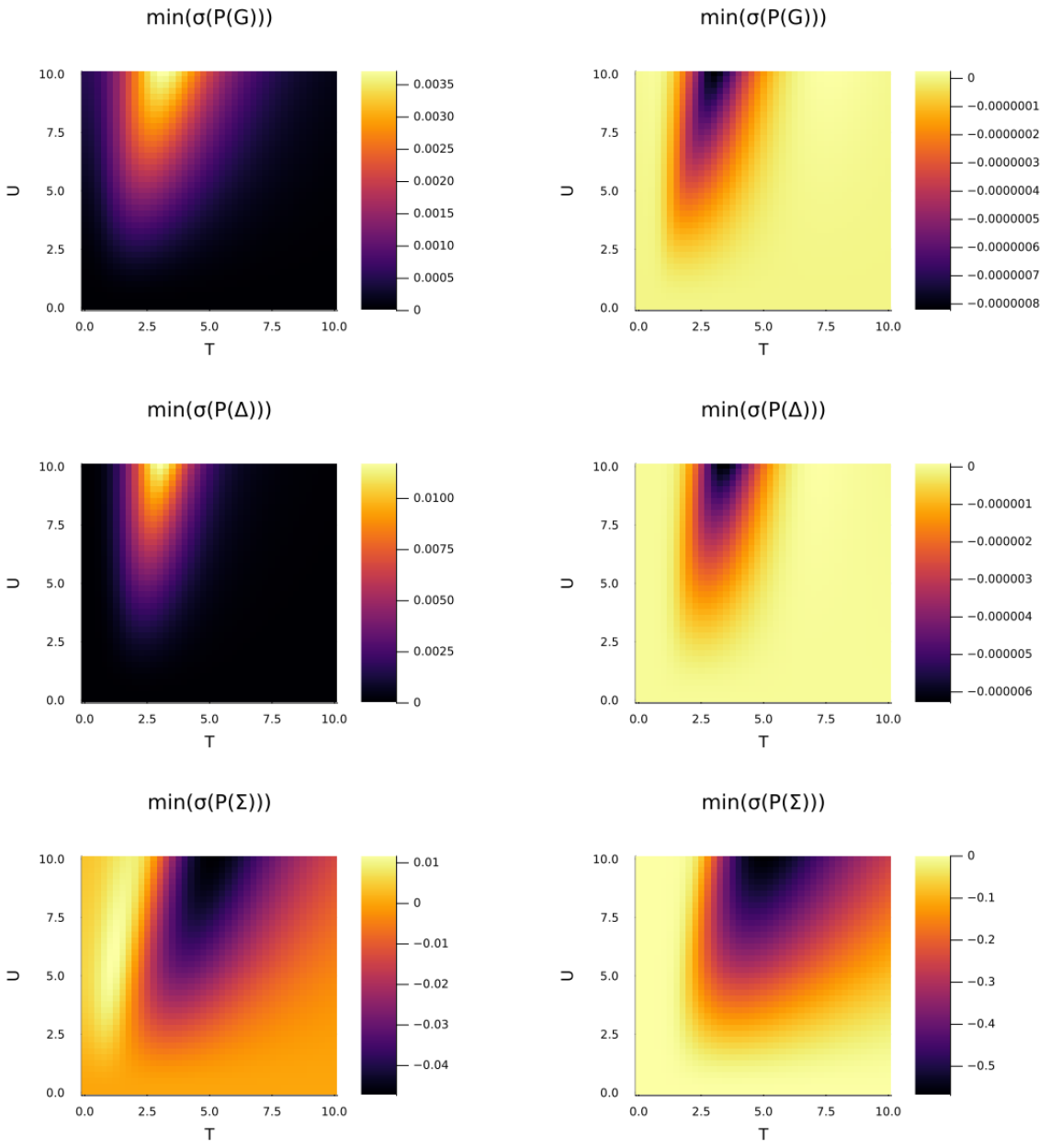
Remark IV.5.3. *Even if it is not clear on Figure IV.6, it is numerically observed that when N_ω increases, the range of parameters for which the Pick criterion is satisfied expands. We conjecture that for fixed (T, U), in the limit $N_\omega \to \infty$, the Pick criterion is satisfied for $\mathbf{\Delta}$ and $\mathbf{G}$ and that the solutions to the interpolation problem, when there is one, converge to the solution of the continuous problem studied in [CKP26].*

IV.5.3 Some insights on linear convergence

In this section, we provide some insights on the convergence of the fixed point algorithm presented in Theorem IV.3.5: we study the convergence of the algorithm given by

$$\mathbf{\Delta}^{(0)} \in \mathfrak{D}_t^{N_\omega}, \text{ and } \forall k \in \mathbb{N}, \quad \mathbf{\Delta}^{(k+1)} = F^{\mathrm{DMFT}}_{N_\omega}(\mathbf{\Delta}^{(k)}). \tag{IV.5.7}$$

More precisely, we focus on the case $N_\omega = 0$, which has been analyzed in Theorem IV.3.5, and consider purely imaginary iterations $\mathbf{\Delta}^{(k)} = -iy^{(k)}$, $y^{(k)} \in \mathbb{R}_+$ (recall that if $y(0) \in$



(a) $N_\omega = 2$ (b) $N_\omega = 5$

Figure IV.6: Test of the Pick criterion (lowest eigenvalue of the Pick matrix) of a solution to the IPT-DMFT equation, for the Nevanlinna-Pick criterion computed for $-\mathbf{G}$ (top), $-\mathbf{\Delta}$ (middle), $-\mathbf{\Sigma}$ (bottom), for two values of N_ω (left: $N_\omega = 2$, right: $N_\omega = 5$) at fixed $\beta = 1$. For a range of parameters (T, U), the solution to the MaF-discretized IPT-DMFT equations leads to analytic interpolation problems that have no solutions.

$\mathbb{R}_+$, then for all $k \leq 1, y^{(k)} \in \mathbb{R}_+$, according to Theorem IV.4.1). In this setting, the iterative algorithm can be rewritten as

$$y^{(0)} \in \mathbb{R}_+, \quad \forall k \in \mathbb{R}, y^{(k+1)} = f_{t,u}(y^{(k)}), \quad \text{with } f_{t,u} : y \mapsto t^2 \left(\pi + \frac{3u^2}{(\pi+y)^3}\right)^{-1}. \tag{IV.5.8}$$

Equation $f_{t,u}(y) = y$ corresponds to $P(y) = 0$ for a certain polynomial P of degree 4. Denoting by δ the discriminant of this polynomial, it can be checked that

- if $\delta < 0$, $f_{t,u}$ has a unique positive fixed point y_1. In this case, the algorithm converges linearly toward this solution, at the rate $f'_{t,u}(y_1)$;
- if $\delta > 0$, $f_{t,u}$ has three positive fixed points $y_1 < y_2 < y_3$. In this case, the algorithm also converges linearly:
  - toward y_1 if $y^{(0)} \in [0, y_2)$, at a rate given by $f'_{t,u}(y_1)$,
  - toward y_3 if $y^{(0)} \in (y_2; +\infty)$, at a rate given by $f'_{t,u}(y_3)$.

Let us compare these results with the ones in Theorem IV.3.5. The value of $s = tu$ must be small enough for the theorem to be valid (in which case $\delta < 0$). Under this assumption (recall that $t_{N_\omega} = +\infty$ for $N_\omega = 0$), the rate of convergence is bounded by a theoretical rate $r_{\text{th}} := 2\left(\frac{tu}{\pi}\right)^2 L_{N_\omega} = 2\left(3tu/\pi^3\right)^2$, with $N_\omega = 0$ (recall that $\deg(\mathcal{G}_H) = 2$ for the Hubbard dimer).

In Figure IV.7, we present the computed rate of convergence, together with the lower bound of this quantity given by Theorem IV.3.5. This is done for several values of the product $s = tu$. The convergence rate is computed by linear regression, and defined by $r = \lim_{k\to\infty} \frac{\|\Delta^{(k+1)} - \Delta^\infty\|_\infty}{\|\Delta^{(k)} - \Delta^\infty\|_\infty}$.

Note that, for all $y \in \mathbb{R}_+$,

$$f'_{t,u}(y) = \frac{9s^2}{\left(\pi + \frac{3u^2}{(\pi+y)^3}\right)^2 (\pi+y)^4}. \tag{IV.5.9}$$

There are clearly 3 distinct regimes for small enough values of $s = tu$, and the observed rate of convergence can be explained as follows:

- in the limit $u \to 0$, we see from (IV.5.8) (and Theorem IV.4.2) that $y_1 \sim \pi^{-1}s^2u^{-2}$, so that the convergence rate is $r = f'_{t,u}(y_1) \sim 9\pi^2u^8s^{-6}$. This accounts for the slope and offset between the lines for different values of s in the small u regime.
- in the limit $u \to \infty$, we can deduce still from (IV.5.8) that $y_1 \sim \pi^3 3^{-1}s^2u^{-4}$, so that the convergence rate is $r \sim \pi^2u^{-4}s^2$. This also accounts for the slope and the offset obtained for several values of s.
- for $s \ll 1$, we observe a third intermediate regime, in which the rate precisely corresponds to the bound provided by Theorem IV.3.5. Indeed, for $s/\sqrt{3} \ll u \ll$

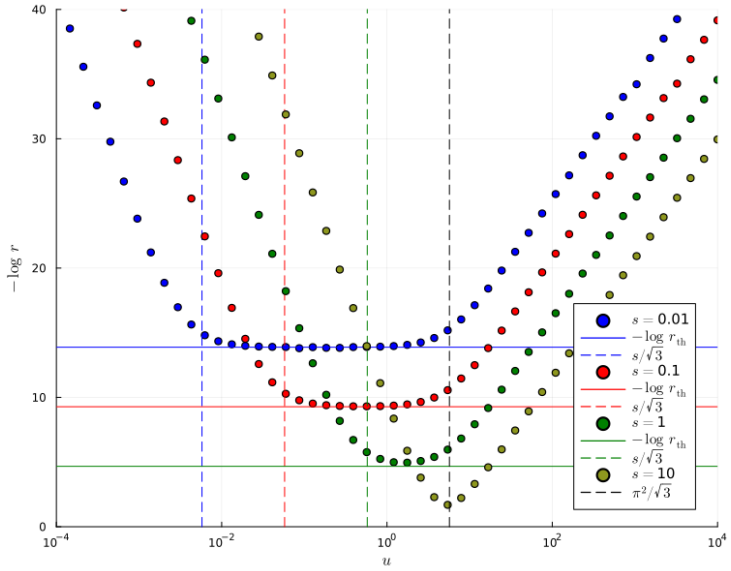


Figure IV.7: Convergence rate r of the numerical scheme depending on t and u for several fixed values of $s = tu$ and different values of u. The horizontal colored lines show the expected convergence rate r_{th} given by the bound of Theorem IV.3.5 for the value of s with corresponding color. The value $s = 10$ is too large, there is no theoritical convergence rate in this case. The vertical dashed lines show the boundaries of the intermediate regime for a fixed value of s: $s/\sqrt{3} < u < \pi^2/\sqrt{3}$.

$\pi^2/\sqrt{3}$, we still have $y \sim \pi^3 3^{-1} s^2 u^{-4} \ll \pi$, and therefore $3u^2/(\pi + y)^3 \ll \pi$, so that $r \simeq 9s^2\pi^{-6}$. This accounts for the observed plateaus, and the offset for the different values of s.

Note that the case $\pi^2/\sqrt{3} \ll u \ll s/\sqrt{3}$ is not studied and corresponds to the case not covered by the condition (IV.3.10) of Theorem IV.3.5.

In this specific example, we see that the rate of convergence estimated by Theorem IV.3.5 is valid for small enough values of s and for an intermediate range of values of u. Based on the two limit cases $u = 0$ and $t = 0$ presented in Section IV.3 where the iterative algorithm converges in one step (in other words the convergence rate is infinite), we expect a similar result to hold for general graphs and values of N_ω.

Acknowledgements

The authors thank Fabian Faulstich, Michel Ferrero, Antoine Georges, David Gontier, and Fatemeh Mohammadi for useful discussions.

IV.A Green's functions formalism and the IPT-DMFT equations

This appendix, greatly inspired from [CKP26, App. A], provides an introduction to the IPT-DMFT equations. In addition, we present the properties of the Hubbard model defined on a bipartite graph: such a model exhibits a so-called particle-hole symmetry, and we are able in this setting to prove that the Green's function also satisfies a symmetry condition. This implies that the values of the Green's function are anti-Hermitian on the Matsubara's frequencies, and we make use of this property in Section IV.4 where we investigate on the MaF-discretized IPT-DMFT equations defined on a bipartite graph.

IV.A.1 Derivation of the IPT-DMFT equations

To make this article as self-contained as possible, we present in this section a concise mathematical introduction to DMFT and the IPT approximation, and recall the derivation of Equations (IV.1.1)-(IV.1.2). Given $\mathcal{X}$ a Hilbert space, we denote by $\mathcal{L}(\mathcal{X})$ the space of bounded linear maps on $\mathcal{X}$, and by $\mathcal{S}(\mathcal{X})$ the space of self-adjoint operators on $\mathcal{X}$.

Second-quantization formalism. Consider a fermionic quantum system with one-particle state space $\mathcal{H}$ which we assume finite-dimensional. We denote respectively by

$$\mathcal{F} = \bigoplus_{n=0}^{\dim(\mathcal{H})} \bigwedge^n \mathcal{H}, \qquad \left(\widehat{a}_\phi^\dagger, \widehat{a}_\phi\right)_{\phi\in\mathcal{H}} \in \mathcal{L}(\mathcal{F}), \qquad \widehat{H} \in \mathcal{S}(\mathcal{F})$$

the Fock space, the creation and annihilation operators (parametrized by a one-particle state $\phi \in \mathcal{H}$), and the Hamiltonian of the system in second quantization formalism (i.e. defined on the whole Fock space). Recall that the creation and annihilation operators satisfy the Canonical Anti-commutation Relations (CAR):

$$\forall \phi, \phi' \in \mathcal{H}, \quad \{\widehat{a}_\phi^\dagger, \widehat{a}_{\phi'}^\dagger\} = 0, \quad \{\widehat{a}_\phi, \widehat{a}_{\phi'}\} = 0, \quad \{\widehat{a}_\phi, \widehat{a}_{\phi'}^\dagger\} = \langle \phi, \phi' \rangle_{\mathcal{H}}, \tag{IV.A.1}$$

and that the mapping $\phi \mapsto \widehat{a}_\phi^\dagger$ (resp. $\phi \mapsto \widehat{a}_\phi$) is linear (resp. antilinear).

In our setting, we restrict ourselves to *steady states*, i.e. states $\mathcal{L}(\mathcal{F}) \ni \mathcal{O} \mapsto \mathrm{Tr}(\widehat{\rho}\mathcal{O})$ with a density matrix $\widehat{\rho} \in \mathcal{S}(\mathcal{F})$ satisfying $\rho \geq 0$, $\mathrm{Tr}\,(\widehat{\rho}) = 1$ and $[\widehat{\rho}, \widehat{H}] = 0$ (in other words, $\widehat{\rho}$ is a stationary solution to the quantum Liouville equation). In practice, $\widehat{\rho}$ can be a projector on an eigenstate of $\widehat{H}$ (generally an N-particle ground state), a Gibbs state in the N-particle sector of the Fock space (canonical equilibrium state), or a Gibbs state for the modified Hamiltonian $\mathcal{K} = \widehat{H} - \varepsilon_F \widehat{N}$ (grand canonical equilibrium state), where ε_F is a chemical potential and $\widehat{N}$ the number operator. The last case is the setting for which we will define the quantity of interest in DMFT. Note that in all these cases, $\widehat{\rho}$ commutes

with $\widehat{N}$, an assumption we make from now on, together with that $\widehat{H}$ is particle-number conserving, i.e. $[\widehat{H}, \widehat{N}] = 0$.

One-body Green's function. The time-ordered one-body Green's function associated to a quantum system with one-particle state space $\mathcal{H}$ and a steady state $\widehat{\rho}$ of a Hamiltonian $\widehat{H}$ is the $\mathcal{L}(\mathcal{H})$-valued map $\widetilde{G} : \mathbb{R} \to \mathcal{L}(\mathcal{H})$ defined, for all time $t \in \mathbb{R}$, by

$$\forall \phi, \phi' \in \mathcal{H}, \quad \langle \phi, i\widetilde{G}(t)\phi' \rangle_{\mathcal{H}} := \mathrm{Tr}\left(\widehat{\rho}\, \mathcal{T}\left(\widehat{a}_{\phi}, \widehat{a}^{\dagger}_{\phi'} \right)(t, 0) \right), \tag{IV.A.2}$$

where $\mathcal{T}$ is the time-ordering operator and $\widehat{a}_{\phi'}(\cdot)^{\dagger}$, $\widehat{a}_{\phi}(\cdot)$ the creation/annihilation operators in the Heisenberg picture, that is

$$\langle \phi, i\widetilde{G}(t)\phi' \rangle_{\mathcal{H}} = \begin{vmatrix} \mathrm{Tr}\left(\widehat{\rho}\, \widehat{a}_{\phi}(t) \widehat{a}^{\dagger}_{\phi'} \right), & \text{if } t \geq 0, \\ -\mathrm{Tr}\left(\widehat{\rho}\, \widehat{a}^{\dagger}_{\phi'} \widehat{a}_{\phi}(t) \right), & \text{if } t < 0, \end{vmatrix} \quad \text{with} \quad \widehat{a}_{\phi}(t) = e^{it\widehat{H}} \widehat{a}_{\phi} e^{-it\widehat{H}}. \tag{IV.A.3}$$

In many applications (and in particular in DMFT), it is more relevant from a physical point of view to work in the frequency domain and focus instead on the time Fourier transform of $\widetilde{G}$. However, this object has singularities (it is in general an order-1 tempered distribution) and it is more convenient to consider the Generalized Fourier Transform $G : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H})$ of $\widetilde{G}$ in the upper half-plane. Indeed, it turns out that the function G has very interesting mathematical properties: it is the negative of an operator-valued Pick function [Nev19; Pic15], i.e. an analytic map on $\mathbb{C}_+$, satisfying $\Im(-G(z)) := -\frac{G(z)-G(z)^{\dagger}}{2i} \geq 0$ for all $z \in \mathbb{C}_+$. It readily follows from basic properties of analytic functions that for any non-constant Pick function f, $\Im(f(z)) > 0$ for all $z \in \mathbb{C}_+$. This implies that for all $z \in \mathbb{C}_+$, $\Im(-G(z))$ is in fact positive definite, so that $G(z)$ is invertible.

In addition, $G(z)$ vanishes when $\Im(z) \to +\infty$, so that it follows from the Nevanlinna-Riesz representation theorem [GT00] that

$$\forall z \in \mathbb{C}_+, \quad G(z) = \int_{\mathbb{R}} \frac{d\mathcal{A}(\varepsilon)}{z - \varepsilon},$$

where $\mathcal{A}$ is a Positive Operator-Valued Borel Measure (POVM) on $\mathbb{R}$ such that

$$\forall B \in \mathcal{B}(\mathbb{R}), \quad \mathcal{A}(B) \in \mathcal{S}(\mathcal{H}), \quad 0 \leq \mathcal{A}(B) \leq 1, \text{ and } \mathcal{A}(\mathbb{R}) = I_{\mathcal{H}},$$

where $\mathcal{B}(\mathbb{R})$ is the set of Borel subsets of $\mathbb{R}$. The POVM $\mathcal{A}$ is called the spectral function of the system. Another useful expression of G is its Källén-Lehmann representation, obtained by expanding the operators $\widehat{\rho}$ and $\widehat{H}$ in an orthonormal basis (Ψ_n^N) diagonalizing both of them (with respective eigenvalues ρ_n^N, E_n^N). We indeed have, for all $z \in \mathbb{C}_+$,

$$\langle \phi, G(z)\phi' \rangle_{\mathcal{H}} = \sum_{N=0}^{\dim(\mathcal{H})-1} \sum_{n=0}^{\binom{\dim(\mathcal{H})}{N}} \sum_{n'=0}^{\binom{\dim(\mathcal{H})}{N+1}} \frac{f_n^N + f_{n'}^{N+1}}{z - (E_{n'}^{N+1} - E_n^N)} \langle \Psi_n^N, \widehat{a}_{\phi} \Psi_{n'}^{N+1} \rangle_{\mathcal{F}} \langle \Psi_{n'}^{N+1}, \widehat{a}^{\dagger}_{\phi'} \Psi_n^N \rangle_{\mathcal{F}}.$$

The Källén-Lehmann representation reveals that G is a rational function of z, the poles of which are some of the excitation energies of the system. For example, if $\widehat{\rho}$ projects on a N-particle ground state, the poles of G are the excitation energies $E_n^{N+1} - E_0^N$ (one-particle affinities) and $E_0^N - E_n^{N-1}$ (one-particle ionization energies).

Beyond non-interacting particles: self-energy. In the special case of a non-interacting system, i.e. if the Hamiltonian is the second quantization of a one-body Hamiltonian $\widehat{H}^0 = d\Gamma(H^0)$ for some self-adjoint operator $H^0 \in \mathcal{S}(\mathcal{H})$, the Green's function and the spectral function are given by

$$G^0(z) = (z - H^0)^{-1}, \quad \mathcal{A}^0 = \Pi^{H^0},$$

where Π^{H^0} is the spectral projector of H^0. In particular, G^0 is independent of $\widehat{\rho}$ (in contrast with the interacting case) and is simply the restriction to $\mathbb{C}_+$ of the resolvent of H_0.

Let us now consider a more general Hamiltonian $\widehat{H} \in \mathcal{S}(\mathcal{F})$ of the form

$$\widehat{H} = d\Gamma(H^0) + \widehat{H}^{\mathrm{I}} \qquad \text{(non-interacting part + interacting part)},$$

and $G : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H})$ the Green's function of $\widehat{H}$ for some steady state $\widehat{\rho}$. The self-energy is the function $\Sigma : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H})$ defined by

$$\Sigma(z) := G^0(z)^{-1} - G(z)^{-1} \qquad \text{or, equivalently,} \qquad G(z) = \big(z - \big(H^0 + \Sigma(z)\big)\big)^{-1}.$$

Note that $-\Sigma$ is also an operator-valued Pick function. For later purposes, we introduce the functional g by

$$[g(\Sigma)](z) = (z - (H^0 + \Sigma(z)))^{-1} \in \mathcal{L}(\mathcal{H}),$$

which is well-defined on the set of negatives of operator-valued Pick functions from $\mathbb{C}_+$ to $\mathcal{L}(\mathcal{H})$ and such that $G = g(\Sigma)$.

Anderson Impurity Models. Anderson Impurity Models (AIM) are fermionic quantum many-body models with a specific structure. The one-particle space $\mathcal{H}_{\mathrm{AIM}}$ can be decomposed as

$$\mathcal{H}_{\mathrm{AIM}} = \mathcal{H}_{\mathrm{imp}} \oplus \mathcal{H}_{\mathrm{bath}},$$

where the interactions between particles are localized in the impurity component. In mathematical terms, the AIM Hamiltonian is of the form

$$\widehat{H}_{\mathrm{AIM}} = d\Gamma(H^0_{\mathrm{AIM}}) + \widehat{H}^{\mathrm{I}}_{\mathrm{imp}} \otimes 1_{\mathcal{F}_{\mathcal{H}_{\mathrm{imp}}}}, \qquad H^0_{\mathrm{AIM}} = \begin{pmatrix} H^0_{\mathrm{i}} & H^0_{\mathrm{ib}} \\ H^{0\,\dagger}_{\mathrm{ib}} & H^0_{\mathrm{b}} \end{pmatrix}$$

(recall the exponential law $\mathcal{F}_{\mathcal{H}_{\mathrm{AIM}}} = \mathcal{F}_{\mathcal{H}_{\mathrm{imp}}} \otimes \mathcal{F}_{\mathcal{H}_{\mathrm{bath}}}$). It can be shown that, for finite-dimensional bath [LL20; CKP26], the self-energy $\Sigma_{\beta,\mu}$ of the Gibbs state $\widehat{\rho} = Z^{-1} e^{-\beta(\widehat{H}_{\mathrm{AIM}} - \mu \widehat{N}_{\mathrm{frag}})}$ of such model has the form in the

$$\Sigma_{\beta,\mu}(z) = \begin{pmatrix} \Sigma_{\mathrm{i},\beta,\mu}(z) & 0 \\ 0 & 0 \end{pmatrix} \qquad \text{and} \qquad \Sigma_{\mathrm{i},\beta,\mu} = \Sigma^{\mathrm{AIM}}_{\mathrm{imp},\beta,\mu}\left(\mathcal{H}_{\mathrm{imp}}, H^0_{\mathrm{i}}, \Delta, \widehat{H}^{\mathrm{I}}_{\mathrm{imp}}\right),$$

where $\Delta : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H}_{\mathrm{imp}})$ is the *hybridization function*

$$\Delta(z) := H^0_{\mathrm{ib}}(z - H^0_{\mathrm{b}})^{-1} H^{0\,\dagger}_{\mathrm{ib}}.$$

Note that Δ is also the negative of an operator-valued Pick function. It is assumed in the physics literature that this sparsity pattern of the self-energy of AIMs also holds true for infinite-dimensional baths. For later purposes, we introduce the functional $g^{\mathrm{AIM}}_{\mathrm{imp}}$ defined by

$$[g^{\mathrm{AIM}}_{\mathrm{imp}}(\mathcal{H}_{\mathrm{imp}}, H^0_{\mathrm{i}}, \Delta, \Sigma_{\mathrm{i}})](z) := \big(z - \big(H^0_{\mathrm{i}} + \Delta(z) + \Sigma_{\mathrm{i}}(z)\big)\big)^{-1} = \left(z - \left(H^0_{\mathrm{AIM}} + \begin{pmatrix} \Sigma_{\mathrm{i}}(z) & 0 \\ 0 & 0 \end{pmatrix}\right)\right)^{-1}\Big|_{\mathcal{H}_{\mathrm{imp}}},$$

which is well-defined for $H^0_{\mathrm{i}} \in \mathcal{S}(\mathcal{H}_{\mathrm{imp}})$ and Σ_{i}, Δ negatives of operator-valued Pick functions from $\mathbb{C}_+$ to $\mathcal{L}(\mathcal{H}_{\mathrm{imp}})$. The impurity block of the exact Green's function of the AIM for the considered Gibbs state is given by

$$G_{\mathrm{AIM}}|_{\mathcal{H}\mathrm{imp}} = g^{\mathrm{AIM}}_{\mathrm{imp}}(\mathcal{H}_{\mathrm{imp}}, H^0_{\mathrm{i}}, \Delta, \Sigma_{\mathrm{i}}).$$

Dynamical Mean-Field Theory. DMFT is a quantum embedding method in which the system, say a Hubbard model defined on a finite graph for the simpler case, is decomposed in non-overlapping fragments. Each of these fragments are considered as an impurity, embedded in a bath accounting for the presence of the other fragments. In mathematical terms, the system state space $\mathcal{H}$ is decomposed as a direct sum of pairwise orthogonal impurity subspaces:

$$\mathcal{H} = \mathcal{H}^1_{\mathrm{imp}} \oplus \cdots \oplus \mathcal{H}^M_{\mathrm{imp}}$$

and we denote by

$$H^0 = \begin{pmatrix} H^0_j & H^0_{j\bar{j}} \\ H^{0\,\dagger}_{j\bar{j}} & H^0_{\bar{j}} \end{pmatrix}, \qquad \widehat{H}^{\mathrm{I}} = \widehat{H}^{\mathrm{I}}_j + \widehat{H}^{\mathrm{I}}_{j\bar{j}} + \widehat{H}^{\mathrm{I}}_{\bar{j}}$$

the block representation of H^0 according to the decomposition $\mathcal{H} = \mathcal{H}^j_{\mathrm{frag}} \oplus (\mathcal{H}^j_{\mathrm{frag}})^{\perp}$ and the corresponding decomposition of the interaction Hamiltonian $\widehat{H}^{\mathrm{I}}$.

As other quantum embedding methods, such as DMET, DMFT can be seen as a fixed-point problem on a set of low-level descriptors of the state of the whole system. In DMFT, this set can be chosen as a set of block-diagonal approximations of the global self-energy

$$\mathfrak{S}_{\mathrm{DMFT}} := \left\{ \Sigma = \begin{pmatrix} \Sigma_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \Sigma_M \end{pmatrix},\ \Sigma_j : \mathbb{C}_+ \to \mathcal{L}(\mathcal{H}^j_{\mathrm{imp}}), -\Sigma_j \text{ op.-valued Pick function} \right\}.$$

The fixed-point iteration goes in two steps:

1. AIM construction: given $\Sigma^{\mathrm{in}} \in \mathfrak{S}_{\mathrm{DMFT}}$, for each j, construct an AIM such that

$$g\left(\Sigma^{\mathrm{in}}\right)\big|_{\mathcal{H}^j_{\mathrm{imp}}} = g^{\mathrm{AIM}}_{\mathrm{imp}}\left(\mathcal{H}^j_{\mathrm{imp}}, H^0_j, \Delta^{\mathrm{out}}_j, \Sigma^{\mathrm{in}}_j\right).$$

   This does not uniquely define the AIM, but this does uniquely define Δ^{out}_j:

$$\Delta^{\mathrm{out}}_j(z) = H^0_{j\bar{j}}\left(z - \left(H^0_{\bar{j}} + \bigoplus_{k\neq j}\Sigma^{\mathrm{in}}_k(z)\right)\right)^{-1} H^{0\,\dagger}_{j\bar{j}};$$

   In physical terms, the hybridization function Δ_j encapsulates the effects of the environment of the j-th cluster and allows one to consider the latter as an impurity embedded in a bath.

2. Impurity solver: for each of these AIMs, compute an updated $\Sigma^{\mathrm{out}} \in \mathfrak{S}_{\mathrm{DMFT}}$ by

$$\Sigma^{\mathrm{out}}_j := \Sigma^{\mathrm{AIM}}_{\mathrm{imp},\beta,\mu}\left(\mathcal{H}^j_{\mathrm{imp}}, H^0_j, \Delta^{\mathrm{out}}_j, \widehat{H}^{\mathrm{I}}_j\right).$$

   In physical terms, for each cluster, compute the self-energy of the associated AIM, and concatenate the so-obtained solutions in the updated low-level descriptor Σ^{out}.

We now restrict ourselves to DMFT within the translation-invariant, single-site, and paramagnetic setting at half-filling applied to a Hubbard model defined on a vertex-transitive graph with $L < \infty$ vertices. In this case

$$M = L, \quad \mathcal{H}^1_{\mathrm{imp}} \cong \cdots \cong \mathcal{H}^L_{\mathrm{imp}} \cong \mathbb{C}^2, \quad \Delta_1 = \cdots = \Delta_L = \Delta\mathbb{1}_2, \quad \Sigma_1 = \cdots = \Sigma_L = \Sigma\mathbb{1}_2. \tag{IV.A.4}$$

In addition, we assume that the AIMs are solved by Iterative Perturbation Theory:

$$[\Sigma^{\mathrm{AIM-IPT}}_{\mathrm{imp},\beta}(\Delta, U)](z) = U^2[F^{\mathrm{IPT}}_\beta(\Delta)](z)\mathbb{1}_2, \tag{IV.A.5}$$

where F^{IPT}_β is derived by perturbation theory at second-order in the on-site interaction parameter U. As mentioned in Section IV.2, $F^{\mathrm{IPT}}_\beta(\Delta)$ can be conveniently characterized by its values on the imaginary Matsubara frequencies.

IV.A.2 Particle-hole transform and Green's function symmetry

The Hubbard model defined on a bipartite graph $\mathcal{G}_H$ inherits a specific *particle-hole symmetry*, as detailed in [Lie04]. Let $\Lambda = \Lambda_A \sqcup \Lambda_B$ be a partition of the vertices such that

$$\{i,j\} \in E \implies i \in \Lambda_A \text{ and } j \in \Lambda_B \text{ (or the converse)},$$

and introduce the corresponding particle-hole transform $\mathcal{B}_{A,B}$ which is the Bogoliubov transform from $\mathcal{L}(\mathcal{F})$ to itself, defined by

$$\mathcal{B}_{A,B}(\widehat{a}_{i,\sigma}) = \begin{cases} \widehat{a}^\dagger_{i,\sigma} & \text{if } i \in A, \\ -\widehat{a}^\dagger_{i,\sigma} & \text{if } i \in B. \end{cases} \tag{IV.A.6}$$

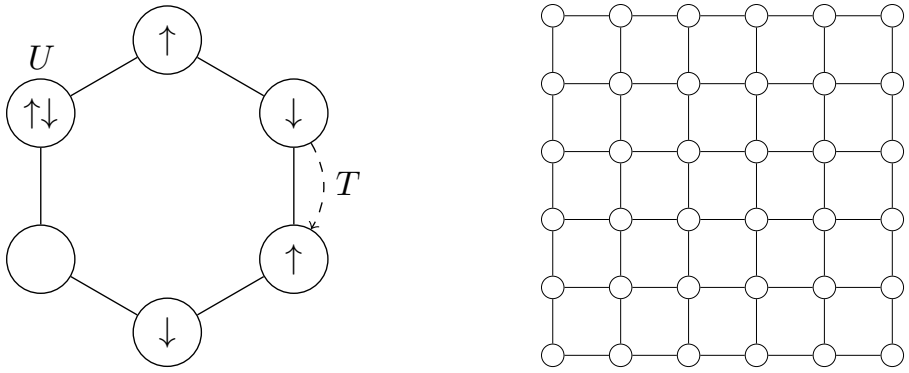


Figure IV.8: Left: Pariser–Parr–Pople model of benzene C_6H_6. Right: supercell model $(\mathbb{Z}/6\mathbb{Z})^2$.

One can check that the Hubbard Hamiltonian $\hat{H}_H$ (see [CKP26, Definition 2.5]) is modified under this transformation as

$$\mathcal{B}_{A,B}(\hat{H}_H) = \hat{H}_H + U(|\Lambda| - \hat{N}), \tag{IV.A.7}$$

so that the grand canonical Hubbard Hamiltonian $\hat{H}'_H = \hat{H}_H - \varepsilon_F \hat{N}$ is invariant under this transform for the half-filled case $\varepsilon_F = U/2$, with ε_F denoting the chemical potential. We stick to this case when using the IPT approximation, as discussed more in details in [CKP26, Section 2.5.2].

As a consequence, the Gibbs state is also invariant under this transformation: $\mathcal{B}_{A,B}(e^{-\beta \hat{H}'_H}) = e^{-\beta \hat{H}'_H}$. From this, one proves that the time-ordered one-body Green's function $\widetilde{G}$ associated with the Gibbs state $Z^{-1}e^{-\beta \hat{H}'_H}$ is anti-Hermitian $\widetilde{G} = -\widetilde{G}^\dagger$: indeed, given $t \geq 0$ and for all one-particle basis vector $\phi = \phi_{i,\sigma}$, with $i \in \Lambda, \sigma = \uparrow, \downarrow,$, we have

$$\begin{aligned} Z\langle \phi, i\widetilde{G}(t)\phi\rangle &= \mathrm{Tr}\left(e^{-\beta \hat{H}'_H} e^{it\hat{H}'_H} \widehat{a}_\phi e^{-it\hat{H}'_H} \widehat{a}^\dagger_\phi\right) = \overline{\mathrm{Tr}\left(\mathcal{B}_{A,B}(\widehat{a}^\dagger_\phi e^{-it\hat{H}'_H} \widehat{a}_\phi e^{it\hat{H}'_H} e^{-\beta \hat{H}'_H})\right)} \\ &= \overline{\mathrm{Tr}\left(e^{-\beta \hat{H}'_H} e^{it\hat{H}'_H} \widehat{a}_\phi e^{-it\hat{H}'_H} \widehat{a}^\dagger_\phi\right)} = Z\overline{\langle \phi, i\widetilde{G}\phi\rangle} = -Z\langle \phi, i\widetilde{G}^\dagger \phi\rangle, \end{aligned}$$

where we make use of the cyclicity of the trace and (IV.A.7). The result for $t < 0$ can be proved in a similar way.

Therefore, its Generalized Fourier transform G satisfies the following property: for all $z \in \mathbb{C}_+$,

$$G(z) = -G(-\bar{z})^\dagger. \tag{IV.A.8}$$

From this, one infers that the Nevanlinna-Riesz measure of G (so to say the spectral function, see [CKP26, Section 3]) is *symmetric*, or equivalently, that the Green's function G is anti-Hermitian on the imaginary axis.

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Thèse de doctorat soutenue le 30 septembre 2026 par

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Étude mathématique de modèles de champ moyen pour les matériaux moirés

Résumé de la thèse

Ce manuscrit traite de l'étude des matériaux moirés en physique quantique, d'un point de vue mathématique. Il considère deux approches, chacune associée à une méthode. La première consiste à modéliser un système physique expérimental, comme une bicouche de graphène 2D encapsulée entre deux électrodes, grâce à ses spécificités géométriques. Ceci fait apparaître les notion d'ergodicité et d'opérateur stationnaire, et permet de formuler un modèle de Théorie de la Fonctionnelle de la Densité, pour lequel ce travail démontre qu'il existe un état fondamental associé à la minimisation de la fonctionnelle d'énergie, pour le modèle de Hartree-Fock réduit.
La seconde approche consiste à changer d'échelle et à considérer une méthode qui s'appliquerait à l'échelle du moiré : la Théorie du Champ Moyen Dynamique. Cette méthode permet de calculer une approximation de la fonction de Green à un corps d'un système tel qu'un modèle de Hubbard sur le réseau moiré et donc d'investiguer les corrélations électroniques du système. Cette thèse apporte une formalisation mathématique de cette théorie lorsqu'elle est couplée à un solveur d'impureté particulier (la Théorie Perturbative Itérative), une démonstration de l'existence d'une solution aux équations qui en découlent, ainsi que l'analyse numérique de la méthode, en pointant ses limites.